

claude-windows / 068560d3-1ca2-4387-981a-d2d6bca55414

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414.jsonl`  
- SHA-256: `9901d68cef2a235d0383046b56537c99f5bc6ad556b7886ad525dd406902e317`  
- Source modified: `2026-06-22T09:03:48+00:00`  
- Imported at: `2026-07-05T16:48:17+00:00`  
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`  
- session_id: `068560d3-1ca2-4387-981a-d2d6bca55414`
```

Transcript

1. user (2026-06-15T16:58:29.374Z)

Four pre-existing tests fail on origin/master (unrelated to any current work) because PR #177 added a WSL readability preflight (`wsl\_source\_unreadable\_reason`) to `WasbRunner.\_stage\_video` in `backend/pipeline/detectors/wasb\_runner.py`. That preflight issues an extra `\_wsl\_bash` call running `test -r ... && echo READABLE` BEFORE the `cp`. These tests mock `\_wsl\_bash` to return a single CompletedProcess without "READABLE" in stdout, so the preflight now reports the source unreadable and `\_stage\_video` returns None, failing the test.

The sibling file `tests/test\_tracknet\_runner.py` was already updated for this (see `test\_stage\_video\_success` / `test\_stage\_video\_failure` there: they use `side\_effect=[\_proc(0, stdout="READABLE"), \_proc(0, stdout="OK")]`). The following WASB tests were NOT updated and need the same treatment (make the mock return "READABLE" for the preflight call, or patch `wsl\_source\_unreadable\_reason`/`\_stage\_video` directly):

1. `tests/test\_wasb\_runner.py::test\_stage\_video\_success\_and\_failure`
2. `tests/test\_wsl\_tilde\_quoting.py::test\_wasb\_stage\_video\_cmd\_has\_no\_quoted\_tilde`
3. `tests/test\_wsl\_tilde\_quoting.py::test\_wasb\_run\_predict\_resolves\_relative\_out\_dir`
4. `tests/test\_wsl\_tilde\_quoting.py::test\_wasb\_run\_predict\_inference\_cmd\_has\_no\_quoted\_tilde`

Reproduce: `python -m pytest tests/test\_wasb\_runner.py tests/test\_wsl\_tilde\_quoting.py -q` (4 failures, all with the "video source is not readable from WSL" ERROR log). Fix the mocks so each test's `fake\_wsl`/_wsl_bash returns stdout containing "READABLE" for the `test -r` preflight call (the call whose command contains `test -r`), then the cp/inference calls as before. Verify the whole suite is green afterwards and run `ruff check` on the changed test files. Branch from origin/master explicitly, one focused PR, do not merge.

2. assistant (2026-06-15T16:58:36.477Z)

I'll start by following the repo-freshness and branching-safety conventions. First, let me understand the current state ? note that the two test files I need to fix already show as modified in the working tree, so I must inspect that before touching anything.

3. user (2026-06-15T16:58:39.174Z)

On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
 (use "git add <file>..." to update what will be committed)
 (use "git restore <file>..." to discard changes in working directory)
 modified: tests/test_wasb_runner.py
 modified: tests/test_wsl_tilde_quoting.py

Untracked files:
 (use "git add <file>..." to include in what will be committed)
 artifacts/

```
no changes added to commit (use "git add" and/or "git commit -a")
=== current branch ===
master
=== HEAD ===
46ad601 fix(detectors): fail fast with an actionable error when a video source is unreadable
from WSL (#177)
```

4. assistant (2026-06-15T16:58:43.138Z)

The two target test files already have uncommitted changes. Per the global working-agreement, I must inspect these before touching anything ? they could be a sibling session's or the owner's WIP. Let me look at the diff.

5. user (2026-06-15T16:58:43.983Z)

```
diff --git a/tests/test_wasb_runner.py b/tests/test_wasb_runner.py
index 8e32336..a43aaab 100644
--- a/tests/test_wasb_runner.py
+++ b/tests/test_wasb_runner.py
@@ -38,9 +38,13 @@ def test_healthcheck_no_wsl_binary():

def test_stage_video_success_and_failure():
    r = WasbRunner(WasbConfig())
-   with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout="OK")):
+   # Two WSL calls now: the readability preflight (echo READABLE) then the cp (echo OK).
+   with patch.object(r, "_wsl_bash",
+       side_effect=[_proc(0, stdout="READABLE"), _proc(0, stdout="OK")]):
        assert r._stage_video(r"C:\v\clip.mp4", "clip.mp4") == "~/clips/clip.mp4"
-   with patch.object(r, "_wsl_bash", return_value=_proc(1, stderr="No such file")):
+   # Source readable, but the cp itself fails -> None.
+   with patch.object(r, "_wsl_bash",
+       side_effect=[_proc(0, stdout="READABLE"), _proc(1, stderr="disk full")]):
        assert r._stage_video(r"C:\v\clip.mp4", "clip.mp4") is None

diff --git a/tests/test_wsl_tilde_quoting.py b/tests/test_wsl_tilde_quoting.py
index ceb08dc..c98521f 100644
--- a/tests/test_wsl_tilde_quoting.py
+++ b/tests/test_wsl_tilde_quoting.py
@@ -44,14 +44,18 @@ def test_wasb_stage_video_cmd_has_no_quoted_tilde():

def fake_wsl(cmd):
    captured.append(cmd)
-   return _proc()
+   return _proc(stdout="READABLE") if "test -r" in cmd else _proc()

with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
    dest = r._stage_video("C:\\vids\\clip.mp4", "clip__abc.mp4")
    assert dest == "~/clips/clip__abc.mp4"
-   assert len(captured) == 1
-   assert _no_quoted_tilde(captured[0]), captured[0]
-   assert "mkdir -p ~/clips" in captured[0]
+   # captured[0] = readability preflight (test -r), captured[1] = the staging cp.
+   stage_cmd = next(c for c in captured if c.startswith("mkdir -p"))
+   assert _no_quoted_tilde(stage_cmd), stage_cmd
+   assert "mkdir -p ~/clips" in stage_cmd

def test_tracknet_stage_video_cmd_has_no_quoted_tilde():
@@ -60,11 +61,12 @@ def test_tracknet_stage_video_cmd_has_no_quoted_tilde():

def fake_wsl(cmd):
    captured.append(cmd)
```

```

-         return _proc()
+         return _proc(stdout="READABLE") if "test -r" in cmd else _proc()

    with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
        r._stage_video("C:\\vids\\clip.mp4", "clip_abc.mp4")
-    assert captured and _no_quoted_tilde(captured[0]), captured
+    stage_cmd = next(c for c in captured if c.startswith("mkdir -p"))
+    assert _no_quoted_tilde(stage_cmd), stage_cmd

def test_wasb_run_predict_resolves_relative_out_dir(tmp_path, monkeypatch):
    """The hybrids pass a RELATIVE 'output/wasb'. to_wsl_mnt_path passes relative
@@ -78,6 +80,8 @@ def test_wasb_run_predict_resolves_relative_out_dir(tmp_path, monkeypatch):

    def fake_wsl(cmd):
        cmds.append(cmd)
+    if "test -r" in cmd:
+        return _proc(stdout="READABLE")
        if "python wasb_infer.py" in cmd:
            (tmp_path / "out" /
"clip_abcdefabcdef_wasb.csv").write_text("Frame,Visibility,X,Y\n")
            return _proc()
@@ -100,6 +104,8 @@ def test_wasb_run_predict_inference_cmd_has_no_quoted_tilde(tmp_path):

    def fake_wsl(cmd):
        cmds.append(cmd)
+    if "test -r" in cmd:
+        return _proc(stdout="READABLE")
        if "wasb_infer.py" in cmd:
            # simulate the run producing the expected CSV
            (tmp_path / "clip_abcdefabcdef_wasb.csv").write_text("Frame,Visibility,X,Y\n")

```

6. assistant (2026-06-15T16:59:02.667Z)

The working tree already contains a fix that matches exactly what the task describes ? this is likely WIP from a prior session (mine or a sibling). Per the working-agreement, I will ****not clobber**** it; I'll verify it's correct and complete, then turn it into the focused PR.

Let me first confirm my understanding of the source preflight, then verify the tests actually pass with these changes (and fail without).

7. user (2026-06-15T16:59:05.183Z)

From <https://github.com/avidullu/badminton-highlight-indexer>
* branch master -> FETCH_HEAD
Already up to date.

8. user (2026-06-15T16:59:05.293Z)

```

81-
82-     def _stage_video(self, video_win_path: str, dest_name: str) -> Optional[str]:
83-         src_mnt = to_wsl_mnt_path(video_win_path)
84-         reason = self.wsl_source_unreadable_reason(src_mnt)
85-         if reason:
86-             logger.error("Cannot stage video into WSL: %s", reason)
87-             return None

```

9. user (2026-06-15T16:59:12.165Z)

Found 3 files
backend\pipeline\detectors\base.py
backend\pipeline\detectors\tracknet_runner.py
backend\pipeline\detectors\wasb_runner.py

10. user (2026-06-15T16:59:16.522Z)

```

backend\pipeline\detectors\base.py:140: def wsl_source_unreadable_reason(self, src_mnt: str)
-> Optional[str]:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-141-
"""Actionable message if ``src_mnt`` is NOT readable from inside WSL, else None.
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-142-
backend\pipeline\detectors\base.py-143-         Google Drive (``G:``/Drive File Stream), UNC and
many network/virtual
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-144-
filesystems are invisible to WSL's ``/mnt`` ? staging a video from such a path
backend\pipeline\detectors\base.py-145-         fails with a cryptic ``cp: cannot stat`` and the
whole run silently produces no
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-146-
trajectory (i.e. "0 rallies", a wasted run that looks like a quality result).
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-147-
A cheap ``test -r`` preflight turns that into one clear, fixable error BEFORE any
backend\pipeline\detectors\base.py-148-         frame extraction. Best-effort: a flaky/timed-out
probe returns None so the real
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-149-
``cp`` still gets its chance (we never block a genuinely-readable source).
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-150-
"""
backend\pipeline\detectors\base.py-151-         try:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-152-
res = self._wsl_bash(f"test -r {shlex.quote(src_mnt)} && echo READABLE")
backend\pipeline\detectors\base.py-153-         except (subprocess.TimeoutExpired, OSError):
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-154-
return None
backend\pipeline\detectors\base.py-155-         if "READABLE" in (res.stdout or ""):
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-156-
return None
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-157-
return (
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-158-
f"video source is not readable from WSL at '{src_mnt}'. Google Drive, UNC, "
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-159-
f"and other network/virtual filesystems are invisible to WSL's /mnt ? copy "
backend\pipeline\detectors\base.py-160-         f"the video to a local disk (e.g. C:\\ or the
WSL filesystem) and retry."
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-161-
)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-162-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-163-
backend\pipeline\detectors\base.py-164- class DetectorRunner(ABC):
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py-165-
"""Common interface for shuttle/ball trajectory detectors (WSL or otherwise)."""

```

11. assistant (2026-06-15T16:59:21.989Z)

My understanding is confirmed: the preflight calls `\_wsl\_bash("test -r ... && echo READABLE")` and looks for "READABLE" in stdout. The working-tree fix handles both the `side\_effect` list cases and the `fake\_wsl` function cases correctly. Let me read the full test files (with changes applied) and the sibling tracknet reference to confirm completeness.

12. user (2026-06-15T16:59:22.956Z)

```

1      """WasbRunner WSL-glue coverage (the LIVE detector path; previously only an ABC-
conformance
2      test existed while the parallel TrackNetRunner had ~21). WSL is mocked."""
3      import types
4      from unittest.mock import patch
5
6      from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
7      from backend.pipeline.detectors.base import DetectorRunner
8
9

```

```

10     def _proc(rc=0, stdout="", stderr=""):
11         return types.SimpleNamespace(returncode=rc, stdout=stdout, stderr=stderr)
12
13
14     def test_is_a_detector_runner():
15         assert isinstance(WasbRunner(WasbConfig()), DetectorRunner)
16
17
18     def test_healthcheck_ok():
19         r = WasbRunner(WasbConfig())
20         with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout="torch 2.6 cuda
True")):
21             ok, msg = r.healthcheck()
22             assert ok and "torch" in msg
23
24
25     def test_healthcheck_env_failure():
26         r = WasbRunner(WasbConfig())
27         with patch.object(r, "_wsl_bash", return_value=_proc(1, stderr="ModuleNotFoundError:
torch")):
28             ok, msg = r.healthcheck()
29             assert not ok and "failed" in msg.lower()
30
31
32     def test_healthcheck_no_wsl_binary():
33         r = WasbRunner(WasbConfig())
34         with patch.object(r, "_wsl_bash", side_effect=FileNotFoundError()):
35             ok, msg = r.healthcheck()
36             assert not ok and "wsl" in msg.lower()
37
38
39     def test_stage_video_success_and_failure():
40         r = WasbRunner(WasbConfig())
41         # Two WSL calls now: the readability preflight (echo READABLE) then the cp (echo
OK).
42         with patch.object(r, "_wsl_bash",
43                             side_effect=[_proc(0, stdout="READABLE"), _proc(0, stdout="OK")]):
44             assert r._stage_video(r"C:\v\clip.mp4", "clip.mp4") == "~/clips/clip.mp4"
45             # Source readable, but the cp itself fails -> None.
46             with patch.object(r, "_wsl_bash",
47                             side_effect=[_proc(0, stdout="READABLE"), _proc(1, stderr="disk
full")]):
48                 assert r._stage_video(r"C:\v\clip.mp4", "clip.mp4") is None
49
50
51     def test_run_predict_returns_none_when_infer_script_stage_fails(tmp_path):
52         r = WasbRunner(WasbConfig())
53         with patch.object(r, "_stage_infer_script", return_value=False):
54             assert r.run_predict(r"C:\v\clip.mp4", str(tmp_path), video_id="abc123abc123")
is None
55
56
57     # --- cache hygiene (keep_frames / min_free_gb / shared WSL helpers) -----
58
59     def test_from_indexing_cfg_reads_cleanup_knobs():
60         cfg = WasbConfig.from_indexing_cfg({"wasb": {"keep_frames": True, "min_free_gb":
25}})
61         assert cfg.keep_frames is True and cfg.min_free_gb == 25.0
62         default = WasbConfig.from_indexing_cfg({})
63         assert default.keep_frames is False and default.min_free_gb == 0.0
64
65
66     def test_wsl_rm_rf_builds_rm_command():
67         r = WasbRunner(WasbConfig())
68         with patch.object(r, "_wsl_bash", return_value=_proc(0)) as wb:

```

```

69         r._wsl_rm_rf(["~/clips/a_frames", "~/clips/a.mp4"])
70         inner = wb.call_args[0][0]
71         assert inner.startswith("rm -rf ") and "a_frames" in inner and "a.mp4" in inner
72
73
74     def test_wsl_rm_rf_noop_on_empty():
75         r = WasbRunner(WasbConfig())
76         with patch.object(r, "_wsl_bash") as wb:
77             r._wsl_rm_rf([])
78             wb.assert_not_called()
79
80
81     def test_wsl_free_gb_parses_bytes_else_none():
82         r = WasbRunner(WasbConfig())
83         with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout=str(50 * 1024 ** 3) +
84 "\n")):
85             assert abs(r._wsl_free_gb("~/clips") - 50.0) < 0.01
86             with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout="garbage")):
87                 assert r._wsl_free_gb("~/clips") is None
88
89     def _run_predict_to_success(cfg, tmp_path, vid="abc123abc123"):
90         """Drive run_predict to its success branch with WSL fully mocked. Returns (rm_spy,
91 key)."""
92         r = WasbRunner(cfg)
93         key = f"clip_{vid[:12]}"
94         (tmp_path / f"{key}_wasb.csv").write_text("Frame,Visibility,X,Y\n") # success = CSV
95         exists
96         with patch.object(r, "_stage_infer_script", return_value=True), \
97             patch.object(r, "_stage_video", return_value=f "~/clips/{key}.mp4"), \
98             patch.object(r, "_wsl_bash", return_value=_proc(0)), \
99             patch.object(r, "_wsl_rm_rf") as rm:
100             out = r.run_predict(r"C:\v\clip.mp4", str(tmp_path), video_id=vid)
101             assert out is not None
102             return rm, key
103
104     def test_run_predict_cleans_frames_and_staged_video_on_success(tmp_path):
105         rm, key = _run_predict_to_success(WasbConfig(), tmp_path)
106         rm.assert_called_once()
107         purged = rm.call_args[0][0]
108         assert f "~/clips/{key}_frames" in purged and f "~/clips/{key}.mp4" in purged
109
110     def test_run_predict_keeps_frames_when_keep_frames_true(tmp_path):
111         rm, _ = _run_predict_to_success(WasbConfig(keep_frames=True), tmp_path)
112         rm.assert_not_called()
113

```

13. assistant (2026-06-15T16:59:26.027Z)

Now the tilde-quoting test file and the sibling tracknet reference:

14. user (2026-06-15T16:59:26.856Z)

```

1     """WSL tilde-quoting regression tests.
2
3     shlex.quote('~/clips/x.mp4') single-quotes the whole path, so bash never
4     expands ~ and staging fails with "cannot create regular file '~/clips/...'".
5     This broke the live trajectory-hybrid path for the DEFAULT ~/clips stage dir
6     (found by the 2026-06-11 wasb_hybrid smoke run; masked until then because
7     trajectories were generated via manual WSL scripts and tests mock _wsl_bash).
8     """
9     from unittest.mock import MagicMock, patch
10

```

```

11 import pytest
12
13 from backend.pipeline.detectors.base import wsl_tilde_quote
14 from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
15 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, TrackNetConfig
16
17
18 @pytest.mark.parametrize("path,expected", [
19     ("~/clips/x.mp4", "~/clips/x.mp4"),          # safe chars: quote is a no-op, ~
expandable
20     ("~/clips/has space.mp4", "~/clips/has space.mp4"),
21     ("~", "~"),                                  # bare home
22     ("/mnt/c/Users/x/out", "/mnt/c/Users/x/out"), # shlex.quote no-op for safe paths
23     ("/mnt/c/has space", "'/mnt/c/has space'"),   # ordinary quoting preserved
24 ])
25 def test_wsl_tilde_quote(path, expected):
26     assert wsl_tilde_quote(path) == expected
27
28
29 def _proc(rc=0, stdout="OK", stderr=""):
30     p = MagicMock()
31     p.returncode = rc
32     p.stdout = stdout
33     p.stderr = stderr
34     return p
35
36
37 def _no_quoted_tilde(cmd: str) -> bool:
38     return "'~" not in cmd and "'~'" not in cmd
39
40
41 def test_wasb_stage_video_cmd_has_no_quoted_tilde():
42     r = WasbRunner(WasbConfig()) # defaults: wsl_stage_dir="~/clips"
43     captured = []
44
45     def fake_wsl(cmd):
46         captured.append(cmd)
47         return _proc(stdout="READABLE") if "test -r" in cmd else _proc()
48
49     with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
50         dest = r._stage_video("C:\\vids\\clip.mp4", "clip_abc.mp4")
51         assert dest == "~/clips/clip_abc.mp4"
52         # captured[0] = readability preflight (test -r), captured[1] = the staging cp.
53         stage_cmd = next(c for c in captured if c.startswith("mkdir -p"))
54         assert _no_quoted_tilde(stage_cmd), stage_cmd
55         assert "mkdir -p ~/clips" in stage_cmd
56
57
58 def test_tracknet_stage_video_cmd_has_no_quoted_tilde():
59     r = TrackNetRunner(TrackNetConfig())
60     captured = []
61
62     def fake_wsl(cmd):
63         captured.append(cmd)
64         return _proc(stdout="READABLE") if "test -r" in cmd else _proc()
65
66     with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
67         r._stage_video("C:\\vids\\clip.mp4", "clip_abc.mp4")
68         stage_cmd = next(c for c in captured if c.startswith("mkdir -p"))
69         assert _no_quoted_tilde(stage_cmd), stage_cmd
70
71 def test_wasb_run_predict_resolves_relative_out_dir(tmp_path, monkeypatch):
72     """The hybrids pass a RELATIVE 'output/wasb'. to_wsl_mnt_path passes relative
73     paths through unchanged, so WSL resolved --out against the inference cwd
74     (~/models/WASB-SBDT/src) and the CSV landed inside WSL while windows polled

```

```

75     the repo path (second find of the 2026-06-11 live smoke). run_predict must
76     abspath the out dir so the WSL-side --out is absolute."""
77     monkeypatch.chdir(tmp_path)
78     r = WasbRunner(WasbConfig())
79     cmds = []
80
81     def fake_wsl(cmd):
82         cmds.append(cmd)
83         if "test -r" in cmd:
84             return _proc(stdout="READABLE")
85         if "python wasb_infer.py" in cmd:
86             (tmp_path / "out" /
"clip__abcdefabcdef_wasb.csv").write_text("Frame,Visibility,X,Y\n")
87             return _proc()
88
89     with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
90         out = r.run_predict("C:\\\\vids\\clip.mp4", "out", video_id="abcdefabcdef0000")
91         assert out is not None
92         infer = next(c for c in cmds if "python wasb_infer.py" in c)
93         out_arg = infer.split("--out ")[1].split(" --")[0]
94         assert not out_arg.startswith("out/"), f"--out is still relative: {out_arg}"
95         assert out_arg.startswith(("/", "'/")), f"--out must be absolute on the WSL side:
{out_arg}"
96
97
98     def test_wasb_run_predict_inference_cmd_has_no_quoted_tilde(tmp_path):
99         """The full inference invocation (video/frames/weights/out) must keep ~
100         expandable too ? quoting any of them reintroduces the staging bug one
101         step later."""
102         r = WasbRunner(WasbConfig()) # weights default lives under ~/models
103         cmds = []
104
105         def fake_wsl(cmd):
106             cmds.append(cmd)
107             if "test -r" in cmd:
108                 return _proc(stdout="READABLE")
109             if "wasb_infer.py" in cmd:
110                 # simulate the run producing the expected CSV
111                 (tmp_path /
"clip__abcdefabcdef_wasb.csv").write_text("Frame,Visibility,X,Y\n")
112                 return _proc()
113
114         with patch.object(r, "_wsl_bash", side_effect=fake_wsl):
115             out = r.run_predict("C:\\\\vids\\clip.mp4", str(tmp_path),
video_id="abcdefabcdef0000")
116             assert out is not None
117             infer_cmds = [c for c in cmds if "wasb_infer.py --video" in c or "python
wasb_infer.py" in c]
118             assert infer_cmds, cmds
119             assert _no_quoted_tilde(infer_cmds[0]), infer_cmds[0]
120

```

15. assistant (2026-06-15T16:59:58.477Z)

The working-tree fix is complete and correct ? it covers all 4 listed tests plus the sibling `test\_tracknet\_stage\_video\_cmd\_has\_no\_quoted\_tilde` (which the task didn't list because, without the fix, it passed for the wrong reason: the early `None` return left only the no-tilde `test -r` cmd captured). This matches the task's intent exactly. I'm confident this is the intended work product from a prior interrupted session.

Now let me verify rigorously. First, run the targeted tests with the fix applied (should pass):

16. user (2026-06-15T17:00:02.499Z)

.....

[100%]

21 passed in 0.47s

17. assistant (2026-06-15T17:00:07.643Z)

21 passed. Now let me reproduce the original 4 failures on the pristine committed version to confirm the fix actually addresses them. I'll stash only these two files, run, then pop.

18. user (2026-06-15T17:00:10.196Z)

```
No local changes to save
=== stashed; running pristine ===
..... [100%]
21 passed in 0.15s
```

19. assistant (2026-06-15T17:00:15.077Z)

"No local changes to save" ? the working-tree changes are gone. This is exactly the shared-checkout concern. Let me check the current state immediately.

20. user (2026-06-15T17:00:16.331Z)

On branch master
Your branch is up to date with 'origin/master'.

```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    artifacts/
    docs/KHELSUTRA_PRODUCT_OVERVIEW.md
```

```
nothing added to commit but untracked files present (use "git add" to track)
=== HEAD ===
a596b34 fix(tests): update WASB stage/run_predict mocks for the WSL readability preflight (#177 follow-up) (#179)
46ad601 fix(detectors): fail fast with an actionable error when a video source is unreadable from WSL (#177)
5fe9beb Merge pull request #176 from avidullu/docs/next-session-handoff
=== diff ===
=== stash list ===
```

21. assistant (2026-06-15T17:00:30.422Z)

A concurrent session committed and merged the exact fix as **PR #179** (`a596b34`) while I was verifying ? the classic shared-checkout race from my memory, but this time nothing was clobbered: the work I was asked to do is now committed and merged. Let me confirm the merged commit contains exactly the fix I reviewed, and that it's truly on the remote.

22. user (2026-06-15T17:00:34.156Z)

```
=== local vs origin/master ===
a596b34 fix(tests): update WASB stage/run_predict mocks for the WSL readability preflight (#177 follow-up) (#179)
=== commit content ===
commit a596b34f59073029ee03ecec61d583476829a41b
Author: Avi <avi.dullu@gmail.com>
Date: Mon Jun 15 22:29:25 2026 +0530
```

```
fix(tests): update WASB stage/run_predict mocks for the WSL readability preflight (#177 follow-up) (#179)
```

```
PR #177 added a `test -r` readability preflight to the shared WSL `_stage_video`, which adds one `_wsl_bash` call before the cp. Four tests in test_wasb_runner.py and test_wsl_tilde_quoting.py mocked `_wsl_bash` with a blanket "OK" return, so the preflight saw no "READABLE" -> `_stage_video` returned None -> assertions failed. These were missed at #177 merge time because only test_tracknet_runner.py was run, not the full suite (CI is
```

down on the GitHub Actions billing outage, so there was no safety net).

Fix: the mocks now answer the `test -r` preflight with "READABLE" and target the actual staging command (mkdir/cp) rather than positional captured[0]. No product code change.
Full suite now green: 690 passed, 5 skipped.

Co-authored-by: Claude Fable 5 <noreply@anthropic.com>

```
tests/test_wasb_runner.py      | 8 ++++++--
tests/test_wsl_tilde_quoting.py | 18 ++++++-----
2 files changed, 18 insertions(+), 8 deletions(-)
=== PR #179 ===
{"mergedAt":"2026-06-15T16:59:25Z","number":179,"state":"MERGED","title":"Fix WASB test mocks
for the #177 WSL readability preflight (full suite
green)","url":"https://github.com/avidullu/badminton-highlight-indexer/pull/179"}
```

23. assistant (2026-06-15T17:00:44.861Z)

The fix is **already merged** as [PR #179](https://github.com/avidullu/badminton-highlight-indexer/pull/179) (`a596b34`) by a concurrent session ? the commit content matches exactly what I reviewed (same 2 files, same changes), and `origin/master == local master`. Nothing was clobbered.

Per my verification discipline, I won't trust the PR's "690 passed" claim ? let me confirm the merged state independently: run the full suite and ruff on the changed files locally.

24. user (2026-06-15T17:01:47.816Z)

```
=====
Full test suite: C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest
-q
obsreport (optional reporting dep): available
=====
..... [ 10%]
..... [ 20%]
..... [ 31%]
..... [ 41%]
..... [ 51%]
..... [ 62%]
..... [ 72%]
.....SSSSS..... [ 82%]
..... [ 93%]
..... [100%]
690 passed, 5 skipped in 59.74s
=====
RESULT: PASS - all tests green
=====
```

25. user (2026-06-15T17:01:48.080Z)

All checks passed!

26. assistant (2026-06-15T17:02:08.261Z)

Done ? but with a twist worth flagging.

What happened

The fix you described was **already in the working tree as uncommitted WIP** when I started. Per the global working-agreement I inspected it rather than clobbering it, and confirmed it was exactly the right fix (all 4 listed tests + the sibling `test\_tracknet\_stage\_video\_cmd\_has\_no\_quoted\_tilde`, which the task didn't list because without the fix it passed for the wrong reason ? the early `None` return left only the no-tilde `test -r` cmd captured).

While I was verifying, a **concurrent session committed and merged it** as [PR #179](https://github.com/avidullu/badminton-highlight-indexer/pull/179) (`a596b34`, authored by Claude Fable 5). This is the documented [shared-checkout collision gotcha](memory/gotcha-shared-checkout-branch-collisions.md) playing out live ? my `git stash` reported "No local changes to save" because the WIP had just been committed out from under the working tree. **Nothing was clobbered**, and the merged commit content matches exactly what I reviewed.

So there is **no work left to do** and no PR for me to open ? opening one would duplicate #179.

Independent verification (not trusting the PR's claims)

I confirmed the merged state on `master` (`a596b34`, == `origin/master`) myself:

```
| Check | Result |
|---|---|
| Target tests `test_wasb_runner.py` + `test_wsl_tilde_quoting.py` | 21 passed |
| Full suite (`python run_all_tests.py`) | 690 passed, 5 skipped ? green |
| `ruff check` on both changed test files | All checks passed |
```

The 690/5 matches the PR description exactly. The fix is purely in the test mocks (preflight answers `test -r` with `READABLE`, then the cp/inference as before); no product code changed.

Note the merge message flags that #177 slipped this regression in because **GitHub Actions CI is currently down (billing outage)** so there was no safety net ? worth keeping in mind that the local full-suite run is the only gate right now.

I did **not** touch memory/handoff files ? the concurrent session that landed #179 is clearly active on this same checkout, and racing to edit `MEMORY.md`/`current-state.md` would risk exactly the kind of collision I just witnessed. I'll leave that to whoever owns that thread, unless you'd like me to update it.

27. user (2026-06-16T02:19:58.886Z)

Can you make a design plan by which the more golden set evals we add, the repository in github should have the json or sml or any structured files which can be reused to run any regression tests without the availability of the video for refactoring and other non-quality PRs. We can extract that info from the current golden labeled set and say in another clone on another machine if someone wants to send some fix or small change to me for review, that the repository test suite can ensure that its not going to have side effects.

28. assistant (2026-06-16T02:21:53.340Z)

This is a meaty design task ? let me first scout the actual code/doc landscape so I can ground the multi-agent design workflow in real paths, then fan out.

29. user (2026-06-16T02:21:54.117Z)

```
backend\eval\__init__.py
backend\eval\classifier.py
backend\eval\training.py
backend\eval\gemini_refine.py
backend\eval\audio\__init__.py
backend\eval\audio\e1_probe.py
backend\eval\export_wasb_segments.py
backend\eval\metrics.py
backend\eval\regression.py
backend\eval\annotations.py
backend\eval\calibrate_wasb.py
backend\eval\calibration.py
backend\eval\eval_partial_labels.py
backend\eval\rally_seq_proto.py
backend\eval\calibrate_local.py
backend\eval\gt_loader.py
backend\eval\harness.py
```

backend\eval\eval_stats.py
backend\eval\score_calibration.py
backend\eval\segmentation_metrics.py
backend\eval\fusion_audio.py
backend\eval\fusion_compare.py
backend\eval\experiment.py
backend\eval\batch.py
backend\eval\fusion_golden.py
backend\eval\per_user_gate.py
backend\eval\per_user_eval.py
backend\eval\per_user_regression.py
backend\eval\ablation.py
backend\eval\served_gate_a.py
backend\eval\distill_local.py
backend\eval\promotion.py
backend\eval\serve_contrast.py
backend\eval>windowing.py
backend\eval\splits.py

30. user (2026-06-16T02:21:54.406Z)

docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\PMF_MARKET_RESEARCH.md
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\HOSTING_PLAN.md
docs\UI_PROPOSAL.md
docs\UI_ALPHA_EXECUTION_PLAN.md
docs\FIELD_VISIT_ANSWER_SHEET.md
docs\FIELD_VISIT_PLAYBOOK.md
docs\I18N_PLAN.md
docs\PMF_FIELD_SIGNALS.md
docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md
docs\VIDEO_LOCALITY_MODEL.md
docs\GOLDEN_SET_VENDORING.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\STORAGE_SHARING_MODEL.md
docs\PER_VIDEO_OUTPUT_LAYOUT.md
docs\QUALITY_ITERATIONS.md
docs\ABLATION_LAYER.md
docs\ANALYZERS_AND_RECONCILERS.md
docs\BACKLOG.md
docs\CODE_AUDIT_AND_TEST_HARDENING.md
docs\COMMERCIALIZATION.md
docs\DEFERRED_HARDENING_PLAN.md
docs\EVALUATION.md
docs\EXECUTION_POLICY.md
docs\EXPERIMENT_HARNESS.md
docs\GOLDEN_DATA_SHARING.md
docs\HARDENING_LOOP_HANDOFF.md
docs\MULTI_SIGNAL_FUSION_PLAN.md
docs\NEXT_STEPS.md
docs\PERSONALIZATION_PLAN.md
docs\PLATFORM_ARCHITECTURE.md
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
docs\README.md
docs\ROADMAP.md
docs\ROORA_LABELING_GUIDELINES.md
docs\TRACKNET_WSL_SETUP.md
docs\VIDEO_RECORDING_GUIDELINES.md
docs\KHELSUTRA_PRODUCT_OVERVIEW.md
docs\CODE_MAP.md

31. user (2026-06-16T02:21:55.831Z)

Found 35 files

tests\test_config.py
tests\test_wsl_tilde_quoting.py
tests\test_api_endpoints.py
tests\test_async_jobs.py
tests\test_base.py
tests\test_job_worker_loop.py
tests\test_training_gen0.py
tests\test_validators_base.py
tests\test_windowing.py
tests\test_serve_contrast.py
tests\test_promotion.py
tests\test_served_gate_a_regression.py
tests\test_database.py
tests\test_personalization_serving.py
tests\test_ablation.py
tests\test_per_user_regression.py
tests\test_per_user_eval.py
tests\test_fusion_golden_telemetry.py
tests\test_variant_tuning.py
tests\test_run_telemetry.py
tests\test_experiment_honest.py
tests\test_fusion_compare.py
tests\test_audio_features.py
tests\test_logging_setup.py
tests\test_yolo_hybrid.py
tests\test_experiment.py
tests\test_watermark.py
tests\test_upload_download.py
tests\test_eval_regression.py
tests\test_gt_loader.py
tests\test_eval_harness.py
tests\test_automated_provider.py
tests\test_export_wasb_segments.py
tests\test_calibrate_wasb.py
tests\test_annotation_providers.py

32. assistant (2026-06-16T02:23:49.117Z)

This repo already has substantial eval/golden/regression machinery ?
`backend/eval/regression.py`, `gt\_loader.py`, `fusion\_golden.py`, `export\_wasb\_segments.py`, and docs like `GOLDEN\_DATA\_SHARING.md`. So the right design *builds on* what exists rather than reinventing it. Given ultracode is on, I'll run a multi-phase design workflow: map the existing infrastructure exhaustively, generate independent design proposals, synthesize, then adversarially critique ? so the plan is grounded and honest about its limits.

33. user (2026-06-16T02:24:52.585Z)

Workflow launched in background. Task ID: wlorsh1wx

Summary: Design a plan for committed, video-free golden-set regression fixtures + tests

Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_b3a30c42-346

Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-regression-fixtures-design-wf_b3a30c42-346.js

(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:

"C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-regression-fixtures-design-wf_b3a30c42-346.js"} to iterate without resending the script.)

Run ID: wf_b3a30c42-346

To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-regression-fixtures-design-wf_b3a30c42-346.js", resumeFromRunId: "wf_b3a30c42-346"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

34. assistant (2026-06-16T02:25:02.217Z)

The workflow is running in the background; I'll be notified when it completes. Rather than poll, let me use the time to read the most pivotal artifacts myself ? `docs/GOLDEN\_DATA\_SHARING.md` (its name suggests it may already cover this exact idea) and the existing `regression.py` + its test ? so I can critically verify the workflow's synthesis instead of trusting it.

35. user (2026-06-16T02:25:02.534Z)

```
1      # Golden / GPU-Extractor Data Sharing ? GPU box ? dev sessions
2
3      **Status:** convention + plan (2026-06-15). Tracked in **#175**.
4      **Purpose:** a single cloud-synced location for the GPU-extracted golden artifacts
(trjectory CSVs, `*_golden_features.json`, manifests, baseline snapshots) so the **GPU box**
and **CPU dev/agent sessions** share the same files ? without committing large data to git.
5
6      ## Why this exists
7      The rally-quality measurement is split across two machines by capability:
8      - **`backend/eval/fusion_golden.py`** (shuttle+person feature **extraction**) needs the
**videos**, the **trajectory CSVs**, the **`.rallies.csv` golden labels**, and a **GPU/WSL**
stack ? only the GPU box can run it.
9      - **`backend/eval/fusion_compare.py`** / **`backend/eval/ablation.py`** (the LOVO
**litmus** : ?F1, bootstrap CI, paired sign test) are **pure-CPU re-scoring** that read only the
small `*_golden_features.json` ? any session (incl. a CPU dev container) can run them.
10
11     Today those artifacts sit in gitignored `output/` (per-machine) and scattered local
folders (`...\Annotation Setup\Collect\Trajectories\`, `...\Golden Labelled\`). Nothing syncs
them between machines, so a CPU session cannot see the GPU box's extraction. This wires that gap
with OneDrive.
12
13     ## Canonical shared folder
14     `%USERPROFILE%\OneDrive\rally-annotator\golden-data\` ?
`C:\Users\avidu\OneDrive\rally-annotator\golden-data\`
15
16     OneDrive is present + signed-in on both the GPU box and the dev machine under the same
account, so it auto-syncs. Layout (date/size-tagged so a re-extraction is a new folder, never an
overwrite):
17
18     ```
19     golden-data/
20     README.md
21     features/<corpus-tag>/      *_golden_features.json      (small ? the key shared
artifact)
22     trajectories/<corpus-tag>/  *_predict.csv / *_wasb.csv (medium ? reusable across
runs)
23     manifests/                  *_lovo_manifest.json
24     baselines/                  leaderboard / regression snapshots (or keep tracked in
eval_baselines/)
25     ```
26
27     `<corpus-tag>` = e.g. `2026-06-15-n6` (date + corpus size). **Do NOT** put raw videos
here (tens of GB) ? keep those local to the GPU box.
28
29     ## The GPU ? CPU workflow
30     1. **GPU box** (videos + CSVs + labels present):
31     ```
32     python -m backend.eval.fusion_golden --manifest <manifest> \
33     --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6"
34     ```
35     (or extract to local `output/` then copy the `*_golden_features.json` into the shared
folder).
36     2. OneDrive syncs the JSONs to the dev machine automatically (small files, seconds).
```

```

37     3. CPU dev/agent session points the analysis at the shared folder:
38     ``
39     python -m backend.eval.fusion_compare --features-dir "...golden-
data/features\2026-06-15-n6" --record
40     python -m backend.eval.ablation --features-dir "...golden-
data/features\2026-06-15-n6" --granularity group
41     ``
42     ? real person-feature ?F1 / CI / sign test. Copy the JSONs into the repo's `output/`
to also light up the `skipif` golden-litmus tests
(`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`).
43
44     ## Optional code ergonomics (in #175, not required)
45     - A `RALLY_GOLDEN_DIR` env var (default `output/`) honored by `fusion_golden` /
`fusion_compare` / `ablation` so neither machine hardcodes the path. (The CLI `--out-dir` /
`--features-dir` flags already make this a convention; the env var is just ergonomics.)
46     - A tiny `scripts/sync_golden.py` to copy `output/*_golden_features.json` ? the shared
folder.
47
48     ## Guardrails
49     - The shared folder lives outside the repo ? never committed. Keep `output/`
gitignored and do not symlink the shared folder into a tracked path.
50     - The features derive from the Proprietary corpus ? keep the OneDrive folder
private (no public share links).
51
52     See also: `docs/NEXT_STEPS.md` (the "Next-session pickup" anchor),
`docs/MULTI_SIGNAL_FUSION_PLAN.md` (P3), `docs/TRACKNET_WSL_SETUP.md` (the GPU/WSL detector
setup).
53

```

36. assistant (2026-06-16T02:25:24.806Z)

This is pivotal. `GOLDEN\_DATA\_SHARING.md` reveals the existing system already has the key artifact ? `\_golden\_features.json` (small, video-free, CPU-rescorable via `fusion\_compare.py`/`ablation.py`) ? but a **deliberate guardrail** keeps it private (OneDrive, out of git) because it derives from a "Proprietary corpus." The user's request to commit these into GitHub directly collides with that decision. There are also already-existing `skipif` golden-litmus tests that *would* run if the artifacts were committed. Let me ground myself on `regression.py` and whether a tracked baseline dir exists, then wait for the workflow.

37. user (2026-06-16T02:25:25.942Z)

```

1     """Regression runner ? verify the production tool against golden-set ground truth.
2
3     Implements the explicit ask: "given a video, get a golden set created and verify
4     its precision and recall against the production tool." This is deliberately a thin
5     wrapper over machinery that already exists:
6
7     ground truth ? AnnotationProvider (file / vendor / oracle) [backend.annotations]
8     predictions ? the production pipeline's stored DB rows [backend.eval.harness]
9     comparison ? interval matching + P/R/F1 [backend.eval.metrics]
10    baseline ? a per-video snapshot of "known good" numbers [this module]
11
12    A "regression" is a precision/recall drop beyond `tolerance` versus the snapshotted
13    baseline. Tiers map to annotation providers: e.g. tier "file" ? FileProvider (CI),
14    later tier "vendor" ? HumanVendorProvider (authoritative), tier "auto" ? an
15    independent-lineage oracle (fast smoke alarm). See docs/GOLDEN_SET_IMPLEMENTATION.md.
16    """
17    import os
18    import json
19    import hashlib
20    import logging
21    from dataclasses import dataclass, asdict
22    from typing import Dict, List, Optional
23
24    from backend.infrastructure.database import Database

```

```

25     from backend.eval import metrics
26     from backend.eval.harness import get_predictions
27     from backend.annotations import annotation_registry
28
29     logger = logging.getLogger(__name__)
30
31     # Tracked location (NOT under the gitignored output/), so a fresh checkout / CI run can
32     # actually load a committed baseline to compare against.
33     DEFAULT_BASELINE_PATH = os.path.join("eval_baselines", "regression_baseline.json")
34     DEFAULT_TOLERANCE = 0.05
35
36
37     def gt_hash(ground_truth: List[metrics.Interval]) -> str:
38         """Stable short hash of the GT intervals ? detects a baseline taken against
different labels."""
39         norm = sorted((round(float(s), 3), round(float(e), 3)) for s, e in ground_truth)
40         return hashlib.sha1(json.dumps(norm).encode()).hexdigest()[:12]
41
42
43     @dataclass
44     class RegressionResult:
45         video_id: str
46         stage: str
47         iou_threshold: float
48         precision: float
49         recall: float
50         f1: float
51         # Baseline values compared against (None when no baseline existed yet).
52         baseline_precision: Optional[float]
53         baseline_recall: Optional[float]
54         tolerance: float
55         regressed: bool
56         reasons: List[str]
57         # True only when a baseline existed to compare against. Distinguishes a genuine PASS
58         # from "nothing to compare" so the CLI can refuse to silently green-light the
latter.
59         has_baseline: bool = False
60         gt_hash: Optional[str] = None
61
62         def as_dict(self) -> dict:
63             return asdict(self)
64
65
66     # ----- baseline store
67
68     def load_baselines(path: str = DEFAULT_BASELINE_PATH) -> Dict[str, dict]:
69         """Load the per-video baseline snapshot file (returns {} if absent)."""
70         if not os.path.isfile(path):
71             return {}
72         with open(path) as f:
73             return json.load(f)
74
75
76     def _baseline_key(video_id: str, stage: str) -> str:
77         return f"{video_id}::{stage}"
78
79
80     def save_baseline(video_id: str, stage: str, precision: float, recall: float,
81                      f1: float, path: str = DEFAULT_BASELINE_PATH,
82                      iou_threshold: Optional[float] = None, detector: Optional[str] = None,
83                      gt_fingerprint: Optional[str] = None) -> None:
84         """Snapshot a video/stage's current numbers as the new 'known good' baseline.
85
86         Records the iou_threshold, detector, and a GT fingerprint alongside the metrics so a
87         later run computed under different settings (or against different labels) is

```



```

detected
88         as incommensurable rather than silently compared.
89         """
90         baselines = load_baselines(path)
91         baselines[_baseline_key(video_id, stage)] = {
92             "video_id": video_id, "stage": stage,
93             "precision": precision, "recall": recall, "f1": f1,
94             "iou_threshold": iou_threshold, "detector": detector, "gt_hash": gt_fingerprint,
95         }
96         os.makedirs(os.path.dirname(path) or ".", exist_ok=True)
97         with open(path, "w") as f:
98             json.dump(baselines, f, indent=2, sort_keys=True)
99         logger.info("Saved baseline for %s (precision=%.3f recall=%.3f)",
100 _baseline_key(video_id, stage), precision, recall)
101
102 # ----- the runner
103
104 def regress(db: Database, video_id: str, ground_truth: List[metrics.Interval],
105            stage: str = "final", iou_threshold: float = 0.5,
106            detector: Optional[str] = None,
107            tolerance: float = DEFAULT_TOLERANCE,
108            baseline_path: str = DEFAULT_BASELINE_PATH) -> RegressionResult:
109     """Score stored predictions for one video against ground truth and compare to
baseline.
110
111     `ground_truth` is a list of (start, end) intervals ? typically obtained from an
112     AnnotationProvider (see `regress_with_provider`). A regression is flagged when
113     precision OR recall falls more than `tolerance` below the snapshotted baseline.
114     If no baseline exists yet, `regressed` is False (nothing to compare to) and the
115     caller can choose to snapshot the current numbers via `save_baseline`.
116     """
117     preds = get_predictions(db, video_id, stage, detector=detector)
118     s = metrics.score(preds, ground_truth, iou_threshold)
119     fp = gt_hash(ground_truth)
120
121     baselines = load_baselines(baseline_path)
122     b = baselines.get(_baseline_key(video_id, stage))
123
124     reasons: List[str] = []
125     regressed = False
126     has_baseline = b is not None
127     base_p = base_r = None
128     if b is not None:
129         base_p, base_r = b["precision"], b["recall"]
130         # Incommensurable comparison guard: a baseline taken at a different IoU/detector
or
131         # against different labels is not a valid reference ? flag it instead of
trusting it.
132         b_iou, b_det, b_hash = b.get("iou_threshold"), b.get("detector"),
b.get("gt_hash")
133         if b_iou is not None and abs(float(b_iou) - iou_threshold) > 1e-9:
134             regressed = True
135             reasons.append(f"baseline IoU {b_iou} != run IoU {iou_threshold}
(incommensurable)")
136         if b_det is not None and b_det != detector:
137             regressed = True
138             reasons.append(f"baseline detector {b_det!r} != run detector {detector!r}
(incommensurable)")
139         if b_hash is not None and b_hash != fp:
140             regressed = True
141             reasons.append(f"baseline GT hash {b_hash} != run GT hash {fp} (labels
changed)")
142         if s.precision < base_p - tolerance:
143             regressed = True

```

```

144         reasons.append(f"precision {s.precision:.3f} < baseline {base_p:.3f} -
{tolerance}")
145         if s.recall < base_r - tolerance:
146             regressed = True
147             reasons.append(f"recall {s.recall:.3f} < baseline {base_r:.3f} -
{tolerance}")
148         else:
149             reasons.append("no baseline recorded for this video/stage ? nothing to compare")
150
151         return RegressionResult(
152             video_id=video_id, stage=stage, iou_threshold=iou_threshold,
153             precision=s.precision, recall=s.recall, f1=s.f1,
154             baseline_precision=base_p, baseline_recall=base_r,
155             tolerance=tolerance, regressed=regressed, reasons=reasons,
156             has_baseline=has_baseline, gt_hash=fp,
157         )
158
159
160     def regress_with_provider(db: Database, video_id: str, video_ref: str,
161                             tier: str = "file", stage: str = "final",
162                             iou_threshold: float = 0.5,
163                             detector: Optional[str] = None,
164                             tolerance: float = DEFAULT_TOLERANCE,
165                             baseline_path: str = DEFAULT_BASELINE_PATH,
166                             guideline: Optional[str] = None,
167                             sport: str = "badminton",
168                             provider_kwargs: Optional[dict] = None) -> RegressionResult:
169         """End-to-end: obtain ground truth from the `tier` annotation provider, then
regress.
170
171         `tier` is an annotation-provider registry key ("file" today; "vendor"/"auto"
172         once those providers land). `video_ref` is whatever that provider expects (a CSV
173         path for FileProvider, a video_id/URL for a vendor). `video_id` keys the stored
174         predictions in the DB (the file MD5).
175         """
176         provider = annotation_registry.get(tier, **(provider_kwargs or {}))
177         gt = provider.annotate(video_ref, guideline=guideline, sport=sport)
178         return regress(db, video_id, gt, stage=stage, iou_threshold=iou_threshold,
179                       detector=detector, tolerance=tolerance, baseline_path=baseline_path)
180
181
182     def format_result(r: RegressionResult) -> str:
183         verdict = "REGRESSED" if r.regressed else "ok"
184         head = (f"[{verdict}] {r.video_id} ({r.stage}) "
185               f"precision={r.precision:.3f} recall={r.recall:.3f} f1={r.f1:.3f}")
186         if r.baseline_precision is not None:
187             head += f" | baseline p={r.baseline_precision:.3f} r={r.baseline_recall:.3f}
tol={r.tolerance}"
188         return head + " ".join(f"\n    - {why}" for why in r.reasons)
189

```

38. user (2026-06-16T02:25:26.066Z)

```

eval_baselines\heuristic_lovo_n6.json
eval_baselines\gemini_wrapper_2026-06-10.json
eval_baselines\experiment_leaderboard.json

```

39. assistant (2026-06-16T02:26:01.406Z)

I've now verified the key facts myself while the workflow runs. The existing infrastructure is remarkably close to what you're asking for ? which sharpens both the design and a real tension I need to flag:

```

**What already exists (verified):**
- **`eval_baselines/`** is a *tracked* (committed-to-git) directory of JSON baseline snapshots ?

```

`regression\_baseline.json`, `heuristic\_lovo\_n6.json`, etc. So "committed structured regression files in the repo" is already an established pattern.

- `backend/eval/regression.py` loads a committed baseline and flags a P/R/F1 drop beyond tolerance ? but it pulls `*predictions` from the DB*, which still requires the pipeline to have run on the video. So it is `**not yet video-free**` on the prediction side.
- `golden_features.json` (from `fusion_golden.py`) `*is*` the video-free seam ? `fusion_compare.py`/`ablation.py` re-score it on pure CPU with no video. But per `GOLDEN_DATA_SHARING.md` (#175) `**these are deliberately kept *private* on OneDrive, out of git, because they derive from a "Proprietary corpus."**`
- There are already `**skipif*` golden-litmus tests `**that would run *if* those features were committed ? they currently skip for lack of committed fixtures.`

`**The crux your plan has to resolve:**` your request (commit structured files so anyone on another clone can run regression) directly collides with the existing guardrail that keeps proprietary-derived features out of git. That's an owner-gated licensing call, and the design will route it explicitly rather than assume it.

The design workflow (map ? 3 proposals ? synthesize ? adversarial+completeness critique) is running in the background; I'll synthesize the final plan against what I've independently verified once it completes.

40. user (2026-06-16T02:38:27.480Z)

fair point but I was more concerned about the `*content*` being leaked to public, I am not sure if `*derived content*` should also have the same guardrails. Feel free to educate me on this topic assuming that we do not want someone to know attribute the structured data to the exact video, the video may be hidden and the tests depend completely on the structured info which by itself are useless to any public user. what if we store all the structured information in google's protobuf format and hookup some secret file etc mechanism so that the users can run the test without actually knowing how it is being tested? Its a very tricky problem and I am fine if you push back. Its a lot of design and I am on a 20x Max plan so I would appreciate a deep research! If you pull this through it is an amazing Byzantine solution!

41. assistant (2026-06-16T02:43:09.950Z)

This is a genuinely deep problem and you're right to push on it ? let me both `**educate**` and `**research/design**` it properly. First, the key conceptual correction, then I'm launching a dedicated deep-research + design effort.

The pushback you invited (the part I'm confident about now)

`**1. "Secret file so users can run tests without knowing how it's tested" ? this is partially impossible, and recognizing why points to the *right* solution.**` If a contributor on another machine must `*run*` the suite, the test process must read the fixture data in plaintext at some moment ? which means the data `**and**` any decryption key are both on their machine. Anyone who can run the test can, with effort, dump what it read. This is the DRM/"analog hole" problem: you cannot hand someone a locked box `*and*` the key and expect the box to stay locked. Encryption only hides data from parties `**without**` the key ? i.e., the public browsing a repo, a leaked clone, or a CI log ? `**not**` from the trusted contributor running the tests.

`**2. But that's fine ? because your real goal isn't secrecy, it's *non-attributability*.**` You said it yourself: "we do not want someone to attribute the structured data to the exact video... by itself useless to any public user." That is a `*much*` more achievable and well-defined property than secrecy, and it's the correct framing:

- `**Unlinkability**` ? strip everything that ties the numbers to a specific match/video/person (notably: `*don't*` key fixtures by `video_id = MD5(file)` ? that's a fingerprint of the source file; use opaque salted IDs; drop titles, timestamps, URLs, player names).
- `**Minimization**` ? commit only the numeric streams the test needs (trajectory + signal columns + GT intervals + expected output); never frames, thumbnails, audio, or person crops.
- `**Uselessness-in-isolation**` ? anonymous `*(frame, x, y)*` numbers reveal neither your corpus nor your method.`

`**3. Protobuf ? secrecy.**` Protobuf is a `*great*` choice for compactness, schema-versioning, and deterministic serialization ? but a binary blob is trivially decoded. It buys you schema

discipline, not protection. Protection (against the no-key public) is a *separate* layer: ``git-crypt` / SOPS+`age` / out-of-repo fetch`. So the architecture is naturally *tiered*: a public, anonymized, unencrypted tier for the logic/refactor tests everyone can run, and an optional encrypted/key-gated tier for richer or more sensitive data.

****4.** The decisive question is provenance. ****** Whether derived data even carries the source's constraints splits hard on: **did you record the corpus yourself (your own IP ? concern is trade-secret/privacy)** vs **is it third-party broadcast/YouTube (copyright + ToS)**. And if the source is a *public* video, an adversary could re-derive trajectories from it and correlate ? so non-attributability is far stronger for a *private, owner-recorded* corpus. This is the crux the research must resolve.

I'm launching a deep legal/IP/privacy research + crypto-design workflow now (covers IN/US/EU, copyright-of-facts, EU database right, ToS, GDPR/DPDP, biometrics, re-identification, plus the tiered protobuf/encryption architecture), and asking three gating questions in parallel so the final synthesis is grounded in your actual situation rather than my guesses.

42. user (2026-06-16T02:44:47.446Z)

Workflow launched in background. Task ID: w3pn6s2mo

Summary: Deep research: IP/privacy of committing DERIVED structured data publicly + a non-attributable encrypted fixture architecture

Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9

Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js

(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:

"C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js"} to iterate without resending the script.)

Run ID: wf_47057438-9c9

To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js", resumeFromRunId: "wf_47057438-9c9"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

43. user (2026-06-16T02:55:36.857Z)

Your questions have been answered: "What's the provenance of the golden-corpus videos whose derived data we'd commit?"="Self-recorded / owned by me", "Is the GitHub repo where fixtures would live public or private?"="Private now, may open-source later", "Do the corpus videos contain identifiable people, and could any be minors?"="Adults, consent not formalized". You can now continue with these answers in mind.

44. assistant (2026-06-16T02:56:08.674Z)

The first workflow (video-free fixture design) just completed, and your three answers are in ? both materially shape the final plan. The second workflow (legal + Byzantine architecture) is still running; I'll synthesize everything when it lands. Let me pull the detailed synthesis from the completed workflow while I wait.

45. user (2026-06-16T02:56:09.943Z)

```
1      {
2        "summary": "Design a plan for committed, video-free golden-set regression fixtures +
tests",
3        "agentCount": 11,
4        "logs": [
5          "Understanding: 5/5 areas mapped"
6        ],
```

```
7         "result": {
8             "understanding": [
```

46. user (2026-06-16T02:56:15.485Z)

```
658:     "synthesis": {
698:         "rolloutPhases": [
730:[Omitted long matching line]
731:         "openOwnerQuestions": [
750:         "critiques": [
806:[Omitted long matching line]
852:[Omitted long matching line]
```

47. user (2026-06-16T02:56:20.284Z)

```
658         "synthesis": {
659             "title": "Committed video-free golden-set regression ? unified design (frozen
trajectory-CSV + feature substrate, dual gates, synthetic-first)",
660             "goal": "Make per-video rally-segmentation regression runnable end-to-end from a
clean `git clone` / CI with ZERO video, GPU, WSL, DB, or network ? so a contributor on another
machine refactoring the deterministic pipeline gets a red/green signal, and any intentional
behaviour change shows up as a reviewable snapshot diff. Achieve this by committing a tiny,
version-tagged, numeric-only per-video fixture corpus and TWO complementary auto-discovered
pytest gates (a characterization/snapshot gate for refactor side-effects, and a quality-floor
gate vs golden labels tied to the ablation track), reusing the entire shipped eval stack
verbatim with essentially no new scoring math. This closes the concrete gap that
`tests/test_served_gate_a_regression.py` and the golden-litmus tests SKIP in CI today because
the corpus lives only in gitignored `output/` / private OneDrive.",
661             "buildsOnExisting": [
662                 "backend/eval/metrics.py::score/match_intervals/interval_iou ? greedy,
deterministic (sort key (-iou,pi,gi)), empty->0.0 not NaN; the ONLY scoring math, reused
verbatim by both gates",
663                 "backend/eval/segmentation_metrics.py::f1_at_overlaps/segment_count_ratio ? the
over-seg signal tIoU hides, frozen in the snapshot",
664                 "backend/eval/experiment.py::VideoCandidates/Variant/CONTROL/predict/evaluate_va
riant/compare/honest_summary ? the pure LOVO scoring engine; predict()=gate->learned-
filter->merge_overlaps is the deterministic re-score both gates call",
665                 "backend/eval/fusion_compare.py::load_videos ? already globs
*_golden_features.json into VideoCandidates; the quality-floor gate reuses it pointed at the
COMMITTED dir instead of gitignored output/",
666                 "backend/eval/windowing.py::WindowingPreset/SERVED/PROPOSAL/build_windows/windows_to_intervals ?
the single-sourced operating point that closed the eval!=serve Gate-A trap; the fixture pins
which preset it was frozen at",
667                 "backend/eval/distill_local.py::build_candidates + FEATURE_NAMES (the 5 shuttle
feats) ? trajectory CSV -> (intervals, shuttle feature rows) under a named preset; the exact
transform the extractor and characterization gate call",
668                 "backend/pipeline/detectors/tracknet_runner.py::parse_trajectory_csv
(Frame,Visibility,X,Y) + trajectory_to_action_windows ? the pure windowing 'brain' under
characterization",
669                 "backend/pipeline/detectors/rally_gate.py (net_crossings Gate A) +
fusion_features.py (PERSON_FEATURE_NAMES, default-OFF) ? pure per-signal feature producers",
670                 "backend/eval/regression.py::gt_hash/save_baseline/load_baselines/regress/RegressionResult +
DEFAULT_BASELINE_PATH=eval_baselines/regression_baseline.json + the exit 0/1/3
incommensurability-guard contract (IoU/detector/gt_hash mismatch -> flagged, not silently
compared)",
671                 "backend/eval/eval_stats.py::paired_delta_ci/mean_with_ci/sign test ?
deterministic (default seed=0); the 'NUMBERS INFORM' honest-stats the quality-floor gate emits
and the snapshot freezes",
672                 "backend/eval/fusion_golden.py::SHUTTLE_FEATURE_NAMES/PERSON_FEATURE_NAMES/extract_video ? the
SAME extractor the owner already runs per golden video; emits the exact committed shape, so 'add
a video' needs no new producer",
673                 "tests/test_served_gate_a_regression.py ? the manifest-driven, per-video, skip-
```

```

if-absent template (env-overridable MANIFEST, _golden_specs() presence filter, _skip decorator,
Path(__file__)-anchored repo root); the new gates mirror it but RUN in CI because fixtures are
committed; its _PROPOSAL=(0.005,1.0,0.5) numbers are the proof the snapshot must record
windowing provenance",
674     "eval_baselines/{heuristic_lovo_n6.json, experiment_leaderboard.json,
gemini_wrapper_2026-06-10.json} ? the proven TRACKED home + provenance-carrying JSON convention
(name/description/`computed` repro cmd + label_set_hash)",
675     "backend/infrastructure/database.py PRAGMA user_version migration ladder (v5) +
the detector-agnostic 'common columns + stats JSON keyed by detector string' pattern ? mirrored
by the fixture schema_version + provenance design",
676     "run_all_tests.py + pytest.ini (testpaths=tests, no conftest, inline-tmp_path
idiom) + .github/workflows/ci.yml (ruff/mypy/import-smoke/full-suite, --cov-fail-under=70) ? the
suite the new parametrized gates slot into with no new CI job required for the core goal"
677 ],
678     "chosenSeams": [
679         "PRIMARY FREEZE SEAM = the post-detector trajectory CSV `Frame,Visibility,X,Y`
per video (Proposal 3's choice, adopted over Proposal 1's feature-JSON-only seam). RESOLUTION OF
CONFLICT: Proposal 1 froze only the *_golden_features.json (one transform DOWNSTREAM of
windowing) ? smaller but blind to the windowing 'brain' (trajectory_to_action_windows) and
Gate-A net_crossings derivation, the exact code most likely to be refactored. The trajectory CSV
is the true GPU/WSL->pure-Python handoff; freezing here characterizes the WIDEST deterministic
surface and lets the gate re-derive features (so a feature-math regression is also caught). It
is still tiny (~20-25 B/row; a deliberately SHORT fixture clip is sub-200KB). We freeze at the
CSV and DERIVE features at test time rather than committing them as inputs.",
680         "REQUIRED SIDE-INPUTS frozen alongside the CSV in provenance: (fps,
frame_width). NON-NEGOTIABLE ? trajectory_to_action_windows normalizes velocity by frame_width
and converts frame->sec by fps, and _probe_fps_width's SILENT fallbacks (fps<=0->30.0,
width<=0->1280.0) would mask real regressions if a replay forgets them. All three proposals
flagged this; we pin them explicitly.",
681         "WINDOWING-PRESET SEAM = a named backend.eval.windowing.WindowingPreset (default
SERVED, single-sourced from IndexingConfig defaults) stamped into every fixture and snapshot.
This neutralizes the Gate-A eval!=serve cautionary tale: the snapshot records WHICH windowing it
was frozen at, so a deliberate retune trips the incommensurability guard rather than silently
passing or regressing.",
682         "GT SEAM = a per-video golden rally labels.csv in gt_loader conventions
(start,end | start_time,end_time | positional; decimal or MM:SS), loaded via
load_gt_intervals(strict=True) so a malformed committed label fails loudly with file:line.",
683         "PERSON SEAM (optional, default-OFF) = an optional per-video
person_detections.json {frames:{idx:[(tid,bbox)]}, fps, frame_width, frame_height} captured from
the SAME decode as the trajectory, so frame indices align 1:1. Synthetic tier hand-authors it
for determinism; real tier carries a codec-seek caveat in provenance.",
684         "EXCLUDED SEAMS (deliberately, with honest reasons): the AI/provider handoff
(provider.analyze_video ? network/non-deterministic; CONTROL emits windows verbatim / skip_ai
path, so we freeze the windows that ENTER the handoff, never provider output); the ffmpeg stitch
(VideoCompiler reel bytes are NOT reproducible across libx264/NVENC builds ? we assert the
merge_overlaps time_pairs, never bytes); DB autoincrement id + created_at (compare on
(start,end,stats) not id); the gemini and motion segmenters (no candidate stage; motion
fabricates random metadata ? only the trajectory-hybrid rally-quality family is freezable).",
685     ],
686     "fixtureLayout": "NEW tracked dir `tests/fixtures/golden_regression/` (under
tests/ so pytest auto-discovers and it survives a clean checkout; NOT output/ which is
gitignored ? that is exactly why test_served_gate_a_regression skips in CI). One subdir per
video keyed by a stable slug (synthetic slug for tier-0; video_id[:12] for owner-cleared real
tier-1, matching the production MD5 key):
`tests/fixtures/golden_regression/<slug>/{trajectory.csv, labels.csv, expected_features.json,
expected_segments.json, provenance.json}` (+ optional `person_detections.json`). A single top-
level repo-relative index `tests/fixtures/golden_regression/manifest.json`: a JSON array of
`{slug, trajectory, labels, features, segments, provenance, fps, frame_width, windowing_preset,
kind:"synthetic"|"cleared_real"}` ? the committed, repo-relative INVERSE of today's
gitignored absolute-pathed output/human_lovo_manifest.json. All paths resolved via
`Path(__file__).parent` (NEVER cwd ? agent/test cwd is unstable). A `README.md` in the dir
states the minors/licensing guardrail + the freeze/refresh commands. The per-video quality-floor
baseline lives at the reserved `eval_baselines/regression_baseline.json` (today referenced but
absent ? a fresh clone hits exit-3; this design finally creates it), keyed `<slug>::final` with
iou/detector/gt_hash stamps. Rationale for splitting the two homes: fixtures are TEST INPUTS

```

(belong under tests/, CODE_MAP \$0 'a test -> tests/'); the floor baseline is a no-regression SCORE artifact (belongs in eval_baselines/, CODE_MAP \$0 'survives a clean checkout; the no-regression gate reads it'). This honors BOTH established conventions rather than overloading eval_baselines/ with raw fixtures (Proposal 1's choice) ? adopted from Proposals 2/3.",

687 "artifactSchema": "Every committed JSON carries a top-level `schema\_version` (int, starts at 1) + `kind`, mirroring the DB PRAGMA user_version discipline and eval_baselines' provenance convention. (1) trajectory.csv: fixed header `Frame,Visibility,X,Y`, one row/frame, ints+floats only (parse_trajectory_csv contract). (2) labels.csv: gt_loader-conformant. (3) expected_features.json: {schema_version, slug, fps, frame_width, windowing_preset:{name,velocity_thresh,merge_gap,min_window_duration}, candidates:[{start,end,shuttle:{duration,peak_velocity,mean_velocity,active_ratio,net_crossings}, person:{players_in_court,two_sided_motion,mean_player_motion,in_court_ratio,shuttle_player_colocation}, label:int}}, gts:[[s,e]]} ? i.e. the EXACT fusion_golden schema (so fusion_compare.load_videos still reads it unchanged) plus only a windowing_preset stamp + schema_version. (4) expected_segments.json (the characterization snapshot): {schema_version, slug, windowing_preset, variant_spec_hash, iou_used, seed, segments:[{start,end}], num_gt, score_vs_gt:{precision,recall,f1,mean_iou,tp,fp,fn}, segmentation:{f1@10,f1@25,f1@50,segment_count_ratio}} ? the deterministic experiment.predict(CONTROL)->merge_overlaps + metrics.score output, floats stored rounded to 6 dp for byte-stable diffs. (5) provenance.json: {schema_version, slug, video_id(real only), detector, fps, frame_width, windowing:{vt,merge_gap,min_window}, gt_hash, label_set_hash, frozen_at(ISO), freeze_command, source:{license:\"owned|ShuttleSet|synthetic\", owned_or_synthetic, minors_free:true, consent_ref}, tool_versions:{python}}. The 'common fields + named-feature blocks + version+provenance stamp' design mirrors the DB's detector-agnostic 'columns + stats JSON keyed by detector'. The fingerprints (gt_hash, label_set_hash, Variant.spec_hash, WindowingPreset name+tuple) travel with the snapshot so any comparison against changed labels/recipe/windowing is flagged incommensurable, not silently trusted ? reusing the shipped regression.py/experiment.py guard contract rather than inventing a parallel scheme.",

688 "extractionTool": "A new `backend/eval/freeze\_fixture.py` CLI (thin composition over SHIPPED functions ? NO new transforms), used both to FREEZE and to REFRESH. `python -m backend.eval.freeze\_fixture --manifest <local> --slug <slug> --out tests/fixtures/golden\_regression/<slug> [--preset served] --license owned --minors-free --consent-ref <doc>`: (1) copies/normalizes trajectory CSV -> trajectory.csv and golden labels -> labels.csv (projecting the rich Roora <video_id>.csv to the minimal gt_loader start,end shape, recording the projection in provenance); (2) calls distill_local.build_candidates(traj,fps,fw,preset) for intervals+shuttle features, rally_gate for net_crossings, and (only --with-person, video present) fusion_golden.extract_video for the person 5-col -> writes expected_features.json; (3) computes the snapshot via experiment.predict(CONTROL, VideoCandidates)->merge_overlaps at the pinned preset + metrics.score + segmentation_metrics -> expected_segments.json with variant_spec_hash + preset + iou + seed stamps; (4) emits provenance.json with regression.gt_hash(labels) + label_set_hash + the exact freeze_command + the licensing/minors assertion. CODE-ENFORCED GATE: it REFUSES to write a real fixture unless --minors-free is passed AND a --license is given (gives a code gate where the consent ledger is still only convention/BACKLOG). A `--refresh-snapshot [--all]` mode re-derives ONLY expected_features.json/expected_segments.json from the committed trajectory+labels after an INTENTIONAL behaviour change (no re-detection ? the trajectory is frozen), the characterization analogue of run_regression --update-baseline. CRITICAL: the extractor and BOTH gate tests call ONE shared helper in a new `backend/eval/golden\_fixtures.py` (load_manifest/load_fixture/replay_fixture, schema_version dispatch, Path(__file__)-relative resolution, CURRENT_SCHEMA const) so the generated artifact and the asserted artifact are computed by identical code. The extractor is itself unit-tested (synthetic in -> deterministic artifacts out).",

689 "characterizationGate": "`tests/test\_golden\_characterization.py` (NEW). Parametrized over manifest.json via `@pytest.mark.parametrize(\"slug\", sorted(load\_manifest(), ids=...))`. Per fixture: parse_trajectory_csv(trajectory.csv) -> windowing.build_windows(preset) -> re-derive shuttle features (distill_local.build_candidates) + net_crossings (rally_gate) into a VideoCandidates -> recompute segments via experiment.predict(CONTROL)->merge_overlaps + metrics.score; assert the recomputed expected_features.json AND expected_segments.json match the committed files. NEEDS NO LABELS for the windows/features assertions ? it catches ANY side-effect of a refactor to the windowing brain / feature math / merge, even one that does NOT move F1 (the kind of silent drift a quality-floor gate alone would miss). Comparison policy (from the determinism contract): EXACT equality on structural/integer fields (window counts, candidate count, net_crossings as counts, tp/fp/fn, segment count) since those carry the behavioral signal; pytest.approx(abs=1e-6) on floats (velocity, F1, IOU) so last-ULP cross-OS drift does

not flap. A label_set_hash/spec_hash/preset guard asserts the snapshot's stamps match what is recomputed, so a fixture edited without re-freezing fails LOUDLY as incommensurable. Module skips (skipif) if the fixtures dir is empty ? but UNLIKE test_served_gate_a_regression it RUNS in CI because the synthetic tier is committed. This is the refactor-side-effect detector and the reviewer-trust affordance: a green snapshot = deterministic post-trajectory behaviour unchanged; a changed expected_*.json in the PR diff = an explicit, reviewable intentional change.",

690 "qualityFloorGate": "`tests/test\_golden\_quality\_floor.py` (NEW). Load the committed substrate via the fusion_compare.load_videos shape; score CONTROL (and the shuttle_only/fusion variants) vs the committed labels with metrics.score per video and experiment.compare for the honest aggregate; assert per-video AND mean F1 >= the committed floor in `eval\_baselines/regression\_baseline.json` (created via the freeze CLI + regression.save_baseline, keyed `<slug>::final` with iou/detector/gt_hash stamps so a label or operating-point change surfaces as regression.py's incommensurability flag = a test failure with a clear message, mirroring the exit-3 semantics). Floor breach uses regression.DEFAULT_TOLERANCE (0.05). The assertion MESSAGE emits eval_stats.paired_delta_ci + the exact sign test (NUMBERS INFORM, FOUNDATIONS DECIDE ? never a bare mean) and the floor is tied doc-side to QUALITY_ITERATIONS / the ablation track. STRUCTURAL GUARD: Gate-A-style structural cues are never auto-demoted on a small-n null (weak-signals-retained directive). HONEST LIMIT stated in the gate's docstring: on the SYNTHETIC tier this measures pipeline CORRECTNESS (the math floors), not field quality; the real field-quality floor still needs the owner-box run_regression against the proprietary DB OR an owner-cleared tier-1 fixture. The two gates are complementary: characterization catches behaviour-changing refactors at constant F1; quality-floor catches a true quality drop.",

691 "versioning": "schema_version:int on every committed JSON (starts at 1), mirroring the DB PRAGMA user_version migration-ladder. The shared loader (backend/eval/golden_fixtures.py) dispatches on schema_version with ordered `if version < N` upgrade shims that normalize an old fixture into the current in-memory shape (additive fields default safely ? same as experiment._feature_column defaulting an absent column to 0.0). A unit test asserts every committed fixture's schema_version <= CURRENT_SCHEMA (drift fails CI, exactly as the DB version assertion does). Schema changes are additive-only (new feature column, new provenance field) ? never a breaking rename ? so the characterization gate stays valid across versions; a genuinely breaking change re-freezes all fixtures in the same PR (freeze --refresh-snapshot --all) and bumps CURRENT_SCHEMA. LABEL versioning: the rich Roora guideline is at v8; provenance records the guideline version + the rich->minimal projection so a relabel under a new guideline version is a new gt_hash -> flagged incommensurable, forcing a deliberate re-baseline. CORPUS versioning: corpus-tag (e.g. 2026-06-n6, matching GOLDEN_DATA_SHARING's convention) lets an expanded corpus preserve old frozen numbers for reproducibility of prior published results (ties the paper-coauthor empirical-backbone aim). Inputs (trajectory.csv/labels.csv) are immutable per slug once committed ? a new behavioral scenario is a NEW slug dir, not an edit, so a diff to an existing slug is always either a rare reviewed input change or a snapshot re-freeze (the intentional-change signal).",

692 "repoSizeStrategy": "Numeric-only, comfortably git-sized, NO git-lfs (it is installed v3.5.1 but intentionally unused ? no .gitattributes; this design keeps the repo's 'derived numbers OK in git, frames/weights/DBs never' line). The trajectory CSV is the only sizeable artifact (~20-25 B/row, one row/frame); a deliberately SHORT fixture clip (synthetic ~30-90s, or real clips trimmed to a few rallies) keeps each CSV well under ~200KB; expected_*.json/provenance.json are KB-scale. A 2-3 video synthetic tier-0 corpus is <~500KB; a full owner-cleared n=6 numeric set is <~2MB. GUARD AGAINST CREEP: a unit test asserts each committed trajectory.csv row-count <= a cap (e.g. 12000 rows) so nobody commits a full 4K-match trajectory, and the freeze CLI warns past the cap. Prefer 3-5 SHORT cases over long matches ? characterization wants behavioral coverage of code branches (a quiet trajectory, a clean high-velocity rally, an occlusion gap that resets the track, a held-shuttle stretch that exercises Gate A, a multi-window video), not footage volume. output/ stays gitignored and the OneDrive sync remains the home for the FULL proprietary corpus; the repo holds only the small committed regression subset. Fixtures are tests/-only (pyproject packages=[backend,training]) so they are not packaged into the wheel ? correct, tests aren't shipped.",

693 "licensingHandling": "This is the load-bearing constraint and an OWNER-GATED reversal of GOLDEN_DATA_SHARING.md (which keeps proprietary-derived per-video artifacts in private OneDrive, never committed; guardrail 'features derive from the Proprietary corpus -> keep the OneDrive folder private', 'do not symlink the shared folder into a tracked path'). RECONCILIATION (a TWO-TIER design that lands the framework WITHOUT waiting on the decision): TIER-0 SYNTHETIC fixtures (hand-authored Frame,Visibility,X,Y trajectories + synthetic labels, no real footage, no players) are licensing/privacy-clean BY CONSTRUCTION and ship with NO owner gate ? they make both gates CI-real on day one and are sufficient for CHARACTERIZATION (we lock code behaviour, not validate real-world accuracy). TIER-1 real fixtures are added ONLY on

explicit owner sign-off, ONLY as DERIVED de-identified NUMERIC streams (frame index, pixel coords, scale/translation-invariant scalar features, interval labels) ? never frames/imagery, so no biometric/identifiable content; sourced ONLY from owned/licensed MINORS-FREE footage (founder collection / ShuttleSet, never end-user uploads, never broadcast scraping), each carrying provenance.json {license, owned_or_synthetic, minors_free:true, consent_ref}. The freeze CLI's --license/--minors-free refusal is the code-enforced gate. Detectors producing the streams (WASB/TrackNet) are MIT; NO paid-LLM (Gemini) output is committed as a training/data source (labels are Roora work-for-hire or synthetic). The PR introducing tier-1 must cite the owner approval and reconcile GOLDEN_DATA_SHARING.md (committed regression SUBSET vs the OneDrive full corpus). HONEST CAVEAT: even derived trajectories encode something about proprietary footage ? the synthetic-first default sidesteps this entirely and is the recommended path.",

694 "determinismHandling": "Freeze ONLY at pure-Python seams BEFORE every non-deterministic step. (1) FLOATS (velocity=(dist/frame_width)*(fps/dframes); rally_gate medians; normalized person motion; F1/IoU) are IEEE-754 double ops, platform-stable but asserted with pytest.approx(abs=1e-6), never exact ==; structural/integer fields (window counts, net_crossings counts, tp/fp/fn, segment counts) asserted EXACT since they carry the behavioral signal. (2) fps/frame_width are NOT in the CSV (probed from video via cv2 with SILENT 30.0/1280.0 fallbacks) -> PINNED in provenance + every feature/segment file's windowing block, so a replay never hits the fallback. (3) windowing operating point pinned to a named WindowingPreset (default SERVED, single-sourced from IndexingConfig) and stamped into expected_segments.json ? neutralizes the Gate-A eval!=serve trap (+0.074 proposal vs -0.008 served); a deliberate retune trips the incommensurability guard. (4) bootstrap CI + sign test are seeded (eval_stats default seed=0) so the honest-stats are reproducible and snapshot-able; seed is stamped in the snapshot and passed explicitly. (5) EXCLUDED entirely: provider/Gemini output + ThreadPool ordering (seam stops before the AI handoff ? CONTROL emits windows verbatim); DB autoincrement id + created_at (compare on (start,end,stats)); ffmpeg/codecs reel bytes (assert time_pairs + merge_overlaps, never bytes). (6) LINE ENDINGS: add a .gitattributes entry pinning tests/fixtures/golden_regression/** to LF (text eol=lf) so CSV/JSON are byte-identical on Windows, WSL, and Linux CI ? this is the one new (and contained) .gitattributes use and the cross-OS linchpin; parse_trajectory_csv already opens with newline='' so CRLF/LF never changes parsed rows. (7) gt_loader reads utf-8-sig (BOM-tolerant) ? keep fixtures UTF-8. RESIDUAL RISK (stated plainly): pytest.approx(abs=1e-6) is a small bounded blind spot for sub-tolerance float drift ? chosen far below any behaviorally meaningful change in seconds/F1, and integer/structural fields stay exact, so it is negligible. Person per-frame detections (cv2 frame-seek ±frames by codec) are hand-authored for the synthetic tier (fully deterministic); real person fixtures carry a codec-seek caveat in provenance and stay default-OFF.",

695 "ciIntegration": "CORE GOAL NEEDS ZERO NEW CI WIRING: both gates are plain pytest auto-discovered by the existing `testpaths = tests` / run_all_tests.py lane that already runs in .github/workflows/ci.yml under the 70% coverage floor, and they run in the no-GPU/no-cv2/no-network lane because the committed fixtures need none of those. KEY: import only the pure windowing/metrics/experiment path (parse_trajectory_csv, trajectory_to_action_windows, rally_gate, distill_local.build_candidates, experiment, metrics, segmentation_metrics) so the gates run even without cv2 ? preferred over pytest.importorskip('cv2') so they DO execute in CI (the whole point: test_served_gate_a_regression skips today). The synthetic tier-0 guarantees fixtures exist in CI. OPTIONAL (second PR, redundant with the pytest assertion which already gates merges): a dedicated `golden\_regression` CI job for a clean red/green signal, and/or a standalone `python -m backend.eval.freeze\_fixture --check` exit 0/1/3 CLI mirroring run_regression.py for a pre-PR local gate. SEPARATE owner-box-only workflow (out of PR CI): the existing run_regression.py against the real proprietary DB for the full corpus ? the DB path stays the owner-box gate. No custom pytest marker is registered (rely on parametrize + path-glob + skipif, matching the zero-conf test/zero-custom-marker convention) to avoid PytestUnknownMarkWarning.",

696 "contributorWorkflow": "CONTRIBUTOR (any machine, no video/GPU): clones, runs `python run\_all\_tests.py`. A behaviour-preserving refactor -> both gates green -> push with confidence. A failure shows exactly which slug/field changed (e.g. 'held_shuttle expected_segments f1 0.157->0.142' or 'kushagra net_crossings count changed'). They then decide: (a) unintended side-effect -> fix code until the snapshot matches; or (b) intended change -> run `python -m backend.eval.freeze\_fixture --refresh-snapshot --all` (regenerates expected_features/expected_segments from the committed trajectory+labels, NO re-detection) and, if quality moved, `run\_regression.py --update-baseline`/re-run the floor write ? BOTH land in the SAME PR as the code change, the diff reviewed. The PR description states which gate moved and why (a characterization diff is mechanical; a floor change cites the QUALITY_ITERATIONS ablation). REVIEWER: if expected_*.json is unchanged, the change is provably behaviour-preserving across the frozen seam (within documented limits); if changed, the precise before/after numbers are in the diff to judge. Rollback of a bad change = revert the code;

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fixtures/baseline are the source of truth, never hand-edited. ADD a video: freeze_fixture --slug
<s> --license owned --minors-free --consent-ref <doc> -> writes the 5 artifacts + appends to
manifest.json -> commit explicit paths. BRANCH HYGIENE (shared-checkout safety, per CLAUDE.md):
git fetch origin; git checkout -b <name> origin/master; stage explicit paths only (the fixtures
+ tool + tests + baseline ? NEVER git add -A, and avoid the untracked artifacts/ dir + weights);
re-verify HEAD before commit; one PR per phase, no stacking; do not open a PR until the owner
asks. README.md in the fixtures dir documents this loop + the minors/licensing guardrail so a
future contributor cannot freeze an unvetted video.",
697     "growthStory": "The corpus auto-grows with near-zero marginal effort because the
SAME fusion_golden extractor the owner already runs per golden video emits the committed shape ?
'add a video' is freeze_fixture --slug <s> + commit, and the new video is instantly a CI-
enforced regression case (both gates pick it up via the manifest glob). As the Roora golden
corpus grows (n=6 -> larger), each new owner-cleared video is one more parametrized case; the
honest-stats (bootstrap CI + sign test, seed=0) tighten as n rises, so the quality-floor gate's
verdicts get more trustworthy automatically. NEW CUES extend without regression: a future cue
(audio, shuttle-drop) is just a new named feature block in expected_features.json + a Variant
feature_name ? the same gates score it default-OFF with zero regression (the weak-signals-
retained directive: non-significant cues stay retained-as-columns, never deleted on a small-n
null). NEW WINDOWING/operating points are new presets stamped into new fixtures, preserving old
frozen numbers under their corpus-tag (reproducibility of prior published results ? the paper-
coauthor backbone). NEW DETECTORS reuse the same trajectory-CSV seam (the schema is detector-
agnostic via the detector string in provenance). The schema_version ladder absorbs additive
growth so old fixtures keep loading. Net: the framework is built once on synthetic data, then
scales by accretion ? every golden video the vendor pipeline produces becomes a permanent,
video-free, CI-runnable regression anchor.",
698     "rolloutPhases": [
699         {
700             "phase": "PR-1: Framework + synthetic tier-0 (NO owner gate ? ships
immediately)",
701             "deliverable": "tests/fixtures/golden_regression/ + README.md (guardrails +
freeze/refresh loop); backend/eval/golden_fixtures.py
(load_manifest/load_fixture/replay_fixture, schema_version dispatch, Path(__file__)-relative,
CURRENT_SCHEMA=1); a .gitattributes LF pin for the fixtures dir; 3-5 hand-authored SYNTHETIC
tier-0 fixtures (clean rally / held-shuttle Gate-A cut / occlusion break / quiet trajectory /
multi-window), tool-generated expected_*.json + provenance.json; manifest.json.",
702             "prScope": "One PR branched from origin/master. Establishes the committed-
fixture convention + the shared loader on licensing-clean synthetic data. No scoring math added.
Stage explicit paths only."
703         },
704         {
705             "phase": "PR-2: The freeze/refresh extractor + its unit tests",
706             "deliverable": "backend/eval/freeze_fixture.py composing
distill_local.build_candidates + rally_gate + experiment.predict(CONTROL)+merge_overlaps +
regression.gt_hash; --license/--minors-free refusal gate; --refresh-snapshot[--all]; --check
exit 0/1/3 (optional). Unit test: synthetic in -> deterministic artifacts out; re-running
--refresh-snapshot is idempotent.",
707             "prScope": "One PR. The tool is the single source of truth for both generation
and the gates' replay (shared helper). Re-generates PR-1's synthetic fixtures so they are tool-
produced, not hand-typed."
708         },
709         {
710             "phase": "PR-3: Characterization gate",
711             "deliverable": "tests/test_golden_characterization.py ? parametrized over
manifest.json, recompute features+segments from the frozen trajectory under the pinned preset,
assert exact-on-structural + approx(1e-6)-on-float match vs
expected_features.json/expected_segments.json; incommensurability/preset/hash guard; assert no
DB-id/timestamp/byte fields touched. Plus the schema_version-drift test and the trajectory row-
count cap test.",
712             "prScope": "One PR. The refactor-side-effect detector. Runs in CI on the
synthetic tier; confirm green + coverage >=70% locally."
713         },
714         {
715             "phase": "PR-4: Quality-floor gate + the first committed
regression_baseline.json",
716             "deliverable": "tests/test_golden_quality_floor.py ? load substrate via

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fusion_compare.load_videos shape, score CONTROL/shuttle_only/fusion vs labels, assert per-
video+mean F1 >= committed floor; emit eval_stats CI + sign test in the failure message. Create
eval_baselines/regression_baseline.json (finally fills the reserved-but-absent path) keyed
<slug>::final via regression.save_baseline with iou/detector/gt_hash stamps.",
717     "prScope": "One PR. Ties the floor to QUALITY_ITERATIONS/ablation. Synthetic-
tier floors measure correctness; documented as such."
718     },
719     {
720         "phase": "PR-5 (DOCS): document the framework + update CODE_MAP",
721         "deliverable": "New docs/GOLDEN_REGRESSION_FIXTURES.md (or a section in
EVALUATION.md) covering what the gates DO and do NOT guarantee (false-safety limits), the
freeze/refresh workflow, the reviewer read-the-diff responsibility, the two-tier licensing
reconciliation; cross-link from EXPERIMENT_HARNESS.md/GOLDEN_DATA_SHARING.md; add the
tests/fixtures/golden_regression/ row to CODE_MAP $0 file-placement table.",
722         "prScope": "One docs PR (or folded into PR-1/PR-3). Keeps live docs honest."
723     },
724     {
725         "phase": "PR-6 (OWNER-GATED, separate): tier-1 owner-cleared real fixtures",
726         "deliverable": "On explicit owner sign-off: add owner-cleared, minors-free,
de-identified real NUMERIC fixtures via freeze_fixture (citing the approval), and reconcile
GOLDEN_DATA_SHARING.md (committed regression subset vs OneDrive full corpus). Lights up the
real-footage edge cases (occlusion, codec jitter) the synthetic tier cannot.",
727         "prScope": "One PR, owner-approved. Purely additive to the framework already
shipped in PR-1..5. Do not land unilaterally ? reverses a product/privacy convention."
728     }
729 ],
730     "recommendedDocHome": "Create a NEW doc `docs/GOLDEN_REGRESSION_FIXTURES.md` as
the canonical home (the framework is distinct enough ? committed video-free fixtures + dual
gates + freeze tool ? that bolting it into EVALUATION.md would bury it). It must: (1) state
plainly what the gates DO and do NOT guarantee (the characterization false-safety limit ? green
= deterministic POST-trajectory behaviour unchanged for the frozen cases, NOT 'correct' and NOT
covering the detector/video-IO/provider/stitch); (2) document the freeze/refresh contributor
loop and the reviewer's read-the-diff responsibility; (3) record the two-tier licensing
reconciliation with GOLDEN_DATA_SHARING.md. CROSS-LINK it from: docs/EVALUATION.md (the scoring
mechanics), docs/EXPERIMENT_HARNESS.md (the Variant/compare engine it reuses + the P3 CI-
promotion-gate it complements), docs/GOLDEN_DATA_SHARING.md (the private-OneDrive convention it
partially reverses for the committed subset), and QUALITY_ITERATIONS.md (the ablation the
quality-floor ties to). ALSO add the `tests/fixtures/golden_regression/` row to the CODE_MAP $0
file-placement table and a TEST MAP entry. Update docs/README.md doc index + the memory current-
state/session-handoff at the PR-1 checkpoint.",
731     "openOwnerQuestions": [
732         "PRIMARY (product/privacy, gates tier-1): Approve reversing
GOLDEN_DATA_SHARING.md for a TINY committed regression SUBSET of DERIVED, de-identified, minors-
free, owned-footage NUMERIC streams (trajectory rows + interval labels + scalar features ? never
frames)? The framework (PR-1..5) ships on synthetic data without this; only tier-1 real fixtures
need sign-off.",
733         "Which n of the owned n=6 golden videos (if any) are cleared as minors-free +
license-clean for a committed numeric subset, and is ShuttleSet's license compatible with
committing DERIVED trajectory/feature numbers to a (public?) repo?",
734         "Is this repo public or private? If public, even derived numeric streams from
proprietary footage are a stronger disclosure than if private ? changes the tier-1 risk calculus
and possibly the synthetic-only decision.",
735         "Should the committed quality-floor numbers
(eval_baselines/regression_baseline.json) be tied to the SERVED windowing preset only, or also
track the PROPOSAL preset (where Gate-A's +0.074 lives)? Recommendation: freeze SERVED as the
floor (the operating point users get), record PROPOSAL as a documented control ? confirm.",
736         "Confirm the floor tolerance (regression.DEFAULT_TOLERANCE=0.05) and the
trajectory row-count cap (~12000) are acceptable, or set preferred values.",
737         "Should tier-1 real fixtures key by video_id[:12] (the production MD5 key, so a
fixture maps to a real DB row) or by an opaque de-identification slug (stronger privacy)? Trade-
off: traceability vs disclosure."
738     ],
739     "honestLimits": [
740         "CHARACTERIZATION = behaviour-locking, NOT correctness. A green snapshot proves
only that DETERMINISTIC POST-TRAJECTORY behaviour is unchanged for the FROZEN cases; it encodes

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CURRENT behaviour including any current bugs. 'Passing' means 'behaviour unchanged', never 'behaviour correct'. Easy to misread as a quality gate ? the docs must say this loudly.",

741 "The gates do NOT cover: the WASB/TrackNet detector itself (needs video/GPU), video-IO/decode (_probe_fps_width, frame seeking, ffmpeg), the AI/provider handoff (excluded/stubbed), reel-byte output (excluded), or any code path the fixtures do not exercise. A refactor that breaks decoding or the detector passes these gates green. The owner-box run_regression against the real DB remains the only check of the full chain.",

742 "Coverage is only as good as the fixture cases ? an untested windowing/Gate-A branch has zero protection; adding a case when a new code path appears is a discipline requirement, not enforced.",

743 "--refresh-snapshot can be abused: a contributor can rubber-stamp a real regression by regenerating the snapshot without justification. The mitigation is social/review (the diff is visible and must be explained), not technical ? the gate's strength depends on the reviewer actually reading snapshot diffs.",

744 "Synthetic tier-0 exercises the PLUMBING faithfully but will NOT reproduce real-footage failure modes (occlusion breaks, codec frame-seek jitter, real velocity distributions); tier-0 quality-floor numbers are about pipeline correctness, not field quality. Real field quality needs owner-cleared tier-1 or the owner-box DB run.",

745 "Two sources of truth still coexist: the DB-backed run_regression (owner-box, full chain) and these file-backed gates (CI, post-trajectory only). This design picks the trajectory/feature substrate and records its provenance, but a contributor must understand the DB path is a separate, owner-box-only gate ? they are not interchangeable.",

746 "splits.assert_disjoint is NOT yet wired into this path; if a committed fixture ever both trains a pinned scorer AND serves as a regression baseline, train/eval leakage is possible until that guard is connected (recommend wiring it when/if a learned variant is pinned against these fixtures).",

747 "pytest.approx(abs=1e-6) is a small bounded blind spot for sub-tolerance float drift (mitigated by exact-on-structural-fields). And the dev container here cannot run the full pipeline (no GPU/torch/cv2/numpy) ? the synthetic-tier gates are verifiable here, but tier-1 extraction and the owner-box DB regression were NOT verifiable in this environment."

748]

749 },

750 "critiques": [

751 {

752 "lens": "Hostile senior reviewer hunting for FALSE CONFIDENCE: a green suite that an external contributor on a different machine/OS trusts, but which either (a) flickers red on correct code, or (b) stays green while real behavior broke. I verified claims against the actual code (classifier.py uses numpy/OpenBLAS DYNAMIC_ARCH gradient descent; metrics.py greedy matcher; rally_gate net_crossings; windowing presets; eval_stats bootstrap; regression.py guard contract) and the live environment (repo is PRIVATE; core.autocrlf=true; no .gitattributes; numpy 2.4.6 present). The design is unusually self-aware (its honestLimits are mostly correct) but it materially understates two things: cross-machine non-determinism of the LEARNED-variant path, and the false-safety surface.",

753 "criticalIssues": [

754 {

755 "issue": "DETERMINISM ? FATAL for the quality-floor gate (and the characterization gate's structural-exact assertions when learned variants are involved). The design repeatedly asserts the frozen seam is 'pure-Python IEEE-754, platform-stable' and dismisses sub-1e-6 float drift as 'negligible, mitigated by exact-on-structural-fields.' This is FALSE for the path it actually exercises. qualityFloorGate scores experiment.compare on the shuttle_only and shuttle_person_fusion variants (fusion_compare.py), both of which are is_learned() (they set feature_names). experiment.predict -> LogisticRegression.fit/predict_proba (backend/eval/classifier.py) runs 1000 epochs of gradient descent built on numpy matmul (Xs @ self.w), Xs.T @ err, and X.std(axis=0). I checked the installed wheel: numpy 2.4.6 links OpenBLAS 0.3.31 with DYNAMIC_ARCH ? the BLAS kernel (and thus float summation order) is chosen at runtime per CPU microarch (Haswell vs Zen vs AVX-512 vs Apple Accelerate on ARM). Over 1000 GD iterations these tiny per-op differences COMPOUND. The resulting proba values sit right at the 0.5 gate (I measured 0.490 / 0.520 / 0.583 on synthetic data). predict() then does `proba[i] >= thr` ? a hard cliff. A ~1e-7 drift in the logit flips a candidate's keep/drop, which changes the kept set, changes merge_overlaps output, and changes the INTEGER tp/fp/fn. So the gate that the design says is protected by 'EXACT equality on structural/integer fields' is precisely the one that will flip red on a contributor's different-CPU machine even though their code is byte-identical and correct. This is a false-RED generator that will erode all trust in the suite the first week. The bootstrap CI in the failure message (eval_stats seed=0) is fine; the model training is not.",

```

756         "severity": "critical",
757         "recommendation": "Do NOT freeze any snapshot that flows through
LogisticRegression across machines. Three concrete options, in order: (1) BEST ? keep the
characterization gate STATIC-ONLY: freeze and assert only the CONTROL variant path (gate ->
merge_overlaps -> metrics.score), which is genuinely pure-Python/stdlib (no numpy), plus
net_crossings (pure). Move all learned-variant scoring out of the cross-machine CI gate. (2) If
learned variants MUST be in CI, pin the threshold decision away from the 0.5 cliff is not enough
? instead assert learned-variant outputs ONLY with interval-level tolerance and NEVER assert
exact tp/fp/fn for them; and pin OPENBLAS_NUM_THREADS=1 + set a documented BLAS, accepting it
still won't be bit-stable across ISA. (3) Freeze the *_golden_features.json substrate (already
the fusion_compare seam) AND a frozen pinned classifier_model JSON, then score with the SHIPPED
model (no per-fold fit) so at least predict_proba is one matmul not 1000 GD steps ? still BLAS-
sensitive at the threshold, so combine with (1)'s tolerance rule. Whichever path: replace every
`>= threshold` exact-equality structural assertion on a learned variant with a recorded-proba-
margin check, and state in the doc that learned-variant numbers are owner-box-only, not CI-
frozen."
758     },
759     {
760         "issue": "FALSE SAFETY ? the characterization gate's central selling point
('green = behaviour-preserving refactor, changed snapshot = the only intentional-change signal')
is materially oversold and will lull reviewers. The frozen seam starts at the trajectory CSV, so
the gate is BLIND to: the WASB/TrackNet detector, _probe_fps_width and all cv2 frame-seeking,
the AI/provider handoff, chunk extraction/re-encode, the ffmpeg stitch, _select_for_handoff, DB
persistence/migrations, and the API filter (stitching.min_shot_count). The design lists these in
honestLimits, which is good ? but the contributorWorkflow section then tells contributors 'both
gates green -> push with confidence' and tells reviewers 'unchanged expected_*.json => provably
behaviour-preserving across the frozen seam.' Those two framings fight each other: a refactor of
trajectory_to_action_windows that ALSO touches _probe_fps_width, or that changes how
fps/frame_width are read, passes green because the fixture PINS fps/frame_width and never calls
the prober. The most likely real-world refactor (someone 'cleaning up' the velocity
normalization or the occlusion-reset branch) is covered ONLY if a fixture happens to exercise
that exact branch ? and coverage is admittedly a discipline, not enforced.",
761         "severity": "high",
762         "recommendation": "Make the doc's headline claim symmetric with the limits:
'green means the FROZEN POST-TRAJECTORY transforms are unchanged for the committed cases ? it
does NOT certify a refactor that also touches video IO, fps/width probing, the detector,
persistence, selection, the provider, or the stitch.' Add a hard, enumerated list of the code
symbols the gate DOES cover (parse_trajectory_csv, trajectory_to_action_windows,
rally_gate.gate_windows, distill_local.build_candidates, merge_overlaps, metrics.score,
segmentation_metrics) and a CI line-coverage assertion that those modules stay >= a floor under
the gate, so 'an untested branch has zero protection' becomes visible rather than implicit.
Remove 'push with confidence' from the contributor instructions; replace with 'green means the
frozen transforms are unchanged; you still need the owner-box DB run for the full chain.'"
763     },
764     {
765         "issue": "DETERMINISM ? the .gitattributes LF pin is necessary but the
design treats it as the whole cross-OS story, and it interacts dangerously with the
contributor's environment. I confirmed there is NO .gitattributes today and the owner's git has
core.autocrlf=true globally. The pin `tests/fixtures/golden_regression/** text eol=lf` only
governs files UNDER that path. But the freeze tool (backend/eval/freeze_fixture.py) and the
loader write/read JSON via json.dump/open ? if a contributor on Windows with autocrlf=true
regenerates a snapshot via --refresh-snapshot, the NEW file content is produced by Python (LF in
the bytes json.dump writes) but git's checkout/normalization and any editor touch can still
reintroduce CRLF on non-pinned siblings, and the row-count cap test / schema test read with
default text mode. parse_trajectory_csv opens newline='' (verified line 240) so trajectory
parsing is safe, but the EXPECTED json files are compared by the design via 'match the committed
files' ? if that comparison is a byte/string compare rather than a parsed-structure compare,
CRLF vs LF will false-RED on Windows.",
766         "severity": "high",
767         "recommendation": "Never byte-compare committed JSON. The characterization
gate must json.load both sides and compare PARSED structures (with the documented exact-on-int /
approx-on-float policy), so line endings and key insertion order are irrelevant. Keep the
.gitattributes LF pin AND add `*.json text eol=lf` / `*.csv text eol=lf` scoped to the fixtures
dir, AND add a CI assertion (or a pre-commit) that each committed fixture file contains no CRLF
bytes, so a contributor who commits CRLF fails loudly with a clear message instead of

```

flickering. Add a test that round-trips freeze -> load -> assert on the SAME machine to catch the refresh-introduced-CRLF case."

48. assistant (2026-06-16T02:56:27.315Z)

Excellent depth ? the first workflow's adversarial critic caught a genuinely sharp issue (LogisticRegression gradient descent over OpenBLAS `DYNAMIC\_ARCH` isn't bit-stable across CPUs ? would false-RED for external contributors). Let me read the rest of the critiques.

49. user (2026-06-16T02:56:30.140Z)

```
768         },
769         {
770             "issue": "FALSE SAFETY ? --refresh-snapshot is a rubber-stamp hole the
design acknowledges but under-mitigates, and it is WORSE than stated because of issue #1. The
design says a contributor could regenerate the snapshot to mask a regression and the mitigation
is 'social/review.' But combine with the BLAS non-determinism: a contributor whose machine
legitimately produces different learned-variant numbers will be TOLD by the red gate to run
--refresh-snapshot --all (that is literally the documented remedy for an 'intended change').
They will refresh in good faith, committing THEIR machine's BLAS-specific numbers, which then
false-RED on the next contributor / on CI. The 'intentional-change signal' becomes noise: every
cross-machine PR carries a snapshot diff that is actually just float/threshold churn, so
reviewers learn to ignore snapshot diffs ? destroying the one property the design is built to
provide.",
771             "severity": "high",
772             "recommendation": "This is the same root cause as #1 and is fixed by the
same move: keep learned-variant numbers OUT of the cross-machine frozen snapshot. Once only
CONTROL/pure-stdlib transforms are frozen, a snapshot diff genuinely means a behavior change and
--refresh-snapshot is safe and meaningful. Additionally: make --refresh-snapshot REQUIRE a
reason flag recorded into provenance.frozen_reason and have the gate print the prior vs new
values, so a refresh is self-documenting in the diff. Consider a CI guard that fails if
expected_*.json changed in the same commit as a refactor WITHOUT a provenance reason update."
773         },
774         {
775             "issue": "LICENSING/PRIVACY ? the tier-1 reconciliation is sound in spirit
but the design lets the FRAMEWORK ship (PR-1..5) in a way that quietly normalizes committing
proprietary-derived numbers, and the ShuttleSet question is unanswered. The repo is PRIVATE (I
confirmed isPrivate=true), which the design correctly flags as lowering risk ? but private today
is not private forever (a repo can be open-sourced; the owner's stated paper-coauthor aim
implies eventual publication of methods/possibly data). Committed git history is FOREVER: if a
tier-1 trajectory from owned/ShuttleSet footage lands and the repo later goes public, the
derived stream is permanently in history even after a delete. The design's own
GOLDEN_DATA_SHARING.md guardrail ('never commit proprietary-derived per-video artifacts') is
being reversed for tier-1, and the open question 'is ShuttleSet's license compatible with
committing DERIVED trajectory/feature numbers to a (public?) repo' is exactly the kind of thing
that must be answered BEFORE the framework conditions everyone to `freeze_fixture --slug
<real>`.",
776             "severity": "medium",
777             "recommendation": "Land PR-1..5 SYNTHETIC-ONLY exactly as designed (this is
genuinely clean and the right call). For tier-1, do not rely only on the freeze CLI's --minors-
free/--license refusal (a flag a human asserts). Add: (1) a written ShuttleSet redistribution-
of-derivatives determination in the PR that introduces tier-1, citing the actual license text ?
derived numeric features are usually fine but trajectory coordinates re-tracing the original are
closer to a derivative; get this in writing. (2) Default tier-1 keys to an OPAQUE de-
identification slug, NOT video_id[:12] ? the MD5 prefix is a re-identification handle against
the proprietary DB and offers no benefit in a video-free fixture. (3) Add a note in the doc:
'committing to git is permanent even if later deleted; treat tier-1 as a public-disclosure-
equivalent decision regardless of current repo visibility.'"
778         },
779         {
780             "issue": "SCHEMA/LABEL DRIFT ? the additive-only + schema_version ladder is
a reasonable mirror of the DB migration discipline, but the incommensurability guard has a gap
that produces SILENT staleness rather than a loud failure. The design pins WindowingPreset by
name+tuple and SERVED is single-sourced from IndexingConfig() defaults (windowing.py
_served_preset_from_defaults reads idx.wasb.velocity_threshold etc.). If a future owner changes
```

a served default in config/models.py (entirely plausible ? that's what config defaults are for), SERVED's TUPLE changes but its NAME stays 'served'. A fixture frozen with windowing_preset.name=='served' will still match by name, but the recomputed windows now differ -> characterization gate red. That is correct (loud). BUT the quality-floor baseline keyed `<slug>::final` stamps only iou/detector/gt_hash (regression.save_baseline signature, verified) ? it does NOT stamp the windowing tuple. So a served-default change is caught by characterization but the floor baseline's gt_hash/detector/iou are unchanged -> regression.regress does NOT flag incommensurable -> the floor silently compares numbers measured at a DIFFERENT operating point. That is the exact eval!=serve Gate-A trap the windowing module was built to prevent, re-introduced at the floor layer.",

```

781         "severity": "medium",
782         "recommendation": "Extend the floor baseline record (or its key) to include
the windowing fingerprint, not just iou/detector/gt_hash. Either add a windowing_hash to
save_baseline's stored dict and assert it in the quality-floor gate's incommensurability check,
or encode the preset into the stage key (`<slug>::final@served:<tuplehash>`). Add a unit test
that mutating IndexingConfig's velocity default trips BOTH gates as incommensurable, not just
characterization. Also add the schema_version<=CURRENT_SCHEMA drift test the design promises AND
a test that an old (schema_version=1) fixture still loads through the upgrade shim, or the
'additive-only' claim is untested."
783     },
784     {
785         "issue": "DETERMINISM ? rally_gate net_crossings is asserted as an
'integer/structural exact' field, but its VALUE depends on float-derived deadband and
median/spread estimates over the whole trajectory, which are sort-based and stable in pure
Python ? EXCEPT the design re-derives net_crossings at test time (chosenSeams: 'freeze at the
CSV and DERIVE features at test time') via rally_gate.gate_windows with estimate=None, which
calls estimate_play_axis -> _median/_spread over ALL visible points. _points_in (rally_gate line
87-88) selects window points by `start*fps <= p.frame <= end*fps` ? a float comparison against
integer frames. The window start/end come from trajectory_to_action_windows as `group[0][0]/fps
+ chunk_start` (float division). So start*fps is float(frame/fps)*fps, which is NOT guaranteed
to equal the original integer frame exactly (e.g. 100/30.0*30.0 may be 99.99999999999999). A
point exactly on the boundary frame can be included on one machine's rounding and excluded on
another's, changing the integer crossing count. The design asserts net_crossings EXACT.",
786         "severity": "medium",
787         "recommendation": "This one is low-probability on a single machine but is a
real cross-platform boundary hazard for an EXACT integer assertion. Either (a) freeze and assert
net_crossings as a recorded value with the windows it was computed from, computing it ONCE at
freeze time and NOT re-deriving the play-axis at test time (store the (axis, net, span) estimate
in provenance and pass it via the gate_windows `estimate=` param the code already supports), or
(b) make _points_in robust to the float round-trip (compare on rounded frames) before relying on
EXACT equality. Add a fixture whose rally boundary lands exactly on an integer frame to actually
exercise this."
788     }
789 ],
790     "missingPieces": [
791         "A concrete answer to 'what does the characterization gate compare and HOW' ?
the design says 'assert the recomputed expected_*.json match the committed files' but never
specifies parsed-structure comparison vs string/byte comparison. This is load-bearing: byte
comparison reintroduces the CRLF and float-formatting-stability problems the design claims to
have solved. Specify: json.load both sides, walk the structure, exact on declared int keys,
approx on declared float keys.",
792         "Float SERIALIZATION stability is unaddressed. The design stores floats
'rounded to 6 dp for byte-stable diffs' but Python's round() + json.dump can still emit 0.157 vs
0.15700000000000001-class artifacts depending on the repr path, and round(x,6) is itself a float
(not a decimal). If the gate byte-compares, this flaps; if it parses-and-approx-compares, the
6dp rounding is pointless. Pick one coherent story (recommend: store full-precision floats,
compare parsed with approx; drop the 6dp rounding theater).",
793         "No statement of which numpy/BLAS the frozen learned-variant numbers were
computed against, nor any pin (OPENBLAS_NUM_THREADS, thread count affects reduction order too).
Even if learned variants are kept (against my recommendation), thread-count non-determinism
within ONE machine is unaddressed.",
794         "No test that the freeze tool is actually idempotent on the SAME machine
(freeze then refresh produces byte-identical output). The design claims '--refresh-snapshot is
idempotent' as a PR-2 unit test deliverable but idempotency on one machine does not imply cross-
machine stability ? the doc should not let the idempotency test be read as a determinism

```

guarantee.",

795 "The design never says whether the synthetic tier-0 trajectories will actually EXERCISE the learned-variant code at all. If tier-0 has too few candidates/videos, experiment.compare with honest LOVO raises ExperimentError ('LOVO needs >=2 videos') and _train raises on zero candidates. The quality-floor gate on a 3-5 video synthetic tier must guarantee >=2 videos each with >=1 labeled positive candidate or it errors rather than measures. No fixture-validity precondition test is specified.",

796 "No migration/repo-history size accounting. The design caps trajectory rows at 12000 and estimates <2MB for n=6, which is fine ? but git stores every REFRESH of every expected_*.json forever. With learned-variant numbers churning per-machine (issue #1/#4), refresh-driven history growth is unbounded in a way the static-only design would avoid. Worth a sentence: prefer few refreshes; static-only minimizes churn."

797],

798 "falseSafetyRisks": [

799 "GREEN-BUT-BROKEN: a refactor to _probe_fps_width, cv2 frame-seeking, or the WASB runner ships green because the fixtures pin (fps, frame_width) and never invoke the prober or the detector ? yet production windowing silently shifts. The gate cannot see it; the contributor is told 'push with confidence.'",

800 "GREEN-BUT-BROKEN: trajectory_to_action_windows treats visible==False as a hard track break (line 285) and silently clamps fps<=0->30 / width<=0->1280 (lines 275-278). A refactor that changes the clamp or the occlusion-reset is invisible unless a fixture specifically has a degenerate fps/width or an occlusion gap on the exact boundary that matters. Coverage of these branches is a discipline, not enforced ? so 'green' overstates safety.",

801 "GREEN-BUT-BROKEN at the floor layer: a change to a served windowing default (config/models.py) is NOT flagged incommensurable by the quality-floor baseline (it stamps only iou/detector/gt_hash, not windowing) ? the floor compares numbers from two different operating points and can pass, re-creating the eval!=serve Gate-A trap the project already got burned by.",

802 "RED-BUT-CORRECT (trust-destroying): the learned-variant path (shuttle_only/fusion through numpy OpenBLAS DYNAMIC_ARCH gradient descent) produces threshold-cliff-sensitive proba near 0.5; an external contributor on a different CPU ISA gets different integer tp/fp/fn on byte-identical correct code. The gate goes red, the contributor refreshes the snapshot as instructed, and now the snapshot is THEIR machine's numbers ? every cross-machine PR carries spurious snapshot churn until reviewers stop reading snapshot diffs. The intentional-change signal is destroyed.",

803 "FALSE 'NO SIDE EFFECTS' READING: the design asserts (PR-3) 'no DB-id/timestamp/byte fields touched' but the gates never touch the DB at all, so this assertion is vacuous ? it proves nothing about the production persistence path (add_candidate_segment / add_play_segment / migrations), which a reviewer might assume is covered because the design talks about mirroring the DB schema discipline.",

804 "SYNTHETIC-FLOOR MISREAD: tier-0 quality-floor numbers measure pipeline plumbing correctness, not field quality; the design says so in honestLimits but the gate's mere existence ('per-video AND mean F1 >= floor') invites a contributor to read a green floor as 'detection quality is fine,' which it categorically is not on synthetic data."

805],

806 "verdict": "REJECT AS SPECIFIED; APPROVE A NARROWED VERSION. The core instinct is right and the seam choice (trajectory CSV) is correct, but the design as written will manufacture false confidence in two directions and must be cut down before it ships. THE ONE FATAL FLAW: the design's central determinism claim ('pure-Python IEEE-754, freeze at CSV, structural fields EXACT, float drift negligible') is FALSE for the learned-variant path it routes through both gates. I verified backend/eval/classifier.py runs 1000 epochs of numpy gradient descent on a wheel that links OpenBLAS with DYNAMIC_ARCH (kernel chosen per CPU microarchitecture), and the resulting proba land on the 0.5 threshold cliff ? so an external contributor on a different ISA gets different INTEGER tp/fp/fn on correct code, and the documented remedy (--refresh-snapshot) bakes their machine's numbers into the 'source of truth.' That is the exact opposite of the goal: a regression suite that flickers red on correct refactors and whose snapshot diffs become noise reviewers learn to ignore. FIX: keep the cross-machine frozen seam STATIC-ONLY ? CONTROL variant (gate -> merge_overlaps -> metrics.score, genuinely stdlib) plus the pure net_crossings/window features; assert learned-variant numbers only on the owner box, never as CI-frozen snapshots, or freeze them with interval tolerance and an explicit 'BLAS-sensitive, owner-box-authoritative' caveat. THE SECOND PROBLEM: false safety is real and partly mitigated only in the honestLimits section while the contributor/reviewer prose oversells ('push with confidence', 'provably behaviour-preserving'); the floor baseline's incommensurability guard is missing the windowing fingerprint, silently re-opening the eval!=serve Gate-A trap. WHAT'S GOOD AND SHOULD SHIP: synthetic tier-0 (licensing-clean by

construction), the two-home split (fixtures under tests/, floor in eval_baselines/), reuse of the shipped scoring/guard stack verbatim, the .gitattributes LF pin (necessary ? autocrlf=true is set on this machine and there is no .gitattributes today), schema_version ladder, and the owner-gated tier-1 deferral. NET: land PR-1.5 with (a) the cross-machine gate restricted to the pure/stdlib CONTROL path, (b) parsed-structure (never byte) JSON comparison, (c) a windowing fingerprint added to the floor baseline's incommensurability check, (d) the contributor/reviewer docs rewritten to match the honestLimits, and (e) tier-1 ShuttleSet-derivative licensing answered in writing before any real fixture. Without (a) and (b) this design ships false confidence; with them, it is a genuine and valuable refactor-safety net for the deterministic post-trajectory transforms ? which is all it should ever claim to be."

```
807     },
808     {
809         "lens": "Verified the design's load-bearing claims against actual code
(regression.py, experiment.py, windowing.py, distill_local.py, fusion_golden.py,
fusion_compare.py, gt_loader.py, served_gate_a test, ci.yml, CODE_MAP section 0). The seam
choice and reuse story are sound and largely accurate. Reviewed for missing pieces across CI
gating during the billing outage, snapshot-update workflow, fixture regeneration if the detector
improves, Roorra v8 label-version tracking, alignment with QUALITY_ITERATIONS /
EXPERIMENT_HARNESS / weak-signals-retained, metrics-vs-base-branch comparison, multi-detector
handling, contributor onboarding, test-runtime growth, and file-placement. Weighted by whether
each blocks the stated goal (red/green on a clean clone) vs. degrades trust over time; flagged
two concrete code-vs-design discrepancies.",
810         "criticalIssues": [
811             {
812                 "issue": "LOADER MISMATCH. The GT seam says labels.csv loads via
gt_loader.load_gt_intervals(strict=True) so a malformed label fails loudly with file:line. But
the replay path the design reuses (distill_local.load_video, fusion_golden.extract_video,
fusion_compare.load_videos) loads labels through calibrate_wasb.load_golden_gts, the LENIENT
loader that skips bad rows and only logs. A malformed or silently-shrunk committed label would
NOT fail the quality-floor gate; it would flatter precision. The two gates disagree about label
validity.",
813                 "severity": "high",
814                 "recommendation": "Re-point the freeze tool AND both gates GT load to
gt_loader.load_gt_intervals(strict=True), or add a freeze-time validation in freeze_fixture.py
that strict-loads every labels.csv and refuses to write on any malformed row, recording the
gt_hash. State in GOLDEN_REGRESSION_FIXTURES.md which loader each gate uses so they cannot
diverge."
815             },
816             {
817                 "issue": "CHARACTERIZATION SNAPSHOT IS NOT COMPUTED THROUGH THE WINDOWING
BRAIN IT CLAIMS TO LOCK. experiment.predict(CONTROL, vc) operates on the already-built
vc.intervals and only applies gate, learned-filter, merge; it never calls parse_trajectory_csv
or trajectory_to_action_windows. As the schema (item 4) and extractionTool (step 3) text reads,
expected_segments.json computed via predict(CONTROL)+merge_overlaps freezes windows passed IN,
not re-derived, so the windowing brain (the code most likely refactored) is UNPROTECTED. The
protection only exists if the gate first runs parse_trajectory_csv then windowing.build_windows
then distill_local.build_candidates. characterizationGate says it does; the freeze/schema text
implies predict alone. Internally inconsistent.",
818                 "severity": "high",
819                 "recommendation": "Funnel freeze and both gates through ONE replay_fixture()
helper with the contract parse_trajectory_csv then windowing.build_windows(preset) then
distill_local.build_candidates plus rally_gate net_crossings then experiment.predict(CONTROL)
then merge_overlaps then metrics.score. Add a unit test asserting that perturbing a trajectory
row changes expected_segments, proving windows are re-derived."
820             },
821             {
822                 "issue": "CI GATE DURING THE ACTIONS BILLING OUTAGE is unaddressed. The
design claims zero new CI wiring because the existing full-suite lane runs the gates, but that
lane is exactly what is down now. With Actions disabled there is no automated red/green; both
gates would land unverified-by-CI like every PR this period. No interim gate is specified.",
823                 "severity": "medium",
824                 "recommendation": "Document an explicit interim local gate while Actions is
down: every PR pastes output of run_all_tests.py with --cov=backend --cov=training --cov-fail-
under=70 plus ruff check . and mypy backend training (the four ci.yml jobs), plus a pytest run
of the two new gate files. Per the owner pre-LGTM gate the reviewer reproduces these locally.
```

When billing is restored the gates auto-run unchanged, so the zero-wiring claim then holds; optionally add the discrete golden-regression job at that point."

```
825         },
826         {
827             "issue": "NO METRICS-VS-BASE-BRANCH COMPARISON. The floor asserts F1 at-or-
above the committed value with DEFAULT_TOLERANCE 0.05 absolute, but the owner pre-LGTM gate
requires no regression versus base branch and the project norm is paired delta vs a reference. A
change dropping every video F1 by 0.04 passes the floor yet regresses vs base; nothing compares
this branch to origin/master frozen numbers.",
828             "severity": "medium",
829             "recommendation": "The base-branch numbers are in git (the committed
baseline). Add a gate mode that diffs working-tree recomputed scores against the baseline at
HEAD and emits the paired delta-F1 CI plus exact sign test (eval_stats seed 0). Require the PR
template to paste this paired-delta summary so no-regression-vs-base is an explicit honest-stats
check, not a bare absolute floor. The snapshot diff already shows per-field before/after; lean
on it."
830         },
831         {
832             "issue": "DETECTOR-IMPROVEMENT FIXTURE REGENERATION is the biggest blind
spot. The trajectory CSV is frozen, so refresh-snapshot CANNOT capture a better WASB/TrackNet
detector; it only re-derives windows/features from the OLD trajectory. If the detector improves,
the committed trajectory is stale and the floor measures re-scoring of outdated detector output,
diverging from served quality, with no staleness signal.",
833             "severity": "medium",
834             "recommendation": "Document a distinct owner-box RE-DETECT workflow
(GPU/WSL): re-run WASB on the source clip, re-freeze trajectory.csv as a reviewed INPUT change
with new provenance and corpus-tag, then refresh-snapshot. Record the detector VERSION and
weights hash in provenance.json (not just the detector string) so a detector upgrade is visibly
incommensurable via regression.py guard. State that synthetic-tier trajectories are hand-
authored and never track real detector gains."
835         }
836     ],
837     "missingPieces": [
838         "SNAPSHOT-UPDATE WORKFLOW lacks a tested IDEMPOTENCE contract: refresh-
snapshot run twice with no code change must produce a byte-identical (empty) git diff, else
unchanged-snapshot-equals-unchanged-behaviour is untrustworthy. Do the 6dp rounding inside the
shared writer; PR-2 must assert refresh is a fixed point. Document the exact command, expected
diff shape, and the never-hand-edit-expected-files rule in the fixtures README and
GOLDEN_REGRESSION_FIXTURES.md.",
839         "ROORA v8 LABEL-VERSION TRACKING is thin: no explicit guideline_version field,
and the lossy rich-to-minimal start,end projection means a v8-to-v9 relabel that does not move
the intervals leaves gt_hash unchanged and passes silently as the same labels. Add
label_guideline_version (e.g. roora-v8) to provenance.json and assert the baseline
guideline_version matches the fixture, flagging a bump as incommensurable even when boundaries
are unchanged, forcing a deliberate re-baseline.",
840         "MULTI-DETECTOR (WASB/TrackNet) is asserted detector-agnostic via the detector
string but not designed through: a slug maps to ONE trajectory, so there is no story for both
detectors on the same clip, and regression.py would (correctly) flag a WASB baseline vs a
TrackNet run as a detector mismatch with no way to hold both. Encode the detector in the slug
(clip-wasb, clip-tracknet) as separate fixtures sharing source provenance, and key
regression_baseline.json by slug-plus-detector-plus-stage (candidate_segments already supports
per-detector stats). Document each detector as its own anchor; cross-detector comparison is out
of scope.",
841         "CONTRIBUTOR ONBOARDING beyond the loop: no first-run quickstart for the
stated audience (a contributor refactoring the deterministic pipeline on another machine)
covering what green/red looks like and the next action, plus an explicit note that
cv2/torch/numpy are NOT needed for these gates (the whole point), and a pointer from the repo-
root README or a CONTRIBUTING stub cross-linked from CODE_MAP section 8 TEST MAP. Contributors
will not find the fixtures-dir README first.",
842         "FILE-PLACEMENT vs CODE_MAP section 0: the placement table has NO row for
tests/fixtures/ (only tests/test_*.py); committing data under tests/ is a new convention. Beyond
adding the row (design says this), PR-5 must confirm pytest does not collect the fixture dir as
tests (no test_*.py inside it), ruff and coverage ignore it as data, and that golden_fixtures.py
and freeze_fixture.py fit the existing eval-logic-to-backend/eval convention.",
843         "TRAIN-EVAL LEAKAGE wiring: the quality-floor gate ALREADY trains learned
```

variants (shuttle_only, fusion) per-fold on the committed fixtures via honest LOVO. If a future learned scorer is PINNED (classifier_model) against these same fixtures and then regression-tested on them, that is train-equals-eval leakage producing a falsely-green floor. Assert no pinned classifier_model is trained on the fixture corpus, and wire splits.assert_disjoint over the fixture slugs the moment a pinned learned variant enters the gate; document the LOVO-honest-but-pinned-is-not distinction in the gate docstring.",

844 "TEST WALL-CLOCK GROWTH: repoSizeStrategy caps trajectory rows but says nothing about runtime. The quality-floor honest LOVO is $O(n^2)$ re-train across videos for the learned variants and can become the slowest test in a fully-mocked suite as the corpus grows. Set a per-suite wall-clock budget in the doc; collapse the quality-floor LOVO to ONE corpus-level test (compare once over all videos) rather than parametrizing the loop per slug; keep characterization parametrized per-slug ($O(n)$, precise failure attribution). Re-measure when tier-1 lands."

845],

846 "falseSafetyRisks": [

847 "A GREEN CHARACTERIZATION GATE WILL BE READ AS CORRECT or SHIPPABLE.

honestLimits says this, but the risk is structural: the gate freezes current behaviour including current bugs and, on the synthetic tier, validates plumbing not field quality. Docs-say-so is weak against habit. Make the gate PASS and FAIL messages literally print: behaviour UNCHANGED for N frozen cases, NOT a correctness or field-quality check, the owner-box DB regression is the only full-chain gate, so the caveat rides with the signal.",

848 "THE FLOOR IS ON THE SERVED PRESET BUT THE +0.074 GATE-A LIFT LIVES ON THE PROPOSAL PRESET. served_gate_a whole cautionary tale is eval-not-equal-serve. If fixtures freeze SERVED (the recommendation) and a contributor reads a green floor as Gate A validated, they repeat the prior mistake; the +0.074 is NOT what these fixtures assert. Clearly label which preset each committed number is under; if PROPOSAL is recorded as a documented control, restate that a SERVED-floor pass says nothing about a PROPOSAL-measured cue lift.",

849 "REFRESH-SNAPSHOT ENABLES RUBBER-STAMPING A REAL REGRESSION (design states this). Beyond reviewer-must-read-the-diff, keep the quality-FLOOR update (regression_baseline.json) a separate deliberate update-baseline action distinct from the mechanical refresh-snapshot, so a characterization refresh cannot silently lower the quality floor. Two updates, two review burdens.",

850 "SYNTHETIC-TIER FLOOR NUMBERS MAY BE LAUNDERED INTO QUALITY_ITERATIONS or the paper backbone as field results. growthStory ties the floor to the ablation track and paper aim; synthetic F1 floors are correctness checks, not field quality, and must never be cited as rally-detection performance. Tag every synthetic fixture kind synthetic in provenance and require any QUALITY_ITERATIONS entry sourced from these gates to carry that tag."

851],

852 "verdict": "Well-grounded and accurate in reusing the shipped eval stack; the trajectory-CSV seam, synthetic-first two-tier licensing reconciliation, and tests/fixtures plus eval_baselines split are correct and code-verified. APPROVABLE in principle, but two HIGH code-vs-design discrepancies must be fixed before PR-2/PR-3 or the gates silently under-protect: (1) the replay path uses the LENIENT load_golden_gts, not the strict gt_loader the GT seam claims, so malformed labels will not fail loudly on the quality-floor path; (2) experiment.predict(CONTROL) does NOT re-derive windows, so the characterization snapshot only locks the windowing brain if a shared replay helper first runs parse_trajectory_csv then build_windows then build_candidates, otherwise the windowing brain is unprotected. Both fixes funnel freeze and both gates through ONE tested replay_fixture() helper (strict label load, CSV-to-windows re-derivation, idempotent 6dp rounding). Top missing pieces: an explicit interim gate for the Actions billing outage (paste the four ci.yml job outputs locally per the pre-LGTM gate; the zero-CI-wiring claim only holds once Actions returns); a metrics-vs-base-branch paired-delta check (the committed baseline makes this free, not a 0.05 absolute floor); a detector-improvement RE-DETECT workflow distinct from refresh-snapshot with a weights/detector version in provenance; a multi-detector slug-plus-detector baseline keying; an explicit roora-v8 guideline_version field so a relabel that does not move intervals still flags incommensurable; and a wall-clock budget plus a corpus-level (not per-slug) LOVO quality-floor. The honest-limits section is candid and correct; the residual false-safety risk is a green characterization gate read as correct/shippable, so put that caveat in the test own pass/fail message, not only the docs. With the two HIGH fixes plus the interim-gate, base-branch-delta, and re-detect-workflow additions, this lands as a durable, CI-runnable regression anchor."

853 }

854]

855 }

856 }

50. assistant (2026-06-16T02:57:27.046Z)

I now have the full first workflow (the fixture **mechanics**), and your three answers are locked in. The second workflow ? the legal memo (IN/US/EU) plus the non-attributable/encryption architecture and red-team ? is still running; I'll deliver the ****single unified written plan**** when it lands. Here's the high-signal interim, because two findings already in hand are genuinely important and shape everything.

Your answers ? a clean convergence

- ****Owned corpus**** ? this is the **easy** legal fork: no third-party copyright/ToS infringement; it's your IP. The concern collapses to trade-secret/competitive-moat (your call to publish) + player privacy. And non-attributability is ****strongest**** here ? a private corpus has **nothing** public to correlate against*, so the fingerprinting/re-identification attack (re-run the detector on a public video, match the trajectory) simply has no reference and fails.
- ****Private now, may open-source later**** ? git history is permanent, so we design the public tier to be ****public-grade from the first commit**** (no scrubbing later).
- ****Adults, consent not formalized**** ? person/pose-derived streams stay ****out of the public tier**** (derived-personal-data risk under DPDP/GDPR without consent); the public tier is ****shuttle-only****.

****The elegant part:**** the privacy-driven choice (publish shuttle-only, keep person streams private) lands on the **exact same** minimal tier that the determinism analysis independently demands ? see below. Privacy and correctness point at one design. That's the spine of the "Byzantine" solution.

The killer finding from the mechanics workflow

The design's natural seam is right ? ****freeze the post-detector trajectory CSV**** (``Frame,Visibility,X,Y``) and replay the deterministic pipeline from there, reusing the **entire shipped eval stack** (``metrics.py``, ``experiment.py``, ``windowing.py``, ``regression.py``, ``eval_baselines/``), with two gates: a ****characterization/snapshot**** gate (catches any refactor side-effect) and a ****quality-floor**** gate (catches quality drops). Synthetic fixtures ship first (licensing-clean by construction); owned real fixtures are owner-gated.

But the adversarial critic caught a ****fatal determinism flaw****, code-verified: the **learned** variants (``shuttle_only``, ``fusion``) run 1000 epochs of numpy gradient descent on OpenBLAS compiled ``DYNAMIC_ARCH`` ? the float-summation order is chosen per CPU microarchitecture, the resulting probabilities sit right on the 0.5 keep/drop cliff, so an external contributor on a ****different CPU gets different integer tp/fp/fn on byte-identical correct code****. The gate goes red, they're told to "refresh the snapshot," they commit **their** machine's numbers, and snapshot diffs become noise reviewers learn to ignore ? destroying the one property you're building this for.

****Fix (and the convergence):**** the cross-machine gate must be ****STATIC-ONLY**** ? the pure-stdlib ``CONTROL`` path (gate ? merge ? score) plus pure shuttle features ? with learned/person scoring kept ****owner-box-only****. That's **exactly** the shuttle-only, person-excluded tier privacy already requires. Two other code-verified must-fixes: route freeze ****and**** both gates through one shared ``replay_fixture()`` helper (or the snapshot doesn't actually exercise the windowing "brain" it claims to lock), and ****parse-compare JSON, never byte-compare**** (CRLF/float-format would false-RED on Windows ? your ``core.autocrlf=true`` is set).

What's still cooking

The legal memo (copyright-of-facts, EU database right, India DPDP, ToS, biometrics, re-ID) and the tiered protobuf/encryption + key-management architecture + red-team are mid-flight. When they complete I'll merge all three into one plan and ****write it to `docs/GOLDEN\_REGRESSION\_FIXTURES.md`**** for your review (I won't commit or open a PR ? per your conventions). I'll flag this to you the moment it's ready.

51. user (2026-06-16T03:46:15.623Z)

any logs whether it is stuck?

52. assistant (2026-06-16T03:46:26.057Z)

Let me check the live status of the running workflow.

53. user (2026-06-16T03:46:27.557Z)

No files found

54. user (2026-06-16T03:46:33.708Z)

<retrieval_status>not_ready</retrieval_status>

<task_id>w3pn6s2mo</task_id>

<task_type>local_workflow</task_type>

<status>running</status>

55. user (2026-06-16T03:46:34.815Z)

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56. assistant (2026-06-16T03:46:49.007Z)

Status: **running, not stuck** ? 17 of the 18 planned agents have already spawned. Let me confirm it's actively progressing (not hung) by checking the workflow journal and how recently the newest agent file was written.

57. user (2026-06-16T03:46:50.773Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-7 of 34 total (91938 tokens, cap 25000). Call Read with offset=8 limit=7 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

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1      {"type":"started","key":"v2:9db7a107f14296d0277ec1742abfa6763d766c62724729f6c3ef7d67840c
b427","agentId":"a466d9415a36b1f58"}
2      {"type":"started","key":"v2:d338e029cb401cfbd219295a0513f4dc3c2513a4fa178cc034c6dd89d3a9
9ca6","agentId":"a0d609f7d61330587"}
3      {"type":"started","key":"v2:ece338c433c1ac8d56826b441904f1b0a9e1805a195156acc31728a090ac
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4      {"type":"started","key":"v2:f5d1c648c8d5bde1b5e127ac374208c463197deec47f113f64bc31a743abb
bd70","agentId":"a4f131c7921bd5ef5"}
5      {"type":"started","key":"v2:1a701f35759fa846c96423447bd0e49e186c0c5288ce6a2ddf40a2580ed4
703e","agentId":"a6202440835be7bf1"}
6      {"type":"started","key":"v2:1d108a270608117d5faa6a052eb127658f13c4af965a567600a48574d389
e6e5","agentId":"a1f8480b6899537f9"}
7      {"type":"result","key":"v2:f5d1c648c8d5bde1b5e127ac374208c463197deec47f113f64bc31a743abb
d70","agentId":"a4f131c7921bd5ef5","result":{"question":"Under the EU sui generis database right
(Directive 96/9/EC) and copyright, does extracting and redistributing DERIVED match data
(coordinates / event timestamps) from a broadcast infringe?","scope":"EU law only (Directive
96/9/EC as interpreted by the CJEU). Covers: (1) sui generis database right ? \"obtaining vs
creating\" data and \"extraction/re-utilisation of a substantial part\"; (2) database copyright
(originality); (3) where re-utilisation occurs (targeting); (4) contract overriding the
directive's user exceptions for UNPROTECTED databases. Focus = independently DERIVED numeric
streams (shuttle/player coordinates, rally/event timestamps) computed by our own CV/AI from a
broadcast video, then redistributed. NOT advice on: broadcast/film copyright in the video signal
itself, performers'/sports-organiser commercial rights, unfair-competition/parasitism under
national law, or post-Brexit UK law (UK retained a parallel but now-diverging regime). Not legal
advice; consult EU IP counsel before relying.","webUsed":true,"findings":[{"claim":"The sui
generis database right protects only investment in OBTAINING (seeking out and collecting pre-
existing independent materials), VERIFYING, or PRESENTING contents ? NOT investment in CREATING
the data. Resources spent generating new data do not count toward the 'substantial investment'
that triggers the right.","support":"In C-203/02 the CJEU held that 'investment in ? the
obtaining ? of the contents refers to the resources used to seek out existing independent
materials and collect them' and that resources used to CREATE the data (e.g. fixing the list of
runners) fall outside this and do not count toward substantial investment under Art 7(1). The
parallel Fixtures Marketing line (C-46/02, C-338/02, C-444/02, decided the same day) reached the
same conclusion for football fixture lists: investment in generating the data does not
qualify.","sourceName":"EUR-Lex ? Case C-203/02 British Horseracing Board v William Hill
(judgment, 9 Nov 2004)","sourceUrl":"https://eur-lex.europa.eu/legal-
content/EN/TXT/?uri=celex%3A62002CJ0203","confidence":"high"},{"claim":"This creating-vs-
obtaining distinction is the crux for sports data: a party who is the SOURCE of dynamically-
generated sports data (fixtures, race entries, and by extension live match events) often holds
NO sui generis right in it, because its spend was on creation, not obtaining. Only investment in
collecting/verifying pre-existing materials qualifies.","support":"Commentary on the 2004
quartet (C-46/02, C-203/02, C-338/02, C-444/02) notes the CJEU 'seriously restricted the scope
of sui generis protection by admitting that investments in the creation of a database's contents
cannot count as investment for obtaining'; the football-fixture and horseracing-runner-list
producers therefore FAILED to establish a sui generis right.","sourceName":"Wolters Kluwer /
Kluwer Copyright Blog ? analysis of the Database Directive sports-data
cases","sourceUrl":"https://legalblogs.wolterskluwer.com/copyright-blog/databases-sui-generis-
protection-and-copyright-protection/","confidence":"high"},{"claim":"'Extraction' and 're-
utilisation' are construed broadly: any unauthorised appropriation/transfer or making-available
of all or part of a protected database's contents, and they do NOT require direct access to the
database ? indirect copying (e.g. via a third party who copied from the maker) still
counts.","support":"C-203/02 ruling: extraction and re-utilisation 'must be construed as
referring to any unauthorised act of appropriation and ? distribution to the public of the whole
or a part of the contents of a database', and 'those terms do not imply direct access to the
database concerned.'","sourceName":"EUR-Lex ? Case C-203/02 British Horseracing Board v William
Hill","sourceUrl":"https://eur-lex.europa.eu/legal-
content/EN/TXT/?uri=celex%3A62002CJ0203","confidence":"high"},{"claim":"A 'substantial part' is
judged QUANTITATIVELY (volume taken vs. total contents) OR QUALITATIVELY (scale of the
investment in obtaining/verifying/presenting the taken portion, regardless of its volume). The
right protects the investment, not the data as such; taking data whose creation ? but not
obtaining ? was costly does not infringe.","support":"C-203/02 ruling defines quantitative
substantiality as 'the volume of data extracted ? assessed in relation to the total volume of
the contents' and qualitative substantiality as 'the scale of the investment in the obtaining,
verification or presentation of the contents ? regardless of whether that subject represents a
quantitatively substantial part.' Because BHB's investment was in CREATING the data, William
Hill's takings were neither quantitatively nor qualitatively substantial.","sourceName":"EUR-Lex
? Case C-203/02 British Horseracing Board v William Hill","sourceUrl":"https://eur-
lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A62002CJ0203","confidence":"high"},{"claim":"Art
7(5) anti-circumvention: repeated and systematic extraction/re-utilisation of INSUBSTANTIAL
parts can still infringe, but only where the cumulative effect would reconstitute and/or make
available a substantial part WITHOUT authorisation, thereby seriously prejudicing the maker's
```

investment. If no protected substantial part exists, systematic small takings still do not infringe.", "support": "C-203/02 ruling: Art 7(5) prohibits acts 'which, because of their repeated and systematic character, would lead to the reconstitution of the database as a whole or ? a substantial part ? and/or ? make available to the public ? a substantial part,' i.e. acts that 'seriously prejudice the investment.'", "sourceName": "EUR-Lex ? Case C-203/02 British Horseracing Board v William Hill", "sourceUrl": "https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A62002CJ0203", "confidence": "high"}, {"claim": "Database COPYRIGHT (Art 3) protects only the STRUCTURE ? an original SELECTION or ARRANGEMENT that is the author's own intellectual creation. It never protects the data/contents themselves, and the intellectual effort/skill spent CREATING the data is irrelevant to copyright eligibility. Mere significant labour and skill ('sweat of the brow') cannot ground copyright absent originality in selection/arrangement.", "support": "C-604/10 ruling: a database is protected by copyright 'provided that the selection or arrangement of the data ? amounts to an original expression of the creative freedom of its author'; 'the intellectual effort and skill of creating that data are not relevant'; and 'significant labour and skill ? cannot as such justify such protection if they do not express any originality in the selection or arrangement.' The Directive also precludes national copyright on conditions different from Art 3(1).", "sourceName": "EUR-Lex ? Case C-604/10 Football Dataco v Yahoo (judgment, 1 Mar 2012)", "sourceUrl": "https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX:62010CJ0604", "confidence": "high"}, {"claim": "Jurisdiction/localisation: sending data extracted from a protected database from a server in Member State A to users in Member State B is an act of 're-utilisation' occurring in B ? but only where there is evidence the operator INTENDED TO TARGET the public in B (language, currency, local match content, targeting of local customers). Locating a server outside the EU/a Member State does not by itself escape liability.", "support": "C-173/11 ruling (18 Oct 2012): re-utilisation occurs in the Member State of reception where 'there is evidence ? of an intention on the part of [the sender] to target members of the public in [that] Member State', considering factors such as data on that country's matches, contracts with that market's betting firms, and the language used.", "sourceName": "IPCuria ? Case C-173/11 Football Dataco v Sportradar (judgment, 18 Oct 2012)", "sourceUrl": "https://ipcuria.eu/case?reference=C-173/11", "confidence": "high"}, {"claim": "Ryanair v PR Aviation: if a database enjoys NEITHER copyright NOR the sui generis right, the Directive does not apply to it at all ? so the mandatory user-protection provisions (Arts 6(1), 8, 15, incl. the no-contracting-out rule) do NOT apply, and the owner is free to impose CONTRACTUAL limits (e.g. a screen-scraping/automated-extraction ban in website T&Cs) enforceable under national contract law. So even where IP fails, contract can still bind a party who agreed to the terms.", "support": "C-30/14 ruling (15 Jan 2015): the Directive 'is not applicable to a database which is not protected either by copyright or by the sui generis right ?, so that Articles 6(1), 8 and 15 of that directive do not preclude the author of such a database from laying down contractual limitations on its use by third parties.'", "sourceName": "EUR-Lex ? Case C-30/14 Ryanair v PR Aviation (judgment, 15 Jan 2015)", "sourceUrl": "https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:62014CJ0030", "confidence": "high"}], "bottomLine": "For our use case ? numeric streams (shuttle/player coordinates, rally/event timestamps) that we COMPUTE ourselves from a broadcast via CV/AI and then redistribute ? the EU IP exposure on the database-right/copyright axis is LOW, but it is NOT zero, and the real risk lives in contract, not IP. Why low on IP: (1) Sui generis right ? under BHB v William Hill (C-203/02) and the Fixtures Marketing line, the right protects investment in OBTAINING/verifying/presenting pre-existing data, not in CREATING it. Coordinates and event timings we derive frame-by-frame are data we CREATE by our own measurement; we are not seeking out and copying a substantial part of someone else's collected dataset. We extract from the VIDEO (a film/broadcast work), not from the rights-holder's official stats database, so 'extraction of a substantial part' of a protected database is hard to make out, and Art 7(5) systematic-takings liability also requires an underlying protected substantial part. (2) Copyright ? under Football Dataco v Yahoo (C-604/10), copyright protects only an original selection/arrangement (structure), never the raw numbers, and never sweat-of-the-brow; our derived coordinates are facts, not a copied original structure. Where it CAN bite: (a) Contract ? Ryanair v PR Aviation (C-30/14) is the live danger: even with no IP right at all, the source's terms of use / broadcast or feed licence can lawfully ban automated extraction and redistribution and bind us if we accepted them; this is the most likely route to liability and is governed by the licence/T&Cs you click through, not the Directive. (b) If you ever ingest an OFFICIAL data feed/stats database rather than self-deriving from pixels, the obtaining-vs-creating analysis flips toward the rights-holder and substantial-part infringement becomes plausible. (c) Football Dataco v Sportradar (C-173/11) means you cannot dodge EU exposure by server location if you target EU users. Separately, the broadcast VIDEO itself is a protected film/signal ? copying or redistributing the footage (as opposed to derived numbers) is a different, real infringement; our highlight reels live on that axis, not this one.

Practical guardrail consistent with the project's licensing discipline: self-derive numeric data from sources you are licensed to process, do not scrape official stats feeds, and read every source's T&Cs/licence for automated-extraction and redistribution clauses before ingesting.", "caveats": ["Not legal advice ? these are CJEU holdings synthesized from authoritative sources; get EU IP counsel before relying, especially before any commercial data product.", "The favourable IP analysis assumes we DERIVE the numeric data ourselves from video (creation), and do NOT copy a substantial part of an official/third-party stats database (obtaining). If the input is a licensed/scraped official feed, the conclusion can reverse.", "Redistributing the broadcast VIDEO/footage itself (highlight clips) raises separate film/broadcast copyright and related-rights issues NOT analysed here ? this question was scoped to derived numeric streams only.", "Contract (Ryanair C-30/14) is the dominant residual risk and is fact-specific to each source's terms of use / licence; a click-through or feed licence banning automated extraction or commercial redistribution can bind us even with zero IP right in the data.", "National-law theories beyond the Directive (e.g. unfair competition / parasitism in some Member States, and protections the Directive expressly leaves to national law per Ryanair) are out of scope and may add exposure.", "Post-Brexit UK has retained a parallel database-right regime that is beginning to diverge from the EU; UK-specific advice is out of scope here.", "The C-173/11 operative-ruling detail was verified via IPCuria rather than the primary EUR-Lex text; substance is corroborated across multiple commentaries but the exact wording should be confirmed against the EUR-Lex judgment (CELEX 62011CJ0173) before quoting verbatim."]]}

58. user (2026-06-16T03:46:53.998Z)

Name	LastWriteTime
AgeSec KB	
----	-----
--	
agent-a99f0842bd56f8647.jsonl	6/16/2026 9:15:28 AM 86.00 333.00
journal.jsonl	6/16/2026 8:38:45 AM 2288.00 236.00
agent-a49314b51fd5d1fb8.jsonl	6/16/2026 8:38:45 AM 2288.00 159.00
agent-aa13cea5f0da98a00.jsonl	6/16/2026 8:38:40 AM 2293.00 210.00

59. assistant (2026-06-16T03:47:31.178Z)

Here's the honest status ? **it's running, not hung, but one agent is a slow straggler.**

What the files show:

- **17 of 18 agents have spawned**, and most have already completed (the journal has their results ? e.g. the EU database-right research agent already returned a fully-cited answer).
- The newest transcript, `agent-a99f0842?`, was **written 86 seconds ago** (and is 333 KB) ? an agent is **actively producing output right now**. So: not deadlocked.
- But the workflow **journal** hasn't logged a completion in ~38 min, which means that one agent has been running ~38 minutes straight. It's the straggler gating the whole run (wall-clock = slowest single chain).

Let me peek at what that straggler is actually doing ? productive work vs. a fetch loop:

60. user (2026-06-16T03:47:37.650Z)

TOTAL LINES: 85

```
--- [assistant] {"parentUuid":"2b0e1f67-4def-43f7-9e16-cd318dfc83a6","isSidechain":true,"agentId":"a99f0842bd56f8647","message":{"model":"claude-opus-4-8","id":"msg_01762qEvQHdpoxXs6Xn3woo","type":"message","role":"assistant","content":[{"type":"text","text":"The empty calls keep failing. I wil"}]}
--- [assistant] {"parentUuid":"0eaba9a0-c9b9-48a9-a166-64f98934eab7","isSidechain":true,"agentId":"a99f0842bd56f8647","message":{"model":"claude-opus-4-8","id":"msg_01762qEvQHdpoxXs6Xn3woo","type":"message","role":"assistant","content":[{"type":"tool_use","id":"toolu_01YHTGvuLQKD4H2iz1DP3GL7"}]}
--- [user] {"parentUuid":"5739ef19-e6fd-4ec0-bee3-8e0c26cd7514","isSidechain":true,"promptId":"b1044f3a-6a37-4ccf-aaf4-aba586eaf235","agentId":"a99f0842bd56f8647","type":"user","message":{"role":"user","content":[{"type":"tool_result","content":"Output does not match required schema: root:"
```

```

--- [assistant] {"parentUuid":"3bfb7b0f-9581-4be7-a942-
ed91fa795721","isSidechain":true,"agentId":"a99f0842bd56f8647","message":{"model":"claude-opus-
4-
8","id":"msg_01JZVTntBPMewc5dxWtk1oT","type":"message","role":"assistant","content":[{"type":"t
ool_use","id":"toolu_01DQZ6BbGUMRb2T9TejvR32U"},
--- [user] {"parentUuid":"c0edd07f-8869-4623-88d0-
88404f41254d","isSidechain":true,"promptId":"b1044f3a-6a37-4ccf-aaf4-
aba586eaf235","agentId":"a99f0842bd56f8647","type":"user","message":{"role":"user","content":[{"
type":"tool_result","content":"Output does not match required schema: root:
--- [assistant] {"parentUuid":"75ae1236-d987-47d4-877a-
e2189710ee5c","isSidechain":true,"agentId":"a99f0842bd56f8647","message":{"model":"claude-opus-
4-
8","id":"msg_01DeLNBbpobC3v1G1qeMx2b5","type":"message","role":"assistant","content":[{"type":"t
ext","text":"I've been calling the tool without

```

61. assistant (2026-06-16T03:48:35.851Z)

The peek is decisive: that straggler is **stuck** in a structured-output validation loop, not doing useful work. Its latest messages read **"The empty calls keep failing"** and **"I've been calling the tool without?"**, interleaved with repeated **"Output does not match required schema"** rejections. It's been retrying a schema it can't satisfy for ~38 min, which is blocking the final phase. Let me map exactly which agent/phase it is and what's already completed before deciding how to recover.

62. user (2026-06-16T03:48:41.746Z)

Completed (result in journal): 16 agents

----- per-agent label / done? -----

running/failed	a06cc22597593cd53.meta
running/failed	a0d609f7d61330587.meta
running/failed	a1bea65cdab0c3b6e.meta
running/failed	a1f8480b6899537f9.meta
running/failed	a466d9415a36b1f58.meta
running/failed	a49314b51fd5d1fb8.meta
running/failed	a4f131c7921bd5ef5.meta
running/failed	a61cd05f22268e81e.meta
running/failed	a6202440835be7bf1.meta
running/failed	a69b45904982be16e.meta
running/failed	a7a372b69f23b997e.meta
running/failed	a82c1547810fe50e2.meta
running/failed	a90f484e70c865137.meta
running/failed	a99f0842bd56f8647.meta
running/failed	aa13cea5f0da98a00.meta
running/failed	ac7c49b8f877090e1.meta
running/failed	afe0ef4bb67f34e89.meta

63. user (2026-06-16T03:49:33.448Z)

```

118     }
119
120     const WEB = "\n\nMETHOD: Use WebSearch and WebFetch to find and cite AUTHORITATIVE
sources (statutes, case law, official regulator guidance, reputable legal commentary). Load them
first if needed via ToolSearch with query 'select:WebSearch,WebFetch'. Cite real URLs. If you
cannot fetch a source, mark that claim's confidence 'low' and set webUsed=false. Do NOT
fabricate citations ? an honest 'unverified' beats an invented case name or URL."
121
122     // ----- legal research
123     phase('Legal Research')
124     const LEGAL_Q = [
125         { key: 'copyright-facts', prompt:
126             "RESEARCH QUESTION: Is algorithmically-DERIVED structured data ? per-frame shuttle
(x,y) coordinates, rally start/end timestamps, audio-onset feature values ? extracted from a
copyrighted sports VIDEO itself protected by copyright, or is it uncopyrightable facts/data, or
a 'derivative work'? Cover: the idea/expression dichotomy; Feist Publications v. Rural Telephone

```

```

(facts not copyrightable; sweat-of-the-brow rejected); originality requirement in
selection/arrangement; whether mechanical extraction yields a 'derivative work' under 17 USC
101; intermediate / non-expressive copying and fair use (Google v. Oracle, Authors Guild v.
Google, Sega v. Accolade). Bottom line: does the derived NUMERIC data inherit the video's
copyright?" },
127     { key: 'eu-database-right', prompt:
128         "RESEARCH QUESTION: Under the EU sui generis DATABASE RIGHT (Directive 96/9/EC) and
copyright, does extracting and redistributing DERIVED match data (coordinates / event
timestamps) from a broadcast infringe? Cover: British Horseracing Board v. William Hill
(C-203/02); Football Dataco v. Sportradar (C-173/11) and Football Dataco v. Yahoo (C-604/10);
the 'obtaining vs creating' data distinction; 'extraction/re-utilization of a substantial part';
Ryanair v. PR Aviation (C-30/14) on contracts overriding DB-right exceptions. Bottom line for
our derived numeric streams in the EU." },
129     { key: 'india-law', prompt:
130         "RESEARCH QUESTION (PRIMARY JURISDICTION ? owner is in Bangalore, India): Under the
Indian Copyright Act 1957 ? derivative works, the originality 'modicum of creativity' standard
(Eastern Book Co. v. D.B. Modak), fair dealing s.52; the ABSENCE of an EU-style sui generis
database right in India; and the Digital Personal Data Protection (DPDP) Act 2023 ? definition
of personal data, treatment of derived/anonymized data, and children's data rules ? is
committing DERIVED numeric data from videos lawful? Note any Indian case law on data
scraping/extraction. Bottom line under Indian law." },
131     { key: 'contract-tos-tradeseecret', prompt:
132         "RESEARCH QUESTION: The CONTRACT / Terms-of-Service and TRADE-SECRET angles, which
can bind even when copyright does NOT. Cover: typical YouTube / broadcaster ToS clauses
prohibiting extraction, scraping, automated access, and building datasets; whether
redistributing a derived dataset breaches platform terms; the scraping-law landscape (hiQ v.
LinkedIn; Van Buren v. United States). CONVERSELY: if the corpus is the OWNER'S OWN recordings,
the analysis shifts to trade-secret / competitive-moat (their data to publish or protect).
Bottom line: when is committing derived data a CONTRACTUAL/ToS problem independent of copyright,
and how does owner-recorded vs third-party-sourced change it?" },
133     { key: 'privacy-biometrics', prompt:
134         "RESEARCH QUESTION: PRIVACY / biometric exposure of derived data. When does data
DERIVED about people in a video become 'personal data' (GDPR Art.4, Recital 26 on
anonymization/pseudonymization), or trigger Art.9 special-category / biometric protection, or
Art.8 children's rules ? and the equivalent under India's DPDP Act 2023 (children's data,
verifiable parental consent)? Crucial distinction: SHUTTLE coordinates (about an inanimate
object, not a person) vs PERSON bounding boxes vs POSE keypoints / GAIT (potentially personal
and/or biometric). Bottom line: which derived streams are privacy-safe to PUBLISH, which are
not, and how the presence of MINORS changes everything." },
135     { key: 'deidentification-reid', prompt:
136         "RESEARCH QUESTION: DE-IDENTIFICATION and RE-IDENTIFICATION risk for a derived
numeric TIMESERIES. What standards make data 'anonymous'/non-attributable (GDPR Recital 26
'means reasonably likely to be used'; k-anonymity intuition; NIST SP 800-188 / ISO de-
identification guidance)? Specifically assess: can an anonymized trajectory/feature timeseries
be RE-LINKED to a specific source video by an adversary who runs the same detector on candidate
PUBLIC videos and correlates the streams (fingerprinting attack)? How does the source being
PRIVATE/owner-recorded (never published, nothing to correlate against) vs PUBLIC
(broadcast/YouTube) change the achievable non-attributability? Bottom line: what non-
attributability is actually achievable, and under what conditions." },
137     ]
138     const findings = (await parallel(LEGAL_Q.map(q => () =>
139         agent(q.prompt + WEB, { label: `research:${q.key}`, phase: 'Legal Research', schema:
LEGAL_FINDING_SCHEMA })))
140         .filter(Boolean)
141         .log(`Legal research: ${findings.length}/${LEGAL_Q.length} questions returned`))
142
143     // ----- adversarial verify
(barrier: needs findings)
144     phase('Verify')
145     const verifications = (await parallel(LEGAL_Q.map((q, i) => () =>
146         agent(
147             "You are a SKEPTICAL second lawyer. Try to REFUTE or materially QUALIFY the bottom
line below using authoritative sources. Look for the strongest counter-authority, jurisdictional
exceptions, and ways an over-confident reading would get the client in trouble. Default to
'qualified' or 'refuted' if there is genuine doubt." + WEB +

```

```

148         "\n\nQUESTION KEY: " + q.key +
149         "\nBOTTOM LINE TO ATTACK: " + ((findings[i] && findings[i].bottomLine) || "(no
finding returned)") +
150         "\nSUPPORTING FINDINGS (JSON): " + JSON.stringify((findings[i] &&
findings[i].findings) || []),
151         { label: `verify:${q.key}`, phase: 'Verify', schema: VERIFY_SCHEMA })
152     })).filter(Boolean)
153
154     // ----- legal memo (barrier:
needs findings + verifications)
155     phase('Legal Memo')
156     const memo = await agent(
157         "You are senior IP/privacy counsel writing an internal memo (general information, NOT
a substitute for retained counsel) for an owner in Bangalore, India, who wants to COMMIT derived
numeric data (shuttle trajectories, rally time-intervals, audio/person/court feature columns)
from a labeled video corpus into a GitHub repo so external contributors can run regression tests
WITHOUT the videos. Synthesize the research findings and the skeptical verifications below into
ONE coherent, cited memo. Be explicit about the OWNED-corpus vs THIRD-PARTY-sourced fork, split
the analysis by jurisdiction (India primary; US and EU because the product/repo are global), and
state concrete, testable CONDITIONS under which a public commit of derived data is defensible
(and which streams must NOT be published ? e.g. anything person/pose/biometric, anything
involving minors). List the questions that genuinely need a retained lawyer. Use only citations
that appear in the findings/verifications; do not invent any." +

```

64. assistant (2026-06-16T03:49:31.196Z)

Diagnosis is now precise:

- **16 of 17 agents completed** ? all 6 legal-research agents, 6 verifications, the legal memo, and the 3 crypto-design agents are **done and on disk**.
- The **single stuck agent is the Tech-Synthesis step** (the 17th), looping on its structured-output schema (`TECH_SYNTH_SCHEMA`` has 15 required fields including a nested `rolloutPhases`` object array ? the model keeps producing output the validator rejects). The Red-Team step (18th) depends on it, so it never started.

So it's not productively working and won't self-recover soon. The clean fix: **stop it, relax that one schema, and resume** ? the 16 finished agents return from cache instantly, and only the synthesis + red-team re-run (~2-4 min). Let me load the stop tool and read the script's schema section to edit precisely.

65. user (2026-06-16T03:49:38.376Z)

```

{"message":"Successfully stopped task: w3pn6s2mo (Deep research: IP/privacy of committing
DERIVED structured data publicly + a non-attributable encrypted fixture
architecture)","task_id":"w3pn6s2mo","task_type":"local_workflow","command":"Deep research:
IP/privacy of committing DERIVED structured data publicly + a non-attributable encrypted fixture
architecture"}

```

66. user (2026-06-16T03:49:41.199Z)

```

80     mechanism:{type:'string'},
81     tradeoffs:{type:'array', items:{type:'string'}},
82     integrationWithExistingInfra:{type:'string'},
83 },
84 }
85 const TECH_SYNTH_SCHEMA = {
86     type: 'object', additionalProperties: false,
87     required: ['architectureName','threatModelTable','tieredModel','antiAttributionMeasure
s','protobufSchemaSketch','serializationDeterminism','encryptionMechanism','keyManagement','ciIn
tegration','contributorWorkflow','harnessIntegration','honestLimits','rolloutPhases','openOwnerQ
uestions'],
88     properties: {
89         architectureName:{type:'string'},
90         threatModelTable:{type:'string'},

```

```

91     tieredModel:{type:'string'},
92     antiAttributionMeasures:{type:'array', items:{type:'string'}},
93     protobufSchemaSketch:{type:'string'},
94     serializationDeterminism:{type:'string'},
95     encryptionMechanism:{type:'string'},
96     keyManagement:{type:'string'},
97     ciIntegration:{type:'string'},
98     contributorWorkflow:{type:'string'},
99     harnessIntegration:{type:'string'},
100    honestLimits:{type:'array', items:{type:'string'}},
101    rolloutPhases:{type:'array', items:{type:'object', additionalProperties:false,
102      required:['phase','deliverable','prScope'],
103      properties:{ phase:{type:'string'}, deliverable:{type:'string'},
prScope:{type:'string'} }},
104    openOwnerQuestions:{type:'array', items:{type:'string'}},
105  },
106  }
107  const REDTEAM_SCHEMA = {
108    type: 'object', additionalProperties: false,
109    required: ['attacks','overallVerdict','mustFixBeforeShipping'],
110    properties: {
111      attacks:{type:'array', items:{type:'object', additionalProperties:false,
112        required:['vector','severity','doesDesignHold','fix'],
113        properties:{ vector:{type:'string'}, severity:{type:'string',
enum:['low','medium','high','critical']},
114        doesDesignHold:{type:'string'}, fix:{type:'string'} }},
115      overallVerdict:{type:'string'},
116      mustFixBeforeShipping:{type:'array', items:{type:'string'}},
117    },

```

67. assistant (2026-06-16T03:50:31.083Z)

Workflow stopped cleanly. The culprit is clear: `TECH\_SYNTH\_SCHEMA` has 14 required fields plus a nested `rolloutPhases` object-array with `additionalProperties:false` ? too strict, so the synthesis agent's output kept getting rejected at `root`. I'll relax both that schema and the (not-yet-run) red-team schema, then resume so the 16 finished agents return from cache and only the last two run.

68. user (2026-06-16T03:50:39.796Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js has been updated successfully. (file state is current in your context ? no need to Read it back)

69. user (2026-06-16T03:50:46.577Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js has been updated successfully. (file state is current in your context ? no need to Read it back)

70. assistant (2026-06-16T03:50:50.647Z)

Both schemas relaxed. Now resuming from the saved run ? the 16 finished agents return from cache; only the synthesis and red-team re-run with the simpler schemas.

71. user (2026-06-16T03:50:51.749Z)

Workflow launched in background. Task ID: wgpct1wbd

Summary: Deep research: IP/privacy of committing DERIVED structured data publicly + a non-attributable encrypted fixture architecture

Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9

Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js

(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:

"C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js"} to iterate without resending the script.)

Run ID: wf_47057438-9c9

To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\derived-data-legal-and-byzantine-fixtures-wf_47057438-9c9.js", resumeFromRunId: "wf_47057438-9c9"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

72. assistant (2026-06-16T03:51:06.681Z)

Answered and fixed. Here's the full picture:

****Was it stuck?**** Yes ? one agent, not the whole run. Diagnosis from the logs:

- 16 of 17 agents had ****completed**** (all legal research ? including a fully-cited EU database-right analysis ? plus the 6 verifications, the legal memo, and 3 crypto-design agents).
- The 17th, the ****Tech-Synthesis**** agent, was caught in a structured-output retry loop (~38 min): its transcript literally said ****"The empty calls keep failing"*** against repeated ``Output does not match required schema: root:`` rejections. Its schema demanded 14 required fields + a nested object-array ? too strict. The Red-Team step depends on it, so the run couldn't finish.

****What I did:****

1. Stopped the stuck run (``TaskStop``).
2. Relaxed the two offending schemas (``TECH_SYNTH_SCHEMA`` ? 6 required fields, ``rolloutPhases`` now a string array; ``REDTEAM_SCHEMA`` ? dropped the strict enum/``additionalProperties``).
3. ****Resumed from the same run ID**** ? the 16 finished agents return instantly from cache; only synthesis + red-team re-run with the easier schemas (~2-4 min).

New background task is ``wgpct1wbd``; I'll be notified when it lands. No action needed from you.

For transparency on cause: this was a schema-strictness bug in ***my*** workflow script, not a problem with your question or the research ? the legal findings already on disk are high quality (e.g., the BHB v. William Hill "obtaining vs creating" analysis came back clean and cited). When the resume completes I'll merge both workflows into the single written plan at ``docs/GOLDEN_REGRESSION_FIXTURES.md`` (draft only ? no commit/PR).

73. user (2026-06-16T03:56:51.235Z)

Can you formalize this as a project doc which we can iterate upon, track progress and then mark as completed with clear deliverables? Seems that can be possible but feel free to push back if not possible

74. assistant (2026-06-16T03:57:11.597Z)

Yes ? this absolutely should be a tracked project doc; it fits how this repo already works (``docs/*_PLAN.md`` with a status header, phases, and deliverables). Let me first check whether the resumed workflow has finished so I can fold in the legal memo + architecture, rather than writing a doc with a hole in it.

75. user (2026-06-16T03:57:12.079Z)

<retrieval_status>not_ready</retrieval_status>

<task_id>wgpct1wbd</task_id>

<task_type>local_workflow</task_type>

<status>running</status>

76. user (2026-06-16T03:57:17.347Z)

journal result events: 17

Name	AgeSec	KB
-----	-----	--
agent-ab96e60cc8743fbbf.jsonl	78.00	213.00
agent-a8362c56b32d882b3.jsonl	155.00	325.00
journal.jsonl	155.00	268.00

77. assistant (2026-06-16T03:58:13.338Z)

Good news on status: the resume worked. The Tech-Synthesis agent **completed** with the relaxed schema (journal now shows 17 results), and the **Red-Team** is the only step left ? actively writing 78 seconds ago, so it's a minute or two from done. I'll fold its findings into the doc's risk section rather than write an incomplete draft, then save the file and flag you.

Yes, this is very doable and it's the right format. Here's the structure I'll write to **docs/GOLDEN_REGRESSION_FIXTURES.md** ? a living doc, so take a look and tell me if you want the *shape* changed before I populate it:

...

Golden Regression Fixtures ? video-free, non-attributable regression suite

Status: DRAFT ? owner-gated · Owner: Avi · Last updated: 2026-06-16
Tracking: docs/NEXT_STEPS.md + memory current-state · Supersedes nothing (extends GOLDEN_DATA_SHARING.md)

1. Problem & goal

(run the rally regression from committed files, no video/GPU)
2. Decisions locked

(owned corpus · private-now/maybe-OSS · adults, consent not formalized)
3. Legal foundation

(IN/US/EU verdict on derived data + the must-ask-counsel list) ? from workflow 2 me
4. Architecture

4a. Freeze seam

(post-detector trajectory CSV; reuse the shipped eval stack)

4b. Tiered model

(Tier-0 public shuttle-only/CONTROL-only · Tier-1 key-gated person streams)

4c. Non-attributability

(opaque salted IDs, minimization, no MD5 key, uselessness-in-isolation)

4d. Protobuf + encryption

(serialization vs secrecy; git-crypt/age; key mgmt; the DRM honesty note)

4e. The two gates

(characterization/snapshot + quality-floor)
5. Honest limits

(what a green suite does NOT prove) ? workflow 1 + workflow 2 red-teams
6. Deliverables & rollout

(PR-1..N as checkboxes, each small + independently shippable)
7. Progress tracker

(? todo / ? in-progress / ? done, per phase, with PR links)
8. Open owner/counsel questions
9. Definition of done

(acceptance criteria to mark the project COMPLETE)
- ...

How progress-tracking will work in practice (matching your repo conventions):

- **Section 7** is the single source of truth ? a checklist where each row is a concrete PR/deliverable with a status marker and, once merged, a PR link. Iteration = edit the row.
- I'll add a one-line pointer in **docs/NEXT_STEPS.md** and the **memory current-state** file so a future session resumes from the tracker, not from this chat.
- "Mark as completed" = every Section-6 deliverable is ? and Section-9 acceptance criteria are met; then the `Status:` header flips `DRAFT ? IN PROGRESS ? DONE` and it moves to `docs/archives/` per the archive policy.

I'll write it as a **draft only** ? no commit, no PR (your conventions). I'll populate Sections 3 and 5 from the research the moment the red-team lands and ping you with the file.

78. user (2026-06-16T04:04:01.424Z)

```
<task-notification>
<task-id>wgpct1wbd</task-id>
<tool-use-id>toolu_01AEmgopxK7Y1nZv5mgQ5wGj</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\wgpct1wbd.output</output-file>
<status>completed</status>
<summary>Dynamic workflow "Deep research: IP/privacy of committing DERIVED structured data
publicly + a non-attributable encrypted fixture architecture" completed</summary>
<result>{"findings":[{"question":"Is algorithmically-derived structured data ? per-frame shuttle
(x,y) coordinates, rally start/end timestamps, audio-onset feature values ? extracted from a
copyrighted sports video itself protected by copyright (uncopyrightable facts/data, or a
\"derivative work\")? Does the derived numeric data inherit the video's copyright?\",\"scope\":\"US
copyright law analysis covering: idea/expression dichotomy (17 USC 102(b)); Feist (facts not
copyrightable; sweat-of-the-brow rejected; originality in selection/arrangement); the 17 USC 101
\"derivative work\" definition; and intermediate/non-expressive copying + fair use (Google v.
Oracle, Authors Guild v. Google, Sega v. Accolade, plus 2025 AI-training cases). This is legal-
research synthesis of statutes, case law, and the U.S. Copyright Office Fair Use Index ? NOT
formal legal advice. Focus is the copyright status of the OUTPUT data itself; the separate
question of whether the EXTRACTION ACT (transiently copying the video) infringes is addressed
via the intermediate-copying line. Foreign sui generis database rights (e.g., the EU Database
Directive) are out of scope but flagged.\",\"findings\":{\"claim\":\"Facts are categorically not
copyrightable because they are discovered, not authored, and thus lack the originality the
Constitution and Copyright Act require; effort ('sweat of the brow') does not earn copyright.
This is the foundational holding most directly governing extracted numeric
data.\",\"support\":\"Feist Publications v. Rural Telephone Service Co., 499 U.S. 340 (1991): the
Court held 'facts are not copyrightable' because they 'do not owe their origin to an act of
authorship' and so are not original, and expressly held that 'originality, not sweat of the
brow, is the touchstone of copyright protection.' Copyright 'rewards originality, not effort.'
Per-frame (x,y) shuttle positions, rally start/end timestamps, and audio-onset values are
observable physical facts about events recorded in the video ? discovered measurements, not
authored expression ? so they fall squarely within the unprotected 'facts' category regardless
of the computational effort to derive them.\",\"sourceName\":\"Feist Publications, Inc. v. Rural
Telephone Service Co., 499 U.S. 340 (1991) ? Justia U.S. Supreme Court Center\",\"sourceUrl\":\"http
s://supreme.justia.com/cases/federal/us/499/340/\",\"confidence\":\"high\"},{\"claim\":\"Facts and
statistics GENERATED BY A SPORTING EVENT are not copyrightable even though the audiovisual
BROADCAST of that event is ? the closest factual analogue to extracting shuttle coordinates and
rally timings from a match video.\",\"support\":\"National Basketball Ass'n v. Motorola, Inc., 105
F.3d 841 (2d Cir. 1997): the court held 'Sports events are not authored in any common sense of
the word' and that the underlying scores/statistics are uncopyrightable facts ('purely factual
information which any patron of an NBA game could acquire from the arena'), while only the
broadcast ? the creative work of directors, camera operators, editors ? is protected. By direct
analogy, shuttle positions, rally boundaries, and onset values are facts ABOUT the game that any
spectator could in principle observe; they are not original to whoever created the video, so
they do not inherit the video's copyright.\",\"sourceName\":\"National Basketball Ass'n v. Motorola,
Inc., 105 F.3d 841 (2d Cir. 1997) ? Wikipedia case summary\",\"sourceUrl\":\"https://en.wikipedia.or
g/wiki/National_Basketball_Ass%27n_v._Motorola,_Inc.\",\"confidence\":\"high\"},{\"claim\":\"Copyright
never extends to any idea, procedure, process, system, method of operation, concept, principle,
or discovery ? only to original expression. Measured coordinates and timestamps are
'discoveries'/facts about the recorded event and fall in the §102(b) exclusion.\",\"support\":\"17
U.S.C. § 102(b): 'In no case does copyright protection for an original work of authorship extend
to any idea, procedure, process, system, method of operation, concept, principle, or discovery,
regardless of the form in which it is described, explained, illustrated, or embodied in such
work.' This is the statutory codification of the idea/expression dichotomy and confirms that the
factual/discovered content extracted from the video (the where-and-when of the shuttle and
rallies) is outside the scope of the video's copyright.\",\"sourceName\":\"17 U.S.C. § 102 ? BitLaw
/ U.S. Code\",\"sourceUrl\":\"https://www.bitlaw.com/source/17usc/102.html\",\"confidence\":\"high\"},{\"c
laim\":\"A 'derivative work' is a work that recasts, transforms, or adapts a preexisting work's
EXPRESSION (e.g., translation, motion-picture version, abridgment). Stripping out non-expressive
numeric facts is not 'recasting' the audiovisual expression, so the coordinate/timestamp/feature
data is not a derivative work that inherits the video's copyright.\",\"support\":\"17 U.S.C. § 101
defines a derivative work as 'a work based upon one or more preexisting works, such as a
translation, musical arrangement, dramatization, fictionalization, motion picture version, sound
recording, art reproduction, abridgment, condensation, or any other form in which a work may be
```


recast, transformed, or adapted.' Each enumerated example reuses the original's protected EXPRESSION. A table of (x,y) positions and timestamps reproduces none of the video's expressive elements (framing, editing, commentary, imagery); it captures only facts, which §102(b)/Feist place outside copyright. A derivative-work claim therefore fails because there is no protected expression carried into the output.", "sourceName": "17 U.S.C. § 101 Definitions ? Legal Information Institute (Cornell)", "sourceUrl": "https://www.law.cornell.edu/uscode/text/17/101", "confidence": "high"}, {"claim": "A compilation/database of facts can earn a THIN copyright, but only in the original SELECTION, COORDINATION, OR ARRANGEMENT ? never in the underlying facts. So a creatively-organized dataset might get limited protection for its structure, but a competitor remains free to copy the raw numbers themselves.", "support": "17 U.S.C. § 101 defines a 'compilation' as data 'selected, coordinated, or arranged in such a way that the resulting work as a whole constitutes an original work of authorship,' and Feist holds copyright in a compilation 'extends only to those components of the work that are original to the author, not to the facts themselves.' Practical upshot: any copyright the extractor obtains is in the dataset's expressive arrangement (if creative), not in the shuttle coordinates, rally timestamps, or onset values, which remain free for others to use.", "sourceName": "Feist Publications, Inc. v. Rural Telephone Service Co., 499 U.S. 340 (1991) ? Justia", "sourceUrl": "https://supreme.justia.com/cases/federal/us/499/340/", "confidence": "high"}, {"claim": "The act of EXTRACTION (transiently copying the video to analyze it) is a distinct question from the status of the OUTPUT data ? and intermediate copying CAN infringe even when the end product does not reproduce protected expression, so the extraction step must independently rely on a license or fair use.", "support": "Sega Enterprises Ltd. v. Accolade, Inc., 977 F.2d 1510 (9th Cir. 1992): 'intermediate copying of computer object code may infringe the exclusive rights granted to the copyright owner in section 106 ... regardless of whether the end product of the copying also infringes those rights.' But the same court held that copying to access UNPROTECTED functional/ideational elements is fair use where it is the only means of access and there is a legitimate reason, because §102(b) elements 'are not protected by copyright.' Applied here: even though the numeric output is unprotectable, ingesting the video to compute it is itself a reproduction that needs a lawful basis (license to the source video, or a fair-use justification for the analytic copying).", "sourceName": "Sega Enterprises Ltd. v. Accolade, Inc., 977 F.2d 1510 (9th Cir. 1992) ? Bitlaw", "sourceUrl": "https://www.bitlaw.com/source/... (truncated 267202 chars, full result in C:\\Users\\avidu\\AppData\\Local\\Temp\\claude\\C--Users-avidu-Projects-badminton-highlight-indexer\\068560d3-1ca2-4387-981a-d2d6bca55414\\tasks\\wgpct1wbd.output)"}</result><usage><agent_count>18</agent_count><subagent_tokens>261531</subagent_tokens><tool_uses>44</tool_uses><duration_ms>789624</duration_ms></usage></task-notification>

79. assistant (2026-06-16T04:04:09.020Z)

The research workflow is complete (18 agents, the resume worked). Now let me harvest the key sections ? the legal memo, the unified architecture, and the red-team ? so the doc is grounded in the actual outputs rather than my paraphrase.

80. user (2026-06-16T04:04:11.407Z)

```
711:   "memo": {
712:[Omitted long matching line]
713:[Omitted long matching line]
742:     "conditionsForSafePublicCommit": [
1037:    "designs": [
1113:    "techSynth": {
1114:      "architectureName": "SEALED-FIXTURES: a tiered (Tier-0 public / Tier-1 key-gated),
de-attributed-at-source, encrypted-where-needed regression-fixture architecture for the existing
video-free eval seam",
1145:      "openOwnerQuestions": [
1164:      "redteam": {
1237:[Omitted long matching line]
```

81. user (2026-06-16T04:04:18.945Z)

```
711     "memo": {
712       "executiveSummary": "INTERNAL MEMO ? general legal information, NOT legal advice;
retained India + EU/UK counsel sign-off required before any public commit. QUESTION: can the
```

project commit DERIVED numeric data (per-frame shuttle (x,y) coordinates, rally start/end time-intervals, audio/person/court feature columns, ground-truth intervals) into a public GitHub repo so external contributors can run the rally-segmentation regression suite WITHOUT the videos? BOTTOM LINE: Yes, but only under conditions, and the analysis FORKS on data provenance. The OUTPUT data itself is, across all three jurisdictions, essentially uncopyrightable FACT/measurement ? it does not inherit the source video's copyright (US: Feist + 17 USC 102(b) + NBA v. Motorola; India: EBC v. D.B. Modak, no sui generis right; EU: Football Dataco v. Yahoo ? copyright protects only original selection/arrangement, never the numbers). So a thin compilation copyright at most attaches to a creative schema, never to the coordinates. BUT copyright is the easy axis and NOT where the risk lives. The real exposure is fourfold and provenance- and stream-dependent: (1) the ACT of ingesting/extracting from the source video needs its own lawful basis ? owned/permissioned video is clean; a US fair-use defense for analytic extraction is now CONTESTED, not \"strongly favorable,\" after Thomson Reuters v. Ross (2025) and the US Copyright Office May-2025 report, and unlawful acquisition independently destroys the defense (Bartz v. Anthropic, ~\$1.5B settlement); (2) CONTRACT / Terms-of-Service can bar extraction and redistribution even where copyright cannot reach the facts (ProCD; hiQ v. LinkedIn; YouTube ToS bars scraping AND \"harvest information that might identify a person (for example... faces)\" AND redistribution) ? and the remedy is injunction + forced deletion of the committed dataset, not nominal damages; (3) EU/UK sui generis DATABASE RIGHT may subsist in self-derived match-event/coordinate streams because, on the leading Football Dataco v. Sportradar reasoning, recording pre-existing on-court facts you do not control is \"OBTAINING,\" not \"creating\" ? so the project's comfortable \"we created it\" premise is genuinely contestable; (4) PRIVACY ? person/pose/gait streams and anything involving MINORS are the highest-risk and several should NOT be published at all. The cleanest defensible path: commit ONLY owned-or-permissioned-source, minors-excluded, person/pose/gait-EXCLUDED (shuttle + abstract rally-interval + non-biometric feature) data, stripped of every source-attribution field (no filename, no video_id MD5, no match metadata), under a no-extraction-from-third-party-source and no-redistribution hygiene posture, with optional encryption only as a barrier against the no-key public (it cannot bind a contributor who runs the tests).\",

713 \"derivedDataCoreVerdict\": \"The derived numeric data DOES NOT inherit the source video's copyright in any of the three jurisdictions ? this is the one firmly-settled conclusion. Per-frame shuttle (x,y) coordinates, rally start/end timestamps, and audio/person/court feature values are uncopyrightable FACTS/measurements about a recorded physical event: discovered, not authored. US: Feist (facts never copyrightable; \"originality, not sweat of the brow, is the touchstone\"), 17 USC 102(b) (no copyright in any \"discovery\"), NBA v. Motorola (facts generated by a sporting event are uncopyrightable \"purely factual information which any patron... could acquire from the arena\" even though the broadcast is protected). A coordinate/timestamp table is also NOT a 17 USC 101 \"derivative work\": a derivative work recasts the original's protected EXPRESSION (framing, editing, imagery, commentary), and a number table carries none of it. India: EBC v. D.B. Modak ? copyright needs \"skill and judgment,\" protects only original arrangement, not facts; no EU-style sui generis right exists. EU: Football Dataco v. Yahoo (C-604/10) ? database copyright protects only an original selection/arrangement; \"the intellectual effort and skill of creating that data are not relevant\" and sweat-of-the-brow cannot ground it. CEILING ON PROTECTION: the MOST that can attach is a THIN compilation copyright in a genuinely creative SELECTION/ARRANGEMENT (e.g., a curated schema) ? never in the raw numbers, and a mechanical/exhaustive layout (every frame, every rally) likely has too little originality even for that (Feist's \"garden-variety listing\" problem). CRITICAL SEPARATION: the STATUS of the output data (settled: unprotected facts) is a different question from the ACT of extraction (transiently copying the video to compute the data), which can independently infringe (Sega v. Accolade: intermediate copying can infringe even when the end product reproduces no protected expression) and which needs its own license or fair-use basis. NON-COPYRIGHT FENCES survive precisely BECAUSE the data is uncopyrightable: enforceable no-extraction ToS (ProCD), narrow hot-news misappropriation (NBA v. Motorola), and the EU/UK sui generis database right. So \"the numbers are free facts\" answers the copyright question and ONLY the copyright question.\",

714 \"byJurisdiction\": [
715 {
716 \"jurisdiction\": \"India (PRIMARY ? owner is Bangalore-based, commercialization intended)\",
717 \"copyright\": \"Derived facts are NOT copyrightable as such: EBC v. D.B. Modak (2008 SC) requires 'skill and judgment' and protects only original arrangement, not facts; India has NO EU-style sui generis database right. BUT this is not as clean as it looks: Indian High Courts protect DATABASES/COMPILATIONS as 'original literary works' under s.2(o) on a low 'skill, labour and judgment' threshold ? Burlington Home Shopping v. Rajnish Chibber, V. Govindan v. E.M. Gopalakrishna Kone, and the OLX v. Padawan order itself (which held a scraped dataset to be

a protectable 'proprietary database'). So if numbers are LIFTED from a third party's structured dataset/feed, copying its selection/arrangement can infringe even though the raw facts are free. Self-derived-from-pixels data avoids this; copying an official stats compilation does not.",

718 "databaseRight": "No sui generis database right exists in India ? a genuine permissiveness advantage over the EU/UK. Protection runs only through copyright (compilation, above) or contract/IT-Act.",

719 "contractToS": "The LIVE risk, not copyright. Breach of anti-scraping ToS plus IT Act s.43 (Government has affirmatively argued public-data scraping 'may be in breach of section 43') and s.66 layering CRIMINAL exposure (up to 3 years / Rs 5 lakh). Enforceability of anti-scraping ToS is untested in Indian courts. OLX v. Padawan was decided AGAINST the scraper; its 'private use' line is obiter, not a safe harbor. Trade-secret/confidentiality for the owner's OWN corpus runs NOT through any US-style DTSA (India has no trade-secret statute) but through the Indian Contract Act 1872, equity, common-law breach of confidence, and IT Act s.72 ? and publication still forfeits the moat under that doctrine.",

720 "privacy": "DPDP Act 2023 governs personal data of identifiable natural persons; genuinely anonymized numbers fall outside it, but the s.3(c)(ii) 'publicly available data' carve-out is CONTESTED (MeitY/Minister has said scrapers must still meet DPDP consent obligations). MINORS are stricter than the EU: 'child' = under 18 (s.2(f)); s.9(1) requires VERIFIABLE parental consent (DPDP Rules 2025 Rule 10 ? real ID/age verification, not a checkbox); s.9(2) bars detrimental processing; s.9(3) PROHIBITS tracking/behavioural monitoring/targeted ads of children ? subject to narrow Rule 12/Fourth-Schedule class/purpose exemptions that do NOT cover an amateur highlight indexer, so the ban DOES bite this project. A persistent per-player pose/gait signature is, on the conservative reading, prohibited behavioural monitoring; 'tracking' is undefined in the Act, so this is the defensible-conservative, not the certain, reading.",

721 "netVerdict": "MORE PERMISSIVE than the EU on the database-right axis, but NOT 'lawful on its face.' Defensible to publish ONLY genuinely-anonymized, independently-derived (not lifted from a third-party compilation) numbers, with NO minors tracked and accepting that anti-scraping-ToS + IT Act s.43/s.66 exposure for any third-party-sourced ingestion is unresolved. Owned-source + minors-excluded + no-person/pose/gait = clean."

722 },

723 {

724 "jurisdiction": "United States (repo + product are global; US fair-use & contract law are the most-developed)",

725 "copyright": "Output data does not inherit the video's copyright (Feist; 102(b); NBA v. Motorola); not a 101 derivative work; thin compilation copyright at most. The EXTRACTION act is the open question: fair use for analytic, non-expressive copying was historically favorable (Authors Guild v. Google; Google v. Oracle) but is now CONTESTED ? Thomson Reuters v. Ross (D. Del. 2025) REJECTED the 'copy an expressive work just to reach unprotected facts/data' theory, distinguished the functional-CODE cases (Sega/Oracle), found the use non-transformative, and found Factor-4 harm in a derivative 'data to train AIs' market the plaintiff had not even entered; the US Copyright Office (May 2025) rejected 'non-expressive therefore inherently transformative' and backs a market-dilution harm theory. If the dataset competes with a rights-holder's own data/analytics market, fair use may FAIL. Unlawful source acquisition independently defeats the defense regardless of how transformative the end use is (Bartz v. Anthropic, 2025; ~\$1.5B settlement).",

726 "databaseRight": "No sui generis database right in the US (Feist rejected sweat-of-the-brow). Competitors stay free to copy raw facts ? subject to the non-copyright fences below.",

727 "contractToS": "The dominant US risk. Post-Van Buren the CFAA no longer reaches mere purpose-violations of accessible/public data (though footnote 8 RESERVED whether contract/policy limits can still ground CFAA liability ? the federal-crime route is narrowed, not dead), so platforms enforce via CONTRACT. ToS is an enforceable contract that bites where copyright/CFAA fail (ProCD; hiQ v. LinkedIn ? breach-of-contract summary judgment + \$500k consent judgment + injunction + ORDER TO DESTROY scraped-derived data). Binding turns on assent: a logged-in/account/API-key or genuine clickwrap user is bound (hiQ; any YouTube-API user); a purely logged-out scraper under browswrap-only may not be (Meta v. Bright Data) ? and thin 'sign-in wrap' flows can fail for want of conspicuous notice. YouTube ToS verbatim bars automated access, 'harvest any information that might identify a person (for example... faces)', and redistribution/commercial reuse. REDISTRIBUTION (committing the dataset publicly) is the single highest-risk act and the remedy is injunction + forced deletion, not nominal damages.",

728 "privacy": "GDPR-style analysis below applies to global users; US state biometric statutes (e.g., Illinois BIPA ? not separately briefed here) and publicity/personality rights are additional fences for face crops and identifiable-player gait. Faces in any committed imagery = identifying personal data; do not publish.",

```

729         "netVerdict": "Output data publishable as facts ONLY if (i) source video was
owned/permissioned (or extraction has a defensible fair-use basis you do NOT over-rely on), (ii)
no enforceable ToS bars extraction/redistribution of the ingested source, and (iii) no
person/pose/gait/face/minor stream is included. Treat third-party-ToS-sourced derived data as
NOT publicly committable."
730     },
731     {
732         "jurisdiction": "EU / UK (product/repo global; English-language highlight
market makes UK a likely target)",
733         "copyright": "Database copyright protects only an original
selection/arrangement, never the numbers, and never sweat-of-the-brow (Football Dataco v. Yahoo,
C-604/10). Self-derived coordinates are facts, not a copied original structure ? so copyright is
a low axis.",
734         "databaseRight": "The genuinely contestable axis. The sui generis right
protects investment in OBTAINING/verifying/presenting pre-existing data, NOT in CREATING it (BHB
v. William Hill C-203/02; Fixtures Marketing) ? which the project reads as 'we create
coordinates frame-by-frame, so we are clear.' THAT READING IS OVERCONFIDENT: in Football Dataco
v. Sportradar the live match-event data (goals/scorers) was held to be OBTAINED, because the
recorder captured pre-existing facts ('events taking place on the pitch') it did NOT control. A
badminton rally/shuttle position is the same kind of pre-existing on-court fact recorded rather
than authored, so a rights-holder can argue the right SUBSISTS in our streams. Infringement is
now governed by CV-Online Latvia (C-762/19, 2021): even a substantial-part taking is actionable
only where it adversely affects the maker's ability to recoup investment, balanced against
information flow/competition ? a fact-intensive harm test, not a clean 'no protected part =
safe' rule. Sportradar (C-173/11) also defeats the server-location dodge for EU-targeted re-
utilisation. UK retained its own sui generis right and pre-2021 CJEU case law, so UK-targeted
products carry AT LEAST equal, arguably greater, risk; a non-UK/EEA maker may not even hold the
right defensively, which does nothing to reduce exposure as a potential infringer.",
735         "contractToS": "Ryanair v. PR Aviation (C-30/14): where a database enjoys
NEITHER copyright NOR the sui generis right, the Directive's user-protections (incl. no-
contracting-out) do not apply, so the source's T&Cs/licence can lawfully BAN automated
extraction and redistribution and bind anyone who agreed ? the dominant residual risk. Trap: the
licence that legitimises INGESTION of the footage is often the very instrument that
contractually bans extraction/redistribution of derived data.",
736         "privacy": "GDPR: shuttle/object coordinates are NOT personal data in
isolation (Art.4(1)) ? but coordinates published as part of an identifiable named match are part
of a personal-data operation. Person bounding boxes that single out a human are ordinary
personal data, NOT Art.9 biometric UNLESS used to uniquely identify (EDPB Guidelines 3/2019).
Pose/gait is the danger zone: gait re-identifies even with faces blurred and skeleton
'anonymization' leaves gait identity intact (Weiss et al., Sensors 2025) ? Art.9 special-
category the moment used to uniquely identify; explicit-consent basis required. 'Anonymous' is
RELATIVE post EDPS v. SRB (CJEU C-413/23 P, 4 Sept 2025): a stream can be non-attributable to a
recipient lacking the source yet remains PERSONAL DATA to the holder who can re-run the
detector. Minors: Art.8 (consent floor 13?16); any biometric/special-category processing of a
child stacks Art.9.",
737         "netVerdict": "Do NOT assume self-derivation defeats the database right ?
treat self-derived event/coordinate streams as potentially 'obtained' data and avoid taking in a
way that could reconstitute a rights-holder's dataset or harm their feed's monetisation. Read
every source's T&Cs for extraction/redistribution clauses. Keep the footage-copyright analysis
(highlight reels) strictly separate. Person/pose/gait/face = do not publish without an Art.9
explicit-consent basis. Get EU + UK jurisdiction-specific advice before redistributing numeric
streams at scale."
738     }
739 },
740     "provenanceFork": "PROVENANCE IS THE DECISIVE FORK ? the legal posture of an
identical numeric file flips entirely on where the source video came from. FORK A ? OWNED /
OWNER-RECORDED (or fully-permissioned/licensed-for-this-use) SOURCE: This is the clean path and
the only one safe to commit publicly. No third-party copyright in the source bars the
extraction; no third-party ToS governs the owner's own recording (with two caveats below); the
EU/UK database-right 'obtained vs created' debate still applies in theory but there is no
adverse rights-holder to assert it against the owner's own footage. The analysis collapses to:
(1) a TRADE-SECRET / competitive-moat CHOICE (US framing: 18 USC 1839 DTSA requires secrecy
measures + value-from-secrecy; the OPERATIVE framing for this Bangalore owner is Indian breach-
of-confidence / Contract Act 1872 / IT Act s.72, NOT the DTSA) ? publishing the data forfeits
whatever moat it carried, which is the owner's strategic call to make, weighed against the

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stated community/benchmark/paper-co-authorship aims; and (2) PRIVACY ? the owner's own footage still contains identifiable people, so DPDP/GDPR and the minors rules below still bind, and person/pose/gait/face streams are still off-limits. TWO CAVEATS that narrow Fork A: (a) 'owner-recorded' is clean ONLY for the owner's OWN non-ticketed/private sessions; footage the owner SHOT AT a third party's event (tournament, club match) carries live exposure ? a ticket is a contract of admission that routinely bars recording/redistribution, and venue/organiser/broadcaster rights (e.g., BWF/league terms, Rome Convention rebroadcast rights) attach independently of any platform ToS; (b) any third-party-collected footage in the corpus (the project's golden set may mix sources) taints those records and pushes them into Fork B. FORK B ? THIRD-PARTY-SOURCED (broadcast / YouTube / scraped / downloaded) SOURCE: Derived data from this provenance SHOULD NOT be committed to a public repo. Even though the numbers are uncopyrightable facts, (1) the EXTRACTION act needs a license or a now-CONTESTED US fair-use defense, and unlawful acquisition (e.g., scraping against ToS, downloading from a platform that bars it) independently sinks the defense (Bartz v. Anthropic); (2) the source's ToS (YouTube et al.) typically bars automated extraction AND redistribution, and committing the derived dataset publicly IS redistribution ? the highest-risk act, remediable by injunction + forced deletion of the committed data (hiQ); (3) re-utilisation/extraction of a substantial part may infringe the EU/UK sui generis database right if the source ties back to a protected feed. PRACTICAL RULE FOR THE REPO: gate the committed corpus to FORK A only. Tag every golden-set record with a verified provenance flag at ingestion; only owned/permissioned-source, non-event-ticketed, minors-excluded, person/pose/gait-excluded records are eligible for the public commit. This maps directly onto the existing split: eval_baselines/ (aggregate JSON snapshots) is already committed precedent and is provenance-agnostic if it carries no per-source attribution; *_golden_features.json are currently kept private (docs/GOLDEN_DATA_SHARING.md) precisely because they derive from a 'Proprietary corpus' ? the right move is to keep Fork-B-tainted features private and publish only a Fork-A, de-attributed subset.",

741 "privacyVerdict": "PUBLISH/NO-PUBLISH BY STREAM (the project's guardrails already point the right way; this hardens them). PUBLISH-OK (lowest risk): (1) SHUTTLE (x,y) coordinates + Visibility ? describe an inanimate object, NOT personal data on their own (GDPR Art.4(1); WP136 content/purpose/result test) ? PROVIDED they are NOT joined to anything that singles out a named player and NOT published as an identifiable named match (the surrounding personal-data obligations survive even when the coordinate column alone is object-data). (2) Abstract RALLY start/end time-INTERVALS and ground-truth intervals stripped of match/player identity ? facts about event timing. (3) Non-biometric AUDIO-ONSET, COURT-BOUNDS, and aggregate PERSON COUNT/feature columns used ONLY for rally detection and DETACHED from any identity ? ordinary low-risk data with normal hygiene; NOT Art.9 because the purpose is detect/categorize, not uniquely identify (EDPB Guidelines 3/2019). CONDITIONAL ? only with an Art.9 explicit-consent basis (and effectively never for this v1): PERSON bounding-box coordinates that persistently single out and follow a specific player across rallies ? the purpose flips to unique identification and Art.9 attaches; this architecture is PRONE to that flip, so treat per-player tracking as a tripwire. DO NOT PUBLISH (high risk, exclude from any committed stream): (4) POSE keypoints / SKELETON / GAIT signatures ? gait is a behavioural biometric that re-identifies people even with faces blurred, and pose 'anonymization' provably leaves gait identity intact (Weiss et al., Sensors 2025, 80?95% gait-recognition on anonymized data); raw pose/gait tracks are NOT reliably anonymous, are personal data when usable to single out a player, and become Art.9 special-category once used to uniquely identify ? this matches the project's existing 'gait/biometrics strictly FUTURE (GDPR Art.9)' guardrail. (5) Any FACE CROP or identifiable IMAGERY ? identifying personal data under GDPR and DPDP, and barred by YouTube ToS's face-harvesting clause; do not commit imagery at all. (6) Anything that joins a NAMED PLAYER to a gait/pose signature. MINORS FLIP EVERYTHING: under DPDP a 'child' is anyone under 18; processing ANY child's data needs VERIFIABLE parental consent (Rule 10 ? real ID/age check, not a checkbox), must not cause detrimental effect, and s.9(3) PROHIBITS behavioural monitoring/tracking of children (Rule 12/Fourth-Schedule exemptions do not cover this project, so the ban applies). A persistent per-player pose/gait signature is, conservatively, prohibited monitoring. CLEANEST PATH for any footage containing minors: EXCLUDE minors entirely from any persisted derived stream beyond inanimate shuttle/interval data ? consistent with the existing 'EXCLUDE MINORS from the training pool' guardrail ? or obtain Rule-10 verified parental consent and keep only de-identified object/coordinate data. NON-ATTRIBUTABILITY ARCHITECTURE (the 'secret file' question): Non-attributability is a property of the (data, holder, environment) triple, not of the timeseries. A per-video trajectory/feature stream is effectively a FINGERPRINT (de Montjoye: 4 points re-identify 95% of 1.5M people; k-anonymity collapses on rich quasi-identifiers), so a deterministic detector re-run on a candidate video can correlate-match it to its source ? BUT only if a correlatable reference EXISTS. PUBLIC source: a permanent cheap reference exists, re-linkage is 'reasonably likely', so a raw public-sourced stream should be treated as ATTRIBUTABLE; only heavy aggregation, perturbation/differential privacy, synthetic

data, or a non-public enclave (NIST SP 800-188) breaks correlatability. PRIVATE/owner-recorded-and-unpublished source: there is no external reference to correlate against, so source-linkage by an EXTERNAL adversary is not 'reasonably likely' ? BUT it is WRONG to call this 'anonymous by default' (EDPS v. SRB, CJEU C-413/23 P, 4 Sept 2025): to the holder who can re-run the detector, it REMAINS personal data; it is merely RELATIVELY non-attributable to outsiders, and that protection is conditional, revocable (publish or leak the source and the link snaps back), and forward-looking (regulators say assume the data environment grows). HARD ENGINEERING IMPLICATION: the committed fixtures must NOT carry the source filename, match metadata, or the video_id (which is the MD5 of the source file ? using it as a public key would FINGERPRINT/confirm the source the instant anyone holds the file). Use an opaque, random, non-reversible surrogate id per record. Strip the 'name' field in *_golden_features.json. The lead's established constraint is correct: a contributor who RUNS the tests can necessarily read the plaintext fixtures and any key on their machine (DRM/analog-hole limit), so secrecy-from-the-contributor is impossible; the achievable goal is NON-ATTRIBUTABILITY + uselessness-in-isolation, with encryption useful ONLY as a barrier against the no-key public, never against a running contributor.",

742 "conditionsForSafePublicCommit": [

743 "PROVENANCE GATE: commit ONLY derived data from owner-recorded / fully-permissioned source video (Fork A). Tag every golden-set record with a verified provenance flag at ingestion; exclude any third-party-sourced (broadcast/YouTube/scraped/downloaded) record from the public commit. For owner-SHOT footage of a third party's event, first clear the admission ticket / venue / organiser / broadcaster terms ? only owner's own non-ticketed sessions are unconditionally clean.",

744 "SOURCE-LICENCE/ToS CHECK: for every source that touched the committed corpus, confirm in writing that its licence/ToS does NOT bar automated extraction OR redistribution of derived data. Do not rely on US fair use for the extraction step ? it is now contested (Thomson Reuters v. Ross; USCO May-2025) and unlawful acquisition independently defeats it (Bartz v. Anthropic). Redistribution (the public commit) is the single highest-risk act.",

745 "MINORS EXCLUSION: exclude all footage containing minors (under 18, per DPDP s.2(f)) from any persisted/committed derived stream beyond inanimate shuttle/interval data ? unless Rule-10 verifiable parental consent is held AND only de-identified object/coordinate data is kept. s.9(3)'s behavioural-monitoring ban means a persistent per-player pose/gait signature on a minor is prohibited.",

746 "BIOMETRIC EXCLUSION: do NOT publish pose keypoints, skeleton tracks, gait signatures, face crops, identifiable imagery, or any named-player-to-movement-signature mapping. Restrict the committed feature columns to shuttle (x,y)+Visibility, abstract rally/GT time-intervals, and non-biometric audio/court/aggregate-person features used only for rally detection and detached from identity.",

747 "DE-ATTRIBUTION: strip every source-identifying field from the committed fixtures ? source filename, match/event/player metadata, capture dates, and especially the video_id (MD5 of the source file). Replace with an opaque, random, non-reversible surrogate record id. The 'name' field in *_golden_features.json must be removed/aliased.",

748 "NON-ATTRIBUTABILITY POSTURE: because per-video trajectories are fingerprints correlatable to a source video, achieve non-attributability by keeping the SOURCE private (Fork A, unpublished) ? which makes external source-linkage not 'reasonably likely' ? and/or by coarsening/aggregating/perturbing where feasible. Document the (data, holder, environment) assessment and re-test it as the public-video environment grows (ICO/OVIC forward-looking duty). Treat the streams as personal data IN THE OWNER'S OWN HANDS regardless (EDPS v. SRB).",

749 "USELESS-IN-ISOLATION + ENCRYPTION-AS-PUBLIC-BARRIER ONLY: keep the committed fixtures useless without the regression harness + schema (abstract intervals + opaque ids, no imagery, no source pointer). Use encryption only as a barrier against the no-key public; do not represent it as protecting against a contributor who runs the tests (they can read plaintext + key locally ? the lead's established DRM/analog-hole constraint).",

750 "PRECEDENT-CONSISTENT PLACEMENT: publish via the already-committed eval_baselines/ pattern (aggregate, provenance-agnostic JSON snapshots) for headline regression baselines; keep Fork-B-tainted or biometric-bearing *_golden_features.json private (per docs/GOLDEN_DATA_SHARING.md) and publish only a Fork-A, de-attributed, minors-excluded, biometric-free subset of features needed to run the suite.",

751 "DPDP HYGIENE: ensure any person-derived numeric column that could single out a player is genuinely identity-detached; do not rely on the DPDP s.3(c)(ii) 'publicly available data' carve-out for non-anonymized personal data (it is contested). Maintain purpose-limitation: features persisted for rally detection must not be repurposed for player identification.",

752 "COUNSEL SIGN-OFF: obtain retained India counsel (DPDP s.9/Rules, IT Act s.43/s.66, breach-of-confidence) AND EU/UK counsel (sui generis database right 'obtained vs created', CV-Online harm test, Ryanair contract) review of the final committed subset and the

source-provenance log BEFORE the first public commit. This memo is general information, not advice."

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753     ],
754     "redFlags": [
755         "OVERCONFIDENCE TRAP #1 ? 'we created the data, so the EU/UK database right
cannot apply.' Football Dataco v. Sportradar held live match-event data was OBTAINED (pre-
existing on-court facts the recorder did not control), exactly analogous to a recorded badminton
rally/shuttle position. A rights-holder can credibly argue the sui generis right subsists in our
self-derived streams. Do not bank on the 'created' characterisation.",
756         "OVERCONFIDENCE TRAP #2 ? 'analytic extraction is strongly favorable fair use
(US).' Thomson Reuters v. Ross (2025) REJECTED the 'copy an expressive work to reach unprotected
facts' theory and found Factor-4 harm in a derivative data market; the USCO (May 2025) rejected
'non-expressive = inherently transformative.' If the dataset competes with a rights-holder's
data/analytics market, fair use may fail. The extraction step is litigable, not settled.",
757         "OVERCONFIDENCE TRAP #3 ? 'owner's own recordings have no contract problem.'
False for footage shot at a third party's event: the admission ticket is a contract barring
recording/redistribution, and venue/organiser/broadcaster rights (BWF/league/Rome Convention)
attach independently. Only the owner's own non-ticketed sessions are unconditionally clean.",
758         "OVERCONFIDENCE TRAP #4 ? 'private/unpublished source = anonymous by default.'
Per EDPS v. SRB (CJEU 2025) the stream REMAINS personal data to the holder who can re-run the
detector; it is only RELATIVELY non-attributable to an external adversary lacking the source,
and that is conditional, revocable, and forward-looking. Misclassifying it as 'anonymous' is the
exact GDPR error the controller's-perspective holding exists to prevent.",
759         "FINGERPRINTING / video_id leak: the project's video_id is the MD5 of the source
file. Committing it (or any source filename/match metadata) as a public key would let anyone
holding the candidate video CONFIRM the source instantly ? defeating non-attributability. Must
be stripped and replaced with an opaque surrogate.",
760         "REDISTRIBUTION REMEDY is not nominal damages: the operative relief in
scraping/ToS disputes is INJUNCTION + FORCED DELETION of the derived dataset (hiQ ordered
destruction of all scraped-derived code/data). A public commit that breaches a source ToS can be
ordered taken down and deleted ? a real, not theoretical, downside.",
761         "POSE/GAIT mislabeled low-risk: skeleton 'anonymization' leaves gait identity
intact (80?95% re-ID), so any pose/gait stream is NOT reliably anonymous and is Art.9-adjacent.
Excluding it is not optional polish ? it is a guardrail.",
762         "MINORS + behavioural monitoring: DPDP s.9(3) bars behavioural monitoring of
under-18s and the Rule 12/Fourth-Schedule exemptions do not cover this project. A persistent
per-player pose/gait signature on a minor is, conservatively, a prohibited act with criminal-
adjacent regulatory exposure ? exclude minors from any persisted stream beyond inanimate
shuttle/interval data.",
763         "MIXED-PROVENANCE corpus: the golden set may blend owned and third-party
footage. A single Fork-B-tainted record contaminates the public commit. Provenance must be
verified per-record, not assumed corpus-wide.",
764         "WRONG-LAW trade-secret framing: analysing the owner's moat under US 18 USC 1839
(DTSA) is a jurisdictional error ? the Bangalore owner's protection runs through Indian breach-
of-confidence / Contract Act 1872 / IT Act s.72. The publish-forfeits-the-moat intuition
survives, but the operative doctrine is different.",
765     ],
766     "counselQuestions": [
767         "EU/UK sui generis database right: on the Football Dataco v. Sportradar
'obtained vs created' reasoning, does the right subsist in our self-derived shuttle-coordinate /
rally-event streams, and if so does redistributing a de-attributed regression subset
'extract/re-utilise a substantial part' and cause CV-Online-Latvia-type harm to any rights-
holder's feed? (EU AND UK, given likely English-language market.)",
768         "US fair use for the EXTRACTION step, post-Thomson Reuters v. Ross and the USCO
May-2025 report: is transient analytic copying of a source video to compute rally/coordinate
features defensible where the resulting dataset could be seen to compete in a 'data/analytics'
market ? and does it matter that we only ever publish numbers, never the footage?",
769         "India: are anti-scraping/ToS terms enforceable in Indian courts, and what is
the realistic IT Act s.43 (civil) / s.66 (criminal) exposure for ingesting third-party footage
to derive features ? and does the pending ANI v. OpenAI judgment (reserved, ~April 2026) change
the fair-dealing/TDM analysis?",
770         "India DPDP s.9 + Rules 10/12: for a corpus that may contain players under 18,
what verifiable-parental-consent and exemption analysis applies, and does a persistent per-
player pose/gait (or even per-player rally-tracking) feature constitute prohibited 'behavioural
monitoring/tracking' of a child under s.9(3)?",
```

771 "Owner-shot event footage: do the admission tickets / venue / organiser (e.g., BWF, local league) / broadcaster terms for the specific tournaments and clubs in our corpus bar recording, derivation, or redistribution of derived data ? and which records must therefore be excluded from the public commit?",

772 "Personal-data classification under GDPR + DPDP for the SPECIFIC committed feature columns: at what point do identity-detached person/court/coordinate features 'single out' a player (especially in a small known squad/roster), and does our de-attribution + opaque-surrogate-id scheme suffice to keep the published subset outside personal-data obligations for downstream recipients (per EDPS v. SRB)?",

773 "Trade-secret / breach-of-confidence (Indian law): given commercialization and possible paper co-authorship, what is the moat trade-off of publishing the Fork-A subset, and are there confidentiality/Contract-Act 1872 / IT Act s.72 steps to take before forfeiting secrecy on any owner-recorded derived data?",

774 "Source-licence audit: for each source category in the corpus (owner-recorded, permissioned partner, downloaded/scraped), confirm the licence/ToS expressly permits BOTH automated extraction AND public redistribution of derived numeric data ? and where silent or prohibitive, which records are ineligible.",

775 "Encryption + 'secret file' design: does adding encryption that a running contributor can necessarily decrypt create any misrepresentation/false-security or DPDP 'reasonable security safeguards' compliance question, and is non-attributability-via-private-source a defensible posture to assert to regulators given its conditional/revocable nature?",

776 "Cross-border data transfer: does committing any record that is (or may be argued to be) personal data to a global public GitHub repo implicate DPDP cross-border-transfer rules or GDPR international-transfer obligations, independent of the copyright/contract analysis?"

```

777     ],
778     "citations": [
779         {
780             "name": "Feist Publications, Inc. v. Rural Telephone Service Co., 499 U.S. 340
(1991) ? Justia",
781             "url": "https://supreme.justia.com/cases/federal/us/499/340/"
782         },
783         {
784             "name": "National Basketball Ass'n v. Motorola, Inc., 105 F.3d 841 (2d Cir.
1997) ? Wikipedia case summary",
785             "url":
"https://en.wikipedia.org/wiki/National_Basketball_Ass%27n_v._Motorola,_Inc."
786         },
787         {
788             "name": "17 U.S.C. § 102 ? BitLaw / U.S. Code",
789             "url": "https://www.bitlaw.com/source/17usc/102.html"
790         },
791         {
792             "name": "17 U.S.C. § 101 Definitions ? Legal Information Institute (Cornell)",
793             "url": "https://www.law.cornell.edu/uscode/text/17/101"
794         },
795         {
796             "name": "Sega Enterprises Ltd. v. Accolade, Inc., 977 F.2d 1510 (9th Cir.
1992) ? BitLaw",
797             "url": "https://www.bitlaw.com/source/cases/copyright/Sega-Accolade.html"
798         },
799         {
800             "name": "Authors Guild v. Google, Inc., 804 F.3d 202 (2d Cir. 2015) ? Justia",

```

82. assistant (2026-06-16T04:04:24.643Z)

That memo is exactly what the doc needs. Now the unified architecture and the red-team:

83. user (2026-06-16T04:04:27.752Z)

```

1113     "techSynth": {
1114         "architectureName": "SEALED-FIXTURES: a tiered (Tier-0 public / Tier-1 key-gated),
de-attributed-at-source, encrypted-where-needed regression-fixture architecture for the existing
video-free eval seam",
1115         "tieredModel": "TWO TIERS layered onto the EXISTING video-free seam

```


(fusion_golden.py extracts private *_golden_features.json on the GPU box ? fusion_compare.py / ablation.py re-score them on pure CPU). The split is by SENSITIVITY, not by file format.
TIER-0 ? PUBLIC, UNENCRYPTED, NO KEY. A small, de-attributed, Fork-A-only, BIOMETRIC-FREE fixture set committed under a NEW tracked dir `eval\_baselines/fixtures/` (extending the existing committed-baseline precedent ? regression.py:33 already comments that eval_baselines/ is "NOT under the gitignored output/, so a fresh checkout/CI can load a committed baseline"). ANY cloner with no key runs the SHUTTLE-ONLY regression path (experiment.predict/compare LOVO ? F1; ablation.run_ablation; the Gate-A litmus shape; regression-style floor vs GT). Contents are minimized to: opaque `fixture\_id`, `fps`/`frame\_width` scale constants, per-candidate `start`/`end` (time-origin reset), the 5 SHUTTLE columns (duration, peak_velocity, mean_velocity, active_ratio, net_crossings ? inanimate-object, fps/res-invariant), integer `label`, and relative `gts`. The PERSON block and AUDIO block are EXCLUDED from Tier-0 (person = the legal memo's Art.9 "tripwire" / DPDP singling-out risk in a small roster). Tier-0 tests are UNCONDITIONAL ? they run on every PR, converting the currently-skipped litmus into a real public gate.
TIER-1 ? RICHER / SENSITIVE, KEY-GATED, SKIPS CLEANLY WHEN ABSENT. The full n=6+ golden corpus WITH person/audio columns (the authoritative ? F1/CI/sign-test and the exact Gate-A litmus values in test_litmus_reproduces_gate_a). Two delivery options, OUT-OF-REPO is the DEFAULT: (A) OUT-OF-REPO + TOKEN ? the *_golden_features.json stay on the existing private OneDrive store (docs/GOLDEN_DATA_SHARING.md #175); contributors/CI set `RALLY\_GOLDEN\_DIR` to a synced/fetched path; nothing sensitive ever enters git history (also sidesteps the legal "redistribution" act). (B) ENCRYPTED-IN-REPO (SOPS+age) only for Fork-A-clean-but-still-private data where in-repo versioning is genuinely wanted; decrypt to a gitignored path before the Tier-1 run. A Tier-1 test SKIPS (never fails) when the key/data is absent, reusing the proven `\_golden\_present()` skipif pattern (tests/test_ablation.py:366-371) generalized to a shared `\_tier1\_present()` helper. So a fresh no-key clone is always GREEN; coverage never depends on a secret being present.
KEY INSIGHT (why this is elegant, not Byzantine): the two privacy properties come from DIFFERENT layers and must not be conflated. NON-ATTRIBUTABILITY comes from DELETION + RELATIVIZATION at the source (Tier-0/Tier-1 both), so it survives the analog hole ? even a fully-keyed contributor cannot re-identify, because attribution data was never written. USELESSNESS-TO-THE-NO-KEY-PUBLIC comes from minimization (opaque numbers, no schema-external meaning) PLUS optional Tier-1 encryption. Encryption does the smaller job; minimization does the real work.",

1116 "antiAttributionMeasures": [
1117 "OPAQUE SALTED SURROGATE ID ? replace BOTH the `name` field (today = source filename via fusion_golden._derive_video_path) AND the MD5 video_id with fixture_id = first-16-hex of HMAC-SHA256(key=PER-RELEASE-SECRET-SALT, msg=video_id||name). Salt is generated once per release, kept PRIVATE (lives with the Fork-A corpus on OneDrive, NEVER in git), rotated per release. Result: (a) non-reversible without the salt, (b) NOT the MD5 video_id so holding the candidate file cannot confirm the source (kills the memo's 'video_id leak' red flag), (c) unlinkable across releases. Harness-transparent: fusion_compare.py:50 uses d['name'] ONLY as a VideoCandidates label, so aliasing it changes nothing downstream.",

1118 "METADATA STRIP ? drop every source-attribution field: filename, video_id, match/event/tournament/player identity, capture date/clock. fps/frame_width are retained as non-identifying scale constants (shuttle features are already fps/res-invariant); quantize/bucket frame_width if counsel flags it as a small-roster/venue tell.",

1119 "TIME-ORIGIN RESET ? for each fixture compute t0 = min over all candidate.start and gts, subtract t0 from every start/end and every gts pair (or add a random per-fixture offset). PROVABLY ZERO-COST for the current harness: metrics.py interval_iou (line 25: inter/union, both durations) and mean_start/end_error (lines 124-125: |pred-gt|) are pure differences, so a constant per-fixture offset leaves every regression metric BIT-IDENTICAL while severing the absolute match timeline that would align a fixture to a known broadcast clock.",

1120 "STREAM MINIMIZATION (publish-by-stream) ? Tier-0 KEEPS the shuttle block + relative start/end + gts + label; EXCLUDES the entire person block (players_in_court, two_sided_motion, mean_player_motion, in_court_ratio, shuttle_player_colocation) and audio block from the PUBLIC subset. NEVER write pose/skeleton/gait/face ? already absent from the schema; the protobuf has NO field for them by design (gait survives anonymization at 80-95% re-ID per the memo).",

1121 "RAW-TRAJECTORY EXCLUSION ? never commit the per-frame (Frame,Visibility,X,Y) trajectory CSV. Only ~5 aggregate, fps/resolution-invariant shuttle scalars per window are published ? far below the raw (x,y) stream that makes de Montjoye trajectory-linkage work (4 points ? 95% re-ID). The high-resolution fingerprint stays private.",

1122 "PROVENANCE GATE (hard-fail) ? the de-attribution pass HARD-FAILS (refuses to emit a public fixture) on any record lacking an explicit owner-recorded (Fork-A) + minors-excluded + biometric-excluded tag verified at ingestion. A single Fork-B-tainted or minor record contaminates the public commit (memo MIXED-PROVENANCE red flag), so the gate is mechanical, not

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trust-based.",
1123         "COMMITTED-LEAK REMEDIATION (do this in the SAME change) ?
eval_baselines/heuristic_lovo_n6.json currently commits real player/source names as per_video
keys (adarsh_avi_singles, kushagra_singles, mahadevpura_singles, gbaaddy, ...). This is a LIVE
de-attribution bug, not hypothetical. Replace those keys with opaque surrogate ids; keep the
surrogate?source map PRIVATE on OneDrive.",
1124         "PRE-COMMIT + CI LEAK GUARD ? reject committing (i) any *_golden_features.json
outside the de-attributed Tier-0 allowlist, (ii) any tracked file containing a 32-hex MD5 (a
possible video_id) or a known-player token or a 'video_id'/'name' source field that is not an
fx_surrogate, (iii) any SOPS-encrypted file committed in plaintext (sops --verify /
unencrypted-MAC check). compute_video_id (backend/utils/hashing.py) is consumed ONLY as private
HMAC input and is mechanically barred from any committed file."
1125     ],
1126     "rolloutPhases": [
1127         "PHASE 1 (SMALL PR) ? Remediate the live leak + land the guard. Scrub
eval_baselines/heuristic_lovo_n6.json (opaque surrogate keys, private mapping kept off-git); add
the pre-commit + CI leak-scan (MD5/name/known-player-token detector). No new tiers yet. Self-
contained, shippable, closes a real committed-attribution bug.",
1128         "PHASE 2 (SMALL PR) ? De-attribution + minimization pass. Add
scripts/deattribute_golden.py (or backend/eval/publish_fixtures.py): consumes a PRIVATE Fork-A
*_golden_features.json ? emits a de-attributed, minimized, time-origin-reset *public_fixture*
(opaque salted fixture_id, person/audio stripped, metadata stripped). Pure-logic, unit-testable
in the dev container (verify metric-invariance of the origin reset against metrics.py).
Provenance gate HARD-FAILS on un-tagged records.",
1129         "PHASE 3 (SMALL PR) ? Tier-0 public fixtures + unconditional test. Commit a tiny
de-attributed Fork-A subset (and/or a SYNTHETIC characterization set mirroring the existing
_learned_corpus test builder) under eval_baselines/fixtures/; add tests/test_fixtures_tier0.py
that points the EXISTING loaders at the dir and runs UNCONDITIONALLY (no skipif), folded into
run_all_tests.py. Now a fresh clone has real, green, video-free coverage. (Counsel sign-off
gates the REAL-subset commit; the synthetic set can land first with zero provenance risk.)",
1130         "PHASE 4 (SMALL PR) ? Protobuf schema + deterministic serialization (OPTIONAL,
only if cross-OS byte-stability or characterization snapshots are wanted). Add
backend/eval/fixtures/rally_fixture.proto (proto3, package rally.fixture.v1), the committed
_gen/ module + a `make proto` + CI `protoc && git diff --exit-code` guard, a loader.py mirroring
fusion_compare.load_videos, the canonical-sorted/quantized writer, .gitattributes *.pb binary
-text`, and protobuf (Apache-2.0) in requirements. Adds the snapshot+floor replay test. Reuse
the JSON path if the determinism/versioning need does not materialize ? proto is earned, not
assumed.",
1131         "PHASE 5 (SMALL PR) ? Tier-1 key-gated path. Add scripts/fetch_golden.py
(OneDrive token, default) and/or a SOPS+age .sops.yaml + a `sops -d` make target; generalize
_golden_present()?_tier1_present(); wire RALLY_GOLDEN_DIR through
fusion_golden/fusion_compare/ablation; gate the Tier-1 CI job on the secret being present (never
exposed to fork-PR runners). Tier-1 absent ? clean SKIP. Doc note in
docs/GOLDEN_DATA_SHARING.md.",
1132         "PHASE 6 (gating, NOT a code PR) ? Counsel sign-off on the final committed
Tier-0 subset + the source-provenance log BEFORE the first REAL-data public commit. This is a
precondition the architecture assumes, not something it satisfies."
1133     ],
1134     "honestLimits": [
1135         "A GREEN no-key suite PROVES only that the pure-CPU re-scoring DECISION code
(gate ? optional learned filter ? merge ? IoU score) is behaviorally stable on the committed
Tier-0 fixtures. It does NOT prove the GPU EXTRACTION step (video+CSV ? features in
fusion_golden.py) is correct ? that is not video-free and still needs the GPU/WSL box.",
1136         "A green Tier-0 suite does NOT reproduce the authoritative numbers. The person-
cue ?F1 litmus and the exact Gate-A values (test_litmus_reproduces_gate_a) need the person-
bearing Tier-1 corpus; Tier-0 publicly runs only the SHUTTLE-ONLY path. External contributors
get framework-behavior + shuttle characterization coverage, not the headline fusion result.",
1137         "This does NOT replace regression.py's DB-backed, full-pipeline-on-real-video
promotion gate. It is the cheap CPU shadow that catches refactor/quality drift, not the
authoritative serve-path check.",
1138         "The data is NOT 'anonymous' in an absolute sense and the design must NEVER
claim so (EDPS v. SRB / CJEU C-413/23 P). To the OWNER who can re-run the detector on the still-
private source, the streams REMAIN personal data. Non-attributability here is RELATIVE (to
outsiders lacking the source), CONDITIONAL (on Fork-A source staying unpublished), REVOCABLE
(publish/leak the source and the link can snap back), and FORWARD-LOOKING (must be re-assessed

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as the public-video environment grows).",

1139 "Encryption does NOT provide secrecy-from-a-running-contributor ? impossible by the lead's DRM/analog-hole constraint (a contributor reads plaintext + key on their machine). Tier-1 crypto is a barrier against the NO-KEY public ONLY; it must be DOCUMENTED as such and never marketed as protection it does not provide (a DPDP 'reasonable-security-safeguards'/misrepresentation question the memo flags for counsel).",

1140 "Protobuf (or any serialization) provides ZERO secrecy ? it is an encoding, trivially decoded (protoc --decode_raw). A binary fixture format is never a confidentiality control.",

1141 "De-attribution does NOT cure provenance or licensing. A single Fork-B-tainted or minor record contaminates the public commit regardless of how well it is de-attributed; the scheme is necessary-but-not-sufficient and sits BEHIND the per-record provenance gate + counsel sign-off, which it treats as gating preconditions, not as things it satisfies.",

1142 "The salt is the single point of failure: the surrogate fixture_id is non-reversible ONLY while the per-release salt stays private. If the salt is ever committed/leaked alongside the public fixtures, an adversary holding candidate files can recompute the HMAC and confirm sources. Rotation per release limits blast radius to one release.",

1143 "Time-origin reset is free for the CURRENT metrics (translation-invariant) but constrains FUTURE ones: any future scorer using absolute wall-clock, cross-fixture temporal alignment, or duration-since-match-start would break or silently mis-score on origin-reset fixtures. Flag this so an absolute-time metric is not added against public fixtures without revisiting."

1144],

1145 "openOwnerQuestions": [

1146 "COUNSEL (gating): does the EU/UK sui generis database right subsist in our self-derived shuttle-coordinate/rally streams on the Football Dataco v. Sportradar 'obtained vs created' reasoning, and does redistributing a de-attributed Tier-0 subset 'extract/re-utilise a substantial part' causing CV-Online-Latvia harm? (EU AND UK.)",

1147 "COUNSEL (gating): post Thomson Reuters v. Ross + USCO May-2025, is the US fair-use basis for the EXTRACTION step defensible given the dataset could be seen to compete in a 'data/analytics' market ? and does it matter we publish only numbers, never footage?",

1148 "COUNSEL (gating): India ? are anti-scraping ToS enforceable and what is the IT Act s.43/s.66 exposure for any third-party-sourced ingestion; does the pending ANI v. OpenAI judgment (~April 2026) move the analysis? Confirm DPDP s.9/Rule 10 (verifiable parental consent) and whether per-player rally-tracking is prohibited s.9(3) behavioural monitoring of a minor.",

1149 "COUNSEL (gating): sign-off on the FINAL committed Tier-0 subset + the source-provenance log before the first REAL-data public commit; and whether non-attributability-via-private-source is a defensible posture to assert to regulators given its conditional/revocable nature.",

1150 "OWNER (strategic): publishing the Fork-A subset FORFEITS whatever trade-secret/competitive moat it carried (Indian breach-of-confidence / Contract Act 1872 / IT Act s.72 framing, NOT US DTSA) ? weighed against the stated community/benchmark/paper-co-authorship aims. This is the owner's call.",

1151 "OWNER (forward-looking duty): the whole posture rests on the Fork-A source staying UNPUBLISHED. Publishing a highlight reel of that same source makes TM5's correlation attack live for those records. Owner must commit to NOT publishing the source video and its derived fixtures from the same match without re-clearing, and to a periodic re-assessment as the public-video environment grows.",

1152 "OWNER (scope): Tier-0 REAL-subset vs SYNTHETIC-only for the first public commit? Synthetic carries zero provenance risk and still exercises the full framework; a real subset adds value-level characterization but needs the full gate. Recommendation: ship synthetic first, add a counsel-cleared real subset later.",

1153 "OWNER (mechanism): default Tier-1 delivery ? OUT-OF-REPO+token (recommended: keeps git lean, zero sensitive bytes in history, sidesteps redistribution) vs encrypted-in-repo SOPS+age (atomic versioning, offline reproducibility, heavier rotation). And: is Phase-4 protobuf wanted at all, or is canonical-JSON sufficient for the determinism/versioning need?"

1154],

1155 "threatModelTable": "| # | Actor (who/what) | Capability | PROTECTED (not achievable if design holds) | NOT protected (achievable / by-design open) |\n|---|---|---|---|---|\n| TM1 | No-key public cloner | Clone repo, read committed cleartext, brute-force | Map any fixture ? source video/match/event/date/player (every attribution field stripped; id is opaque salted surrogate, not the MD5 video_id); decrypt Tier-1 (the ONE actor encryption defends); confirm a source even holding a candidate file (no MD5/timestamp/name to correlate) | Read Tier-0 shuttle scalars + relative intervals + GT and run the shuttle-only harness. By design ? Tier-0 is public. Data is useless-in-isolation (numbers with no schema-

external meaning, no source pointer). |\n| TM2 | Trusted contributor WITH key, running tests | Read decrypted plaintext + key locally, re-run fusion_compare/ablation | Recover source filename, MD5 video_id, match identity, absolute timeline, face/pose/gait ? NONE are ever written into the fixture (removed at source, not gated by a key they hold). Non-attributability survives the analog hole. | Read every value the suite reads (UNPREVENTABLE per the DRM/analog-hole limit). The design does NOT pretend secrecy-from-this-actor; encryption is explicitly not represented as protecting against them. |\n| TM3 | CI runner (GitHub Actions, no human, no videos) | Decrypt if CI secret holds key; run headless; emit pass/fail + deltas | A leaked CI log/artifact cannot leak attribution it never had ? CI needs no source video, match identity, or person/biometric columns to score interval-IoU. Published subset is self-sufficient (relative intervals + shuttle + GT) so CI is green WITHOUT any stripped field. | Run the suite and read the committed fixtures (intended). Tier-1 CI secret is never exposed to fork-PR runners. |\n| TM4 | Malicious contributor (clone + key + intent; may fetch public videos + re-run our detector) | Everything TM2 can do + active correlation attacks (re-run WASB/TrackNet on candidate public videos, try to match a committed fixture) | A correlation match needs a correlatable REFERENCE. Committed subset is gated to Fork-A (owner-recorded, UNPUBLISHED), so no public reference exists ? external source-linkage 'not reasonably likely'. Even a lucky near-match carries no timestamp/name/video_id to CONFIRM, and ~5 invariant shuttle scalars/window are too coarse to back-match a 4K source. | RESIDUAL, acknowledged: protection is conditional + revocable (EDPS v. SRB / memo trap #4). Publish or leak the Fork-A source later and the link can snap back ? hence the forward-looking re-test duty. |\n| TM5 | Re-identification adversary (outside project, no key needed) | Run same deterministic detector on public videos, correlate-match a stream to its source | NOT achievable against the committed subset because: (a) source is Fork-A/unpublished (no permanent cheap public reference); (b) fixtures minimized to ~5 invariant scalars/window + relative intervals, far below the raw per-frame (x,y) that makes trajectory-linkage work ? the raw CSV is NEVER committed; (c) absolute timestamps reset to a relative origin. | Becomes LIVE if a Fork-A source is ever published ? exactly why provenance gating + the 'data-environment grows' re-assessment (NIST SP 800-188 / ICO motivated-intruder / OVIC) are load-bearing, not optional. |\n| TM6 | Accidental plaintext commit (most likely real-world failure) | A maintainer commits a decrypted *_golden_features.json or a file carrying name/video_id/match metadata | Pre-commit hook + CI scan reject any tracked file matching the private-features pattern, any 32-hex MD5 / known-player token / non-surrogate name field, and any SOPS file committed in plaintext (unencrypted-MAC check). | If the guard is bypassed/disabled, the leak lands. The guard is the mitigation; CI re-scan on every PR is the backstop. |",

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1156      "protobufSchemaSketch": "/* backend/eval/fixtures/rally_fixture.proto (proto3,
package rally_fixture.v1)\n// OPTIONAL layer (Phase 4) ? only if cross-OS byte-stability +
characterization snapshots are wanted.\n// Serializes EXACTLY today's *_golden_features.json
contract (fusion_golden.extract_video) plus\n// version guards, a de-attributed opaque id, and
an expected-output snapshot + quality floor.\n// NO field exists for
pose/skeleton/gait/face/bbox-track ? unrepresentable BY DESIGN.\n\nsyntax = \"proto3\";\npackage
rally_fixture.v1;\n\nmessage RallyFixture {\n  // --- versioning / incommensurability guards
(mirror regression.py + ablation.py) ---\n  uint32 schema_version = 1;    // bump on a non-
additive change; gates the loader migration shim\n  string label_version = 2;    // Roora
guideline version (e.g. \"v8\")\n  string detector_tag = 3;    // e.g. \"wasb_hybrid@<ver>\" ?
the regression.py detector key\n  string gt_hash = 4;    // regression.gt_hash(gts),
recomputed + verified on load\n  float iou_threshold = 5;    // IoU the snapshot/floor were
computed at (default 0.5)\n\n  // --- de-attribution: NO source filename / video_id MD5 / match
/ player ---\n  string fixture_id = 6;    // OPAQUE salted surrogate (HMAC-derived hex).
NEVER the MD5 video_id.\n\n  // --- replay inputs (today's contract; time-origin reset already
applied) ---\n  float fps = 7;    // non-identifying scale constant\n  float
frame_width = 8;    // non-identifying scale constant (quantize if counsel flags)\n  repeated
Candidate candidates = 9;\n  repeated Interval gts = 10;\n\n  // --- expected outputs
for the two assertions ---\n  Snapshot expected = 11;    // blessed output of replaying this
fixture (characterization)\n  QualityFloor floor = 12;    // minimum P/R/F1 vs gts this fixture
must hold\n\n  message Candidate {\n    Interval window = 1;    // {start,end}
seconds, relative to the per-fixture origin\n    // map (not packed array) so a NEW column is
additive and an OLD fixture missing it defaults to\n    // 0.0 ? exactly
experiment._feature_column / ablation.load_videos tolerate today.\n    map<string, double> shuttle
= 2;    // duration,peak_velocity,mean_velocity,active_ratio,net_crossings\n    map<string,
double> person = 3;    // TIER-1 ONLY ? omitted entirely from the Tier-0 public subset\n    map<string, double> audio = 4;    // TIER-1 ONLY ? omitted from Tier-0\n    int32 label = 5;
// IoU-derived 0/1 (training.label_candidates), kept for parity\n    // NOTE: NO
pose/skeleton/gait/face/bbox column exists in this schema by design.\n  }\n\n  message Interval
{ double start = 1; double end = 2; }\n\n  message Snapshot { string variant_spec_hash = 1;
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repeated Interval segments = 2; } // sorted, quantized\nmessage QualityFloor { float
min_precision = 1; float min_recall = 2; float min_f1 = 3; }",
1157     "serializationDeterminism": "Cross-OS byte-stability is enforced at four layers so
a fixture and its snapshot are byte-identical on Windows/Linux/mac and CI cannot flake on float
repr or CRLF: (1) FLOATS ? quantize every coordinate/feature before serialization:
round(feature, 6), round(seconds, 3) (matching regression.gt_hash's round(.,3) on line 39);
store the quantized values so platform float repr can never diverge. (2) FIELD/MAP ORDER ?
protobuf map iteration order is unspecified, so serialize through a canonical writer that sets
deterministic_serialization on the message AND sorts candidates by (start,end), feature-map keys
lexicographically, and snapshot segments by (start,end). (3) NEWLINES ? fixtures are binary .pb
(no newline issue); commit with `*.pb binary -text` in .gitattributes so git never CRLF-mangles
them; any sidecar text/JSON debug dump is normalized to LF. (4) ROUND-TRIP ? a test asserts
serialize(parse(bytes)) == bytes for every committed fixture (canonical bytes are stable). If
the project stays on JSON instead of protobuf, the SAME discipline applies via json.dump(...,
sort_keys=True) + pre-quantized floats + LF ? the existing committed JSON (eval_baselines/*)
already uses sort_keys=True (regression.py:98, save_baseline). The time-origin reset is itself a
determinism aid: it removes large absolute timestamps whose low-order float bits are the most
platform-sensitive.",
1158     "encryptionMechanism": "Encryption is a NARROW, honestly-scoped layer used ONLY
against the no-key public (TM1) ? it is belt, not parachute; the upstream minimization does the
real work. RECOMMENDATION: SOPS + age for any encrypted-in-repo Tier-1 secrets (per-recipient
public-key encryption, git-diff-friendly value-level for JSON, trivial CI/contributor key
handling, an explicit decrypt step so there is no silent-plaintext-commit failure mode, and a
MAC that detects tampering). age over GPG for ergonomics (no keyring/agent ceremony). REJECTED
alternatives: git-crypt (one shared symmetric key ? painful rotation, opaque binary diffs, a
misconfigured checkout silently commits plaintext); git-lfs+encryption (LFS solves large-binary
storage not secrecy ? our feature JSONs are KB-scale, so it buys nothing and reinvents git-crypt
with a server dependency). PREFERRED OVERALL for Tier-1: OUT-OF-REPO + token (the existing
OneDrive store) ? keeps ZERO sensitive bytes in immutable git history (a future key compromise
can't decrypt old commits, and the legal 'redistribution into a public repo' act never happens
for sensitive data) and reuses the #175 RALLY_GOLDEN_DIR seam. Pick per-fixture: Fork-A-clean-
and-stable ? in-repo SOPS is fine; anything Fork-B-tainted or identity-adjacent ? out-of-repo
only. CRYPTO LIBRARY note: the surrogate-id HMAC-SHA256 uses Python's stdlib hmac/hashlib (no
new dependency, license-clean); SOPS+age are external CLI tools invoked from a make target, not
Python deps. DOCUMENT EXPLICITLY in code comments + docs/GOLDEN_DATA_SHARING.md that Tier-1
crypto defends ONLY against the no-key public and that a running contributor necessarily holds
plaintext+key ? stating this plainly heads off the DPDP 'reasonable security
safeguards'/misrepresentation question the memo flags.",
1159     "keyManagement": "TWO secrets, by Tier-1 delivery mode. (A) OUT-OF-REPO+token
(default): a scoped OneDrive read token. CI = repository secret RALLY_GOLDEN_TOKEN, exposed only
to the Tier-1 job on trusted branches, NEVER to fork-PR runners (a malicious fork PR cannot
exfiltrate it). Contributor onboarding = granted scoped read access to the private store.
Rotation = revoke the share / rotate the token. (B) Encrypted-in-repo (SOPS+age): an age keypair
per recipient. CI = the age PRIVATE key as repository secret RALLY_AGE_KEY, Tier-1 job only,
never to fork runners. Contributor onboarding = the contributor runs age-keygen and sends their
PUBLIC key; a maintainer adds it to .sops.yaml and runs `sops updatekeys` ? NO existing secret
is reshared, each recipient decrypts with their own private key. Rotation = edit .sops.yaml
recipients (remove a departed/compromised key, add the new) + `sops updatekeys`; because age is
per-recipient, removing a recipient is a one-command re-key (vs git-crypt's whole-repo re-
encrypt where the old key still decrypts history). THE SALT (separate from the encryption key,
and more important): the per-release HMAC salt that makes fixture_id non-reversible lives with
the PRIVATE Fork-A corpus on OneDrive, NEVER in git, rotated per public release so a single salt
leak does not retro-link all releases. On a suspected DATA leak: rotate the token/re-key AND ?
because non-attributability is revocable ? re-assess whether the underlying Fork-A source must
stay unpublished. A PRIVATE surrogate?source map is kept in the out-of-repo store for owner/CI
debugging and is NEVER committed.",
1160     "ciIntegration": "TWO CI jobs with a hard split so coverage never depends on a
secret: (1) TIER-0 job ? runs on EVERY PR including fork PRs, no key, executes
tests/test_fixtures_tier0.py + the unconditional fixture-replay test against
eval_baselines/fixtures/ (folds into the existing run_all_tests.py / pytest, no new runner).
This is the public always-green gate. (2) TIER-1 job ? gated on the secret being present
(RALLY_GOLDEN_TOKEN or RALLY_AGE_KEY), runs only on trusted branches (never fork-PR runners),
fetches/decrypts the corpus to a gitignored path, runs the full ?F1/CI/sign-test + the exact
Gate-A litmus. When the secret is absent the Tier-1 tests SKIP cleanly via _tier1_present()
(generalizing tests/test_ablation.py:366-371). GUARD JOBS on every PR: (a) the leak scan ?

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reject any committed 32-hex MD5 / known-player token / non-surrogate name field / private-
features filename / plaintext-committed SOPS file; (b) if protobuf is adopted, re-run `protoc`
and `git diff --exit-code` to prove the checked-in _gen/ module matches the .proto. All of this
is offline, pure-CPU, GPU-free ? runs in the dev container (the project's full pipeline cannot,
but this layer can).",
1161     "contributorWorkflow": "END-TO-END, contributor on another clone with NO video
present: (1) git clone ? gets the public repo including eval_baselines/fixtures/ (Tier-0). (2)
pip install -r requirements.txt (+ protobuf if Phase 4 landed) ? pure-CPU deps, no GPU/torch/cv2
needed for this path. (3) python run_all_tests.py ? the UNCONDITIONAL Tier-0 tests run
immediately: the existing loaders (fusion_compare.load_videos / ablation.load_videos, schema-
compatible) read the committed fixtures, replay them through the REAL
experiment.predict/compare/run_ablation + metrics.score code, and assert the shuttle-only LOVO
?F1 framework behavior (and, if Phase 4, a byte-stable snapshot + quality floor). GREEN with no
key, no video, no DB. (4) The Tier-1 tests SKIP cleanly with a clear reason ('Tier-1 golden
corpus absent: set RALLY_GOLDEN_DIR or provide the key'). (5) A TRUSTED contributor
additionally: is granted scoped OneDrive read access (or sends an age public key for SOPS), runs
scripts/fetch_golden.py (or `sops -d` / `make decrypt-golden`) to populate RALLY_GOLDEN_DIR,
then re-runs ? now the full person/audio ?F1, CI, sign test, and the exact Gate-A litmus values
execute. At NO point can any contributor ? keyed or not, trusted or malicious ? recover a source
video, match, date, or player from the fixtures, because that data was deleted at the source,
not encrypted. They CAN read every number the tests read (the analog-hole limit, accepted by
design).",
1162     "harnessIntegration": "Purely ADDITIVE on existing seams ? no registry fork, no
schema fork, no change to experiment.py/regression.py/ablation.py/fusion_compare.py internals.
MODULES TO EXTEND / ADD: (1) backend/eval/fusion_compare.py + backend/eval/ablation.py ? already
read ONLY c['shuttle'][...]/c['start']/c['end']/d['gts'] and use d['name'] as an opaque label
(verified fusion_compare.py:35-52, ablation.py:386-407); the Tier-0 public loader iterates only
SHUTTLE columns (skips PERSON/AUDIO), a small additive code path; the existing skipif litmus
tests keep running against the PRIVATE person-bearing Tier-1 features unchanged. (2)
eval_baselines/ ? the committed-structured-data precedent (regression.py:33); add the tracked
subdir eval_baselines/fixtures/ alongside regression_baseline.json / experiment_leaderboard.json
/ heuristic_lovo_n6.json. (3) backend/eval/regression.py ? reuse gt_hash (line 37) for the
fixture's gt_hash guard and its IoU/detector incommensurability checks (lines 132-141); reuse
the committed-baseline floor pattern. (4) backend/eval/experiment.py ? reuse Variant.spec_hash()
(line 148) as the snapshot's variant key and _feature_column's 0.0 default (line 180) for the
additive-column tolerance; the replay calls predict/compare/run_ablation UNCHANGED. (5)
backend/eval/metrics.py ? the translation-invariance that makes the time-origin reset free is a
property of score()'s pure-difference math (lines 25,124-125); no change needed, just relied
upon. (6) backend/utils/hashing.py::compute_video_id ? the canonical MD5 source, consumed ONLY
as private HMAC input, mechanically barred from any committed file. NEW FILES:
scripts/deattribute_golden.py (the de-attribution+minimization pass),
tests/test_fixtures_tier0.py (unconditional), the leak-scan pre-commit hook; OPTIONAL Phase-4:
backend/eval/fixtures/{rally_fixture.proto, loader.py, _gen/rally_fixture_pb2.py},
scripts/build_fixtures.py, tests/test_fixture_replay.py, .gitattributes, +protobuf in
requirements; OPTIONAL Phase-5: scripts/fetch_golden.py + .sops.yaml. COMPOSES with the
concurrent feat/tier0-rally-seg-eval worktree (currently at master HEAD, no fixtures landed):
that workflow owns the runnable-without-video MECHANICS; THIS layer is the LEGAL/privacy FILTER
deciding which fields are publishable and under what id ? mechanics produce the runnable
fixture, this de-attributes + minimizes it before the public commit."
1163     },
1164     "redteam": {
1165         "attacks": [
1166             {
1167                 "vector": "Unconditional Tier-0 test trains a numpy/BLAS LogisticRegression
(classifier.py:18-81), so 'byte-stable cross-OS snapshot' is false; near threshold 0.5 predict()
can flip Windows vs Linux CI, flaking the always-green job.",
1168                 "severity": "HIGH",
1169                 "doesDesignHold": "NO",
1170                 "fix": "Assert numeric tolerance not byte-equality; restrict Tier-0 to
static/pure-metrics or pin float env (OMP_NUM_THREADS=1); add a windows+ubuntu+mac CI matrix
(ci.yml is ubuntu-only).",
1171             },
1172             {
1173                 "vector": "Leak remediation scrubs only heuristic_lovo_n6.json, but real names
are also in gemini_wrapper_2026-06-10.json, eval_partial_labels.py:23,119 (plus a

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C:/Users/avidu/... local path) and test_cleanup_caches.py, and in git history since PR #77
(d51efe3).",
1174         "severity": "HIGH",
1175         "doesDesignHold": "PARTIALLY",
1176         "fix": "Scan whole tree, fix every file, re-scan to zero, decide on history
(accept vs filter-repo). Keep the scanner name-list private."
1177     },
1178     {
1179         "vector": "Pre-commit hook is mis-cast as a barrier (TM6); no hook infra
exists and --no-verify/fresh clones bypass it; the design never makes the CI scan a required
check.",
1180         "severity": "HIGH",
1181         "doesDesignHold": "NO",
1182         "fix": "Make the CI leak-scan a REQUIRED status check on master (forks too,
diff+full tree); demote the hook to local convenience."
1183     },
1184     {
1185         "vector": "Tier-0 loader still reads person: fusion_compare.py:47-48 and
ablation.py:400-401 read c['person'] unconditionally, so a person-stripped fixture raises
KeyError and Tier-0 goes RED. 'Reuse unchanged' and 'strip person' contradict.",
1186         "severity": "MEDIUM",
1187         "doesDesignHold": "NO",
1188         "fix": "Add a person-optional Tier-0 loader (model on ablation's audio
handling, line 403)."
1189     },
1190     {
1191         "vector": "experiment._feature_column:180-192 silently defaults a missing
column to 0.0 (warn only), so a fixture that accidentally keeps person data, or Tier-1 missing
person data, both pass green; green can't prove stripping.",
1192         "severity": "MEDIUM",
1193         "doesDesignHold": "NO",
1194         "fix": "Add a positive Tier-0 assertion that no candidate has 'person'/'audio'
keys; have the scanner reject those strings."
1195     },
1196     {
1197         "vector": "fixture_id=HMAC(salt, video_id||name) over a tiny convention-named
roster is brute-forceable the instant the salt leaks (and the salt lives with the corpus on
OneDrive).",
1198         "severity": "MEDIUM",
1199         "doesDesignHold": "PARTIALLY",
1200         "fix": "Use a random per-record surrogate (os.urandom) with a private map;
keep any HMAC private-side; store salt/map in a separate trust domain."
1201     },
1202     {
1203         "vector": "Retained fps/frame_width + the per-fixture
(duration,net_crossings,gt-count) vector are a camera/venue/match quasi-identifier at n=6, not
'non-identifying'.",
1204         "severity": "MEDIUM",
1205         "doesDesignHold": "WEAKLY",
1206         "fix": "Quantize/bucket fps/frame_width by default or drop them (shuttle
features are fps/res-invariant); treat the per-fixture vector as a quasi-identifier in the re-
assessment."
1207     },
1208     {
1209         "vector": "Green Tier-0 covers only the shuttle-column path; person/audio
fusion regressions are caught only by key-gated Tier-1 (skipped on forks/no-key clones) - a
false 'no regression' confidence the limits omit.",
1210         "severity": "MEDIUM",
1211         "doesDesignHold": "PARTIALLY",
1212         "fix": "Add the caveat; commit a synthetic person/audio Tier-0 fixture or make
Tier-1 a required check before master advances."
1213     },
1214     {
1215         "vector": "Out-of-repo+token (default Tier-1) grants the whole sensitive
corpus via one broad long-lived token; SOPS+age (rejected) gives finer per-recipient revocable

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control.",
1216         "severity": "LOW",
1217         "doesDesignHold": "MOSTLY",
1218         "fix": "Prefer per-recipient SOPS+age or a scoped short-lived single-folder
token; never echo it; audit grantees."
1219     },
1220     {
1221         "vector": "'Survives the analog hole' overstates TM2/TM4: a keyed insider with
roster/venue knowledge plus the committed experiment_leaderboard.json can map a fixture to a
match.",
1222         "severity": "LOW",
1223         "doesDesignHold": "WEAKLY",
1224         "fix": "Soften to 'non-attributable to a contributor without roster
knowledge'; pair with fps/frame_width quantization."
1225     }
1226 ],
1227     "mustFixBeforeShipping": [
1228         "Numeric tolerance not byte-equality for learned snapshots; pin float env or
restrict Tier-0 to static paths; add a multi-OS CI matrix.",
1229         "Scrub names in ALL files (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json, eval_partial_labels.py, test_cleanup_caches.py) and decide on
git history (PR #77).",
1230         "Make the CI leak-scan a required status check on master; demote the pre-commit
hook to convenience.",
1231         "Add a person/audio-optional Tier-0 loader and a positive assertion that
fixtures have no person/audio keys; don't rely on the silent 0.0-default.",
1232         "Use a random per-record surrogate id (os.urandom) with a private map instead of
HMAC(salt, video_id||name); separate trust domains.",
1233         "Quantize/bucket or drop fps/frame_width; treat the per-fixture interval/scalar
vector as a quasi-identifier.",
1234         "Document that green Tier-0 omits the person/audio fusion path; add a synthetic
fusion fixture or a required Tier-1 check.",
1235         "Keep counsel sign-off, Fork-A-only provenance, and minors/biometric exclusion
as hard preconditions; ship synthetic Tier-0 first."
1236     ],
1237     "overallVerdict": "Defensible foundation, not ship-ready. Right instincts (de-
attribution-at-source does the real work; encryption is a public-only barrier; non-
attributability honestly relative/conditional/revocable; counsel/Fork-A/minors are
preconditions). But three load-bearing technical claims fail against the code: (1) byte-stable
cross-OS determinism is false because the unconditional Tier-0 test trains a numpy/BLAS
LogisticRegression (classifier.py) that drifts near the decision boundary; (2) leak remediation
is too narrow - names are in 4+ files and git history since PR #77; (3) 'reuse loaders unchanged
+ strip person' is contradictory (fusion_compare.py:47-48 / ablation.py:400-401 read person
unconditionally) and _feature_column's silent 0.0-default hides field omission so green can't
prove stripping. Secondary: HMAC salt weaker than a random surrogate for a small roster; pre-
commit hook not enforceable (only a required CI check is); retained fps/frame_width are quasi-
identifiers at n=6; Tier-0 green covers a weaker surface than production. All fixable with the
design's own escape hatches. Fix the HIGH items and privacy-MEDIUMs before any public commit;
keep legal gating intact.</overallVerdict>\n</invoke>\n"
1238     }
1239 }
1240 }

```

84. assistant (2026-06-16T04:04:59.223Z)

I now have both workflows fully harvested. The second red-team corroborated workflow 1's determinism catch *and* surfaced a concrete, checkable claim: real player/source names already committed in `eval\_baselines/` and a couple of source files. Per my honesty rule I'll verify that before asserting it in a doc ? it's the kind of finding that must be true, not plausible.

85. user (2026-06-16T04:05:00.745Z)

```

eval_baselines\heuristic_lovo_n6.json:8:     "adarsh_avi_singles": 0.751,
eval_baselines\heuristic_lovo_n6.json:9:     "kushagra_singles": 0.157,
eval_baselines\heuristic_lovo_n6.json:10:    "largetest_doubles": 0.737,

```



```
eval_baselines\heuristic_lovo_n6.json:11:    "gbaaddy": 0.309,
eval_baselines\heuristic_lovo_n6.json:13:    "mahadevpura_singles": 0.657
eval_baselines\gemini_wrapper_2026-06-10.json:7:    "mahadevpura_singles": {
eval_baselines\gemini_wrapper_2026-06-10.json:12:    "kushagra_singles": {
eval_baselines\gemini_wrapper_2026-06-10.json:19:    "heuristic_lovo": {"mahadevpura_singles":
0.657, "kushagra_singles": 0.157},
eval_baselines\gemini_wrapper_2026-06-10.json:20:    "gen0_learned_lovo":
{"mahadevpura_singles": 0.744, "kushagra_singles": 0.561}
```

86. user (2026-06-16T04:05:01.947Z)

```
23:      --trajectory C:/Users/avidu/AppData/Local/Temp/adarsh_avi_traj.csv \
119:    comp_out = comparison_out or osp.join("output", "adarsh_avi_vs_human.csv")
```

87. assistant (2026-06-16T04:05:37.887Z)

Confirmed ? the leak is real and in tracked files: `eval\_baselines/heuristic\_lovo\_n6.json` and `gemini\_wrapper\_2026-06-10.json` use real player/venue names as keys (`adarsh\_avi\_singles`, `kushagra\_singles`, `mahadevpura\_singles`, `gbaaddy`?), and `eval\_partial\_labels.py` even embeds a personal local path. Private repo today, but it's in git history and would persist if open-sourced. That becomes deliverable #1.

I have everything. Writing the tracked project doc now.

88. assistant (2026-06-16T04:07:52.162Z)

Golden Regression Fixtures ? video-free, non-attributable regression suite

```
> **Status:** `DRAFT` ? owner-gated . **Owner:** Avi . **Created:** 2026-06-16 . **Last
updated:** 2026-06-16
> **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE,
per archive policy)
> **Tracking anchors:** this doc's $7 progress tracker is the source of truth; pointer lives in
`docs/NEXT_STEPS.md` + memory `current-state`.
> **Relation to existing docs:** EXTENDS (does not supersede) `docs/GOLDEN_DATA_SHARING.md`
(#175); reuses the eval stack documented in `docs/EVALUATION.md` / `docs/EXPERIMENT_HARNESS.md`;
quality floor ties to `docs/QUALITY_ITERATIONS.md`.
> **Honesty note:** $3 (Legal) is **researched general information, NOT legal advice** ? it
gates on retained counsel ($8). $2/$4/$7 mark which claims are **[verified]** against code vs
**[design]** proposals.
```

0. TL;DR

Let a contributor on **another machine with no video, GPU, WSL, or DB** run the rally-segmentation regression suite from **committed structured files**, so refactors and non-quality PRs get a red/green side-effect signal. Two layers, built once and grown by accretion:

1. **Mechanics** ? freeze the deterministic pipeline at the **post-detector trajectory seam** and replay it through the **already-shipped** eval stack (`experiment.py` / `metrics.py` / `windowing.py` / `regression.py`), with two auto-discovered gates: a **characterization/snapshot** gate (catches any behaviour change) and a **quality-floor** gate (catches quality drops).
2. **Privacy/legal filter (SEALED-FIXTURES)** ? only **de-attributed, minimized, owner-source, biometric-free** numbers are ever committed; richer/sensitive streams stay key-gated or out-of-repo. Non-attributability comes from **deletion-at-source**, not encryption.

The convergence that makes this clean: the **privacy** answer (publish shuttle-only, drop person/pose streams) and the **determinism** answer (freeze only the pure-stdlib `CONTROL` path, keep the numpy/BLAS learned scorer off the cross-machine gate) point at the **same minimal public tier**. That coincidence is the spine of the design.

****Two findings already actionable, independent of the rest:****
- ? ****Live de-attribution bug**** *(verified)*: real player/venue names are committed today in `eval\_baselines/heuristic\_lovo\_n6.json`, `eval\_baselines/gemini\_wrapper\_2026-06-10.json`, and `backend/eval/eval\_partial\_labels.py` (the last also has a personal `C:/Users/avidu/...` path) ? and in git history. ? ****Deliverable P0.****
- ? ****Determinism trap**** *(verified, corroborated by two independent red-teams)*: the learned variants (`shuttle\_only`, `fusion`) train a numpy/OpenBLAS `DYNAMIC\_ARCH` logistic regression whose decisions flip near the 0.5 threshold across CPUs ? would false-RED on a contributor's machine. ? the cross-machine gate must be ****CONTROL/static-only****.

1. Problem & goal

The rally-quality measurement is split across two machines by capability

(`docs/GOLDEN\_DATA\_SHARING.md`):

- `backend/eval/fusion\_golden.py` ****extracts**** `\_golden\_features.json` ? needs the ****video + trajectory CSV + labels + GPU/WSL**** (GPU box only).
- `backend/eval/fusion\_compare.py` / `backend/eval/ablation.py` ****re-score**** those JSONs on ****pure CPU, no video**** ? any machine.

The CPU re-scoring is **already** video-free, but the inputs live in gitignored `output/` / private OneDrive, so the regression tests that depend on them ****skip everywhere**** (`tests/test\_ablation.py::test\_litmus\_reproduces\_gate\_a`, `tests/test\_fusion\_compare.py`). The committed `eval\_baselines/` baselines exist but `regression.py` still scores DB-stored predictions (needs a pipeline run on the video).

****Goal:**** commit a tiny, version-tagged, ****numeric-only, non-attributable**** per-fixture corpus + auto-discovered gates so a clean `git clone` / CI gets a real red/green regression signal with ****zero video/GPU/DB****, while never leaking attributable or sensitive data.

2. Decisions locked

#	Decision	Source	Implication
D1	**Corpus is owner-recorded / owned**	owner answer 2026-06-16	Legal analysis takes **Fork A** (clean): no third-party copyright/ToS infringement; concern is trade-secret/moat + privacy. Strongest non-attributability (no public reference to correlate against).
D2	**Repo private now, may open-source later**	owner answer 2026-06-16	Design for **public-grade hygiene from day one** ? git history is permanent; nothing scrubable-later.
D3	**Adults, consent not formalized**	owner answer 2026-06-16	**Person/pose/gait streams stay OUT of the public tier** (DPDP/GDPR derived-personal-data risk without consent). Public tier = **shuttle + abstract intervals + non-biometric features only** .
D4	**Freeze seam = post-detector trajectory**	*(design)*	workflow-1 synthesis Widest deterministic surface; reuse the shipped eval stack verbatim; the raw per-frame CSV is **never committed** (only ~5 aggregate, fps/res-invariant shuttle scalars/window).
D5	**Cross-machine gate is CONTROL/static-only**	*(design)*	both red-teams [verified cause] Learned variants (numpy/BLAS) are **owner-box/Tier-1 only** , asserted with tolerance, never as a cross-machine byte-snapshot.
D6	**Non-attributability = deletion-at-source, encryption = public-only barrier**	*(design)*	workflow-2 synthesis A contributor who runs the tests necessarily reads plaintext+key (DRM/analog-hole). So we **delete** attribution, not hide it. Encryption only stops the no-key public.
D7	**Two tiers**	*(design)*	workflow-2 synthesis **Tier-0** public/no-key (shuttle-only, de-attributed). **Tier-1** key-gated/out-of-repo (person/audio, full corpus), **skips cleanly when absent** .

3. Legal foundation (researched ? gates on counsel, §8)

****Core verdict** (settled across IN/US/EU): ****the derived numbers ? per-frame shuttle (x,y), rally**

start/end intervals, audio/court/aggregate-person feature values ? ****do NOT inherit the source video's copyright.**** They are uncopyrightable **facts/measurements**, not authored expression, and not a §101 "derivative work."

Jurisdiction	Copyright of the data	Database right	Contract / ToS	Net
****India**** (primary)	Not copyrightable as facts (**EBC v. D.B. Modak**); but compilations get thin protection on a low "skill/labour" bar (**Burlington**, **OLX v. Padawan**) ? only bites if ****lifted from a 3rd-party dataset****, not self-derived	****None**** (no sui-generis right) ? a real permissiveness edge	Anti-scraping ToS (untested in court) + ****IT Act s.43/s.66**** civil/criminal exposure for 3rd-party ingestion	****Owned-source + minors-excluded + no person/pose = clean****
****US****	Not copyrightable (**Feist**; §102(b); **NBA v. Motorola**)	None (**Feist** killed sweat-of-the-brow)	****Dominant risk.**** ToS enforceable where copyright/CFAA fail (**ProCD**, **hiQ** ? injunction + ****forced deletion****). Remedy for redistribution is takedown, not nominal damages	Publishable as facts ****iff**** owned/permissioned source + no barring ToS + no person/biometric/minor stream
****EU / UK****	Not copyrightable (**Football Dataco v. Yahoo**)	****Genuinely contestable.**** **Football Dataco v. Sportradar**: recording pre-existing on-court facts = ****obtaining****, not "creating" ? our "we created it" premise is ****overconfident****; harm test = **CV-Online Latvia**	*Ryanair v. PR Aviation*: ToS can ban extraction/redistribution even with no IP right	Get EU+UK advice before redistributing numeric streams at scale

****Provenance is the decisive fork.**** ****Fork A** (owner-recorded, unpublished) ? the only path safe to commit publicly; collapses to a trade-secret **choice** (Indian breach-of-confidence / Contract Act 1872 / IT Act s.72 ? ****not**** US DTSA) + privacy. ****Fork B** (broadcast/YouTube/scraped) ? do ****not**** commit; the extraction act needs a license or now-****contested**** US fair use (**Thomson Reuters v. Ross** 2025; USCO May-2025), and unlawful acquisition independently sinks it (**Bartz v. Anthropic**, ~\$1.5B settlement). Caveat: owner footage ****shot at a third party's event**** (tournament/club) carries ticket/venue/organiser/broadcaster terms ? treat as Fork B until cleared.

****Publish-by-stream (privacy):****
- ? ****Publish-OK:**** shuttle (x,y)+Visibility (inanimate object), abstract rally/GT ****intervals**** (identity-stripped), non-biometric audio/court/aggregate-person features detached from identity.
- ?? ****Conditional** (effectively never in v1): ****person bounding boxes that *persistently single out a player* ? flips to GDPR Art.9 the moment it identifies. This architecture is prone to that flip ? treat per-player tracking as a tripwire.**
- ? ****Never publish:**** pose/skeleton/****gait**** (re-identifies at 80?95% even with faces blurred ? **Weiss et al. 2025**), face crops, any named-player?movement mapping. Matches the existing "gait/biometrics strictly future (GDPR Art.9)" guardrail.
- ? ****Minors flip everything:**** DPDP "child" = under 18; s.9(3) ****prohibits behavioural monitoring**** (the Rule-12 exemptions don't cover this project). A persistent per-player signature on a minor is, conservatively, prohibited. ****Exclude minors**** from any persisted stream beyond inanimate shuttle/interval data.

****Non-attributability is relative, not absolute.**** A per-video trajectory is a fingerprint (de Montjoye: 4 points ? 95% re-ID). It's only non-attributable because the ****Fork-A source stays unpublished**** (no public reference to correlate against). Per **EDPS v. SRB** (CJEU C-413/23 P, 2025) it ****remains personal data to the holder**** who can re-run the detector. So: conditional, revocable (publish the source and the link snaps back), forward-looking (re-assess as the public-video environment grows). Never market it as "anonymous."

Key authorities: Feist 499 U.S. 340; NBA v. Motorola 105 F.3d 841; 17 U.S.C. §101/§102(b); Sega v. Accolade 977 F.2d 1510; Football Dataco v. Yahoo (C-604/10) & v. Sportradar (C-173/11); BHB v. William Hill (C-203/02); CV-Online Latvia (C-762/19); Ryanair v. PR Aviation (C-30/14); Thomson Reuters v. Ross (D. Del. 2025); EBC v. D.B. Modak; DPDP Act 2023 s.3/s.9 + Rules 2025; EDPS v. SRB (C-413/23 P). Full URLs in the research artifact (§10).

4. Architecture (design)

4a. Freeze seam + reuse map

Freeze at the **trajectory CSV ? pure-Python replay**. One shared helper ``backend/eval/golden_fixtures.py::replay_fixture()`` is the *single* code path used by both the freeze tool and both gates:

```
`parse_trajectory_csv ? windowing.build_windows(preset) ? distill_local.build_candidates ?
rally_gate net_crossings ? experiment.predict(CONTROL) ? merge_overlaps ? metrics.score`.
Reuses verbatim: `metrics.score/interval_iou`, `segmentation_metrics`,
`experiment.predict/compare/Variant.spec_hash`, `windowing.WindowingPreset` (default SERVED),
`regression.gt_hash/save_baseline + incommensurability guards`, `eval_stats` (seed=0),
`eval_baselines/` committed-baseline precedent.
```

4b. Tiered model (SEALED-FIXTURES)

Tier	Where	Key?	Contents	Test behaviour
0	opaque	fixture_id	fps/frame_width (quantized), relative start/end, 5 shuttle scalars, label, relative gts	No person/audio/pose. Unconditional ? runs on every PR/fork; turns today's skipped litmus into a real public gate
1	OneDrive (default)	or SOPS-encrypted in-repo	token/age key full corpus	with person/audio; authoritative ? F1/CI/sign-test + exact Gate-A litmus Skips cleanly when key/data absent (generalize `golden_present()` ? `tier1_present()`)

A fresh no-key clone is always **green**; coverage never depends on a secret.

4c. Anti-attribution measures (Tier-0)

- **Opaque RANDOM surrogate id** (``os.urandom`` + a private surrogate?source map kept off-git) ? *not* the MD5 ``video_id``, and *not* HMAC(salt, video_id?name) (red-team: brute-forceable over a tiny known roster if the salt leaks). Strip the ``name`` field (today = source filename).
- **Metadata strip** ? no filename, video_id, match/event/player identity, capture date.
- **Time-origin reset** ? subtract per-fixture ``t0`` from all start/end/gts. **Provably metric-invariant** (``metrics.interval_iou`` and start/end-error are pure differences) ? bit-identical scores, severed absolute timeline.
- **Stream minimization** ? Tier-0 keeps shuttle block only; person/audio excluded; pose/gait/face have **no field** in the schema by design.
- **Raw-trajectory exclusion** ? never commit the per-frame ``(Frame,Visibility,X,Y)`` CSV.
- **Quantize fps/frame_width** ? red-team: at n=6 the raw values are a camera/venue quasi-identifier.
- **Provenance hard-fail gate** ? the de-attribution pass refuses to emit a public fixture for any record not tagged owner-recorded(Fork-A) + minors-excluded + biometric-excluded.

4d. The two gates

- **Characterization/snapshot** ? replay fixture, **parse-compare** JSON (never byte-compare ? CRLF/float-format safe; ``core.autocrlf=true`` here), exact on int/structural fields, ``approx(abs=1e-6)`` on floats; stamp + check windowing-preset/label/gt fingerprints (incommensurable ? loud fail). Catches any behaviour change even at constant F1. **CONTROL path only** across machines.
- **Quality-floor** ? score vs committed labels; assert per-video + mean F1 ? floor in ``eval_baselines/regression_baseline.json`` (created here; carries a **windowing fingerprint** so a served-default change is flagged incommensurable, not silently compared); emit ``eval_stats`` paired-delta-CI + sign test (NUMBERS INFORM); add a **metrics-vs-base-branch** paired delta. Synthetic-tier floors measure **correctness**, not field quality (documented).

4e. Serialization & optional protobuf

JSON with ``sort_keys=True`` + pre-quantized floats + LF is sufficient. **Protobuf** is optional (01) ? adopt only if cross-OS byte-stable snapshots are wanted (proto3 ``rally.fixture.v1``, codegen + ``protoc && git diff --exit-code`` guard, ``*.pb`` binary -text). Protobuf is **encoding**, not secrecy (trivially ``--decode_raw``).

4f. Encryption & keys (Tier-1 only)

Default = out-of-repo + token (existing OneDrive ``RALLY_GOLDEN_DIR`` seam ? zero sensitive bytes in git history, sidesteps the legal "redistribution" act). **Alternative** = SOPS + age (per-recipient, revocable, value-level diffs) for Fork-A-clean-but-private data wanting in-repo versioning. Rejected: git-crypt (whole-repo re-key), git-lfs (storage ? secrecy). CI secret exposed **only** to the Tier-1 job on trusted branches, never to fork-PR runners. The **HMAC/surrogate salt** (if used) and the surrogate?source map live with the private corpus,

rotated per release. Document explicitly that Tier-1 crypto defends **only** the no-key public.

5. Threat model (who's protected vs not)

Actor	Protected against	NOT protected (by design / residual)
TM1 no-key public	map fixture?source/match/player; decrypt Tier-1; confirm a source even holding the candidate file read Tier-0 numbers + run shuttle-only harness (intended; useless-in-isolation)	
TM2 keyed contributor	recover filename/video_id/match/face/pose/gait ? never written (survives the analog hole) read every value the suite reads (unpreventable; not pretended otherwise)	
TM3 CI runner	leaked log can't leak attribution it never had runs the suite (intended); Tier-1 secret never on fork runners	
TM4 malicious keyed insider	correlation needs a public reference; Fork-A source is unpublished ? "not reasonably likely" residual: conditional + revocable ? publish/leak the source later and the link can snap back (forward-looking re-test duty)	
TM5 re-ID adversary	source unpublished + only ~5 invariant scalars/window + reset timeline ? no back-match becomes live if a Fork-A source is ever published	
TM6 accidental plaintext commit	required CI leak-scan rejects MD5/known-name/non-surrogate/plaintext-SOPS if the required check is disabled, a leak lands	

6. Honest limits ? what a green suite does NOT prove

- Green proves only that the **deterministic post-trajectory transforms** are unchanged for the **frozen cases**. It does **not** cover: the WASB/TrackNet detector, `_probe_fps_width/cv2` frame-seek, the AI/provider handoff, chunking/re-encode, the ffmpeg stitch, DB persistence/migrations, or the API filter. **A refactor that breaks decoding or the detector passes these gates green.** The owner-box `regression.py` DB run remains the only full-chain check.
- Characterization = **behaviour-locking, not correctness** ? it freezes current behaviour **including current bugs**. "Passing" = "behaviour unchanged", never "behaviour correct". This caveat must ride in the **test pass/fail message**, not only the docs.
- **Synthetic-tier floors** = plumbing correctness, NOT field quality. **Never cite synthetic F1 in `QUALITY_ITERATIONS.md` / the paper as rally-detection performance (tag `kind: synthetic`).**
- Tier-0 public coverage is **shuttle-only**; the person/audio fusion ?F1 and exact Gate-A values need Tier-1 (skipped on no-key clones).
- Data is **not "anonymous"** absolutely (EDPS v. SRB) ? relative, conditional, revocable, forward-looking.
- Encryption gives **no secrecy from a running contributor**; protobuf gives **no secrecy at all**.
- De-attribution does **not** cure provenance/licensing ? it sits **behind** the per-record provenance gate + counsel sign-off.
- `--refresh-snapshot` can rubber-stamp a real regression; mitigation is **review of the visible diff** + a separate, deliberate floor-baseline update. Require a recorded reason.

7. Deliverables & progress tracker ? **source of truth**

Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One **small PR per row**, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.

ID	Deliverable	Depends on	Counsel-gated?	Status	PR
P0	Leak remediation + required CI leak-scan. Scrub real names/paths in <code>eval_baselines/heuristic_lovo_n6.json</code> , <code>gemini_wrapper_2026-06-10.json</code> , <code>backend/eval/eval_partial_labels.py</code> , <code>tests/test_cleanup_caches.py</code> ? opaque surrogate keys + private map; add CI leak-scan (MD5 / known-name / non-surrogate / plaintext-SOPS) as a required status check ; decide on git history (PR #77, accept vs <code>filter-repo</code>). ? No				

```
(owned data hygiene) | ? | ? |
| **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper.**
`eval_baselines/fixtures/` + `backend/eval/golden_fixtures.py`
(`load_manifest`/`load_fixture`/`replay_fixture`, schema_version dispatch,
`Path(__file__)`-relative); 3?5 hand-authored synthetic fixtures (clean rally / held-shuttle
Gate-A / occlusion / quiet / multi-window); `manifest.json`; `.gitattributes` LF pin. | P0 | No
(synthetic) | ? | ? |
| **M2** | **Freeze/refresh extractor.** `backend/eval/freeze_fixture.py` composing shipped fns;
`--refresh-snapshot[--all]`; `--license/--minors-free` refusal gate; idempotence unit test. | M1
| No (synthetic) | ? | ? |
| **M3** | **Characterization gate.** `tests/test_golden_characterization.py` ? parsed-structure
compare, exact-int/approx-float, windowing+label fingerprint guard, **CONTROL-only**; schema-
drift + row-count-cap tests; **multi-OS CI matrix** (win/ubuntu/mac). | M1, M2 | No | ? | ? |
| **M4** | **Quality-floor gate + first `regression_baseline.json`.**
`tests/test_golden_quality_floor.py`; create `eval_baselines/regression_baseline.json` with
**windowing fingerprint**; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for
runtime. | M1, M2 | No | ? | ? |
| **P1** | **De-attribution + minimization pass.** `scripts/deattribute_golden.py`: private
Fork-A `*_golden_features.json` ? de-attributed/minimized/time-origin-reset public fixture;
**random surrogate id**; person/audio stripped + **positive assertion** no person/audio keys;
quantize fps/frame_width; **person-optional Tier-0 loader**; provenance hard-fail gate. | M1 |
No (tooling) | ? | ? |
| **P2** | **Counsel sign-off (gating ? not a code PR).** Retained IN + EU/UK counsel review of
the final committed Tier-0 subset + source-provenance log. | P1 | ? **Yes** | ? | ? |
| **P3** | **Tier-0 REAL subset.** Commit counsel-cleared, de-attributed, minors/biometric-
excluded real Fork-A subset; unconditional test. | P1, P2 | ? **Yes** | ? | ? |
| **O1** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1` proto3,
codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes` binary. Only
if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
| **O2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive token,
default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job gated on
secret, never on fork runners. | M1 | No | ? | ? |
| **DOC** | **Finalize this doc + cross-links.** Add `eval_baselines/fixtures/` row to `CODE_MAP
$0` + TEST MAP; cross-link `EVALUATION.md` / `EXPERIMENT_HARNESS.md` / `GOLDEN_DATA_SHARING.md`
/ `QUALITY_ITERATIONS.md`; update `docs/README.md` index + memory `current-state`/handoff. |
M1..M4 | No | ? | ? |
```

****Must-fix-before-any-public-commit (red-team), folded into the rows above:**** ? numeric-
tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5); ? scrub names in ****all****
files + history decision (P0); ? leak-scan a ****required**** check, hooks are convenience only
(P0); ? person-optional loader + ****positive**** no-person assertion, don't trust the silent `0.0`
default (P1); ? random surrogate id not HMAC-over-roster (P1); ? quantize/drop fps/frame_width
(P1); ? document Tier-0 omits the fusion path (DOC); ? counsel + Fork-A + minors/biometric
exclusion as hard preconditions (P2).

8. Open owner / counsel questions

****Counsel (gating ? block P2/P3):****

1. Does the EU/UK ****sui-generis database right**** subsist in our self-derived shuttle/rally
streams on the **Sportradar** "obtained vs created" reasoning, and does redistributing a de-
attributed Tier-0 subset cause **CV-Online** harm?
2. Post **Thomson Reuters v. Ross** + USCO-2025, is ****US fair use** for the extraction step**
defensible if the dataset could compete in a "data/analytics" market ? does publishing only
numbers (never footage) matter?
3. India: are anti-scraping ToS enforceable; ****IT Act s.43/s.66**** exposure for any 3rd-party
ingestion; does pending **ANI v. OpenAI** (~Apr 2026) move it? Confirm ****DPDP s.9/Rule 10**** and
whether per-player tracking is prohibited s.9(3) monitoring of a minor.
4. Sign-off on the ****final committed Tier-0 subset + provenance log****; is non-attributability-
via-private-source a defensible posture to assert given its conditional/revocable nature?

****Owner (strategic):****

5. Publishing the Fork-A subset ****forfeits the trade-secret moat**** (Indian breach-of-confidence

framing) ? weighed against community/benchmark/paper aims. Your call.

6. **Forward-looking commitment:** do not publish a highlight reel of the *same* source whose fixtures are committed (without re-clearing), and accept a periodic re-assessment as the public-video environment grows.

7. **Scope of first public commit:** synthetic-only (zero provenance risk, recommended) vs a counsel-cleared real subset.

8. **Tier-1 mechanism:** out-of-repo+token (recommended) vs SOPS+age in-repo. And: is protobuf (01) wanted, or is canonical-JSON enough?

9. Definition of done

The project is **DONE** (flip `Status ? DONE``, archive) when:

- [] **P0, M1?M4, P1, DOC** are ? and merged.
- [] A clean `git clone`` on **win + ubuntu + mac** runs `python run_all_tests.py`` and the **Tier-0** gates pass green with no key / no video / no GPU / no DB.
- [] The **CI leak-scan** is a required status check and passes (no MD5 / real-name / non-surrogate / plaintext-SOPS in the tree).
- [] Characterization + quality-floor gates are **parse-compared, CONTROL-only, windowing-fingerprinted**, and demonstrably catch a seeded behaviour change.
- [] `docs/CODE_MAP.md`` §0 + TEST MAP updated; this doc cross-linked; `NEXT_STEPS.md`` + memory `current-state`` point here.
- [] **Either** the public commit is synthetic-only, **or** **P2** counsel sign-off + **P3** are ? for any real-data commit.
- [] §6 honest-limits caveat rides in the test pass/fail messages, not only the docs.

Optional 01/02 may remain ? without blocking DONE (they're earned, not assumed).

10. References

Internal code anchors (verified): `backend/eval/{regression.py,fusion_golden.py,fusion_compare.py,ablation.py,experiment.py,metrics.py>windowing.py,distill_local.py,gt_loader.py,eval_stats.py}`` · `backend/pipeline/detectors/{base.py,tracknet_runner.py,rally_gate.py}`` ·

`backend/utils/hashing.py::compute_video_id`` · `eval_baselines/{heuristic_lovo_n6.json,gemini_wrapper_2026-06-10.json,experiment_leaderboard.json}`` ·

`tests/test_ablation.py::test_litmus_reproduces_gate_a`` · `.github/workflows/ci.yml``.

Internal docs: `GOLDEN_DATA_SHARING.md`` (#175), `EVALUATION.md``, `EXPERIMENT_HARNESS.md``, `QUALITY_ITERATIONS.md``, `MULTI_SIGNAL_FUSION_PLAN.md``, `CODE_MAP.md`` §0.

Legal authorities: see §3 (full URLs in the research workflow artifact). *This section is researched general information, not legal advice.*

Changelog

- **2026-06-16** ? Initial draft synthesized from two multi-agent workflows (mechanics design + 2 critiques; legal memo IN/US/EU + SEALED-FIXTURES architecture + red-team). Verified the live name-leak and the BLAS-determinism cause against code. Status: DRAFT, owner-gated.

89. user (2026-06-16T04:07:52.179Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\GOLDEN_REGRESSION_FIXTURES.md (file state is current in your context ? no need to Read it back)

90. assistant (2026-06-16T04:08:17.095Z)

Draft written to docs/GOLDEN_REGRESSION_FIXTURES.md.

Before I hand it over, let me verify one claim I carried into the P0 deliverable from the red-team but hadn't personally checked ? that `tests/test_cleanup_caches.py`` also contains real names ? so the doc doesn't assert something unverified.

91. user (2026-06-16T04:08:18.829Z)

```
19: assert classify("adarsh_traj.csv", False, "wsl")[2] is False # trajectory
-> keep
24: cat, regen, default_purge = classify("adarsh_traj_wasbcbache", True, "wsl")
55: entries = [_e("GX010125_frames", 90_000, True), _e("kushagra_frames", 5_000, True)]
58: assert [e.name for e in to_del] == ["kushagra_frames"]
```

92. assistant (2026-06-16T04:08:40.588Z)

Verified ? `tests/test\_cleanup\_caches.py` does carry the tokens (`adarsh\_traj.csv`, `kushagra\_frames`), so the P0 deliverable is accurate across all four files.

Done ? the tracked project doc is ready

? docs/GOLDEN_REGRESSION_FIXTURES.md ? **draft**, **uncommitted** (your conventions: no commit/PR unless you ask).

It's structured to be exactly what you asked for ? iterate, track, mark complete:

- **§7 Progress tracker** is the single source of truth: one row per deliverable (P0, M1-M4, P1, P2, P3, O1, O2, DOC), each a small independent PR, with `?/?/?` status + a PR-link column. Iterating = editing a row.
- **§9 Definition of done** is the explicit acceptance checklist; when it's all ticked, the `Status:` header flips DRAFT ? IN PROGRESS ? DONE and the doc moves to `docs/archives/`.
- **§2 Decisions locked** captures your three answers so they're not relitigated; **§8** isolates the counsel-gated vs owner-strategic questions.
- **§3 Legal** is the IN/US/EU memo (cited), clearly marked **researched, not legal advice**; **§5 threat model**, **§6 honest limits**, and the **§7 red-team must-fixes** keep it from overselling.

Two findings in there are real and verified now, independent of the rest:

1. ? **P0 ? live name leak:** real player/venue names are committed in `eval\_baselines/heuristic\_lovo\_n6.json`, `gemini\_wrapper\_2026-06-10.json`, `backend/eval/eval\_partial\_labels.py` (+ a personal `C:/Users/avidu/?` path), and `tests/test\_cleanup\_caches.py` ? plus git history. Private repo today, permanent if open-sourced.
2. ? **D5 ? determinism trap:** both independent red-teams + the code confirm the learned scorer (numpy/OpenBLAS `DYNAMIC\_ARCH`) would false-RED across CPUs ? the cross-machine gate must be CONTROL/static-only. Happily, that's the **same** shuttle-only tier privacy already requires.

Recommended next step: **P0** is the natural first PR ? it's self-contained, needs **no** counsel (it's hygiene on your own data), and fixes a real bug today. Everything legally sensitive (P2/P3) stays gated behind counsel + your scope call (synthetic-first is the low-risk path).

Want me to (a) start **P0** as the first focused PR, (b) wire the pointers in `NEXT\_STEPS.md` + memory `current-state` so a future session resumes from the tracker, (c) commit the doc itself, and/or (d) adjust the doc's shape first? I held off on (b)/(c) to avoid churn and shared-checkout collisions until you've eyeballed it. The full raw research (every legal citation + the architecture/red-team JSON) is on disk if you want me to expand any section.

93. user (2026-06-16T04:08:43.364Z)

love the framing of it! Please freeze this and I guess this could be a good norm to follow throughout

94. assistant (2026-06-16T04:10:12.880Z)

Love it ? let me freeze the doc into a PR and distill the format into a reusable template + convention so it's a norm going forward. First, branching-safety hygiene and reading the files I'll touch so I match repo conventions.

95. user (2026-06-16T04:10:16.971Z)


```

=== status ===
?? artifacts/
?? docs/GOLDEN_REGRESSION_FIXTURES.md
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
=== current branch / HEAD ===
master
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
=== origin/master ===
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)

```

96. user (2026-06-16T04:10:16.998Z)

```

1      - [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md) ? Latest (v8) vendor
labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-driven
court-calibration / singles-doubles additions marked `? ADDED (fusion)`.
2      - [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) ? Foundation doc for
closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
class, golden-corpus + active learning, Gemini-agreement as a shadow signal), an
adversarially-verified, cited external-research synthesis (TAL/TAS boundary heads, racket-
sports rally-state cues, small-data, eval rigor), a prioritized cheap?durable roadmap with
falsifiable success criteria, and ($7) a tracked project plan ? work-item checklist,
sequencing, the single quality number attributed to the effort, and a definition of done
(project completes when all tiers land + the number is recorded). The plan that the
[QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) log executes against; ties together
EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
OWNED_MODEL_IMPLEMENTATION_PLAN.
3      - [EVALUATION.md](EVALUATION.md) ? How we measure and improve quality.
4      - [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) ? Design (P1 SHIPPED) for A/B + shadow
+ SxS experimentation on rally-detection variants with zero production regression ?
experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift,
rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`,
`regression`, `metrics`); implemented in `backend/eval/experiment.py`.
5      - [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) ? Living log of rally-detection
quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-
chronological record; terminal-state snapshots get archived under `archives/quality-
iterations/`.
6      - [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) ? Design (owner-gated,
NOT started): a signal-fusion framework for rally detection ? producers emit confidence-
scored signals at three temporal scopes (frame detectors / window analyzers / session
analyzers), a learned scorer weights them, and an auditable report-first reconciliation
pass lints the decision against the evidence. The spine: no special-casing in the decision
path (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-
up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to
`EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).
7      - [PERSONALIZATION_PLAN.md](PERSONALIZATION_PLAN.md) ? Design (owner-adopted hybrid;
Phase-0 landing) for the per-user personalization feature on a generalization-first
baseline: a thin per-user tuning layer gated by the honest small-N statistics (a min-N
promotion floor + a structural-guard exemption, never a service gate) + a contribution
flywheel (free-tier videos pool into the owner-trained baseline; "a tool that gets better the
more you help it"). Records the locked owner decisions (identity, storage, `min_n`
semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) +
held-out-USER learning curve (#142).
8
9      ## Practical / ongoing
10     - [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
11     - Golden set docs (`GOLDEN_SET`)
12     - Evaluation harness and calibration drivers (in `backend/eval/`)
13     - [GOLDEN_DATA_SHARING.md](GOLDEN_DATA_SHARING.md) ? How the GPU box and CPU dev/agent
sessions share golden artifacts (trajectory CSVs, `_golden_features.json`) via a OneDrive
folder; the GPU-extract ? CPU-litmus split. See ##175.
14

```

```

15     ## Historical documents
16     Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).
17
18     - [CODE_AUDIT_AND_TEST_HARDENING.md](CODE_AUDIT_AND_TEST_HARDENING.md) ? **COMPLETED**
(2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing
/ contract / architectural audit, 626-test infrastructure, positive test plans for the 5
identified issues, multi-layer regression protocol (including ablation/experiment layer), and
pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md`
for the final archived state. (Top-level is now pointer stub per review; full content in
archives/.)
19
20     See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md`
for classification of small tech debt / starter projects.

```

97. user (2026-06-16T04:10:55.808Z)

```

40     - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
41     - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
42       user_version` migration ladder in `backend/infrastructure/database.py`.
43
44     ## Committed guardrails (non-negotiable)
45     - **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
46       training data source** (terms/copyright). Commercial-clean licensing for EVERY
dependency ?
47     code, data, AND weights: **MIT / Apache-2.0 / BSD only** (ratified 2026-06-09, owner-
agreed).
48     - **Owned-model base:** **Qwen3-VL (Apache-2.0)** primary, InternVL3 (MIT) fallback;
**Gemma-4
49     is AMBER / bench-only** (Apache grant likely but Prohibited-Use posture unconfirmed ?
counsel
50     needed; do NOT build on it without sign-off). **Reject Gemma 3** (viral license).
**Exclude
51     minors** from the training pool; per-user training data needs a separate explicit opt-
in.
52     - **v1 = rally + highlight + history + correction loop**; **gait/biometrics strictly
future**
53     (GDPR Art. 9).
54     - **Generic data schema**; `coach ? athlete` is an **optional overlay**, not baked in.
55
56     ## Workflow conventions
57     - **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
58     - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.8+; `ffmpeg`
required.
59     - Never push to the default branch directly; descriptive commit messages.

```

98. assistant (2026-06-16T04:10:52.996Z)

Master moved again (sibling session landed #185) ? I'll branch explicitly from `origin/master` per branching-safety. Note: this repo already has a precedent for this format ?

`RALLY\_QUALITY\_RESEARCH.md` \$7 is a "tracked project plan? definition of done." So the norm is really *standardizing an emergent pattern*, not inventing one. Let me cut the branch and read the files I'll touch.

99. user (2026-06-16T04:10:55.677Z)

branch 'docs/golden-regression-fixtures-plan' set up to track 'origin/master'.

Switched to a new branch 'docs/golden-regression-fixtures-plan'

=== verify HEAD is mine ===

docs/golden-regression-fixtures-plan

90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline (#185)

=== only my commits vs base (should be empty now) ===

100. assistant (2026-06-16T04:11:01.876Z)

Clean branch, no sibling commits. Let me see the rest of the Workflow-conventions block and the top of NEXT_STEPS so I insert pointers cleanly.

101. user (2026-06-16T04:11:02.842Z)

```
56     ## Workflow conventions
57     - **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
58     - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.8+; `ffmpeg`
required.
59     - Never push to the default branch directly; descriptive commit messages.
60     - ?? This dev container lacks GPU / torch / cv2 / numpy ? the **full pipeline can't run
here**.
```

61 Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to
the owner.

62

102. user (2026-06-16T04:11:03.577Z)

```
1     # Next steps (live) ? owned-model + multisport
2
3     **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
4     owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
5     ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
6     `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
```

7 **PR-2

8 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
9 compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
10 coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
11 `trajectory\_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).

12 **Authoritative roadmap:**

13 `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
14 `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.

15 ## ? Next-session / GPU-box pickup (2026-06-15)

16 **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion\_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).

17

18 **Immediate next step ? the real P3 LOVO measurement, split GPU?CPU** (the dev box is
CPU-only: `torch+cpu`/`cv2` present but no GPU, and the corpus/videos/CSVs/labels are not here):

19 1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6"` ? produces `*_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `rallies.csv` labels are owner-side only.)

20 2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-

```

annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](GOLDEN_DATA_SHARING.md)**; tracked in ##175.
21      3. [CPU dev/agent session] `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
22
23      Also queued (owner-gated / pending input):
24      - Ship-gate target calibration ? ##172 (owner decision; needs corpus growth).
25      - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
26      - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
27
28      Leaving (bulky / M2 ? no start without owner go): ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
29
30      ## Where we are (one paragraph)

```

103. assistant (2026-06-16T04:12:00.689Z)

Branch is clean. Now I'll: (1) tweak the doc's tracking-anchors line to match what I'll actually wire, (2) create the reusable template, (3) index both in `docs/README.md`, and (4) add a one-line convention to `CLAUDE.md`. The norm is really **standardizing** the pattern already in `RALLY\_QUALITY\_RESEARCH.md §7`.

104. assistant (2026-06-16T04:12:17.318Z)

```

<!--
PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.

```

A "tracked project doc" is how this repo carries substantial design/project work: it can be ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables, then archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and docs/GOLDEN_REGRESSION_FIXTURES.md.

Rules of thumb:

- §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable PR.
 - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs sign-off), or [design] (proposed). The doc reflects real shipped state, never aspirational.
 - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
 - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model for research- or security/privacy-heavy work). Delete sections that do not apply.
 - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state work).
- ```
-->
```

## <Project / Plan Title>

```

> Status: `DRAFT` ? `<owner-gated? | in progress>` . Owner: `<name>` . Created:
`<YYYY-MM-DD>` . Last updated: `<YYYY-MM-DD>`
> Lifecycle: `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> Tracking anchors: §7 progress tracker is the source of truth; indexed in `docs/README.md`;
pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
> Relation to existing docs: `<extends | supersedes | peer-of ?>`
> Honesty note: mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.

```

---

## 0. TL;DR

<2?5 sentences: the goal, the approach, and the single most important caveat. Call out any

finding that is already actionable on its own.>

1. Problem & goal

<Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a reader must read to get current.>

2. Decisions locked

<Table ? capture answered questions so they are not relitigated.>

| #  | Decision | Source / date | Implication |
|----|----------|---------------|-------------|
| D1 | ?        | ?             | ?           |

3. Foundation ? research / prior art \*(OPTIONAL)\*

<Include when the plan rests on external facts: cited research, legal/compliance, benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this creates.>

4. Design / architecture

<The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing modules/code this builds on so it is clear what is new vs reused.>

5. Threat model / risk table \*(OPTIONAL ? security / privacy / compliance work)\*

<Who/what is protected vs NOT; residual risks stated plainly.>

6. Honest limits ? what this does NOT do

<The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>

7. Deliverables & progress tracker ? \*\*source of truth\*\*

Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. \*\*One small PR per row\*\*, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.

| ID | Deliverable | Depends on | Gated? | Status | PR |
|----|-------------|------------|--------|--------|----|
| P0 | ?           | ?          | No     | ?      | ?  |

8. Open questions ? owner / external \*(split blocking-gates from nice-to-knows; name who decides)\*

1. ?

9. Definition of done

<Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>  
- [ ] ?

10. References

\*\*Internal code anchors `[verified]`:\*\* `<paths>`. \*\*Internal docs:\*\* `<docs>`. \*\*External:\*\* `<cited sources>`.  
---

Changelog

- `<YYYY-MM-DD>` ? `<what changed>`.

105. user (2026-06-16T04:12:17.333Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PROJECT\_DOC\_TEMPLATE.md (file state is current in your context ? no need to Read it back)

106. assistant (2026-06-16T04:12:22.240Z)

Now the three edits ? fix the tracking-anchors line, index both docs in the README, and add the

convention to CLAUDE.md:

#### 107. user (2026-06-16T04:12:24.613Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\GOLDEN\_REGRESSION\_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 108. user (2026-06-16T04:12:28.850Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 109. user (2026-06-16T04:12:31.312Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 110. user (2026-06-16T04:12:33.446Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\CLAUDE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 111. assistant (2026-06-16T04:12:48.785Z)

All edits applied. Now staging **explicit paths only** (never ``-A``), re-verifying the branch state right before commit per branching-safety:

#### 112. user (2026-06-16T04:12:51.843Z)

```
warning: in the working copy of 'docs/GOLDEN_REGRESSION_FIXTURES.md', LF will be replaced by
CRLF the next time Git touches it
warning: in the working copy of 'docs/PROJECT_DOC_TEMPLATE.md', LF will be replaced by CRLF the
next time Git touches it
=== staged ===
CLAUDE.md
docs/GOLDEN_REGRESSION_FIXTURES.md
docs/PROJECT_DOC_TEMPLATE.md
docs/README.md
=== branch / HEAD (mine?) ===
docs/golden-regression-fixtures-plan
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
=== commits vs origin/master (should be none yet) ===
=== unstaged/untracked left behind (should NOT include my 4 files) ===
M CLAUDE.md
A docs/GOLDEN_REGRESSION_FIXTURES.md
A docs/PROJECT_DOC_TEMPLATE.md
M docs/README.md
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
```

#### 113. assistant (2026-06-16T04:13:02.586Z)

Exactly my four files staged; ``artifacts/`` and the KHELSUTRA doc correctly left untracked. Committing with the required trailer, then pushing:

#### 114. user (2026-06-16T04:13:11.274Z)

```
[docs/golden-regression-fixtures-plan 95cd154] docs: freeze Golden Regression Fixtures plan +
adopt tracked-project-doc template
4 files changed, 280 insertions(+)
create mode 100644 docs/GOLDEN_REGRESSION_FIXTURES.md
create mode 100644 docs/PROJECT_DOC_TEMPLATE.md
```

```
=== HEAD now ===
95cd154 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
=== push ===
remote:
remote: Create a pull request for 'docs/golden-regression-fixtures-plan' on GitHub by visiting:
remote: https://github.com/avidullu/badminton-highlight-indexer/pull/new/docs/golden-
regression-fixtures-plan
remote:
branch 'docs/golden-regression-fixtures-plan' set up to track 'origin/docs/golden-regression-
fixtures-plan'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch] docs/golden-regression-fixtures-plan -> docs/golden-regression-fixtures-
plan
```

#### 115. assistant (2026-06-16T04:13:16.050Z)

Committed (only my commit atop `90ab087`) and pushed. Opening the PR:

#### 116. user (2026-06-16T04:13:31.211Z)

<https://github.com/avidullu/badminton-highlight-indexer/pull/186>

#### 117. assistant (2026-06-16T04:13:50.436Z)

PR opened: `**[#186](https://github.com/avidullu/badminton-highlight-indexer/pull/186)** ? docs-only, draft/owner-gated. Now let me make the norm durable in my memory too (the repo side is in the PR via `CLAUDE.md` + the template). Reading the memory index + current-state first so I update them without clobbering a concurrent session.`

#### 118. user (2026-06-16T04:13:51.834Z)

```
1 # Memory Index
2
3 ## ? Start here
4 - [Current state](current-state.md) ? live micro-checkpoint (updated every
checkpoint): open PRs, what just landed, next steps, open decisions. Read this first.
5 - [Session handoff](session-handoff.md) ? current state · next steps · open decisions ·
ramp-up kit. AUTO-refreshed at meaningful checkpoints (PR merged, key decision, phase shift,
session wrap) ? see working-agreement-checkpoints staleness tripwire.
6 - [Code map artifact](code-map-artifact.md) ? for CODE/architecture ramp-up, read
`docs/CODE_MAP.md` FIRST (dense anchor-rich map; replaces a file crawl). How to regenerate it.
7 - [File-placement conventions](file-placement-conventions.md) ? where new files go
(scripts/ vs backend/eval/ vs tools/ vs scratch/ vs training/ vs artifacts/ ?); read before
committing a new file. Authoritative table = `docs/CODE_MAP.md` §0 (PR #138); scratch/
gitignored (#136).
8
9 - [Working agreement: attribution](working-agreement-attribution.md) ? never assert
fabricated facts; NEVER invent a provenance (a discussion/review/source that didn't happen). The
silver-bullet rule.
10 - [Working agreement: checkpoints](working-agreement-checkpoints.md) ? user prefers
frequent small PRs + a Claude-consumable current-state file updated at every micro-checkpoint.
11 - [Feedback: tail long-job logs](feedback-tail-long-job-logs.md) ? proactively tail
progress/summaries from long-running background jobs while they run, not just the final result.
12 - [Working agreement: merging](working-agreement-merging.md) ? owner authorized Claude
to MERGE its own good-engineering-fix PRs on verified-green CI (2026-06-12);
product/strategy/licensing stay owner-gated.
13 - [Gotcha: shared-checkout branch collisions](gotcha-shared-checkout-branch-
collisions.md) ? spawned tasks + sibling sessions SHARE this checkout; branch from
`origin/master` explicitly, re-verify HEAD before commit, stage explicit paths, serialize repo
writes. The #116/#117 collision (owner upset). Global form in ~/.claude/CLAUDE.md.
14 - [Gotcha: long GPU/WSL runs](gotcha-long-runs-gpu-wsl.md) ? running the WASB pipeline
E2E on a big 4K video: Modern Standby kills it (keep-awake), cv2 extraction is the bottleneck
```

(ffmpeg pre-extract), proxy-index/4K-stitch recipe + how to observe each phase.

15 - [Cache-hygiene policy](cache-hygiene-policy.md) ? retention table (purge frames/staged vs keep CSVs/DB/reels) + the shipped tooling: self-cleaning runners (keep\_frames default false), tools/cleanup\_caches.py (dry-run-first), disk guard, SessionStart nudge hook. The `for e in \*` WSL footgun.

16 - [Tool: trim\_video.py](tool-trim-video.md) ? reusable `scratch/trim\_video.py` for keyframe-aware LOSSLESS video trims (drop first/last N sec), `--exact` re-encode with NVENC?libx264 fallback. Reach for it instead of hand-rolling ffmpeg trims.

17 - [Decision-rigor norm](decision-rigor-norm.md) ? **\*\*NUMBERS INFORM, FOUNDATIONS DECIDE\*\*** (owner directive 2026-06-14, set by the Gate-A flip). Every signal/architecture change gets ablation-layer rigor (per-signal marginal ?F1 + honest CI/sign-test) AND a foundational lens (extensible/orthogonal/reversible), never numbers alone. Ties **[[weak-signals-retained]]** + **[[paper-coauthor-aim]]**.

18

19 ## Project: rally segmentation pipeline

20 - [Candidate segments & fine-tuning](candidate-segments-finetuning.md) ? why raw detector candidates are persisted (detector-agnostic table); golden-eval-set goal

21 - [TrackNetV4/WASB via WSL2](tracknet-wasb-wsl2-decision.md) ? self-hosted shuttle trajectory models (not LLM APIs); WASB/TrackNetV3 = MIT, commercial-clear

22 - [Collector?indexer bridge](collector-indexer-bridge.md) ? 4-phase plan to train/eval a generic temporal rally segmenter; Phases 1?2 done, 3 in progress, 4 next

23 - [Golden-set vendoring](golden-set-vendoring.md) ? generate golden rally labels via 3rd-party vendors (Bangalore-local) + extensible regression flow; Roora selected for a 3-sport pilot, guideline + collector ingestion built, GS-1/2/4 landed, GS-3 + gold key pending

24 - [Roora guidelines Drive doc](roora-guidelines-drive-doc.md) ? the CANONICAL vendor-facing Brief+Guide Google Doc (supersedes v3/v5/v6/v7 Drive copies; v7 anomaly RESOLVED 2026-06-12); TODO markers still to fill before sending

25 - [Audio (E1) findings](audio-e1-findings.md) ? quantitative gains/losses of the classical-onset audio cue (recall 100% but discrimination ~2.2x, band-pass null) + reusable lessons for adding the NEXT signal. Audio NOT adopted for fusion; detector kept (PR #35).

26 - [Weak signals retained](weak-signals-retained.md) ? owner directive: non-significant cues (person/audio/court-bounds/future **\*\*shuttle-drop\*\***) stay **\*\*default-OFF + retained\*\*** as scorer feature columns (no special-casing) for a growing (Roora) corpus ? never deleted on a small-n null. Measured: person ?F1 ?0.021, +audio +0.079 (both CIs span 0 at n=6).

27

28 ## Monetization (hosted API)

29 - [Monetization plan](monetization-plan.md) ? hosted video?highlights API on a GPU VM; licensing & per-video cost are first-class; user is Bangalore-based

30 - [Monetization research status](monetization-research-status.md) ? audit done: Ultralytics AGPL swapped out (PR #8); MonoTrack/YOLO-NAS weights avoided; codec=watch; full report in docs/

31

32 ## Forward-looking

33 - [Sibling internship repo](sibling-internship-repo.md) ? khelacharya shot-intelligence layer builds ON TOP of our rally segmentation; kept off master at branch level (NOT a separate GitHub repo ? corrected 2026-06-06)

34 - [Paper co-author aim](paper-coauthor-aim.md) ? **\*\*standing side-aim\*\***: co-author a paper on "local, cheap & efficient rally detection via multiple clever ideas". Keep `docs/QUALITY\_ITERATIONS.md` as the ablation-ready empirical backbone; honest measured results ONLY.

35

## 119. user (2026-06-16T04:13:52.023Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-447 of 2130 total (101240 tokens, cap 25000). Call Read with offset=448 limit=447 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```

1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4 metadata:
5 node_type: memory

```



```

6 type: project
7 originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8 ---
9
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 **Checkpoint:** 2026-06-16 ~03:30 (? RALLY-QUALITY RESEARCH + EVAL-FRAMEWORK FOUNDATION
DOC DONE ? **PR #184 OPEN for
13 owner review**, docs-only, NOT auto-merged). Owner asked for deep research to bridge the
local-vs-Gemini gap with a strong
14 harness/eval framework, checked into the repo as the foundation for quality iterations.
Scope (owner-picked) = research +
15 harness design doc, NO code changes; research focus = all 4 dimensions. **DELIVERED:**
16 - **`docs/RALLY_QUALITY_RESEARCH.md`** (worktree `.claude/worktrees/docs-rally-quality`,
branch `docs/rally-quality-research`,
17 PR #184, **2 commits**): §1 measured gap, §2 code-grounded over-merge diagnosis, §3
harness inventory + 7 gaps, §4
18 strengthened eval framework (segmental Edit/F1@k as primary; golden growth + active
learning; Gemini = SHADOW signal
19 only; one-command eval entrypoint), §5 cited+adversarially-verified external research,
§6 cheap?durable roadmap w/
20 falsifiable criteria, **§7 PROJECT PLAN** (owner asked to "do this as a project") =
the single NUMBER to attribute
21 [multi-court tIoU-F1 baseline 0.157 ? final + gap-closure% vs Gemini 0.66; primary
lens = segmental F1@0.25/Edit/
22 merge-rate], a work-item status checklist (E1/W1/W2/W3/M1/M2/R1/R2 + corpus-growth C)
w/ acceptance criteria +
23 sequencing, and a DEFINITION OF DONE (Tier 0-2 land + number recorded; Tier-3 owned-
model = corpus-gated stretch),
24 §8 constraints. Indexed in docs/README.md. **Opened for review (foundational/strategy
? owner-gated merge).**
25 **2 owner decisions flagged in the PR comment:** (1) the ?50%-gap-closure success bar;
(2) Tier-3 in/out of the DoD.
26 - **Deep research** (`wambqniwk`, 110 agents, 27 sources, 25 claims 3-vote-verified, 23
confirmed/2 refuted). Headlines:
27 add an explicit **boundary head that SPLITS** (ASRF/TriDet ? research bets at n=6-16,
port the idea not the deep model);
28 **See et al.** badminton inactivity end-of-rally rule (?80% no-motion frames, ~95%
acc) + serve classifier = the
29 low-effort win; **TemporalMaxer** ? keep the scorer parameter-light (learned temporal
nets overfit small data);
30 **lead the leaderboard with segmental Edit + F1@k** (tIoU/MoF are blind to over-
segmentation = our exact failure).
31 Full report: `%TEMP%\?wambqniwk.output`. OPEN (research couldn't verify): active-
learning method + Gemini weak-label
32 governance ? flagged in §5.3 as a future grounded pass.
33 - **CORRECTION shipped:** over-merge is NOT fps (velocity already per-second-normalized
via `fps/dframes`); it's
34 "shuttle-moving ? rally" + 2s merge + no cap. Fixed `_LOCAL_vs_GEMINI_REPORT.md` +
`scratch/compare_local_vs_gemini.py`.
35 (The Phase-C checkpoint's "fps not invariant" line is SUPERSEDED by this ? recorded
here for the log.)
36 - ?? NEXT (owner): review/merge #184; then the cheapest testable win = Tier-0 (add
segmental Edit-score to leaderboard) +
37 Tier-1 (bounded windowing: max-window cap + inactivity split) on the 6-video LOVO
harness ? each its own measured PR.
38 Worktree `docs-rally-quality` removable after #184 merges.
39
40 **Checkpoint:** 2026-06-16 02:35 (CONTINUATION ? ? ALL 3 PHASES COMPLETE; #178 + #183
MERGED; Kushagra regen done +
41 verified; local-vs-Gemini report delivered). The expanded overnight mandate is DONE:
42 - **Phase A DONE ? #178 MERGED (`c7df281`).** Fixed the broken streaming fast-path (was:
shimmed cv2.imread but
43 WASB reads via PIL read_image + osp.exists guard + BGR?RGB). Now patches
`dataloaders.dataset_loader.read_image`,

```

```

44 returns PIL RGB. **PROVEN byte-identical** end-to-end (same MD5 CSV, PNG-path vs
stream). Full suite 709 / ruff / mypy
45 green. (Also ~2min faster on a 180s clip; bigger win on long 4K.)
46 - **Phase C FEATURE DONE ? #183 MERGED (`b6a39c46`). ** `IndexingConfig.skip_ai_handoff`
(default OFF) ? trajectory
47 hybrids emit candidate windows AS rallies, no Gemini/key/upload; runner flags `--skip-ai-handoff` + `--wasb-stream`;
48 runner now `load_dotenv()`'s (FIXED a real bug: runner didn't load .env ? silent zero-
rally Gemini runs; parent only
49 worked because key was in its shell). Full suite 713 / ruff / mypy green. Worktree:
`\claude/worktrees/local-noai`.
50 - **Phase B ? DONE + VERIFIED + REPORTED (task `bcwft4e2x`, ~00:37?01:40). ** 6x 1080p
NVENC proxies (4K?1080p HEVC-10bit,
51 59.94fps, ~9.7GB) of the F: GoPro originals ?
`F:\KheISutra_work\kushagra_june12_proxies`. Runner `--segmenter gemini`
52 over proxies ? `G:\...\Kushagra and Yash\June 12 Highlights`. **6/6 reels, 0 failed;
165 rallies / 17:54 play from
53 52:29 footage. ** Per-video: GX010128 39, GX010129 27, GX020125 11, mahadevpura-0 40,
mahadevpura-1 9, mahadevpura-2 39.
54 **`verify_highlights.py` ? ALL REELS DEDUP-CLEAN ?** (0 dup/overlap, log==DB). Report
at `?\June 12 Highlights\REGEN_REPORT.md`.
55 - **Phase C ? DONE ? local-vs-Gemini report DELIVERED.** `wasb_hybrid --skip-ai-handoff
--wasb-stream` on 3 proxies ?
56 isolated `output\local_compare.db`; reels + `_LOCAL_vs_GEMINI_REPORT.md` in
`G:\...\Kushagra and Yash\June 12
57 Local-vs-Gemini`. Generator = `scratch/compare_local_vs_gemini.py` (IoU matching +
computed findings). **KEY FINDING:
58 local WASB does NOT under-detect ? it OVER-MERGES. ** Per video (local vs gemini
rallies): mahadevpura-1 **1 vs 9**
59 (the 1 window = the whole 174s clip!), GX020125 **6 vs 11**, mahadevpura-2 **5 vs
39**. Aggregate Gemini 59 rallies
60 (6:28 play) vs local 12 windows (**17:41** "play" ? mega-windows swallow dead time);
mean window **65s vs 6.1s
61 (10.6x)**; 0 gemini-only (no true misses), 7 merge-groups. **Shuttle tracking is
FINE** (per-frame visibility 98.6%
62 / 72.8% / 78.8%); the bottleneck is `trajectory_to_action_windows` velocity windowing
(vt=0.02/frame, merge_gap=2.0)
63 ? it bridges dead time, esp. at 59.94fps (per-frame velocity is NOT fps-invariant;
thresholds were tuned ~30fps).
64 **Recs (in report):** fps-invariant velocity (per-second), `max_window_duration` cap +
inactivity splitter, per-fps
65 golden calibration, Gate-A, learned segmenter; throughput (WASB ~15fps/1080p60 ? a
10min clip > 30min `timeout_sec`,
66 hit on mahadevpura-2 first pass ? raised to 5400 for the rerun, **config since
reverted to 1800**). Ties
67 [[paper-coauthor-aim]] + [[decision-rigor-norm]] + [[collector-indexer-bridge]].
68 - Cleanup done: `tools.cleanup_caches --apply` freed 45.7G (stale frame caches; golden
CSVs/reels/DB kept).
69 - **Tree state:** main checkout on `master`=`b6a39c4` (#183), clean (config reverted).
Worktree `local-noai`
70 (`feat/local-no-ai-segmenter`, merged, remote branch deleted) clean ? removable.
Proxies on `F:\KheISutra_work\
71 kushagra_june12_proxies` (~9.7GB, regenerable). `output\local_compare.db` = the local-
run DB (for re-analysis).
72 - **?? HOUSEKEEPING FOR OWNER (flagged, not done): ** rotate the Gemini key (owner said
AM); top up GitHub Actions
73 billing to restore CI gating; OPTIONAL: land the local-vs-Gemini finding into
`docs/QUALITY_ITERATIONS.md` +
74 productionize the over-merge fixes (own PRs) ? offered, owner-gated. (Clock: the
"01:15" stamp above was a misread,
75 actual ~00:47.)
76
77 **Checkpoint:** 2026-06-16 00:10 (CONTINUATION ? OWNER EXPANDED MANDATE: #178 merge +
Kushagra/Yash regen + local-vs-Gemini
78 report. "Let this GPU have a party tonight!"). Owner lifted the #178 gate and added a

```

```

big overnight job. **3 PHASES:**
79 **(A) #178 ? FIX + MERGE.** The SxS PROVED #178-as-written is BROKEN (3 defects,
empirically proven max|diff| 0 vs 183):
80 (1) WASB `read_image()` `osp.exists()` guard rejects synthetic stream paths (the crash);
(2) it shims `cv2.imread` but
81 `ImageDataset` reads via PIL `Image.open()` in `read_image` (`dataset_loader.py:14 from
utils import read_image`) ? shim
82 never fires; (3) it'd feed raw BGR, WASB wants RGB (max|diff|=183). FIX IMPLEMENTED in
worktree
83 `agent-a771ff1d85e18f2ed`: `_streaming_read_image_patch` now patches
`dataloaders.dataset_loader.read_image` + returns
84 `Image.fromarray(cvtColor(bgr,BGR2RGB))` (PROVEN byte-identical to disk:
`cv2.imwrite`?PIL-RGB == cvtColor, max|diff|=0).
85 **SxS RE-RUN with the fix RUNNING (`bhywu1hu`)** ? streaming `num_workers=0` is slow
(~10min), GPU active, progressing.
86 TODO after it proves identical: refactor shim for unit-testability (it imports WSL-only
`dataloaders` ? make a testable
87 factory + rewrite the 2 `test_wasb_cache.py` tests that assert old cv2.imread behavior),
rebase `perf/wasb-fast-frames`
88 onto origin/master (behind, missing #179-#182), FULL `run_all_tests.py`+ruff+mypy, then
`gh pr merge --squash --admin`.
89 **(B) KUSHAGRA/YASH REGEN.** Source `F:\GoPro Hero Black 12\KhelSutra\Badminton 12 June
2026` = **6x 4K GoPro originals
90 ~47GB** (GX010128 11.3G, GX010129 8.0G, GX020125 4.5G, mahadevpura-0 11.5G,
mahadevpura-1 2.6G, mahadevpura-2 9.2G).
91 Dest `G:\My Drive\Sharing\Badminton\Kushagra and Yash\June 12 Highlights` (exists).
**DECISION: make 1080p NVENC proxies
92 first**, run the runner `--segmenter gemini` on proxies ? 1080p reels. WHY:
VideoCompiler stitches at SOURCE res w/
93 libx264 (no downscale) ? 4K reels = huge+slow; Gemini already downscales chunks to 1280
(`ai_chunk_scale_width`) so
94 indexing is res-agnostic; proxies also serve Phase C (WSL-readable local-disk inputs);
matches Boxhill 1080p precedent.
95 Clean-slate first (cleanup_caches dry-run?apply; 45.7G reclaimable). Then verify dedup-
clean + run.log + report.
96 **(C) LOCAL-vs-GEMINI.** Disable Gemini ? regenerate 3 videos FULLY LOCAL (best weights
= WASB shuttle). The hybrids'
97 `play_segments` come ONLY from the Gemini Phase-3 handoff (trajectory_hybrid.py) ? NO
no-AI path exists ? **add a
98 `skip_ai_handoff` mode** (focused PR) that emits WASB candidate windows (optionally
Gate-A filtered) directly as rallies.
99 Then detailed report: Gemini vs local rally diffs (count/dur/total-play,
kept/dropped/merged, FP/FN) to guide owned-model
100 ([[paper-coauthor-aim]], [[weak-signals-retained]], [[decision-rigor-norm]]). Gemini key
valid (owner rotates AM). GPU
101 is mine alone ? steamroll. ?? NEXT: await `bhywu1hu` ? finalize+merge #178 ? Phase B
proxies+regen ? Phase C.
102
103 **Checkpoint:** 2026-06-15 (? REGEN DONE ? ALL 5 REELS VERIFIED DEDUP-CLEAN; parent
session). Parent task `b8ecngf9u`
104 exited 0; `python scratch/verify_highlights.py` ? **`ALL REELS DEDUP-CLEAN`** for all
five ? Adarsh v Avi 21r/342MB
105 (**was doubled 37/620MB ? bug fixed**), Adarsh v Vasant 2 17r/328MB, Adarsh v Vasant
16r/228MB, All 1 v 1 45r/599MB,
106 Vasant vs [Avi and Adarsh] 41r/641MB ? every one near_dup_pairs=0, overlap_pairs=0,
reel+run.log present, log
107 rally-count==DB segments. End-to-end proof that #180 (overlap-merge) + per-run
`clear_video_data` + #181 (per-video chunk
108 dir) + the runner's atomic non-destructive publish all hold together on real
4-min?61-min footage. ? **CONCURRENCY:** the
109 spawned continuation session (chip ? its 23:20 checkpoint just below) is ALSO live and
**OWNS the remaining open thread =
110 the WASB #178 GPU side-by-side** (it prepped `%TEMP%\wasb_sxs.py`; #178 branch
`perf/wasb-fast-frames` is behind master ?
111 rebase past #179/#180/#181/#182 before any merge). Parent session (me) is at context

```

```

limit and **STANDING DOWN** ? the
112 continuation drives #178 + ongoing fixes. Repo on master ?`4942d51`.
113
114 **Checkpoint:** 2026-06-15 23:45 (CONTINUATION session ? owns #178; reels DONE+VERIFIED
independently). I'm the spawned
115 overnight continuation; the parent (`b8ecngf9u`) STOOD DOWN (checkpoint just above)
handing me the WASB #178 side-by-side
116 + ongoing fixes. I independently re-ran `scratch/verify_highlights.py` after the regen
finished (23:42:55, poller
117 `bmc5tqqv` exited DONE) ? **`ALL REELS DEDUP-CLEAN`** (matches the parent's numbers;
owner acceptance test PASSED).
118 ***? NOW running step 3 = #178 WASB GPU side-by-side** via `%TEMP%\wasb_sxs.py`
(cwd=`.claude/worktrees/
119 agent-a771ff1d85e18f2ed`): WASB PNG-frame path vs `stream_video=true` on
`output/mahadevpura_smoke_180s.mp4`
120 (90MB/180s, local?WSL-readable) ? byte-compare trajectory CSVs in fresh
`output/wasb_sxs/{png,stream}/`. #178 pure-logic
121 tests PASS (26/26). #178 branch `perf/wasb-fast-frames`@66bf0c6 is BASED ON #177 (behind
master ? missing
122 #179/#180/#181/#182) ? **REBASE before any merge**. Byte-identical ? mergeable (after
rebase + FULL `run_all_tests.py`
123 green); divergent ? file the divergence, leave #178 OPEN.
124
125 **Checkpoint:** 2026-06-15 (HIGHLIGHTS RUNNER landed + gemini concurrency fix ? **PR
#181 + #182 MERGED**); clean reel
126 regen running). Closes the owner's overnight ask: "ensure the memory and docs point to a
runner which can orchestrate
127 such requests and ensure clean execution from next time / when launching a new video
while another is running, the setup
128 doesn't undo other progressed work ? also check and fix in code." **THE RUNNER =
`tools/generate_highlights.py`** (THE
129 canonical batch-reel entrypoint ? don't hand-roll a scratch driver again; CODE_MAP
$2.4). Guarantees: generate LOCALLY ?
130 **atomic publish** (copy?`os.replace`) into the dest ? the destination is only ever
ADDED to, **never bulk-deleted**
131 (fixes the `rm -f Highlights` incident that lost reels + run.logs); per-video
`clear_video_data(vid)` before processing
132 (idempotent ? no dup-rally accumulation, belt-and-suspenders with `_merge_overlapping`
#180); **resumable** (skips a reel
133 already published unless `--force`, so launching a 2nd request never undoes the 1st's
work); per-video `` - run.log`
134 + `_batch_run.log` published additively; WSL detectors localize source first. CLI:
`python -m tools.generate_highlights
135 --videos-dir <dir> --out <dir> --segmenter gemini [--force]`. **PR #181** = the in-code
concurrency fix the owner asked
136 for: Gemini segmenter shared `output/chunks` + `os.rmdir`'d it ? a finishing run could
undo another in-flight video's
137 chunks ? now per-video `output/chunks/<video_id>` (`_chunk_workdir`). Both merged
admin/local-green (CI still red on the
138 billing outage). **Full suite 696 passed / 5 skipped** before each merge (ran the FULL
suite this time ? the discipline
139 the #177?#179 miss taught). ? Clean Gemini regen of all 5 Boxhill reels running via the
runner (`--force`, task
140 b8ecngf9u; per-video chunk dir confirmed live in its log) ? verify reels are dedup-clean
+ run.logs present when done.
141 **As of handoff: 2/5 published + verified clean** via `scratch/verify_highlights.py`
(cold-runnable verifier I added):
142 **Adarsh v Avi = 21 rallies, near_dup_pairs=0, reel 342MB ? the doubling is GONE** (was
37/620MB); Adarsh v Vasant 2 =
143 17, clean. 3 remaining in-flight (Adarsh v Vasant in progress over a stale pre-runner
reel; All 1 v 1 + Vasant-vs pending).
144 Then the WASB #178 GPU side-by-side. Repo on master ?`4942d51`.
145
146 > ***? COLD-START NEXT ACTIONS (resumable ? context was ~full at handoff; do these in
order):**

```

```

147 > 1. **Is the regen done?** `claude /tasks` (or check the runner output) for task
b8ecngf9u. If still running, wait ?
148 > Gemini path is ~6-8 min/video × 5. If it died, just re-run it (idempotent,
resumable, skips finished reels):
149 > `python -m tools.generate_highlights --videos-dir "G:\My
Drive\Sharing\Badminton\Boxhill\June 12 2026" --out "G:\My Drive\Sharing\Badminton\Boxhill\June
12 2026\Highlights" --segmenter gemini` (add `--force` to redo all).
150 > 2. **Verify the reels are dedup-clean** (this is the owner's actual acceptance test ?
the doubled-reel must not recur):
151 > `python scratch/verify_highlights.py` ? expects `ALL REELS DEDUP-CLEAN` ? (per
video: DB segment count = one run's
152 > worth, `near_dup_pairs=0`, `overlap_pairs=0`, reel present + run.log present). Any
`!!` line = investigate that video.
153 > 3. **WASB #178 GPU side-by-side** (gate before merging the streaming speedup): run
WASB on 1-2 videos BOTH ways ?
154 > `indexing.wasb.stream_video=true` vs default PNG-path ? and diff the trajectory
CSVs (`output/wasb/*_wasb.csv`); they
155 > MUST be byte-identical (streaming is meant to be output-preserving). If identical ?
merge #178 (admin/local-green,
156 > after `python run_all_tests.py` is FULLY green). Worktree:
`.claude/worktrees/agent-a771ff1d85e18f2ed`.
157 > 4. **Housekeeping for owner:** rotate the chat-pasted Gemini key (it's only in
gitignored `.env` but was exposed in
158 > transcript); top up GitHub Actions billing to restore CI gating (we've been admin-
merging on local-green).
159
160 **Checkpoint:** 2026-06-15 (DUPLICATE-RALLY-IN-REEL BUG FIXED ? PR #180). Owner saw
"Adarsh v Avi" reel with ~every
161 rally TWICE. DB ground truth: 37 play_segments for that video = TWO near-identical
batches (ids 340-359 + 360-379) =
162 two accumulated runs. **Root cause:** `compile_highlights` stitched its segment list
VERBATIM (no overlap/dedup), and
163 `add_video` is an UPSERT (PR-C "re-ingest must not destroy") so it doesn't clear
children ? the API path prunes via
164 `segment_watermarks`+`prune_segments` but my scratch driver didn't, so the FORCE re-run
accumulated. **PR #180 (merged,
165 admin/local-green): `VideoCompiler._merge_overlapping` ? sorts + merges
overlapping/abutting ranges (collapses exact
166 + near dups + true overlaps; preserves gapped distinct rallies; drops end<=start) before
trimming; logs merge count.
167 6 compiler tests (incl. integration: 5 dup segs ? 3 trims). Full suite **692 passed**.
Belt-and-suspenders: scratch
168 driver now `clear_video_data(vid)` per run. **Verified clean:** re-ran video1 ? 17
rallies / 285MB reel / 17 DB rows
169 (was 37/620MB doubled). Clean Gemini batch re-running all 5 (bcl2ca0tg); stale doubled
reels deleted first.
170 ? My process note: keep running the FULL suite before merges (this + the #177?#179 miss
both would've been caught;
171 CI still down on billing outage = no net).
172
173 **Checkpoint:** 2026-06-15 (BOXHILL HIGHLIGHTS + WASB-speedup + detector-robustness
PRs). Owner's 5 non-Rooraa Boxhill
174 June-12 share videos (1080p copies on Google Drive `G:\?\Boxhill\June 12 2026\`):
175 - **Vasant partial FIXED** earlier: regenerated 4K?1080p from intact F: original,
deleted the truncated `.partial`.
176 - **? WSL-can't-read-Drive** is why GPU WASB highlights failed: `_stage_video` `cp
/mnt/g/?` silently fails ?
177 "0 rallies". **PR #177** added a `test -r` readability preflight (shared
`WslCommandMixin.wsl_source_unreadable_reason`)
178 ? fails fast with an actionable "copy to local disk" message. Merged (admin, CI
billing-outage down).
179 - **? MY MISS on #177:** merged after running only `test_tracknet_runner.py`, NOT the
full suite ? 4 stale WASB mocks
180 broke on master (blanket "OK" `_wsl_bash` mocks didn't return "READABLE" for the new
preflight). Caught by the

```

```

181 spun-off agent. **PR #179** fixed the mocks; **full suite now 690 passed / 5
skipped. ** LESSON (reinforces pre-LGTM
182 gate): run `python run_all_tests.py` (FULL) before every merge, ESPECIALLY during the
CI outage when there's no net.
183 - **WASB perf: PR #178 OPEN (NOT merged)** ? spun-off agent implemented stream-frames-
from-video (cv2.VideoCapture +
184 scoped imread shim) replacing the slow ~46k-PNG dump that caused 30-min timeouts;
default-OFF `indexing.wasb.stream_video`;
185 output-preserving (PNG lossless ? identical uint8 frames). Awaiting **OWNER/me GPU
side-by-side**: stream vs PNG-path
186 trajectory CSV must be identical on 1-2 videos before merge. Worktree:
`.claude/worktrees/agent-a771ff1d85e18f2ed`.
187 - **Highlights delivery = Gemini path** (owner chose: fast, best quality on clean
single-court; WASB too slow w/o #178).
188 Driver `scratch/make_boxhill_highlights.py` (BOXHILL_SEG/BOXHILL_FORCE envs; Gemini
reads Drive directly via host
189 ffmpeg, no WSL localize). **Per-video fine-grained`<name> - run.log` +
`_batch_run.log` written into the Highlights
190 folder** (root cause of earlier coarse progress: driver hadn't called
logging.basicConfig ? segmenter INFO suppressed;
191 fixed). Reels: video1 done (37 rallies, 620MB), batch in progress (~7 min/video,
b84tltc4x). ETA ~25 min for 5.
192 - ? NEXT: when reels done + I run the #178 GPU SxS (streamed==PNG trajectory) ? if
identical, #178 mergeable. Optional:
193 eyeball Gemini-reel vs WASB-reel coverage. Top up GitHub Actions billing to restore CI
gating.
194
195 **Checkpoint:** 2026-06-15 (PRODUCT RAMP-UP DOC for the field associate + alpha/beta
users). **Owner found a field
196 associate** (willing to talk to coaches/serious players); associate's feedback = he
needs to understand the WHOLE TOOL
197 from a PRODUCT perspective before going out. Owner refined scope twice: (1) the doc
should ALSO serve invited alpha/beta
198 academies so they can use the tool well + give feedback; (2) **headline the flywheel** ?
players/coaches use THEIR OWN
199 videos to fine-tune the underlying model; over time + more videos it sharpens AND grows
into shot detection + deeper
200 skill analysis = an **AI Coach**. ? **BUILT** `docs/KHELSUTRA_PRODUCT_OVERVIEW.md`
(canonical source, untracked) + shareable PDF
201 `KhelSutra_Product_Overview.pdf` (repo root, 4pp, reportlab; generator
`scratch/make_product_overview_pdf.py`).
202 Product-level only (no model names/F1/internals); foregrounds the flywheel + AI-coach
roadmap; dual-audience
203 (associate + alpha/beta users) with added "How to use it (alpha)" + "What feedback helps
most" sections. Ran a
204 4-lens adversarial review workflow (`wf_f10e69b6-333`:
overpromise/consent/grounding/clarity) ? **2 important catches
205 incorporated:** (1) removed "finds EVERY rally" + present-tense "model gets better at
YOUR courts/players" (implied
206 live per-user self-tuning that doesn't exist) ? reframed as collective+opt-in+future;
(2) ? **PRIVACY CLAIM
207 CORRECTED ? do NOT advertise "local-first / your video stays yours"**: alpha/beta
uploads to server + Gemini cloud,
208 and `PERSONALIZATION_PLAN.md:145` explicitly forbids advertising data-locality until on-
device ships. Doc now says
209 "processed securely on our infrastructure incl. a trusted cloud AI step today; on-device
on roadmap; not published;
210 not trained on without opt-in; minors' footage never used/published." Watermark string
matched to shipped
211 `"Generated by Khelsutra.guru"` (lowercase s). **FINALIZED per owner:** data-handling
kept deliberately HAZY ?
212 removed all on-device / "never leaves your machine" language (large videos need GPU
processing); now reads "handled
213 securely on our infrastructure to produce your reel" + not-published + opt-in + minors-
never-used (no locality promise

```

```

214 either way). Access wording = "your KhelSutra associate helps you get access" / "...will
help you with that" (owner
215 liked it). **Final PDF written to
`C:\Users\avidu\OneDrive\Documents\KhelSutra_Product_Overview.pdf`** (4pp; stale
216 repo-root copy removed). Complements (? duplicates) `PRODUCT_FIELD_ASSOCIATE_FLYER.md`
(recruiting) +
217 `FIELD_VISIT_PLAYBOOK.md` (visit script). `docs/KHELSUTRA_PRODUCT_OVERVIEW.md` source
NOT committed (per "don't open a
218 PR unless asked"); offer to PR stands.
219 Both repos synced to remote HEAD at session start (indexer 5fe9beb/#176, collector
031e410/#29) ? clean.
220
221 **Checkpoint:** 2026-06-15 (ROORA JUNE-15 BATCH + 3 collector PRs merged + guidelines
enriched). Owner-authorized
222 merging of "process" PRs (prep?verify?transcode?pre-ingest?ingest); billing outage still
red CI account-wide ?
223 `--admin` merged on local-green (precedent: indexer NEXT_STEPS). **Collector master
`031e410`:**
224 - **#27 preflight gate** (`src/core/batch_preflight.py`): prep-annotation-batch now
scans the tree FIRST ? BLOCKS on
225 duplicate videos (size+head+tail fingerprint, never full-hashes 10GB over USB),
surfaces empty dirs / VFR /
226 multi-video dirs; `--dry-run` (scan only) + `--allow-duplicates` (for the A/A test).
Header-only `proxy.probe_light`
227 (no `--count_packets`) ? 3.8s scan on the real 115GB tree. Caught the empty-TT-dir +
Video6/9-byte-dup live.
228 - **#28 --encoder nvenc** (proxy + batch): GPU h264_nvenc (`-rc vbr -cq <crf> -b:v 0`,
preset p5); libx264 still
229 default. Parity is codec-agnostic. encoder_available() guards a missing GPU. **NVENC
litmus VERIFIED** (RTX 2070S,
230 36901 frames @ 59.94 both); laptop stays cool (GPU encode, I/O-bound). cq19 ? 2x
libx264 size (raise --crf to shrink).
231 - **#29 richer schema** (court+confidence+capture-report+IAA ? owner picked ALL 4): per-
rally `court`
232 (primary|secondary, the multi-court gold label for the 0.66 weakness) + `confidence`
(sure|unsure ? `annotator_unsure`
233 review flag, never penalised) carried Brief+Guide+convert-cvat?canonical CSV?round-
trip; capture-report (per-video
234 quality form, doc-only) + IAA ~15% A/A overlap (Brief §4/§9). ? court/confidence
currently STOP at the canonical
235 CSV+review sheet ? **DB/export integration is a tracked FOLLOW-UP before training on
them** (import-local reads core
236 cols only).
237 - **THE BATCH:** `prep-annotation-batch --encoder nvenc` running on F:\GoPro?\Roora June
15 (12 unique CFR videos,
238 ~115GB 4K/5.3K/one 120fps/one 1080p) ? mirrored proxies-only tree at `G:\My
Drive\Sharing\Annotation Videos\Roora
239 Request - June 15` (Google Drive ? shareable). Manifest (private) on
F:\?\roora_june15_proxy_manifest.csv; batch log
240 F:\?\roora_batch_log.txt (? python stdout block-buffers ? watch G: proxy files +
manifest rows, NOT the log's
241 print lines). ~3/12 done at checkpoint; ~3.7GB/proxy at cq19 ? ~30-40GB Drive upload
(iterate to higher --crf if slow).
242 **? CONFIRMED COMPLETE (verified on disk 2026-06-15 later): all 12 done ? log "Proxied
11 . Already-have 1 .
243 Failed 0", preflight all-clear; ffprobe confirms all 12 are 1920x1080 h264 on the G:
tree (source fps preserved;
244 big files = GX010055 61min / GX010124 120fps, not failures). Proxies-only tree ready
to share with the vendor.**
245 - **Roora guidelines Doc** (canonical Google Doc, see [[roora-guidelines-drive-doc]])
now NEEDS regeneration with the
246 new schema ? I can't edit the Doc in place (no Drive write tool); gave owner paste-
ready deltas. ?? owner: fill the
247 Doc TODOs + paste the new sections; A/A duplicate-annotation test deferred (owner
idea: 2 identical proxies ? check

```

248 byte-identical CVAT return). DB integration of court/confidence = next collector PR  
when we train on them.

249

250 **\*\*Checkpoint:\*\*** 2026-06-14 (? GATE-A REGRESSION RESOLVED + eval?serve FOUNDATION FIXED ?  
the "owner decision pending" in the

251 checkpoint below is now DONE). The served Gate-A regression (#146 flip ? ?0.008 served,  
drops ~12 real rallies) is **\*\*REVERTED\*\***

252 (**\*\*#151\*\*** ? served default back to `gemini`, `min\_crossings=0`, verified) + the served  
litmus (**\*\*#148\*\***) + finding docs (**\*\*#152\*\***)

253 merged. The **\*\*foundational eval?serve gap is FIXED\*\*** (**\*\*#154\*\***):  
`backend/eval/windowing.py` single-sources the SERVED preset

254 from production `IndexingConfig` (eval+serve can't drift);  
`backend/eval/promotion.py::promote\_if\_served\_safe` VETOES any cue

255 winning on PROPOSAL windowing but regressing SERVED (Gate-A: proposal +0.0739/6-0 ?  
served ?0.0077/2-2 ? **\*\*promote=False\*\***);

256 `serve\_contrast.py` = the worked example. 21 tests, CI green. **\*\*[[decision-rigor-norm]]**  
updated with EVAL?SERVE\*\* (a served check

257 is REQUIRED before flipping any served default). Owner directive "do not regress + fix  
foundations before any other task" ?

258 **\*\*BOTH DONE.\*\*** ?? **\*\*OPEN/HELD:\*\*** **\*\*#149\*\*** (exec-policy P3a worker self-scheduling loop,  
green, additive) is now UN-BLOCKED but

259 un-merged pending owner go; **\*\*#150\*\*** (per-user store/bridge v4/v5 + `serving.py`) MERGED  
additive/inert (NEXT owner-gated = wire

260 the bridge into `\_select\_for\_handoff`); owned-model Phase-0 PLAN at  
`scratch/OWNED\_MODEL\_PHASE0\_PLAN.md` (item 7, for discussion).

261 Repo clean on `master` ? 4251c9c; worktree `.claude/worktrees/bhi-execpolicy` remains  
for #149. **\*\*Context ~critical ? keep handoffs granular.\*\***

262

263 **\*\*Checkpoint:\*\*** 2026-06-14 (SERVED-SIDE GATE-A CHECK ? the #146 flip's offline win does  
NOT transfer to the production

264 windowing; mild served regression found, owner decision pending). Ran the owner-gated  
follow-up flagged in PR #146: drove the

265 **\*\*live `FusionSegmenter` served seams\*\*** (`\_enrich\_candidate\_stats`?net\_crossings,  
`\_select\_for\_handoff`?the gate) over each

266 golden video's **\*\*cached\*\*** WASB trajectory at the **\*\*production windowing\*\***  
`process\_video` actually uses (vt=0.02, merge\_gap=2.0,

267 min\_win=2.0 from `load\_config()`), no WASB re-run / GPU / Gemini. Repro  
`scratch/served\_gateA\_{verify,sweep}.py`; artifacts

268 `output/gateA\_served\_verify/`. **\*\*a) CONFIRMED:\*\*** at `min\_crossings=0`, fusion\_hybrid  
seeding+handoff is **\*\*byte-identical to**

269 wasb\_hybrid on all 6\*\* (no-regression invariant verified on live served code, not just  
review). **\*\*b) FINDING:\*\*** the +0.074

270 win was measured on the eval harness's recall-oriented \*proposal\* windowing  
(0.005,1.0,0.5; CONTROL over-segments seg-ratio

271 1.68?1.01 = Gate-A cleaning fragmentation). The **\*\*served coarse windowing already merges  
aggressively\*\*** ? Gate-A has little FP

272 to remove and instead trims **\*\*real rallies\*\*** with net\_crossings 0?1. Candidate-handoff  
F1 (IoU0.5, n=6) sweep: **\*\*mc0 0.300 |**

273 **mc1 0.303 (+0.003, 1 real-rally lost) | mc2(SHIPPED) 0.292 (?0.008, 12 real-rallies  
lost) | mc3 0.269 (?0.031, 28 lost)\*\***;

274 worst largetest\_doubles ?0.051. Tiny/within-noise BUT mc2 is **\*\*not\*\*** a served  
improvement and drops genuine rallies (dropped

275 windows never reach Gemini = unrecoverable recall loss; precision "gain" partly  
duplicates Gemini ? served tradeoff ? neutral).

276 **\*\*Root cause = eval?serve windowing\*\*** (harness promotes on a windowing production  
doesn't serve). **\*\*Recorded in**

277 `docs/QUALITY\_ITERATIONS.md`\*\* ? committed as **\*\*PR #152 (merged, `53bc743`)\*\***. **\*\*?  
RESOLVED IN PARALLEL by a SIBLING session:\*\***

278 the same regression was independently found + productionized as a standing litmus  
(**\*\*#148\*\*** `backend/eval/served\_gate\_a.py` +

279 `tests/test\_served\_gate\_a\_regression.py`) and the **\*\*flip was REVERTED in #151\*\***  
(`config.json` default\_segmenter ? `gemini`,

280 `indexing.fusion` block removed; `FusionConfig.min\_crossings` default 2?0 ? Gate A OFF-  
by-default-but-opt-in). My independent

281 numbers MATCH #151's (?0.008, ~12 rallies dropped, 2/6 losses) = good corroboration. So



the owner decision I'd queued is M00T

282 (already reverted). Reconciled my #152 doc entry to point at the revert (#151) + the  
productionized guard (#148) instead of

283 "decision pending". \*\*OPEN follow-up (real lesson):\*\* align the experiment-harness  
windowing with the served windowing before

284 treating a harness lift as a served default. Ties [[decision-rigor-norm]] (foundations:  
eval?serve) + [[weak-signals-retained]]

285 + [[paper-coauthor-aim]]. ? Watch [[gotcha-shared-checkout-branch-collisions]]: this  
whole arc (#148/#151 sibling vs my #152)

286 was concurrent slates on the same finding ? branch-from-origin protocol held.

287

288 \*\*Checkpoint:\*\* 2026-06-14 (PERSONALIZATION Phase-0 LANDED + quality checkpoint).  
\*\*Owner adopted the HYBRID\*\* ? personalization as a

289 flag-gated FEATURE on a generalization-first baseline + a \*\*contribution flywheel\*\* ("a  
tool that gets better the more you help it";

290 free-tier videos pool into the owner-trained baseline). Pivot analysis  
(`scratch/PERSONALIZATION\_PIVOT\_ANALYSIS.md`) recommended

291 hybrid-not-personalization-first; the owner's contribution-flywheel closes the flywheel-  
loss objection. \*\*MERGED this session:\*\*

292 telemetry wire-in \*\*#140\*\* ; per-user promotion gate \*\*#141\*\* (min-N floor + structural-  
guard exemption); held-out-USER learning

293 curve \*\*#142\*\* ("<K videos to superior quality" = falsifiable); per-user no-regression  
guard \*\*#143\*\* ; \*\*`docs/PERSONALIZATION\_PLAN.md`

294 #144\*\* (the checked-in project doc + locked decisions). Gemini-training-source concern =  
\*\*CLEAN\*\* (eval-only, gitignored, M0.3 owner-gated).

295 \*\*LOCKED owner decisions:\*\* identity = north-star-design/current-impl (nullable user\_id,  
device UUID); inline classifier\_json behind a

296 cloud-load interface; \*\*`min\_n` is a PROMOTION-confidence floor, NOT a service gate\*\* (a  
1-video user is always fully segmented by the

297 baseline); free-tier video pooling (consent + minors-exclusion); \*\*Gate-A-only  
structural + `min\_n=5`\*\* . ? \*\*GATE-A MILESTONE DONE:\*\* Gate-A FLIPPED to the served default  
(\*\*#146\*\* , config-reversible ? `default\_segmenter`

298 gemini?`fusion\_hybrid` + `min\_crossings` 0?2; rollback = one flag); the \*\*ablation  
layer\*\* built + \*\*LITMUS PASSED\*\* (\*\*#147\*\* ?

299 `backend/eval/ablation.py` LOS0/LOG0 runner reproduces Gate-A +0.074 / 6-of-6 / p=0.031  
EXACTLY; PDF export ? also in

300 `OneDrive\Documents\Rally\_Ablation\_Report.pdf`); quality-iterations checkpoint  
(\*\*#145\*\*); \*\*[[decision-rigor-norm]]\*\* codified

301 (NUMBERS INFORM, FOUNDATIONS DECIDE). Numbers: Gate-A = the ONLY lever clearing the  
honest bar; audio +0.079 = suggestive lead (CI

302 spans 0); Gate-B inert/trap (court-mask dep). \*\*GPU + Gemini AUTHORIZED for evals\*\*  
(cooler boost on). ?? \*\*OPEN:\*\* served

303 `fusion\_hybrid` path is \*\*mock-tested-only on real video\*\* ? a \*\*served golden  
regression\*\* is recommended (re-score cached WASB ?

304 fusion-default vs golden labels); `reportlab`/`pypdf` ? requirements (CI-exercise the  
PDF). NEXT

305 personalization = Phase 1 (serving bridge: wire `LogisticRegression.load` into  
`\_select\_for\_handoff`) ? owner-gated, see [[weak-signals-retained]] +  
`docs/PERSONALIZATION\_PLAN.md`.

306

307 \*\*Checkpoint:\*\* 2026-06-14 (FUSION P3/P4 MEASURED + extraction COMPLETE + telemetry #134  
merged + Cooler Boost). \*\*THE NUMBERS

308 (honest LOVO n=6, IoU?0.5, C1 bootstrap-CI + sign-test):\*\* shuttle-only F1=\*\*0.307\*\*;  
\*\*+person (P3) ?F1=?0.021\*\* (?6.7%, CI

309 [?0.165,+0.162], sign 2W/4L p=0.688, \*\*CI-clears-0: NO\*\*) ? NO lift, stays default-OFF  
(no court polygons ? `players\_in\_court`

310 counts spectators; per-video wash: testlarge +0.40 / gbaaddy ?0.22). \*\*+person+audio (P4  
incremental) ?F1=+0.079\*\* (+27.7%, F1

311 0.286? \*\*0.366\*\* , CI [?0.054,+0.203], sign 4W/2L p=0.688, \*\*CI-clears-0: NO\*\*) ?  
PROMISING (wins 4/6, largetest/mahadevpura

312 +0.27; endpoint 0.366 > shuttle-only 0.307) but n=6 can't confirm. \*\*VERDICT: neither  
promotable (both CIs span 0); audio is

313 the lead ? need more matches + measure audio on shuttle-only directly (drop the person  
handicap).\*\* Recorded in

314 `eval\_baselines/experiment\_leaderboard.json`; honest negative/uncertain results KEPT

```

315 SESSION:** result logged to `docs/QUALITY_ITERATIONS.md` (#137);
`experiment_leaderboard.json` tracked (#139); **weak-signals
316 audit ?** (person/audio/court-bounds all present, default-OFF, no-special-casing,
retained ? refuted=false ? [[weak-signals-retained]]);
317 7-page paper-style **quality report PDF** at
`OneDrive\Documents\Rally_Detection_Quality_Report.pdf` (gen `scratch/quality_report.py`;
318 conservative academic restyle, strict-reviewer-approved); `FusionGoldenExtract`
scheduled task **unregistered** (cleanup); tree
319 clean on `master`. ?? **NEXT (fusion track):** (1) **wire `run_telemetry` (#134) into
fusion_golden + the ps1 wrapper** (deferred,
320 ready, mergeable); (2) **measure audio on shuttle-only directly** (quick eval, reuses
the 6 `output/*_golden_features.json`, NO new
321 data ? audio's standalone lift without the person handicap); (3) **grow the golden
corpus via Roora** (owner's job ? gives power
322 to confirm the +0.079 audio lead) + **activate court polygons** (re-measure person
properly ? the ?0.021 is partly a
323 `court_polygon=None` artifact); (4) future **shuttle-dropped** cue ([[weak-signals-
retained]]). **OTHER track (per
324 [[session-handoff]] ? the stated IMMEDIATE PICKUP): EXECUTION-POLICY P3 then P4**
(owner-gated; #132/#133/#135 done). **Extraction infra** (all working): detached Scheduled Task
`FusionGoldenExtract`
325 runs `scratch/extract_until_done.ps1` (self-restart loop + in-process keep-awake + GPU-
busy gate: waits if free VRAM<3500MB) ?
326 deaths were `0xC000013A` console-control kills (NOT crash/sleep/TDR/OOM ? verified via
event log), auto-recover via 3-min
327 repetition trigger + restart-on-failure. **Cooler Boost** (MSI Creator Center) dropped
85?67°C, SM ~1100?~1300MHz, velocity
328 ~26?~50 fps (the rest is Max-Q power cap, not thermal).
329
330 **Checkpoint:** 2026-06-14 (EXECUTION-POLICY framework: design + P1 + P2 SHIPPED ? **PR
#132 (design) + #133 (P1) + #135 (P2)
331 MERGED**, self-merged on green CI). Owner asked to harden the GX010125 Gemini hang as a
**policy-driven, per-job,
332 self-scheduling** system (default WAIT_AND_RESUME, ABORT, POLL_EVERY_N_SEC?, **no master
node**) with **very high coverage**
333 (failures cause delays) and **clean poll-logging** (visible but not polluting). **#132**
= design doc `docs/EXECUTION_POLICY.md`
334 (validated the owner's architecture: per-job policy = data on the job; self-schedule via
re-enqueue; no master ? rides the
335 existing single-worker executor + `jobs` table + crash sweep; DB is the coordinator).
#133 (P1, the actual fix) =
336 `backend/infrastructure/execution_policy.py`: `ExecutionPolicy` + `run_bounded` (HARD
`op_timeout_sec` per-attempt ? the
337 hang-killer; retries failures AND hangs; hung attempt's daemon thread abandoned) +
`poll_until` (bounded poll loop). Clocks
338 injected ? **12 tests, 100% module coverage, instant**. Refactored `gemini.py`:
`generate_content`?`run_bounded` (a
339 never-returning generate previously blocked FOREVER = the GX010125 hang; now bounded),
processing-wait?`poll_until` (**per-poll
340 logs at DEBUG, not an INFO line every 10s** ? the no-pollution ask). 23 gemini tests
stay green. **#135 (P2, substrate)** =
341 DB migration **v3** (`jobs.policy` JSON + `next_poll_at`; existing rows survive via
ALTER) + `set_job_waiting(delay_sec)`
342 (self-schedule: park 'waiting' until due) + **`claim_due_job()`** (atomic compare-and-
set claim of queued/due-waiting ?
343 running; concurrent-safe, no master) + `create_job(policy=)`/`get_job` policy.
Additive, no executor change; 10 tests;
344 version asserts bumped 2?3. ? **THIS IS THE OWNER'S REAL GPU MACHINE**
(torch+CUDA/cv2/WSL/Gemini live). ?? **REMAINING
345 (owner-gated, NOT done ? deliberately not rushed at session tail given the high-coverage
bar): P3** = wire the executor
346 (main.py) to a `claim_due_job` self-scheduling loop honoring WAIT_AND_RESUME
(release+reschedule) + **resumable AI handoff**
347 (persist per-window confirmation; resume only unconfirmed ? the GX010125 reel becomes

```

exhaustive across resumes, no wasted

348 Gemini spend); **\*\*P4\*\*** = apply the policy to the WSL/WASB GPU pass + ffmpeg. Each = its own distinct PR. Ties [[paper-coauthor-aim]] indirectly (infra).

349

350 **\*\*Checkpoint:\*\*** 2026-06-14 (ACTED ON FIRST-A/B FINDINGS: Gate-A PROMOTED + learned FIXED + GX010125 run ? **\*\*PR #130 + #131**

351 MERGED\*\* to master `2e5892e`; self-merged on green CI). ? **\*\*THIS IS THE OWNER'S REAL GPU MACHINE\*\*** (torch+CUDA, cv2, WSL

352 Ubuntu, Gemini key in .env all present) ? not the dev container the older CLAUDE.md note assumes. **\*\*#130 = learned-variant**

353 **FIX\*\***: additive/default-OFF `Variant.tune\_threshold` (operating point on the TRAIN fold) + `calibrate` (Platt/isotonic, C2);

354 validated on n=6 golden ? **\*\*mean held-out F1 0.307±0.423 (+0.116), testlarge 0.000±0.300 (un-zeroed), now BEATS CONTROL**

355 0.388\*\*; threshold-in-LOVO is the big lever. **\*\*#131 = Gate-A PROMOTED\*\*** (first cue promoted via the harness; doc trail in

356 **QUALITY\_ITERATIONS** ? measured win dF1 +0.074, 6/6, p=0.031; recommended `indexing.fusion.min\_crossings=2`; **\*\*live served-default**

357 flip stays **OWNER-GATED\*\*** until a real fusion-path run confirms ? fusion P2 is unverified on real video) **\*\*+ new**

358 `docs/PER\_VIDEO\_OUTPUT\_LAYOUT.md`\*\* (one dir per video; fixes the hardcoded `"output"` joins in `trajectory\_hybrid.py`~176`

359 so `--output-dir` truly isolates). **\*\*GX010125\_1080p RUN (offline, no GPU):\*\*** ? key finding ? `wasb\_runner.run\_predict` has

360 **\*\*NO cache-hit early-return\*\***, so the full pipeline would RE-RUN WASB (~tens of min on 1080p) + overwrite the cached CSV; the

361 cached trajectory is reusable ONLY via offline re-score. So ran `scratch/run\_gx010125\_rescore.py` = parse cached

362 `output/wasb/GX010125\_1080p\_89605e80ed01\_wasb.csv` ? 45 candidates ? CONTROL 45 rallies (11.0min) vs **\*\*Gate-A 27\*\*** (9.3min),

363 Gate-A **\*\*dropped 18 low-crossing windows (~1.7min)\*\*** of likely held-shuttle/feed FPs ? isolated `output/GX010125\_run/` (no

364 pollution; no golden labels for this footage ? divergence not F1). A FULL Gemini-confirmed highlight reel = a separate

365 ~tens-of-min GPU+API run (owner kicks off). Repro: `scratch/{run\_gx010125\_rescore,ab\_n6\_lovo,ab\_eval\_demo}.py`. All in

366 isolated worktrees off master (siblings concurrent). ?? NEXT (owner-gated): real fusion-path run to confirm Gate-A served-side

367 + flip the live default; impl `PER\_VIDEO\_OUTPUT\_LAYOUT` L1; Tier-2/3 §13 corrections; analyzers/reconcilers A1?A5/R1?R3. Ties [[paper-coauthor-aim]].

368

369 **\*\*Checkpoint:\*\*** 2026-06-14 (FIRST GOLDEN A/B EXPERIMENT + HARNESS HONEST-REPORTING WIRED ? **\*\*PR #128 + #129 MERGED\*\***

370 to master `f680b8d`, self-merged on green CI per [[working-agreement-merging]]; owner-directed momentum). Ran the

371 **\*\*first real end-to-end A/B on the n=6 golden corpus, 100% OFFLINE\*\*** (re-scored persisted WASB trajectories via

372 `distill\_local.build\_candidates`, no GPU ? the "persist once re-score" thesis proven). One candidate set re-scored

373 three ways: **\*\*Gate-A (net\_crossings?2) is a MEASURED WIN ? ?F1 +0.074, 6/6 matches, sign-test p=0.031, CI**

374 [+0.042,+0.104], seg-count-ratio 1.68±1.01\*\* (over-seg nearly eliminated); the **\*\*window-level 5-feature logistic did**

375 NOT clear the bar\*\* (?0.081, 2W/4L; testlarge zeroed = threshold/calibration artifact ? C2 motivation; **\*\*distinct from**

376 the stronger per-frame `rally\_seq\_proto` 0.583\*\* ? don't conflate). Headline: **\*\*the framework held up ? C1 promoted one**

377 variant and refused another (right call both ways); C4 surfaced over-seg tIoU hid.**\*\*#128\*\*** = archived posterity doc

378 `docs/archives/quality-iterations/2026-06-first-ab-experiment/` (full tables) + archives-index + live **QUALITY\_ITERATIONS**

379 pointer. **\*\*#129 = (b) wired C1+C4 into the CORE `experiment.compare()`\*\*** (additive/default-ON `honest\_stats=True`:

380 `honest` block = per-metric bootstrap CIs + paired ?F1 CI + exact sign test + F1@k + segment\_count\_ratio; `record\_experiment`

```

381 carries honest ?F1; existing fields byte-unchanged, `honest_stats=False` = legacy
shape). Reuses #124 helpers; complements
382 sibling **#127** which wired the same C1 into `fusion_compare` (no dup). 19 existing
experiment tests pass + 3 new;
383 ruff+mypy+CI all green. Repro `scratch/ab_n6_lovo.py` + `scratch/ab_eval_demo.py`
(machine-side; read gitignored
384 manifest/external golden). All done in **isolated worktrees off master** (siblings
merged #121-#127 concurrently). ?? NEXT
385 (owner-gated): promote Gate-A via the no-regression gate (it's a real measured win); fix
the logistic (C2 calibration +
386 threshold-inside-LOVO); Tier-2/3 §13 corrections; analyzers/reconcilers impl A1?A5/R1?R3
(none started). Ties [[paper-coauthor-aim]].
387
388 **Checkpoint:** 2026-06-14 (FUSION P3/P4 MEASUREMENT BUILT + GPU EXTRACTION RUNNING ?
master `d7b1f51`; suite 514 green).
389 The "modeling" track for the person-cue (P3) + audio (P4) fusion lift ? all default-OFF,
no-special-casing, harness-measured.
390 **Merged this run (mine): #121** fusion P3 (`backend/eval/fusion_golden.py` offline
person-feature EXTRACTOR + `fusion_compare.py`
391 LOVO driver ? reuses build_candidates/PersonDetector/fusion_features/experiment.compare,
NO new scoring); **#122** P4
392 (`backend/pipeline/detectors/audio_features.py` + `backend/eval/fusion_audio.py`
incremental audio cue); **#123** progress-logging
393 + resumability (`--fresh`); **#125** single-pass
`PersonDetector.detections_over_windows` (fixed the per-window-peek 0(N)
394 pathology ? ~28s/window ? one sequential decode); **#126** `--smallest-first`; **#127**
honest ?F1 = bootstrap CI + paired
395 sign test on per-video held-out F1 (wires #124's `eval_stats.paired_delta_ci`; "never a
bare mean"). Concurrent sessions
396 merged #119 (analyzers framework), #120 (run.log), #124 (eval-honesty utils:
eval_stats/score_calibration/segmentation_metrics).
397 **? GPU EXTRACTION via DETACHED SCHEDULED TASK `FusionGoldenExtract`** (runs
`scratch/extract_until_done.ps1` ? Task Scheduler
398 owns it, OUTSIDE Claude's job object ? survives turn/context/session boundaries; in-
process keep-awake + self-restart loop until
399 6/6; clean UTF-8 logs to `output/fusion_golden.log` + `scratch/extract_wrapper.log`).
Loops `python -u -m backend.eval.fusion_golden
400 --manifest output/human_lovo_manifest.json --out-dir output --smallest-first` ? 6
`output/<name>_golden_features.json` (GITIGNORED
401 + RESUMABLE: skips any video whose JSON exists). **3/6 DONE** (testlarge, mahadevpura,
largetest); remaining
402 kushagra?gbaaddy?adarsh(last, 84k frames). **MONITOR:** JSON count + `tail
output/fusion_golden.log` + `nvidia-smi`;
403 `Get-ScheduledTask FusionGoldenExtract|Select State`. **RESTART (resumes, skips done):**
`Start-ScheduledTask -TaskName
404 FusionGoldenExtract`. **STOP:** `Stop-ScheduledTask -TaskName FusionGoldenExtract` +
`Get-Process python|Stop-Process -Force`.
405 ?? **WHY DETACHED (debugged 2026-06-14):** Claude-bg-task runs DIED TWICE silently
(largestest 61%, kushagra 6% @137s), NO Python
406 traceback AND ? verified via Windows event log ? NO sleep (Kernel-Power 42), NO TDR
(Display 4101), NO WER crash, NO RADAR-OOM
407 (no event 1003; 18GB free). Traceless vanish = external `TerminateProcess`; common
factor = both were Claude-managed bg tasks
408 (job-object kill-on-close at a context boundary fits death #1 @ the prior context
summary). Scheduled Task = immune; GPU access
409 CONFIRMED in the interactive task session (53?64% util, 3.1GB). GPU util while running =
20?64% bursty ? batch=1 Faster-RCNN
410 latency-bound (small kernels + CPU NMS + per-frame ByteTrack), NORMAL, model IS on CUDA,
NOT decode-bound; batching the forward
411 pass is the next-extraction speed lever (changes features, so NOT mid-run). ~1.5-2h
total. Stale Claude bg keep-awakes stopped. **DEATH #3 + GPU-job check (2026-06-14):** died a
3RD time (gbaaddy 69%)
412 ? `LastTaskResult=0xC000013A` = STATUS_CONTROL_C_EXIT = terminated by a **console-
control event / CTRL+C** (NOT crash/sleep/TDR/OOM
413 ? explains the no-trace vanish; killed the detached task too, so NOT Claude's job

```

```

object). GPU-contention angle CHECKED: **NO
414 competing GPU compute job** (only desktop apps + our 1 python; 5GB free; HW is a
laptop RTX 2070S **Max-Q**, MSI). Mitigations:
415 (1) task re-registered with **3-min repetition trigger + restart-on-failure (count 10,
1-min)** ? auto-recovers within minutes of
416 ANY death; (2) **GPU-busy gate in the ps1** (owner policy = *wait until free, auto-
resume*); before each python launch, if free
417 VRAM <3500MB (another job has the GPU) it waits 30s + re-checks, then launches. **??
WHEN IT COMPLETES ? the remaining steps for
418 the NUMBERS:** (1) `python -m backend.eval.fusion_compare --features-dir output
--record` ? honest P3 LOVO ?F1 (shuttle-only
419 vs +person) w/ CI+sign-test; (2) `python -m backend.eval.fusion_audio --augment
--compare --record` ? P4 incremental audio
420 ?F1; (3) log both to `docs/QUALITY_ITERATIONS.md` I3 (HONEST measured numbers only ?
[[paper-coauthor-aim]]). Honest caveat:
421 NO court polygons ? `players_in_court` counts spectators (3-7 on "single-court"
Mahadevpura); robust signals = motion +
422 `shuttle_player_colocation`. GPU = RTX 2070S, person detection on Windows torch cu124
(NO WSL). Branch protocol held through
423 heavy concurrent merges (#119-127) ? [[gotcha-shared-checkout-branch-collisions]]. On
master, clean.
424 **STORAGE (this session, owner-approved "delete cache + compact WSL"):** deleted the
stale **93GB GX010125 frame cache**
425 (WSL `~/clips` 95?2.3GB; WSL fs used 152?60GB, 896GB free; C: 53?72GB free) ? kept the
2.2GB GX010125 proxy. The WSL
426 `ext4.vhdx` was `--set-sparse true` + `fstrim`d (946GB trimmed) but this WSL build
(2.3.24) did NOT auto-deallocate on
427 shutdown ? the file is still **~221GB allocated on C:**. **DEFERRED to post-extraction:
the physical ~150GB C: reclaim via
428 `wsl --shutdown` then elevated `diskpart` ? `select vdisk
file="...\CanonicalGroupLimited.Ubuntu_79rhkp1fndgsc\LocalState\ext4.vhdx"`
429 ? `compact vdisk`** (heavy C: I/O would fight the GPU run's video reads). [[gotcha-long-
runs-gpu-wsl]] · [[cache-hygiene-policy]].
430
431 **Checkpoint:** 2026-06-14 (RALLY-DETECTION FRAMEWORK doc + LITERATURE REVIEW + eval-
honesty code ? **PR #119 + #124 MERGED**
432 to master via self-merge on green CI per [[working-agreement-merging]]; owner-directed).
PR #119 (`docs/ANALYZERS_AND_RECONCILERS.md`):
433 reframed the analyzers/reconcilers design into a **multi-scope signal-fusion framework**
(frame detectors / window analyzers /
434 **session analyzers**); warm-up `SessionAnalyzer` + shuttle-state frame-detector folded
in) + **§13 = a verified multi-agent
435 literature-review** (8 patterns surveyed, ~40 citations adversarially checked, **ZERO
fabrications**; author-line fixes applied).
436 **Verdict (honest):** the skeleton is STANDARD prior art ? candidate-windows+scorer =
Temporal Action Localization;
437 reconciler-as-refinement = Temporal Action Segmentation post-proc; "signal not verdict"
= late/score-level fusion + stacked
438 generalization (~20 yrs old, already in racket sports). **Only the report-first
reconciler-as-linter bundle rated
439 `novel-combination`**; warm-up-as-soft-feature = thinnest gap. **Cite, don't claim**;
terminology realigned (producer?labeling
440 function, scorer?stacked meta-learner, LOVO?leave-one-group-out, tIoU?F1@k/mAP). Ties
[[paper-coauthor-aim]] (§13 is the honest
441 related-work backbone). **PR #124** implements the **Tier-1 measurement-honesty course
corrections C1?C4** as PURE ADDITIVE /
442 default-OFF helpers (no existing compare/LOVO/metrics behaviour changed):
`backend/eval/eval_stats.py` (C1 `mean_with_ci`/
443 `bootstrap_ci`/`paired_sign_test`/`paired_delta_ci` ? never a bare LOVO mean at n?6; C3
`out_of_fold_predictions`/`grouped_folds`
444 ? no stacking leakage), `score_calibration.py` (C2 Platt+isotonic+brier/ECE ? comparable
cue confidences), `segmentation_metrics.py`
445 (C4 F1@{10,25,50} + mAP@tIoU + segment_count_ratio ? make over-segmentation tIoU is
blind to visible). 24 tests; ruff+mypy clean;
446 all 4 CI lanes green (full-suite coverage floor held). Both PRs done in **isolated git

```

worktrees\*\* (off master) to dodge the  
447 recurring shared-checkout branch-flips (siblings merged #117 fusion-P2, #118, #120, #123  
concurrently; \*\*#116 closed ? its content

#### 120. assistant (2026-06-16T04:14:54.461Z)

Now make the norm durable in memory (the repo side is in PR #186) and log the checkpoint. These  
memory files live outside the repo, so they don't touch the PR.

#### 121. assistant (2026-06-16T04:15:02.504Z)

```

name: working-agreement-tracked-project-docs
description: "Substantial design/project work gets a TRACKED PROJECT DOC (template + §7 progress
tracker + definition-of-done); freeze the plan, track it across sessions, mark it done,
archive."
metadata:
 type: feedback

```

Substantial design/project work (anything bigger than a single small PR ? multi-PR efforts,  
research-backed designs, security/privacy/architecture plans) is captured as a **tracked project**  
**doc** in `docs/`, following `docs/PROJECT\_DOC\_TEMPLATE.md`.

Shape: status header (`DRAFT ? IN PROGRESS ? DONE ? archived`) · decisions-locked table ·  
(optional) research/legal **foundation** · design + **reuse map** · (optional) **threat model** ·  
· **honest-limits** · **§7 progress tracker** (one ??? row per independently-shippable PR ?  
the source of truth) · open owner/counsel questions · **definition-of-done** (acceptance  
checklist) · references · changelog. Tag each claim `[verified]` / `[researched]` / `[design]`.

**Why:** the owner explicitly loved this framing (2026-06-16, the Golden Regression Fixtures  
plan) and asked to make it a norm "throughout" ? it lets a plan be iterated on, its progress  
tracked across sessions, and be marked COMPLETE against clear deliverables (then archived).  
Generalizes the emergent pattern already in `docs/RALLY\_QUALITY\_RESEARCH.md` §7.

**How to apply:** for a substantial ask, write/extend the doc FIRST (draft, owner-gated), index  
it in `docs/README.md` + a memory `current-state` pointer (+ `docs/NEXT\_STEPS.md` once work  
starts); each deliverable = ONE small PR per `[[gotcha-shared-checkout-branch-collisions]]`; flip  
Status and archive when the definition-of-done is met. Codified in repo `CLAUDE.md` (workflow  
conventions) + `docs/PROJECT\_DOC\_TEMPLATE.md` (PR #186). Ties `[[working-agreement-checkpoints]]`  
+ `[[decision-rigor-norm]]` + `[[paper-coauthor-aim]]`.

#### 122. user (2026-06-16T04:15:02.538Z)

File created successfully at: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-  
highlight-indexer\memory\working-agreement-tracked-project-docs.md (file state is current in  
your context ? no need to Read it back)

#### 123. user (2026-06-16T04:15:05.745Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-  
indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ?  
no need to Read it back)

#### 124. user (2026-06-16T04:15:30.234Z)

<tool\_use\_error>File has not been read yet. Read it first before writing to it.</tool\_use\_error>

#### 125. assistant (2026-06-16T04:15:39.263Z)

The partial read didn't register as a full read for edit-tracking. Let me read the top of the  
file explicitly, then prepend the checkpoint.

#### 126. user (2026-06-16T04:15:40.283Z)

```

1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4 metadata:
5 node_type: memory
6 type: project
7 originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8 ---
9
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 **Checkpoint:** 2026-06-16 ~03:30 (? RALLY-QUALITY RESEARCH + EVAL-FRAMEWORK FOUNDATION
DOC DONE ? **PR #184 OPEN for
13 owner review**, docs-only, NOT auto-merged). Owner asked for deep research to bridge the
local-vs-Gemini gap with a strong
14 harness/eval framework, checked into the repo as the foundation for quality iterations.
Scope (owner-picked) = research +

```

### 127. user (2026-06-16T04:15:56.001Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

### 128. assistant (2026-06-16T04:16:13.687Z)

Frozen and made a norm. Summary:

```

Frozen ? [PR #186](https://github.com/avidullu/badminton-highlight-indexer/pull/186) (branch
`docs/golden-regression-fixtures-plan`, one commit atop `90ab087`/#185, docs-only):
- `docs/GOLDEN_REGRESSION_FIXTURES.md` ? the plan, verbatim from what you reviewed.
- `docs/PROJECT_DOC_TEMPLATE.md` ? the reusable skeleton.
- `CLAUDE.md` workflow-conventions line + `docs/README.md` index entries.

```

```

The norm ? rather than invent one, I standardized the pattern this repo already grew
(`RALLY_QUALITY_RESEARCH.md` §7): substantial design/project work ? a tracked project doc with a
status header, decisions-locked, a **§7 progress tracker** (one ?/?/? row per shippable PR =
source of truth), and a **§9 definition-of-done**; archived when DONE. It's now codified three
ways: the template, the `CLAUDE.md` convention, and a memory ([[working-agreement-tracked-
project-docs]]) so I default to it across sessions.

```

```

Process notes (honest):

```

```

- Branching-safety held: branched from `origin/master`, staged only my 4 files (left
`artifacts/` and the untracked KHELSUTRA doc alone), re-verified HEAD was mine before commit.
- I did **not** merge ? this is owner-gated content (strategy/legal-adjacent), and per our
merging agreement I only self-merge verified-green *engineering* fixes. #186 is yours to
review/merge.
- Memory + `current-state` checkpoint updated so a future session resumes from the tracker.

```

When you're ready, \*\*P0\*\* (scrub the real names from `eval\_baselines/\*.json` + `eval\_partial\_labels.py` + `test\_cleanup\_caches.py`, add a required CI leak-scan) is the clean first implementation PR ? self-contained, no counsel needed. Just say the word and I'll take it.

### 129. user (2026-06-16T04:29:54.789Z)

can you make a project local memory to ensure that future projects follow the template and also FYI another session is iterating on the quality metrics for pure local (no gemini) segmentation so would your execution get hindered ? Feel free to make a different branch and start executing. Follow the regular discipline and use CI and gemini to ensure high quality results. Merge the PRs are you go along.

### 130. assistant (2026-06-16T04:33:33.282Z)

Good questions ? let me answer the concurrency one directly, because it's the crux, then act on it.

**\*\*Would the concurrent session hinder me? Yes, if we share one working tree\*\*** ? that's the exact `[[gotcha-shared-checkout-branch-collisions]]` hazard. The other session iterating on local-seg quality metrics will be in ``backend/eval/`` and possibly ``eval_baselines/`` ? which overlaps my plan. Mitigations I'll apply:

- **\*\*Isolate in a git worktree\*\*** (the repo's established ``.claude/worktrees/`` pattern) so my edits never touch their checkout.
- **\*\*Re-sequence to avoid collision:\*\*** start with **\*\*M1\*\*** (the fixture framework ? `*all new, additive files*`: ``backend/eval/golden_fixtures.py``, ``eval_baselines/fixtures/``, new tests).
- \*\*Defer P0\*\*** (the name-scrub) precisely *\*because\** it rewrites keys in ``eval_baselines/*.json`` that their metrics work likely depends on ? doing that concurrently would rug-pull them. P0 lands once their eval churn settles or in coordination.

On the memory: I already created it last turn ? **\*\*[[working-agreement-tracked-project-docs]]\*\*** (auto-memory) **\*\*\*** the convention line in ``CLAUDE.md`` (the *\*project-local, checked-in\** enforcement, riding in PR #186). Merging #186 makes that enforcement live on master. Let me verify state and CI before I merge it and start M1.

### 131. user (2026-06-16T04:33:41.209Z)

```
=== master vs my checkpoint ===
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0) (#187)
=== working tree (other session WIP?) ===
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
=== #186 mergeability + CI ===
{"mergeStateStatus":"CLEAN","mergeable":"MERGEABLE","state":"OPEN"}
--- checks ---
Full test suite (offline, mocked) pass 2m0s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27593679980/job/81579445780
Mypy (lenient baseline) pass 33s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27593679980/job/81579445825
Ruff lint pass 15s https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27593679980/job/81579445782
Syntax sweep + import smoke (no GPU stack) pass 21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27593679980/job/81579445805
=== recent Actions runs (is CI up?) ===
completed success feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge?
CI master push 27593871028 2m17s 2026-06-16T04:19:19Z
completed success feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge
(?F1 +0.139, 6/0) CI feat/tier1-bounded-windowing pull_request 27593854352 2m37s
2026-06-16T04:18:43Z
completed cancelled feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-
merge (?F1 +0.139, 6/0) CI feat/tier1-bounded-windowing pull_request 27593836591
53s 2026-06-16T04:18:10Z
```

### 132. assistant (2026-06-16T04:36:14.266Z)

Key findings from the assessment:

- **\*\*CI is back up\*\*** (billing outage resolved) ? #186's checks all **\*\*pass\*\***, and it's **\*\*MERGEABLE/CLEAN\*\***.
- **\*\*The concurrent session just merged #187\*\*** ? ``feat(windowing): Tier 1 (W1) bounded windowing (?F1 +0.139)``. That's directly relevant: they're actively reshaping ``windowing.py``, the *\*exact\** behavior my characterization gate would freeze.

That sharpens my collision answer. Let me merge #186 (makes the template/convention live = fulfills the memory ask), then set up an isolated worktree off the latest master.

### 133. user (2026-06-16T04:36:22.122Z)

```
=== fetch + new master ===
```



```

1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0)
(#187)
=== create isolated worktree off origin/master ===
Preparing worktree (new branch 'feat/golden-fixtures-m1')
branch 'feat/golden-fixtures-m1' set up to track 'origin/master'.
HEAD is now at 1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc
template (#186)
=== worktree HEAD ===
1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)

```

#### 134. assistant (2026-06-16T04:37:54.101Z)

Excellent position now:

```

- **#186 merged** ? master is `1de2817`; the template + the `CLAUDE.md` convention are **live**.
That's the project-local enforcement of "future projects follow the template" ? done and now
permanent, not just in my memory.
- **Isolated worktree** `feat/golden-fixtures-m1` created off the latest `origin/master`
(includes #187). My edits there can't touch the other session's checkout ? so, concretely: **no,
my execution won't be hindered, and I won't hinder theirs.**

```

One real coordination note I'll act on: since they're actively reshaping `windowing.py` (#187, likely more), I'm starting with **\*\*M1\*\*** (the additive framework: `backend/eval/golden\_fixtures.py` + synthetic fixtures + a CONTROL-only, parse-compared characterization test) and **\*\*deferring P0\*\*** (the `eval\_baselines/\*.json` name-scrub) so I don't rewrite keys their metrics work depends on.

Let me map the exact APIs I need (from the post-#187 worktree, so I build against current behavior) before writing the helper.

#### 135. user (2026-06-16T04:40:38.915Z)

Based on my thorough read of the golden-fixtures worktree, here is the EXACT API mapping you requested:

### 1. EXACT FUNCTION SIGNATURES + RETURN TYPES

#### **\*\*tracknet\_runner.py\*\***

```

```python
parse_trajectory_csv(csv_win_path: str) -> List[TrajectoryPoint]
# Returns sorted list of TrajectoryPoint(frame: int, visible: bool, x: float, y: float)
# Parses CSV with columns: Frame, Visibility (0|1), X, Y
# Silently skips malformed rows (KeyError/ValueError)

trajectory_to_action_windows(
    points: List[TrajectoryPoint],
    fps: float,
    frame_width: float,
    chunk_start: float = 0.0,
    velocity_thresh: float = 0.02,
    merge_gap: float = 2.0,
    min_window_duration: float = 2.0,
    chunk_index: Optional[int] = None,
    max_window_duration: float = 0.0,
    min_split_gap: float = 0.5,
) -> List[Dict[str, Any]]
# Returns: [{
#   "start": float (seconds),
#   "end": float (seconds),
#   "active_frame_count": int,
#   "peak_velocity": float,
#   "mean_velocity": float,
#   "track_id_count": int (always 1),

```

```

# "chunk_index": Optional[int]
# }]
# OCCLUSION: visible==False breaks trajectory (sets prev=None, resets velocity chain)
# CLAMPS: fps<=0 ? 30.0; frame_width<=0 ? 1280.0
# BOUNDED WINDOWING (#187): max_window_duration > 0 triggers _split_overlong_groups()
# ? recursively splits windows > max_window_duration at largest inactivity gap
# (gap >= min_split_gap frames), never merges, only subdivides
...

```

****windowing.py****

```

```python
@dataclass(frozen=True)
class WindowingPreset:
 name: str
 velocity_thresh: float
 merge_gap: float
 min_window_duration: float
 max_window_duration: float = 0.0

 def as_tuple(self) -> Tuple[float, float, float]
 # Returns (velocity_thresh, merge_gap, min_window_duration)

 def to_dict(self) -> Dict[str, Any]

SERVED: WindowingPreset # sourced from IndexingConfig defaults
name="served", velocity_thresh=0.02, merge_gap=2.0,
min_window_duration=2.0, max_window_duration=0.0 (default OFF)

PROPOSAL: WindowingPreset
name="proposal", velocity_thresh=0.005, merge_gap=1.0,
min_window_duration=0.5, max_window_duration=0.0

PRESETS: Dict[str, WindowingPreset] # {"served": SERVED, "proposal": PROPOSAL}

served_windowing_from_config(config: Any, family: str = "wasb") -> WindowingPreset
Reads actual served config (overrides honoured)

build_windows(
 points: List[Any],
 fps: float,
 frame_width: float,
 preset: WindowingPreset
) -> List[Dict[str, Any]]
Thin wrapper: calls TrackNetRunner.trajectory_to_action_windows with preset params

windows_to_intervals(windows: List[Dict[str, Any]]) -> List[Interval]
Returns [(w["start"], w["end"]) for w in windows]
Interval = Tuple[float, float]
...

```

## **\*\*distill\_local.py\*\***

```

```python
FEATURE_NAMES: List[str] = ["duration", "peak_velocity", "mean_velocity", "active_ratio",
"net_crossings"]
PROPOSAL = (0.005, 1.0, 0.5) # velocity_thresh, merge_gap, min_window_duration
BASELINE_LOCAL = (0.02, 2.0, 2.0) # the SERVED windowing

build_candidates(
    trajectory_csv: str,
    fps: float,
    frame_width: float,
    proposal: Tuple[float, float, float] = PROPOSAL
) -> Tuple[List[Interval], List[List[float]]]

```

```

# Returns (candidate_intervals, feature_rows)
# Interval = Tuple[float, float]
# feature_rows: each row = [duration, peak_velocity, mean_velocity, active_ratio,
net_crossings]
# Net crossings sourced from rally_gate.gate_windows(...) ? .crossings field
...

```

****experiment.py****

```

```python
@dataclass
class VideoCandidates:
 name: str
 intervals: List[Interval]
 features: Dict[str, List[float]] = field(default_factory=dict)
 # Column-major: feature_name ? list of per-candidate values
 gts: Optional[List[Interval]] = None
 # Interval = Tuple[float, float]

@dataclass
class Variant:
 name: str
 segmenter: str = "wasb_hybrid"
 config_overrides: Dict[str, Any] = field(default_factory=dict)
 cue_flags: Dict[str, bool] = field(default_factory=dict)
 min_crossings: int = 0 # Gate A (net-crossing threshold)
 min_players: int = 0 # Gate B (player count threshold)
 classifier_model: Optional[str] = None # path to frozen LogisticRegression JSON
 feature_names: Optional[List[str]] = None # which persisted features to use
 threshold: float = 0.5 # classifier probability cutoff
 merge_gap: float = 1.0 # post-filter overlap-merge gap (seconds)
 tune_threshold: bool = False # pick F1-best threshold on train fold
 calibrate: Optional[str] = None # "platt" | "isotonic" | None

 def is_learned(self) -> bool
 # True iff (classifier_model is not None) OR (feature_names is not None)

 def to_dict(self) -> dict

 @classmethod
 def from_dict(cls, d: dict) -> "Variant"

 def spec_hash(self) -> str
 # Stable 12-char SHA1 hash of to_dict() (json sorted)

CONTROL = Variant(name="control")
Baseline: all fields at defaults (no gate, no learned filter, pure merge)
is_learned() ? False
predict(CONTROL, ...) uses ONLY _gate_mask (trivial, all True) + merge_overlaps
PURE STL LIB (no numpy/LogisticRegression)

def predict(
 variant: Variant,
 vc: VideoCandidates,
 model: Optional[LogisticRegression] = None,
 threshold: Optional[float] = None,
 calibrator: Optional[Callable[[float], float]] = None,
) -> List[Interval]
1. _gate_mask(variant, vc) ? boolean keep-mask
- if min_crossings > 0: read "net_crossings" column, apply threshold
- if min_players > 0: read "players_in_court" column, apply threshold
- missing columns default to [0.0] * n_candidates (logs warning)
2. if variant.feature_names:
- load model (supplied, or from variant.classifier_model)
- predict_proba(feature_matrix) ? probs

```

```

- optionally calibrate(proba)
- apply threshold ? additional mask
3. kept = [iv for iv, keep in zip(vc.intervals, mask) if keep]
4. return merge_overlaps(kept, gap_tol=variant.merge_gap)
...

```

## **\*\*metrics.py\*\***

```

```python
Interval = Tuple[float, float]

@dataclass
class Score:
    iou_threshold: float
    num_pred: int
    num_gt: int
    true_positives: int
    false_positives: int
    false_negatives: int
    precision: float
    recall: float
    f1: float
    mean_iou: float
    mean_start_error: float
    mean_end_error: float

    def as_dict(self) -> dict

interval_iou(a: Interval, b: Interval) -> float
# Temporal IoU; returns 0.0 if non-overlapping or degenerate

match_intervals(
    preds: List[Interval],
    gts: List[Interval],
    iou_threshold: float = 0.5
) -> MatchResult
# Greedy one-to-one matching by descending IoU
# Returns MatchResult(matches, unmatched_pred_indices, unmatched_gt_indices)

score(
    preds: List[Interval],
    gts: List[Interval],
    iou_threshold: float = 0.5,
    match_result: Optional[MatchResult] = None
) -> Score
# Computes precision, recall, F1, mean_iou, boundary errors
...

```

****gt_loader.py****

```

```python
load_gt_intervals(csv_path: str, *, strict: bool = True) -> List[Interval]
strict=True: raise ValueError on bad rows (corpus/gate use)
strict=False: skip bad rows, LOG count (legacy tuning-driver behavior)
Accepts both header conventions: "start"/"end" OR "start_time"/"end_time"
Positional fallback if no header: columns 0,1
Timestamps: decimal seconds or MM:SS / HH:MM:SS
Returns sorted by start; warns on overlapping GTs
...

```

## **\*\*rally\_gate.py\*\***

```

```python
@dataclass

```

```

class GatedWindow:
    start: float
    end: float
    crossings: int      # net crossing count
    visible_points: int
    kept: bool          # crossings >= min_crossings

estimate_play_axis(points) -> Tuple[str, float, float]
    # Returns (axis: "x"|"y", net_value: float, span: float)
    # Axis = the image dimension with larger shuttle spread (play axis)
    # net_value = median of that axis (where shuttle clusters near net)
    # span = spread (p95 - p5) for that axis

count_net_crossings(window_points, axis: str, net: float, deadband: float) -> int
    # Counts sign changes of (coord - net) across visible points
    # Ignores points within deadband of the net

gate_windows(
    points,
    windows: List[Interval],
    fps: float,
    min_crossings: int = 2,
    deadband_frac: float = 0.06,
    estimate: Optional[Tuple[str, float, float]] = None
) -> List[GatedWindow]
    # Annotates each window with crossing count and keep decision
    # estimate parameter: precomputed (axis, net, span) to avoid O(P log P) re-estimation
    # deadband = deadband_frac * span
    # Returns sorted by window start time

apply_gate(
    points,
    windows: List[Interval],
    fps: float,
    min_crossings: int = 2,
    deadband_frac: float = 0.06
) -> List[Interval]
    # Convenience: returns only windows where kept=True
...

```

****fusion_compare.py** (loader pattern)**

```

...python
load_videos(features_dir: str) -> List[experiment.VideoCandidates]
    # Loads every *_golden_features.json in features_dir
    # For each: reads c["shuttle"][fn] and c["person"][fn] UNCONDITIONALLY
    # (no .get() fallback ? KeyError if missing)
    # Returns VideoCandidates with features={shuttle_cols + person_cols}
...

```

****ablation.py** (loader pattern)**

```

...python
load_videos(features_dir: str) -> List[VideoCandidates]
    # Loads every *_golden_features.json
    # For shuttle + person: reads unconditionally c["shuttle"][fn], c["person"][fn]
    # For audio: reads c.get("audio") or {} ? .get(fn, 0.0) with fallback to 0.0
    # Older JSONs without audio block: all audio columns default to [0.0]
...

```

2. CONCRETE INTEGRATION RECIPE (CONTROL-only, pure stdlib)

```
```python
```

## Step 1: Load and parse trajectory CSV

```
points: List[TrajectoryPoint] = TrackNetRunner.parse_trajectory_csv(trajectory_csv)
```

## Step 2: Window the trajectory (use SERVED preset from windowing.py)

```
from backend.eval.windowing import SERVED, build_windows
windows: List[Dict[str, Any]] = build_windows(points, fps=30.0, frame_width=1280.0,
preset=SERVED)
```

## Step 3: Build candidates (extract features including net\_crossings via rally\_gate)

```
from backend.eval.distill_local import build_candidates
intervals, X = build_candidates(trajectory_csv, fps=30.0, frame_width=1280.0)
```

**intervals: List[Interval]**

**X: List[List[float]] with 5 columns [duration, peak\_velocity, mean\_velocity, active\_ratio]**

## Step 4: Gate on net-crossings (rally\_gate.gate\_windows is called within build\_candidates)

```
from backend.pipeline.detectors import rally_gate
gated = rally_gate.gate_windows(points, intervals, fps=30.0, min_crossings=2)
```

**gated: List[GatedWindow]**

## Step 5: Score predictions against golden labels

```
from backend.eval.gt_loader import load_gt_intervals
gts: List[Interval] = load_gt_intervals(gt_csv_path, strict=True)
```

## Step 6: Create VideoCandidates structure for experiment harness

```
from backend.eval.experiment import VideoCandidates, CONTROL, predict
vc = VideoCandidates(
 name="test_video",
 intervals=intervals,
 features={}, # CONTROL doesn't use features
 gts=gts
)
```

## Step 7: Predict with CONTROL variant

```
preds: List[Interval] = predict(CONTROL, vc)
```

**predict() for CONTROL:**

- mask = [True] \* len(intervals) (no gates, feature\_names=None)
- kept = intervals (all pass)
- return merge\_overlaps(intervals, gap\_tol=1.0)

## Step 8: Score

```
from backend.eval.metrics import score as score_metrics
result = score_metrics(preds, gts, iou_threshold=0.5)
```

**result.precision, result.recall, result.f1, result.mean\_iou,**

**result.true\_positives, result.false\_positives, result.false\_negatives**

```
```
```

```
**EXACT CALL CHAIN (no branches):**
```
```

```
parse_trajectory_csv(csv)
? List[TrajectoryPoint]
```

```
build_windows(points, fps, fw, SERVED)
```

```

? calls TrackNetRunner.trajectory_to_action_windows(vt=0.02, mg=2.0, mwd=2.0, max_win=0)
? List[Dict]

windows_to_intervals(windows)
? List[Interval]

gate_windows(points, intervals, fps, min_crossings=2)
? List[GatedWindow] (crossings field used by build_candidates)

predict(CONTROL, VideoCandidates)
? _gate_mask(CONTROL, vc) # all True
? merge_overlaps(intervals, gap_tol=1.0)
? List[Interval]

score(preds, gts, 0.5)
? Score object
...

```

### 3. RED-TEAM GOTCHAS (CONFIRMED AGAINST CODE)

```

(a) Do existing loaders read GT via LENIENT vs STRICT?

YES ? INCONSISTENT (NOT FIXED) [gotcha confirmed]
- `calibrate_wasb.load_golden_gts()` ? calls `load_gt_intervals(csv_path, strict=False)`
- **File: backend/eval/calibrate_wasb.py:41**
- Silently skips bad rows, logs count
- `experiment.evaluate_variant()` and `distill_local.load_video()` call
`load_golden_gts(spec["labels"])` (from calibrate_wasb)
- Uses lenient loader, not strict
- The strict loader `load_gt_intervals(csv_path, strict=True)` exists but is **NOT used by the
eval loaders**
- **Impact:** A GT CSV with malformed rows (missing columns, bad timestamps) silently shrinks
the GT set, flattering precision

(b) Does experiment `_feature_column` silently default MISSING features to 0.0?

YES [gotcha confirmed]
- **File: backend/eval/experiment.py:180-192**
```python
def _feature_column(vc: VideoCandidates, name: str) -> List[float]:
    col = vc.features.get(name)
    n = len(vc.intervals)
    if col is None:
        logger.warning("variant references feature %r absent from %s ? treating as 0.0",
            name, vc.name)
        return [0.0] * n # SILENT ZERO FILL
    # ...rest
...
- Logs a warning but DOES NOT raise; missing feature columns silently become 0.0
- This lets a sparse/old fixture still predict (gates + learned scorer work with zeros)

**(c) Does any loader read c['person'] UNCONDITIONALLY (KeyError on stripped fixture)?**

**YES ? CRITICAL** [gotcha confirmed]
- **fusion_compare.py:48** `features[fn] = [float(c["person"][fn]) for c in candls]`
- Direct subscript access, no `.get()` fallback
- If a fixture has NO "person" key in any candidate, **KeyError** on the first loop
- **File: backend/eval/fusion_compare.py:35-52**
- **ablation.py:401** Same pattern: `features[fn] = [float(c["person"][fn]) for c in candls]`
- **File: backend/eval/ablation.py:386-407**
- **ablation.py:403** Audio HAS defensive `.get()`: `c.get("audio") or {}.get(fn, 0.0)`
- Audio defaults to 0.0 if missing; person does NOT
```

```
- **Impact:** A fixture with person-cue stripped (e.g., for privacy) will crash both
fusion_compare and ablation at load time

**(d) Does CONTROL avoid numpy/LogisticRegression entirely (pure stdlib)? Which variants are
is_learned()?**

**YES ? CONTROL is pure stdlib; learned variants use numpy/LogisticRegression** [gotcha
confirmed]
- **CONTROL definition (backend/eval/experiment.py:158)**:
  ```python
 CONTROL = Variant(name="control") # all fields at defaults
  ```

  - `feature_names=None`, `classifier_model=None`, `min_crossings=0`, `min_players=0`
  - `is_learned()` ? `False`
  - `predict(CONTROL, vc)` ? calls `_gate_mask` (stdlib only) + `merge_overlaps` (stdlib only)
  - **NO numpy, NO LogisticRegression**

- **is_learned() definition (backend/eval/experiment.py:123-125)**:
  ```python
 def is_learned(self) -> bool:
 return self.classifier_model is not None or bool(self.feature_names)
  ```

  - Learned: any variant with `classifier_model` set OR `feature_names` set
  - Learned variants call `LogisticRegression.load()` (requires numpy)
  - **File: backend/eval/experiment.py:233** `model =
  LogisticRegression.load(variant.classifier_model)`
  - **File: backend/eval/experiment.py:234** `proba = [float(p) for p in
  model.predict_proba(...)]`

- **Examples of is_learned() variants**:
  - `fusion_compare.shuttle_only`: `feature_names=list(SHUTTLE_FEATURE_NAMES)`
  - `ablation.learned_full_variant()`: `feature_names=[all 13 columns]`
  - Any variant with `classifier_model="path/to/model.json"` set

---
```

SUMMARY

All exact APIs mapped. The CONTROL variant is production-ready stdlib code (pure Python, no ML deps). The integration recipe is deterministic and reproducible: trajectory CSV ? windows ? candidates ? predict(CONTROL) ? score(preds, gts). Three critical loader gotchas confirmed: (a) lenient GT reading flattens precision, (b) missing features silently become 0.0, (c) ****person cue strips cause KeyError on fusion_compare/ablation**** ? this is the highest-risk one for your `replay\_fixture()` if feeding a CSV with no person data. CONTROL avoids all ML entirely.

136. assistant (2026-06-16T04:42:16.775Z)

I have the exact APIs and confirmed the gotchas. Critically, the ****CONTROL path is pure-stdlib**** (no numpy ? cross-machine safe), and I'll ****bypass**** `fusion\_compare`/`ablation` loaders (they read `c['person']` unconditionally ? would crash on person-stripped fixtures) by building `VideoCandidates` directly. That's the person-optional Tier-0 loader the red-team asked for.

I'll implement ****M1**** in the isolated worktree via a focused coding pass, then verify (full suite + ruff + mypy) and merge myself. Spawning it in the background so it can write + run the full suite while I stay responsive:

137. user (2026-06-16T04:42:54.036Z)

Async agent launched successfully.

agentId: a724501ac1d1831fb (internal ID - do not mention to user. Use SendMessage with to: 'a724501ac1d1831fb' to continue this agent.)

The agent is working in the background. You will be notified automatically when it completes. Do not duplicate this agent's work ? avoid working with the same files or topics it is using. Work on non-overlapping tasks, or briefly tell the user what you launched and end your response.

output_file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a724501ac1d1831fb.output
Do NOT Read or tail this file via the shell tool ? it is the full subagent JSONL transcript and reading it will overflow your context. If the user asks for progress, say the agent is still running; you'll get a completion notification.

138. assistant (2026-06-16T04:43:17.892Z)

Status ? here's where things stand after this turn:

****Concurrency: resolved, both directions.**** I'm working in an ****isolated git worktree**** (``feat/golden-fixtures-m1``, the repo's ``.claude/worktrees/`` pattern) off the latest ``origin/master``. My edits there can't touch the other session's checkout and vice-versa. I also ****re-sequenced to avoid semantic collision****: starting with ****M1**** (additive new files) and ****deferring P0**** (the ``eval_baselines/*.json`` name-scrub) so I don't rewrite keys their windowing/metrics work depends on. Bonus: their #187 added the bounded-windowing **capability** but left it ****OFF in the SERVED preset****, and I pinned each fixture's preset params explicitly ? so my snapshots are robust to a SERVED retune (only an algorithm change churns them, and that's a documented one-command ``--refresh-snapshot``).

****The template norm is now LIVE and enforced**** (your memory ask): ****PR #186 is merged**** (``master`` = ``1de2817``). That puts the ``.CLAUDE.md`` workflow convention + ``.docs/PROJECT_DOC_TEMPLATE.md`` on master ? the **project-local, checked-in** enforcement that future projects follow the template. Plus the cross-session memory `[[working-agreement-tracked-project-docs]]`. CI was green for it (the billing outage is resolved).

****Executing M1 now.**** A background coding agent is implementing the framework in the worktree:
- ``backend/eval/golden_fixtures.py`` ? the shared ``replay_fixture()`` (CONTROL/pure-stdlib, ****no numpy**** ? cross-machine stable) + a freeze/refresh CLI.
- 3 ****synthetic**** fixtures (generated by the freeze tool, not hand-authored) + manifest + ``.gitattributes`` LF pin.
- ``tests/test_golden_fixtures.py`` ? parse-compared characterization (exact-int / approx-float), a positive "no person/audio keys" privacy assertion, a perturbation test proving windows are re-derived, and an idempotent-freeze test.

It bypasses the ``fusion_compare`/`ablation`` loaders (which ``KeyError`` on person-free fixtures ? a verified red-team catch) by building ``VideoCandidates`` directly.

When it finishes I'll ****verify myself**** (full ``run_all_tests.py`` + ``ruff`` + ``mypy``, per the pre-LGTM gate), review the diff, and ****merge**** M1 with CI green ? then continue down the S7 tracker (M2 freeze-tool hardening ? M3/M4 gates ? P1 de-attribution tooling), holding P0 until the windowing churn settles and P2/P3 for counsel. I'll flag you as each lands.

139. user (2026-06-16T04:53:14.713Z)

```
<task-notification>
<task-id>a724501ac1d1831fb</task-id>
<tool-use-id>toolu_01UGEHYaZQb6AAGvR51nQaTN</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a724501ac1d1831fb.output</output-file>
<status>completed</status>
<summary>Agent "Implement M1 golden fixtures framework" completed</summary>
<result>M1 is complete and fully verified. Here is the report.
```

Summary

Implemented deliverable ****M1**** (Fixture framework + synthetic Tier-0 + shared replay helper) of the Golden Regression Fixtures project, entirely within the ``golden-fixtures`` worktree. No git commands beyond read-only ``status`` for verification.

Files created

- ``backend/eval/golden_fixtures.py`` ? the shared module: ``CURRENT_SCHEMA=1``, ``FIXTURES_DIR`` (resolved from ``__file__``, `parents[2]``), ``make_synthetic_trajectory``, ``replay_fixture``,

```
`freeze_fixture`, `freeze_synthetic`, `refresh_snapshot`, `load_manifest`, `load_fixture`,
`compare_expected`, and the `__main__` CLI (`--freeze-synthetic` / `--refresh-snapshot
[--slug|--all]` / `--check`).
- `eval_baselines/fixtures/&slug&gt;/{trajectory.csv, labels.csv, expected.json,
provenance.json}` for the 3 slugs (generated by `--freeze-synthetic`, never hand-written).
- `eval_baselines/fixtures/manifest.json` ? array of `{slug, fps, frame_width,
windowing:{vt,mg,mwd,max_win}, kind:"synthetic", schema_version, paths}`.
- `eval_baselines/fixtures/README.md` ? guardrails + freeze/refresh commands + "generated, never
hand-edited" + $6 honest-limits.
- `.gitattributes` (worktree root) ? `eval_baselines/fixtures/** text eol=lf`.
- `tests/test_golden_fixtures.py` ? parametrized over `load_manifest()` slugs.
```

Verify results (verbatim)

```
1. `--freeze-synthetic` ? `froze 3 synthetic fixture(s): clean_single_rally,
multi_rally_with_deadtime, occlusion_break`. `--check` ? all `[ok]`, `EXIT=0`.
2. `python run_all_tests.py` ? **744 passed, 7 skipped** (the pre-existing skips; new file adds
17 tests, all green). With CI coverage flags: **Total coverage 80.36%** ("Required test coverage
of 70% reached"); the new module itself is at 85%.
3. `ruff check .` ? `All checks passed!` (EXIT 0). `mypy backend_training` ? `Success: no issues
found in 103 source files` (EXIT 0). One initial ruff finding (unused `field` import) fixed.
```

The 3 synthetic scenarios (num candidates, num segments, F1)

slug	candidates	segments	F1	what it exercises
clean_single_rally	1	1	1.0	one ~4s in-flight rally ? single window, net_crossings=7
multi_rally_with_deadtime	2	2	1.0	two rallies split by 3s of still (low-velocity) dead-time
occlusion_break	2	2	1.0	two rallies split by 3s of occlusion (Visibility=0)

Hard-constraint compliance

- CONTROL/pure-stdlib only**: builds `VideoCandidates` directly with only the 5 shuttle columns and replays via `experiment.predict(CONTROL, vc)`; never calls `fusion\_compare.load\_videos`/`ablation.load\_videos` (verified `CONTROL.is\_learned()` is False).
- Parse-compare, never byte-compare**: `compare\_expected` does `json.load` on both sides ? exact on int/structural fields (candidate count, `net\_crossings` as int, TP/FP/FN, segment count, slug, schema, gt_hash) and abs-tol 1e-6 on floats.
- Windowing pinned per fixture**: `(vt=0.02, mg=2.0, mwd=2.0, max\_win=0.0)` written into each `provenance.json` and the manifest, read back and passed explicitly to `build\_candidates` ? does not depend on the named `SERVED` constant.
- Determinism**: trajectories from deterministic sine formulae (no `random`); all floats rounded to 6dp; UTF-8 / LF-only / trailing-newline confirmed.
- Privacy**: every fixture carries `kind:"synthetic"`, `minors\_free:true`, `license:"synthetic"`; `expected.json` has no person/pose/gait/face/audio/player fields (positive assertion test scans the raw text and the parsed candidate keys).

Beyond the spec's required tests I added `test\_check\_cli\_passes`, `test\_synthetic\_provenance\_tags`, `test\_control\_is\_not\_learned`, and `test\_manifest\_is\_non\_empty`, and I additionally demonstrated out-of-band that `--check` returns exit 1 on a seeded trajectory perturbation (then restored) ? proving the gate catches behaviour changes, not just echoes the snapshot.

Deviations from spec

- One** minor wording in `test\_perturbing\_trajectory\_changes\_segments`: I perturb by **truncating** the trajectory to its first half (which shrinks the derived window's `end`) rather than deleting an interior hole. Reason: deleting interior rows does not move the segment boundaries ? `merge\_gap=2.0s` coalesces across the gap and the surviving rows keep their original frame indices, so start/end are unchanged. Truncation guarantees a structural change and still proves the windows are re-derived from the CSV (not echoed). No API or fixture-content deviation otherwise.

Note: numpy and cv2 *are* installed in this environment, so the full suite ran here rather than skipping ? but the M1 replay path itself touches no numpy (CONTROL/stdlib), so it remains valid on a clean no-GPU clone.</result>

```
<usage><subagent_tokens>127378</subagent_tokens><tool_uses>40</tool_uses><duration_ms>620623</duration_ms></usage>
</task-notification>
```

140. assistant (2026-06-16T04:53:37.719Z)

M1 reports green, but per the pre-LGTM gate I verify myself ? never trust the report. Let me review the actual files and re-run the suite, ruff, and mypy in the worktree.

141. user (2026-06-16T04:53:39.912Z)

```
=== worktree status ===
?? .gitattributes
?? backend/eval/golden_fixtures.py
?? eval_baselines/fixtures/
?? tests/test_golden_fixtures.py
=== HEAD ===
1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)
=== fixture files ===
?? eval_baselines/fixtures/
```

142. user (2026-06-16T04:53:40.580Z)

```
1      """Golden Regression Fixtures (M1) ? video-free, CONTROL-only characterization harness.
2
3      `docs/GOLDEN_REGRESSION_FIXTURES.md` ($4, $6, $7). This module is the single shared
4      replay code path** used by both the freeze tool and the characterization gate, and it
5      deliberately stays on the pure-stdlib CONTROL path** (D5 in that doc): the learned
6      variants train a numpy/OpenBLAS logistic regression whose near-threshold decisions flip
7      across CPUs, so they can never anchor a cross-machine byte/parse snapshot. Here we
replay
8      only `experiment.CONTROL` (`is_learned() == False`), which keeps all candidate
intervals,
9      merges overlaps, and uses no numpy.
10
11     The replay reuses the already-shipped eval stack VERBATIM::
12
13         parse_trajectory_csv ? distill_local.build_candidates(proposal=pinned)
14         ? VideoCandidates(name, intervals, features={5 shuttle cols},
gts=load_gt_intervals)
15         ? experiment.predict(CONTROL, vc) ? metrics.score(preds, gts, iou)
16
17     Tier-0 / privacy invariants (D3, $4c, $6 of the doc), enforced here by construction:
18
19     * Fixtures are synthetic ? deterministic formulae, NO randomness ? so they carry
zero
20     provenance/licensing risk and are safe to commit publicly (owner Q7 "synthetic-only").
21     * Every fixture is tagged ``kind="synthetic"`, ``minors_free=true`,
``license="synthetic"`.
22     * ``expected.json`` carries only the 5 shuttle feature columns ? there is, by
design, no
23     person / pose / gait / face / audio field anywhere in the schema (positive-assertion
test).
24     * The windowing preset params ``vt, mg, mwd, max_win`` are pinned inside each
fixture's
25     provenance** and passed explicitly to ``build_candidates`` ? we do NOT rely on the
named
26     ``windowing.SERVED`` constant, so a future SERVED retune cannot silently churn
fixtures.
27
28     $6 honest-limits caveat (carried in the CLI / test messages, not only the docs): a green
29     suite proves only that the deterministic post-trajectory transforms are unchanged for
the
30     frozen synthetic cases ? NOT that the detector/decoder/AI handoff/DB are correct, and
```

```

NOT
31     that rally detection is good. Synthetic-tier numbers measure plumbing correctness, never
32     field quality (never cite them in QUALITY_ITERATIONS.md / the paper).
33
34     Pure + offline + stdlib. The replay imports `tracknet_runner` (which is itself stdlib-
only
35     for the parse/window math ? no TF), but pulls in no numpy/cv2 on the CONTROL path.
36     """
37     from __future__ import annotations
38
39     import argparse
40     import json
41     import math
42     import sys
43     from dataclasses import dataclass
44     from pathlib import Path
45     from typing import Any, Dict, List, Optional, Sequence, Tuple
46
47     # --- shipped eval stack, reused verbatim (CONTROL / pure-stdlib path only)
-----
48     from backend.eval import distill_local
49     from backend.eval.experiment import CONTROL, VideoCandidates, predict
50     from backend.eval.gt_loader import load_gt_intervals
51     from backend.eval.metrics import score
52     from backend.eval.regression import gt_hash
53
54     Interval = Tuple[float, float]
55
56     #: Bump when the fixture/expected schema changes shape. Loaders refuse a fixture whose
57     #: ``schema_version`` exceeds this (forward-incompatible), per the schema-drift guard.
58     CURRENT_SCHEMA = 1
59
60     #: Repo-root-relative fixtures dir, resolved from THIS file (never cwd) so the harness
works
61     #: from any working directory / a fresh clone. ``parents[2]`` = backend/eval ? backend ?
root.
62     FIXTURES_DIR: Path = Path(__file__).resolve().parents[2] / "eval_baselines" / "fixtures"
63
64     #: The pinned SERVED windowing the synthetic fixtures freeze at: (velocity_thresh,
merge_gap,
65     #: min_window_duration). Mirrors the historical SERVED values (0.02, 2.0, 2.0) but is a
LITERAL
66     #: here on purpose ? the spec forbids depending on the named ``windowing.SERVED``
constant so a
67     #: future retune of SERVED can never silently re-baseline a committed fixture. Each
fixture also
68     #: re-states these in its provenance; the replay reads them from there.
69     PINNED_VT = 0.02
70     PINNED_MERGE_GAP = 2.0
71     PINNED_MIN_WINDOW = 2.0
72     PINNED_MAX_WINDOW = 0.0 # 0 = bounded-windowing OFF (W1), matching the SERVED default
73
74     #: The 5 fps/resolution-invariant shuttle feature columns produced by distill_local. Re-
stated
75     #: here (not imported as a mutable alias) so a fixture's expected.json is pinned to
exactly these.
76     SHUTTLE_FEATURES: Tuple[str, ...] = (
77         "duration", "peak_velocity", "mean_velocity", "active_ratio", "net_crossings",
78     )
79     assert tuple(distill_local.FEATURE_NAMES) == SHUTTLE_FEATURES, (
80         "distill_local.FEATURE_NAMES drifted from the pinned shuttle feature set"
81     )
82
83     #: Forbidden key substrings ? biometric / identity / non-shuttle streams that MUST NOT
appear

```

```

84 #: anywhere in a Tier-0 fixture (D3 / §4c). The privacy assertion scans for these.
85 FORBIDDEN_KEYS: Tuple[str, ...] = ("person", "pose", "gait", "face", "audio", "player")
86
87 _FLOAT_ROUND_DP = 6 # all written floats rounded to this many decimals (determinism)
88 _FLOAT_TOL = 1e-6 # abs tolerance when parse-comparing recomputed vs committed floats
89
90
91 # =====
92 # Synthetic trajectory generation (deterministic ? NO random)
93 # =====
94
95
96 @dataclass(frozen=True)
97 class RallySpec:
98     """One in-flight rally segment: the shuttle oscillates across the net midline.
99
100     ``start_frame`` ? first frame index; ``n_frames`` ? length; ``period`` ? oscillation
101     period in frames (governs net-crossing count); ``amplitude`` ? px excursion from the
102     net midline along the play axis. Deterministic sine motion ? fixed
103     velocity/crossings.
104     """
105     start_frame: int
106     n_frames: int
107     period: float = 30.0
108     amplitude: float = 400.0
109
110
111 @dataclass(frozen=True)
112 class GapSpec:
113     """Dead-time between rallies ? what separates one window from the next.
114
115     ``kind="still"`` ? shuttle visible but resting (velocity ? 0 ? no active frames);
116     ``kind="occluded"`` ? shuttle not visible (Visibility=0 ? trajectory broken). Both
117     end an active run so the next rally forms a distinct window.
118     """
119     start_frame: int
120     n_frames: int
121     kind: str = "still" # "still" | "occluded"
122
123
124
125 @dataclass(frozen=True)
126 class FixtureSpec:
127     """A full synthetic scenario: a frame-indexed sequence of rallies and gaps + its GT.
128
129     ``gts`` are the hand-authored ground-truth rally intervals (seconds) ? the synthetic
130     "labels.csv". For a synthetic fixture they line up with the rallies by construction,
131     which is exactly what makes the quality-floor a *plumbing-correctness* check, not a
132     field-quality one (§6).
133     """
134     slug: str
135     description: str
136     fps: float
137     frame_width: float
138     frame_height: float
139     segments: Tuple[Any, ...] # ordered (RallySpec | GapSpec)
140     gts: Tuple[Interval, ...]
141     net_midline: float = 640.0
142
143
144
145 def _rally_points(spec: RallySpec, net_midline: float, frame_height: float
146                 ) -> List[Tuple[int, int, float, float]]:
147     """Deterministic in-flight shuttle: sine oscillation across the net midline (play

```

```

axis = x)."""
148     rows: List[Tuple[int, int, float, float]] = []
149     y_mid = frame_height / 2.0
150     for i in range(spec.n_frames):
151         x = net_midline + spec.amplitude * math.sin(2.0 * math.pi * i / spec.period)
152         # A small orthogonal wobble keeps y-spread < x-spread so the play axis is
unambiguously x.
153         y = y_mid + 30.0 * math.sin(2.0 * math.pi * i / (spec.period / 2.0))
154         rows.append((spec.start_frame + i, 1, x, y))
155     return rows
156
157
158     def _gap_points(spec: GapSpec, net_midline: float, frame_height: float
159                    ) -> List[Tuple[int, int, float, float]]:
160         """Dead-time rows: resting-visible (still) or absent (occluded)."""
161         rows: List[Tuple[int, int, float, float]] = []
162         y_mid = frame_height / 2.0
163         for i in range(spec.n_frames):
164             if spec.kind == "occluded":
165                 rows.append((spec.start_frame + i, 0, 0.0, 0.0))
166             elif spec.kind == "still":
167                 rows.append((spec.start_frame + i, 1, net_midline, y_mid))
168             else: # pragma: no cover - guarded by builder
169                 raise ValueError(f"unknown gap kind {spec.kind!r} (use 'still'/'occluded')")
170         return rows
171
172
173     def make_synthetic_trajectory(spec: FixtureSpec) -> List[Tuple[int, int, float, float]]:
174         """Build the deterministic ``(Frame, Visibility, X, Y)`` rows for a synthetic
scenario."""
175         rows: List[Tuple[int, int, float, float]] = []
176         for seg in spec.segments:
177             if isinstance(seg, RallySpec):
178                 rows.extend(_rally_points(seg, spec.net_midline, spec.frame_height))
179             elif isinstance(seg, GapSpec):
180                 rows.extend(_gap_points(seg, spec.net_midline, spec.frame_height))
181             else: # pragma: no cover - guarded by dataclass shape
182                 raise TypeError(f"unknown segment type {type(seg).__name__}")
183         rows.sort(key=lambda r: r[0])
184         return rows
185
186
187     def write_trajectory_csv(rows: Sequence[Tuple[int, int, float, float]], path: Path) ->
None:
188         """Write trajectory rows as the TrackNet CSV ``(Frame,Visibility,X,Y)``, LF + 6dp
floats."""
189         lines = ["Frame,Visibility,X,Y"]
190         for frame, vis, x, y in rows:
191             lines.append(f"{int(frame)},{int(vis)},{_fmt(x)},{_fmt(y)}")
192         path.write_text("\n".join(lines) + "\n", encoding="utf-8", newline="\n")
193
194
195     def write_labels_csv(gts: Sequence[Interval], path: Path) -> None:
196         """Write GT intervals as a ``start,end`` labels CSV (the strict gt_loader format),
LF."""
197         lines = ["start,end"]
198         for s, e in gts:
199             lines.append(f"{_fmt(s)},{_fmt(e)}")
200         path.write_text("\n".join(lines) + "\n", encoding="utf-8", newline="\n")
201
202
203     # =====
204     # Built-in synthetic scenarios (the 3 committed slugs)
205     # =====
206

```

```

207
208     def _fr(seconds: float, fps: float) -> int:
209         return int(round(seconds * fps))
210
211
212     def builtin_specs() -> List[FixtureSpec]:
213         """The committed synthetic scenarios. Deterministic; chosen so each exercises a
distinct
214         windowing behaviour (one window / dead-time split / occlusion split)."""
215         fps = 30.0
216         fw, fh = 1280.0, 720.0
217         net = 640.0
218
219         # 1) clean_single_rally: one ~4s in-flight rally ? exactly 1 candidate window,
crossings>0.
220         single = FixtureSpec(
221             slug="clean_single_rally",
222             description="One continuous ~4s rally; shuttle oscillates across the net ? 1
window.",
223             fps=fps, frame_width=fw, frame_height=fh, net_midline=net,
224             segments=(RallySpec(start_frame=0, n_frames=120),),
225             # Window is [0.033, 3.967]; GT is the rally span the synthetic motion fills.
226             gts=((0.0, 4.0),),
227         )
228
229         # 2) multi_rally_with_deadtime: rally, 3s STILL dead-time, rally ? 2 windows split
by the
230         # low-velocity gap (> merge_gap=2.0s) ? proves dead-time separates windows.
231         r0 = RallySpec(start_frame=0, n_frames=120)
232         gap_still = GapSpec(start_frame=120, n_frames=_fr(3.0, fps), kind="still")
233         r1 = RallySpec(start_frame=120 + _fr(3.0, fps), n_frames=120)
234         multi = FixtureSpec(
235             slug="multi_rally_with_deadtime",
236             description="Two ~4s rallies separated by 3s of still (resting-shuttle) dead-
time ? 2 windows.",
237             fps=fps, frame_width=fw, frame_height=fh, net_midline=net,
238             segments=(r0, gap_still, r1),
239             gts=((0.0, 4.0), (7.0, 11.0)),
240         )
241
242         # 3) occlusion_break: rally, 3s OCCLUDED (Visibility=0), rally ? 2 windows split by
the
243         # visibility break ? proves occlusion (not just low velocity) separates windows.
244         o0 = RallySpec(start_frame=0, n_frames=120)
245         gap_occ = GapSpec(start_frame=120, n_frames=_fr(3.0, fps), kind="occluded")
246         o1 = RallySpec(start_frame=120 + _fr(3.0, fps), n_frames=120)
247         occ = FixtureSpec(
248             slug="occlusion_break",
249             description="Two ~4s rallies separated by 3s of shuttle occlusion (Visibility=0)
? 2 windows.",
250             fps=fps, frame_width=fw, frame_height=fh, net_midline=net,
251             segments=(o0, gap_occ, o1),
252             gts=((0.0, 4.0), (7.0, 11.0)),
253         )
254
255         return [single, multi, occ]
256
257
258     def _spec_by_slug() -> Dict[str, FixtureSpec]:
259         return {s.slug: s for s in builtin_specs()}
260
261
262     # =====
263     # The shared replay (CONTROL / pure-stdlib path)
264     # =====

```

```

265
266
267     def _fmt(x: float) -> str:
268         """Render a float with up to 6dp, no scientific notation, ``-0.0`` normalised to
269         ``0.0``."""
270         r = round(float(x) + 0.0, _FLOAT_ROUND_DP)
271         if r == 0.0:
272             r = 0.0
273         return f"{r:.6f}"
274
275     def _round(x: float) -> float:
276         r = round(float(x) + 0.0, _FLOAT_ROUND_DP)
277         return 0.0 if r == 0.0 else r
278
279
280     def _pinned_proposal(prov: Optional[Dict[str, Any]]) -> Tuple[float, float, float]:
281         """The windowing proposal tuple, read from the fixture provenance when present (the
282         pinned
283         params travel with the fixture), else the module-level pinned defaults. NEVER
284         windowing.SERVED."""
285         if prov and "windowing" in prov:
286             w = prov["windowing"]
287             return (float(w["vt"]), float(w["mg"]), float(w["mwd"]))
288         return (PINNED_VT, PINNED_MERGE_GAP, PINNED_MIN_WINDOW)
289
290     def replay_fixture(slug_or_dir: Any) -> Dict[str, Any]:
291         """Replay one fixture through the shipped CONTROL path; return its recomputed
292         snapshot.
293         Resolves ``slug_or_dir`` (a slug string or a fixture directory ``Path``) to its
294         committed
295         ``trajectory.csv`` + ``labels.csv`` + ``provenance.json``, then:
296         parse_trajectory_csv ? build_candidates(proposal=pinned) ? VideoCandidates(5
297         shuttle
298         cols only) ? predict(CONTROL, vc) ? score(preds, gts, iou).
299         Returns ``{slug, schema_version, iou, candidates:[{start,end,shuttle:{5 names}}],
300         control_segments:[[s,e]], score:{...}, gt_hash}``. CONTROL keeps every candidate
301         then
302         merges overlaps; NO numpy is touched.
303         """
304         fdir = _resolve_dir(slug_or_dir)
305         slug = fdir.name
306         traj_csv = fdir / "trajectory.csv"
307         labels_csv = fdir / "labels.csv"
308         prov = _read_json(fdir / "provenance.json") if (fdir / "provenance.json").is_file()
309         else None
310
311         fps, fw = _fps_fw(prov, slug)
312         iou = float(prov["iou"]) if (prov and "iou" in prov) else 0.5
313         proposal = _pinned_proposal(prov)
314
315         intervals, feature_rows = distill_local.build_candidates(
316             str(traj_csv), fps=fps, frame_width=fw, proposal=proposal)
317
318         # Build VideoCandidates DIRECTLY with ONLY the 5 shuttle feature columns. We never
319         call
320         # fusion_compare.load_videos / ablation.load_videos ? those read c['person']
321         unconditionally
322         # and would KeyError on these person-free fixtures (HARD CONSTRAINT 1).
323         features: Dict[str, List[float]] = {
324             name: [row[i] for row in feature_rows] for i, name in

```



```

enumerate(SHUTTLE_FEATURES)
320     }
321     gts = load_gt_intervals(str(labels_csv), strict=True)
322     vc = VideoCandidates(name=slug, intervals=intervals, features=features, gts=gts)
323
324     preds = predict(CONTROL, vc) # CONTROL: keep all ? merge_overlaps; pure-stdlib, no
numpy
325     sc = score(preds, gts, iou)
326
327     candidates: List[Dict[str, Any]] = []
328     for j, (s, e) in enumerate(intervals):
329         shuttle = {name: _round(feature_rows[j][i]) for i, name in
enumerate(SHUTTLE_FEATURES)}
330         # net_crossings is conceptually an integer count ? keep it exact (int), per
constraint 2.
331         shuttle["net_crossings"] =
int(round(feature_rows[j][SHUTTLE_FEATURES.index("net_crossings")]))
332         candidates.append({"start": _round(s), "end": _round(e), "shuttle": shuttle})
333
334     return {
335         "slug": slug,
336         "schema_version": CURRENT_SCHEMA,
337         "iou": _round(iou),
338         "candidates": candidates,
339         "control_segments": [[_round(s), _round(e)] for s, e in preds],
340         "score": {
341             "iou_threshold": _round(sc.iou_threshold),
342             "num_pred": int(sc.num_pred),
343             "num_gt": int(sc.num_gt),
344             "true_positives": int(sc.true_positives),
345             "false_positives": int(sc.false_positives),
346             "false_negatives": int(sc.false_negatives),
347             "precision": _round(sc.precision),
348             "recall": _round(sc.recall),
349             "f1": _round(sc.f1),
350             "mean_iou": _round(sc.mean_iou),
351             "mean_start_error": _round(sc.mean_start_error),
352             "mean_end_error": _round(sc.mean_end_error),
353         },
354         "gt_hash": gt_hash(gts),
355     }
356
357
358     # =====
359     # Freeze / refresh (expected.json + provenance.json are correct-by-construction)
360     # =====
361
362
363     def _provenance(spec_or_dir: Any, slug: str, fps: float, fw: float, iou: float = 0.5
364                     ) -> Dict[str, Any]:
365         """The committed provenance block ? pins the windowing + privacy/licensing tags
($4c, §5)."""
366         desc = ""
367         if isinstance(spec_or_dir, FixtureSpec):
368             desc = spec_or_dir.description
369         return {
370             "slug": slug,
371             "schema_version": CURRENT_SCHEMA,
372             "kind": "synthetic",
373             "license": "synthetic",
374             "minors_free": True,
375             "description": desc,
376             "fps": _round(fps),
377             "frame_width": _round(fw),
378             "iou": _round(iou),

```

```

379         "windowing": {
380             "vt": PINNED_VT,
381             "mg": PINNED_MERGE_GAP,
382             "mwd": PINNED_MIN_WINDOW,
383             "max_win": PINNED_MAX_WINDOW,
384         },
385         "paths": {
386             "trajectory": "trajectory.csv",
387             "labels": "labels.csv",
388             "expected": "expected.json",
389         },
390         "note": (
391             "Synthetic, deterministic, owned (no real footage). Tier-0 / CONTROL-only. "
392             "Plumbing-correctness fixture ? NOT a measure of rally-detection quality
($6).",
393         ),
394     }
395
396
397     def freeze_fixture(slug: str, fixtures_dir: Path = FIXTURES_DIR,
398                       iou: float = 0.5) -> Dict[str, Any]:
399         """Compute ``expected.json`` + ``provenance.json`` for one fixture from its
COMMITTED
400         ``trajectory.csv`` + ``labels.csv`` via the replay (so expected is correct-by-
construction).
401
402         Idempotent: rewrites the same bytes given the same committed inputs. Requires the
trajectory
403         + labels CSVs to already exist (use ``--freeze-synthetic`` to generate them first).
404         """
405         fdir = fixtures_dir / slug
406         traj_csv, labels_csv = fdir / "trajectory.csv", fdir / "labels.csv"
407         if not traj_csv.is_file() or not labels_csv.is_file():
408             raise FileNotFoundError(
409                 f"{slug}: trajectory.csv/labels.csv missing ? run --freeze-synthetic to
generate them")
410
411         # Provenance first (it pins the windowing the replay reads), then replay ? expected.
412         spec = _spec_by_slug().get(slug)
413         fps, fw = (spec.fps, spec.frame_width) if spec is not None else
_fps_fw_from_prov(fdir, slug)
414         prov = _provenance(spec if spec is not None else fdir, slug, fps, fw, iou)
415         _write_json(fdir / "provenance.json", prov)
416
417         expected = replay_fixture(fdir)
418         _write_json(fdir / "expected.json", expected)
419         return expected
420
421
422     def freeze_synthetic(fixtures_dir: Path = FIXTURES_DIR) -> List[str]:
423         """Generate trajectory.csv + labels.csv for every built-in scenario, freeze
expected.json +
424         provenance.json, and (re)write the manifest. Returns the slugs frozen."""
425         specs = builtin_specs()
426         fixtures_dir.mkdir(parents=True, exist_ok=True)
427         slugs: List[str] = []
428         for spec in specs:
429             fdir = fixtures_dir / spec.slug
430             fdir.mkdir(parents=True, exist_ok=True)
431             rows = make_synthetic_trajectory(spec)
432             write_trajectory_csv(rows, fdir / "trajectory.csv")
433             write_labels_csv(spec.gts, fdir / "labels.csv")
434             freeze_fixture(spec.slug, fixtures_dir=fixtures_dir)
435             slugs.append(spec.slug)
436         _write_manifest(specs, fixtures_dir)

```

```

437         return slugs
438
439
440     def refresh_snapshot(slug: Optional[str] = None, all_slugs: bool = False,
441                          fixtures_dir: Path = FIXTURES_DIR) -> List[str]:
442         """Re-freeze expected.json (+ provenance.json) from the COMMITTED trajectory.csv +
443         labels.csv ?
444         NO regeneration of the trajectory/labels. ? §6: this can rubber-stamp a real
445         regression; the
446         mitigation is reviewing the visible diff + recording a reason (the CLI requires
447         one)."""
448         if all_slugs:
449             targets = [d.name for d in sorted(fixtures_dir.iterdir())
450                        if d.is_dir() and (d / "trajectory.csv").is_file()]
451         elif slug:
452             targets = [slug]
453         else:
454             raise ValueError("refresh_snapshot needs --slug S or --all")
455         for s in targets:
456             freeze_fixture(s, fixtures_dir=fixtures_dir)
457         return targets
458
459
460     # =====
461     # Manifest + loaders
462     # =====
463
464     def _write_manifest(specs: Sequence[FixtureSpec], fixtures_dir: Path) -> None:
465         entries = [{
466             "slug": spec.slug,
467             "fps": _round(spec.fps),
468             "frame_width": _round(spec.frame_width),
469             "windowing": {
470                 "vt": PINNED_VT, "mg": PINNED_MERGE_GAP,
471                 "mwd": PINNED_MIN_WINDOW, "max_win": PINNED_MAX_WINDOW,
472             },
473             "kind": "synthetic",
474             "schema_version": CURRENT_SCHEMA,
475             "paths": {
476                 "dir": spec.slug,
477                 "trajectory": f"{spec.slug}/trajectory.csv",
478                 "labels": f"{spec.slug}/labels.csv",
479                 "expected": f"{spec.slug}/expected.json",
480                 "provenance": f"{spec.slug}/provenance.json",
481             },
482         } for spec in specs]
483         _write_json(fixtures_dir / "manifest.json", entries)
484
485
486     def load_manifest(fixtures_dir: Path = FIXTURES_DIR) -> List[Dict[str, Any]]:
487         """Load the fixtures manifest (returns [] if absent)."""
488         path = fixtures_dir / "manifest.json"
489         if not path.is_file():
490             return []
491         data = _read_json(path)
492         if not isinstance(data, list): # pragma: no cover - corrupt manifest
493             raise ValueError(f"{path}: manifest must be a JSON array")
494         return data
495
496
497     def load_fixture(slug: str, fixtures_dir: Path = FIXTURES_DIR) -> Dict[str, Any]:
498         """Load one fixture's committed artifacts: ``{provenance, expected, dir}``. Refuses
499         a fixture
500         whose ``schema_version`` exceeds ``CURRENT_SCHEMA`` (forward-incompatible)."""

```

```

498     fdir = fixtures_dir / slug
499     prov = _read_json(fdir / "provenance.json")
500     expected = _read_json(fdir / "expected.json")
501     sv = int(expected.get("schema_version", prov.get("schema_version", 0)))
502     if sv > CURRENT_SCHEMA:
503         raise ValueError(
504             f"{slug}: schema_version {sv} > CURRENT_SCHEMA {CURRENT_SCHEMA} ? upgrade
the harness")
505     return {"slug": slug, "dir": fdir, "provenance": prov, "expected": expected}
506
507
508     # =====
509     # Parse-compare (NEVER byte-compare): exact ints/structure, abs-tol floats
510     # =====
511
512
513     def compare_expected(recomputed: Dict[str, Any], committed: Dict[str, Any]) ->
List[str]:
514         """Return a list of human-readable mismatch messages (empty == match).
515
516         Parse-compare per HARD CONSTRAINT 2: structural/int fields (candidate count,
net_crossings,
517         TP/FP/FN, segment count, num_pred/num_gt, slug, schema_version, gt_hash) must be
EXACTLY
518         equal; float fields compared with abs tolerance 1e-6. Never a byte compare ?
CRLF/float-format
519         safe (the repo runs with core.autocrlf=true)."""
520         out: List[str] = []
521
522         def exact(path: str, a: Any, b: Any) -> None:
523             if a != b:
524                 out.append(f"{path}: {a!r} != {b!r} (exact)")
525
526         def approx(path: str, a: Any, b: Any) -> None:
527             try:
528                 if abs(float(a) - float(b)) > _FLOAT_TOL:
529                     out.append(f"{path}: {a!r} != {b!r} (|?| > {_FLOAT_TOL})")
530             except (TypeError, ValueError):
531                 out.append(f"{path}: non-numeric float field {a!r} vs {b!r}")
532
533         exact("slug", recomputed.get("slug"), committed.get("slug"))
534         exact("schema_version", recomputed.get("schema_version"),
committed.get("schema_version"))
535         exact("gt_hash", recomputed.get("gt_hash"), committed.get("gt_hash"))
536         approx("iou", recomputed.get("iou"), committed.get("iou"))
537
538         rc, cc = recomputed.get("candidates", []), committed.get("candidates", [])
539         if len(rc) != len(cc):
540             out.append(f"candidates: count {len(rc)} != {len(cc)} (exact)")
541         else:
542             for i, (r, c) in enumerate(zip(rc, cc)):
543                 approx(f"candidates[{i}].start", r.get("start"), c.get("start"))
544                 approx(f"candidates[{i}].end", r.get("end"), c.get("end"))
545                 rs, cs = r.get("shuttle", {}), c.get("shuttle", {})
546                 if set(rs) != set(cs):
547                     out.append(f"candidates[{i}].shuttle: keys {sorted(rs)} != {sorted(cs)}
(exact)")
548             else:
549                 for k in rs:
550                     if k == "net_crossings":
551                         exact(f"candidates[{i}].shuttle.net_crossings", rs[k], cs[k])
552                     else:
553                         approx(f"candidates[{i}].shuttle.{k}", rs[k], cs[k])
554
555         rseg, cseg = recomputed.get("control_segments", []),

```

```

committed.get("control_segments", [])
556     if len(rseg) != len(cseg):
557         out.append(f"control_segments: count {len(rseg)} != {len(cseg)} (exact)")
558     else:
559         for i, (r, c) in enumerate(zip(rseg, cseg)):
560             approx(f"control_segments[{i}][0]", r[0], c[0])
561             approx(f"control_segments[{i}][1]", r[1], c[1])
562
563     rsc, csc = recomputed.get("score", {}), committed.get("score", {})
564     int_fields = ("num_pred", "num_gt", "true_positives", "false_positives",
565 "false_negatives")
566     float_fields = ("iou_threshold", "precision", "recall", "f1", "mean_iou",
567 "mean_start_error", "mean_end_error")
568     for k in int_fields:
569         exact(f"score.{k}", rsc.get(k), csc.get(k))
570     for k in float_fields:
571         approx(f"score.{k}", rsc.get(k), csc.get(k))
572
573     return out
574
575 # =====
576 # Small JSON / path helpers (deterministic writers: sort_keys, LF, trailing newline)
577 # =====
578
579
580 def _write_json(path: Path, obj: Any) -> None:
581     text = json.dumps(obj, indent=2, sort_keys=True, ensure_ascii=False)
582     path.write_text(text + "\n", encoding="utf-8", newline="\n")
583
584
585 def _read_json(path: Path) -> Any:
586     with open(path, encoding="utf-8") as f:
587         return json.load(f)
588
589
590 def _resolve_dir(slug_or_dir: Any) -> Path:
591     if isinstance(slug_or_dir, Path):
592         return slug_or_dir
593     p = Path(str(slug_or_dir))
594     if p.is_dir():
595         return p
596     return FIXTURES_DIR / str(slug_or_dir)
597
598
599 def _fps_fw(prov: Optional[Dict[str, Any]], slug: str) -> Tuple[float, float]:
600     if prov and "fps" in prov and "frame_width" in prov:
601         return float(prov["fps"]), float(prov["frame_width"])
602     spec = _spec_by_slug().get(slug)
603     if spec is not None:
604         return spec.fps, spec.frame_width
605     raise ValueError(f"{slug}: no fps/frame_width in provenance and not a built-in
spec")
606
607
608 def _fps_fw_from_prov(fdir: Path, slug: str) -> Tuple[float, float]:
609     prov_path = fdir / "provenance.json"
610     prov = _read_json(prov_path) if prov_path.is_file() else None
611     return _fps_fw(prov, slug)
612
613
614 # =====
615 # CLI
616 # =====
617

```

```

618
619     def _cmd_check(fixtures_dir: Path) -> int:
620         manifest = load_manifest(fixtures_dir)
621         if not manifest:
622             print("[golden-fixtures] no fixtures found ? run --freeze-synthetic first",
file=sys.stderr)
623             return 1
624         failures = 0
625         for entry in manifest:
626             slug = entry["slug"]
627             committed = _read_json(fixtures_dir / slug / "expected.json")
628             recomputed = replay_fixture(fixtures_dir / slug)
629             mismatches = compare_expected(recomputed, committed)
630             if mismatches:
631                 failures += 1
632                 print(f"[FAIL] {slug}: {len(mismatches)} mismatch(es) "
633                     f"{characterization = behaviour-locking, NOT correctness ? $6}:")
634                 for m in mismatches:
635                     print(f"    - {m}")
636             else:
637                 sc = recomputed["score"]
638                 print(f"[ok] {slug}: {len(recomputed['candidates'])} candidates, "
639                     f"{len(recomputed['control_segments'])} segments, F1={sc['f1']:.6f}")
640             print(f"[golden-fixtures] {len(manifest) - failures}/{len(manifest)} fixtures match
"
641                 f"(synthetic-tier = plumbing correctness, not field quality ? $6)")
642             return 0 if failures == 0 else 1
643
644
645     def main(argv: Optional[Sequence[str]] = None) -> int:
646         ap = argparse.ArgumentParser(
647             description="Golden Regression Fixtures (M1) ? synthetic, CONTROL-only replay
harness.")
648         g = ap.add_mutually_exclusive_group(required=True)
649         g.add_argument("--freeze-synthetic", action="store_true",
650             help="generate synthetic trajectory+labels for all built-in
scenarios, then "
651                 "freeze expected.json+provenance.json + write manifest.json")
652         g.add_argument("--refresh-snapshot", action="store_true",
653             help="re-freeze expected.json from COMMITTED trajectory+labels (no
regeneration)")
654         g.add_argument("--check", action="store_true",
655             help="replay all fixtures + parse-compare vs committed expected.json
(exit 0/1)")
656         ap.add_argument("--slug", help="single slug for --refresh-snapshot")
657         ap.add_argument("--all", action="store_true", help="all slugs for --refresh-
snapshot")
658         ap.add_argument("--reason", help="recorded reason for --refresh-snapshot (review the
diff!)")
659         ap.add_argument("--fixtures-dir", type=Path, default=FIXTURES_DIR,
660             help="override the fixtures directory (default: repo
eval_baselines/fixtures)")
661         args = ap.parse_args(list(argv) if argv is not None else None)
662
663         fixtures_dir: Path = args.fixtures_dir
664
665         if args.freeze_synthetic:
666             slugs = freeze_synthetic(fixtures_dir)
667             print(f"[golden-fixtures] froze {len(slugs)} synthetic fixture(s): {'
'.join(slugs)}")
668             print(f"    -> {fixtures_dir}")
669             return 0
670
671         if args.refresh_snapshot:
672             if not args.slug and not args.all:

```

```

673         ap.error("--refresh-snapshot requires --slug S or --all")
674     if not args.reason:
675         print("[golden-fixtures] WARNING: no --reason given. --refresh-snapshot can
"
676             "rubber-stamp a real regression ? review the visible diff ($6).",
file=sys.stderr)
677     targets = refresh_snapshot(slug=args.slug, all_slugs=args.all,
fixtures_dir=fixtures_dir)
678     print(f"[golden-fixtures] refreshed snapshot for: {' ', '.join(targets)}"
679         + (f" (reason: {args.reason})" if args.reason else ""))
680     return 0
681
682     # --check
683     return _cmd_check(fixtures_dir)
684
685
686 if __name__ == "__main__":
687     raise SystemExit(main())
688

```

143. assistant (2026-06-16T04:53:48.212Z)

The module is high quality ? CONTROL-only, parse-compare, pinned preset, person-free by construction, deterministic writers. Let me review the test + a generated fixture, then run the verification myself.

144. user (2026-06-16T04:53:49.167Z)

```

1     """Golden Regression Fixtures (M1) ? characterization tests over the committed synthetic
Tier-0.
2
3     Parametrized over ``load_manifest()`` so every committed fixture is exercised. The gate
is
4     **parse-compare** (never byte-compare), **CONTROL-only** (pure stdlib, no numpy on the
replay
5     path), and asserts the Tier-0 privacy invariant positively (no
person/audio/pose/gait/face keys).
6
7     $6 honest-limits caveat: green here proves only that the deterministic post-trajectory
transforms
8     are unchanged for the frozen synthetic cases ? characterization is behaviour-locking,
NOT
9     correctness, and synthetic-tier numbers measure plumbing, not field quality.
10    """
11    import json
12
13    import pytest
14
15    from backend.eval import golden_fixtures as gf
16
17    # Slugs discovered from the committed manifest ? the suite grows by accretion
automatically.
18    MANIFEST = gf.load_manifest()
19    SLUGS = [entry["slug"] for entry in MANIFEST]
20
21
22    def test_manifest_is_non_empty():
23        """A committed manifest with at least one fixture must exist (else nothing is
gated)."""
24        assert SLUGS, "no fixtures in eval_baselines/fixtures/manifest.json ? run --freeze-
synthetic"
25
26
27    @pytest.mark.parametrize("slug", SLUGS)
28    def test_replay_matches_committed_expected(slug):

```

```

29         """Replay the fixture and parse-compare against its committed expected.json.
30
31         Characterization = behaviour-locking, NOT correctness (§6): a non-empty mismatch
list means
32         the deterministic post-trajectory behaviour CHANGED for this frozen case, not that
it is wrong.
33         """
34         committed = gf.load_fixture(slug)["expected"]
35         recomputed = gf.replay_fixture(slug)
36         mismatches = gf.compare_expected(recomputed, committed)
37         assert mismatches == [], (
38             f"{slug}: behaviour changed vs frozen snapshot (characterization, not
correctness ? §6):\n"
39             + "\n".join(f" - {m}" for m in mismatches)
40         )
41
42
43     @pytest.mark.parametrize("slug", SLUGS)
44     def test_schema_version_le_current(slug):
45         """Every committed fixture's schema_version must be <= CURRENT_SCHEMA (forward-
compatible)."""
46         fx = gf.load_fixture(slug)
47         assert int(fx["expected"]["schema_version"]) <= gf.CURRENT_SCHEMA
48         assert int(fx["provenance"]["schema_version"]) <= gf.CURRENT_SCHEMA
49
50
51     @pytest.mark.parametrize("slug", SLUGS)
52     def test_no_person_or_audio_keys(slug):
53         """Positive privacy assertion (D3 / §4c): NO biometric/identity/non-shuttle key
appears
54         anywhere in the committed expected.json ? not trusting a silent 0.0 default."""
55         fdir = gf.FIXTURES_DIR / slug
56         raw = (fdir / "expected.json").read_text(encoding="utf-8").lower()
57         for forbidden in gf.FORBIDDEN_KEYS:
58             assert forbidden not in raw, (
59                 f"{slug}: forbidden key substring {forbidden!r} present in expected.json
(Tier-0 leak)")
60
61         expected = gf.load_fixture(slug)["expected"]
62         for i, cand in enumerate(expected["candidates"]):
63             # The only feature block is the shuttle block, and it holds EXACTLY the 5
shuttle cols.
64             assert set(cand.keys()) == {"start", "end", "shuttle"}, f"{slug}: candidate[{i}]
has extra keys"
65             assert set(cand["shuttle"].keys()) == set(gf.SHUTTLE_FEATURES), (
66                 f"{slug}: candidate[{i}].shuttle is not exactly the 5 shuttle features")
67
68
69     @pytest.mark.parametrize("slug", SLUGS)
70     def test_synthetic_provenance_tags(slug):
71         """Tier-0 hard tags: kind=synthetic, license=synthetic, minors_free=true (constraint
5)."""
72         prov = gf.load_fixture(slug)["provenance"]
73         assert prov["kind"] == "synthetic"
74         assert prov["license"] == "synthetic"
75         assert prov["minors_free"] is True
76         # Windowing pinned in the fixture (not the named SERVED constant) ? constraint 3.
77         assert set(prov["windowing"]) == {"vt", "mg", "mwd", "max_win"}
78
79
80     def test_control_is_not_learned():
81         """The replay must use the pure-stdlib CONTROL variant (no numpy/LogisticRegression
? D5)."""
82         from backend.eval.experiment import CONTROL
83         assert CONTROL.is_learned() is False

```



```

84
85
86     def test_perturbing_trajectory_changes_segments(tmp_path):
87         """Mutating a copy of a trajectory and replaying yields DIFFERENT segments ? proving
the
88         windows are RE-DERIVED from the CSV each run, not merely echoed from
expected.json."""
89         slug = SLUGS[0]
90         src = gf.FIXTURES_DIR / slug
91         work = tmp_path / slug
92         work.mkdir()
93         # Copy the fixture, then TRUNCATE the trajectory to its first half so the derived
window's
94         # `end` shrinks ? a guaranteed structural change in the re-derived segments.
(Deleting an
95         # interior hole would not move start/end, since merge_gap coalesces across it and
the frame
96         # indices of the surviving rows are unchanged.)
97         lines = (src / "trajectory.csv").read_text(encoding="utf-8").splitlines()
98         header, body = lines[0], lines[1:]
99         mutated = [header] + body[: len(body) // 2]
100         (work / "trajectory.csv").write_text("\n".join(mutated) + "\n", encoding="utf-8",
newline="\n")
101         (work / "labels.csv").write_text((src / "labels.csv").read_text(encoding="utf-8"),
encoding="utf-8", newline="\n")
102         (work / "provenance.json").write_text((src /
"provenance.json").read_text(encoding="utf-8"),
103         encoding="utf-8", newline="\n")
104
105         original = gf.load_fixture(slug)["expected"]
106         perturbed = gf.replay_fixture(work)
107         assert perturbed["control_segments"] != original["control_segments"], (
108             "perturbed trajectory produced identical segments ? replay is echoing, not re-
deriving")
109
110
111
112     def test_freeze_is_idempotent(tmp_path):
113         """Re-freezing every built-in fixture into a temp dir produces expected.json
structurally
114         equal to the committed one (correct-by-construction + deterministic)."""
115         slugs = gf.freeze_synthetic(fixtures_dir=tmp_path)
116         assert slugs, "freeze_synthetic produced no fixtures"
117         for slug in slugs:
118             committed = gf.load_fixture(slug)["expected"]
119             refrozen = json.loads((tmp_path / slug /
"expected.json").read_text(encoding="utf-8"))
120             # Structural (parsed) equality ? no float churn, exact same shape/values.
121             assert gf.compare_expected(refrozen, committed) == [], (
122                 f"{slug}: re-frozen expected.json differs from committed (non-idempotent
freeze)")
123             assert refrozen == committed, f"{slug}: re-frozen expected.json not structurally
identical"
124
125
126     def test_check_cli_passes(capsys):
127         """The --check CLI replays all fixtures and exits 0 when they match the committed
snapshots."""
128         rc = gf.main(["--check"])
129         assert rc == 0
130

```

145. user (2026-06-16T04:53:50.102Z)

```

1      {
2      "candidates": [

```

```

3      {
4        "end": 3.966667,
5        "shuttle": {
6          "active_ratio": 1.008475,
7          "duration": 3.933333,
8          "mean_velocity": 1.27678,
9          "net_crossings": 7,
10         "peak_velocity": 1.970041
11       },
12       "start": 0.033333
13     }
14   ],
15   "control_segments": [
16     [
17       0.033333,
18       3.966667
19     ]
20   ],
21   "gt_hash": "6c93801fd607",
22   "iou": 0.5,
23   "schema_version": 1,
24   "score": {
25     "f1": 1.0,
26     "false_negatives": 0,
27     "false_positives": 0,
28     "iou_threshold": 0.5,
29     "mean_end_error": 0.033333,
30     "mean_iou": 0.983333,
31     "mean_start_error": 0.033333,
32     "num_gt": 1,
33     "num_pred": 1,
34     "precision": 1.0,
35     "recall": 1.0,
36     "true_positives": 1
37   },
38   "slug": "clean_single_rally"
39 }
40

```

146. user (2026-06-16T04:53:50.649Z)

```

1   [
2     {
3       "fps": 30.0,
4       "frame_width": 1280.0,
5       "kind": "synthetic",
6       "paths": {
7         "dir": "clean_single_rally",
8         "expected": "clean_single_rally/expected.json",
9         "labels": "clean_single_rally/labels.csv",
10        "provenance": "clean_single_rally/provenance.json",
11        "trajectory": "clean_single_rally/trajectory.csv"
12      },
13       "schema_version": 1,
14       "slug": "clean_single_rally",
15       "windowing": {
16         "max_win": 0.0,
17         "mg": 2.0,
18         "mwd": 2.0,
19         "vt": 0.02
20      }
21     },
22     {
23       "fps": 30.0,
24       "frame_width": 1280.0,

```

```

25     "kind": "synthetic",
26     "paths": {
27         "dir": "multi_rally_with_deadtime",
28         "expected": "multi_rally_with_deadtime/expected.json",
29         "labels": "multi_rally_with_deadtime/labels.csv",
30         "provenance": "multi_rally_with_deadtime/provenance.json",
31         "trajectory": "multi_rally_with_deadtime/trajectory.csv"
32     },
33     "schema_version": 1,
34     "slug": "multi_rally_with_deadtime",
35     "windowing": {
36         "max_win": 0.0,
37         "mg": 2.0,
38         "mwd": 2.0,
39         "vt": 0.02
40     }
41 },
42 {
43     "fps": 30.0,
44     "frame_width": 1280.0,
45     "kind": "synthetic",
46     "paths": {
47         "dir": "occlusion_break",
48         "expected": "occlusion_break/expected.json",
49         "labels": "occlusion_break/labels.csv",
50         "provenance": "occlusion_break/provenance.json",
51         "trajectory": "occlusion_break/trajectory.csv"
52     },
53     "schema_version": 1,
54     "slug": "occlusion_break",
55     "windowing": {
56         "max_win": 0.0,
57         "mg": 2.0,
58         "mwd": 2.0,
59         "vt": 0.02
60     }
61 }
62 ]
63

```

147. assistant (2026-06-16T04:54:09.508Z)

Both the test file and the generated fixture look correct (parametrized over the manifest, positive privacy assertion, perturbation + idempotence tests; `clean\_single\_rally` ? 1 candidate, `net\_crossings:7` as int, F1=1.0, floats at 6dp). Now my own verification gate ? full suite + ruff + mypy + the `--check` CLI, in the worktree:

148. user (2026-06-16T04:56:28.420Z)

```

===== 1. --check CLI =====
[ok] clean_single_rally: 1 candidates, 1 segments, F1=1.000000
[ok] multi_rally_with_deadtime: 2 candidates, 2 segments, F1=1.000000
[ok] occlusion_break: 2 candidates, 2 segments, F1=1.000000
[golden-fixtures] 3/3 fixtures match (synthetic-tier = plumbing correctness, not field quality ?
$6)
CHECK_EXIT=0
===== 2. full suite (coverage, CI floor 70) =====
backend\providers\gemini.py          98      19      81%
backend\reporting\__init__.py        26       2      92%
backend\reporting\harvest.py         111       8      93%
backend\reporting\run_signals.py      14       0     100%
backend\results.py                   22       0     100%
backend\sports\__init__.py           17       1      94%
backend\sports\badminton.py          13       1      92%
backend\sports\base.py               12       3      75%

```

backend\tools__init__.py	0	0	100%
backend\tools\export.py	75	2	97%
backend\tools\rally_slicer.py	104	47	55%
backend\utils__init__.py	0	0	100%
backend\utils\geometry.py	29	0	100%
backend\utils\hashing.py	9	0	100%
backend\utils\logging_setup.py	11	0	100%
backend\utils\paths.py	27	3	89%
backend\utils\run_telemetry.py	126	12	90%
backend\utils\video.py	25	3	88%
training__init__.py	0	0	100%
training\gen0__init__.py	0	0	100%
training\gen0\harness.py	108	11	90%

TOTAL	7552	1483	80%
-------	------	------	-----

Required test coverage of 70% reached. Total coverage: 80.36%

744 passed, 7 skipped, 7 warnings in 125.68s (0:02:05)

PYTEST_EXIT=0

===== 3. ruff =====

All checks passed!

RUFF_EXIT=0

===== 4. mypy =====

backend\annotations\base.py:85: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

backend\sports__init__.py:7: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

backend\providers__init__.py:7: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

Success: no issues found in 103 source files

MYPY_EXIT=0

149. assistant (2026-06-16T04:57:18.660Z)

Independently verified ? all green:

- `--check`: 3/3 fixtures match, exit 0.
- **Full suite: 744 passed, 7 skipped**; coverage **80.36% ? 70%** floor.
- ruff: clean. mypy: clean (103 files).

That clears my pre-LGTM gate. Committing with branching-safety discipline (re-verify HEAD, explicit paths only), then pushing and opening the PR:

150. user (2026-06-16T04:57:23.282Z)

=== origin/master ===

a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)

=== my branch/HEAD ===

feat/golden-fixtures-m1

1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)

warning: in the working copy of '.gitattributes', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'backend/eval/golden_fixtures.py', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'tests/test_golden_fixtures.py', LF will be replaced by CRLF the next time Git touches it

=== staged (explicit paths only) ===

.gitattributes

backend/eval/golden_fixtures.py

eval_baselines/fixtures/README.md

eval_baselines/fixtures/clean_single_rally/expected.json

eval_baselines/fixtures/clean_single_rally/labels.csv

eval_baselines/fixtures/clean_single_rally/provenance.json

eval_baselines/fixtures/clean_single_rally/trajectory.csv

eval_baselines/fixtures/manifest.json

```

eval_baselines/fixtures/multi_rally_with_deadtime/expected.json
eval_baselines/fixtures/multi_rally_with_deadtime/labels.csv
eval_baselines/fixtures/multi_rally_with_deadtime/provenance.json
eval_baselines/fixtures/multi_rally_with_deadtime/trajectory.csv
eval_baselines/fixtures/occlusion_break/expected.json
eval_baselines/fixtures/occlusion_break/labels.csv
eval_baselines/fixtures/occlusion_break/provenance.json
eval_baselines/fixtures/occlusion_break/trajectory.csv
tests/test_golden_fixtures.py
=== nothing else swept ===

```

151. assistant (2026-06-16T04:58:08.684Z)

Staged cleanly ? exactly my files, nothing swept. But note **master** moved again to `a471745` (#188, "Tier 1 W2 net-crossings gate") ? the concurrent session is iterating fast on the eval surface. My fixtures were frozen against `1de2817` (which already includes #187's bounded windowing, since I pinned `max\_win=0` they're robust to that). I must **rebase** onto the latest master and re-verify that #188 didn't change my CONTROL/windowing replay before I merge ? otherwise the post-merge master could fail. Let me commit, rebase, and re-run `--check`.

152. user (2026-06-16T04:58:22.018Z)

```

=== committed ===
8239609 feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness
=== rebase onto origin/master (a471745, incl #187+#188) ===
Rebasing (1/1)

Successfully rebased and updated refs/heads/feat/golden-fixtures-m1.
=== re-verify --check against rebased code ===
[ok]   clean_single_rally: 1 candidates, 1 segments, F1=1.000000
[ok]   multi_rally_with_deadtime: 2 candidates, 2 segments, F1=1.000000
[ok]   occlusion_break: 2 candidates, 2 segments, F1=1.000000
[golden-fixtures] 3/3 fixtures match (synthetic-tier = plumbing correctness, not field quality ?
$6)
CHECK_EXIT=0

```

153. assistant (2026-06-16T04:58:36.647Z)

The rebase onto `a471745` (with #187 **and** #188) still passes `--check` 3/3 ? empirical proof my pinned-preset + CONTROL-only design is **robust** to their windowing/gate churn (their bounded-windowing is OFF in my pinned preset, and #188's gate only affects gated variants, not CONTROL). That's the determinism-safe choice paying off. Let me run the full suite on the rebased branch (now includes their new tests), then push and open the PR.

154. user (2026-06-16T05:00:19.954Z)

```

=== full suite on rebased branch (incl #187/#188 tests + mine) ===
-----
TOTAL                                7567    1482    80%
Required test coverage of 70% reached. Total coverage: 80.41%
746 passed, 7 skipped, 7 warnings in 92.43s (0:01:32)
PYTEST_EXIT=0
=== only my commit vs origin/master ===
84f8428 feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness
=== push ===
remote:
branch 'feat/golden-fixtures-m1' set up to track 'origin/feat/golden-fixtures-m1'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      feat/golden-fixtures-m1 -> feat/golden-fixtures-m1

```

155. assistant (2026-06-16T05:00:35.252Z)

Post-rebase: ****746 passed, 7 skipped, coverage 80.41%**** (their #188 added tests too), and only my single commit sits atop master. Pushed. Opening the PR:

156. user (2026-06-16T05:00:51.442Z)

<https://github.com/avidullu/badminton-highlight-indexer/pull/189>

157. assistant (2026-06-16T05:01:10.835Z)

PR opened: ****[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189)****. Now the real cross-OS determinism test ? I froze the fixtures on Windows; CI replays on Linux. If a float drifts beyond the 1e-6 tolerance, CI catches it. Let me watch the checks before merging.

158. user (2026-06-16T05:03:18.501Z)

```
Mypy (lenient baseline) pass      36s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271762
Ruff lint                pass      8s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271701
Full test suite (offline, mocked) pending 0      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271726
Syntax sweep + import smoke (no GPU stack) pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271785
Full test suite (offline, mocked) pass    2m9s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271726
Mypy (lenient baseline) pass      36s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271762
Ruff lint                pass      8s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271701
Syntax sweep + import smoke (no GPU stack) pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271785
Full test suite (offline, mocked) pass    2m9s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271726
Mypy (lenient baseline) pass      36s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271762
Ruff lint                pass      8s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271701
Syntax sweep + import smoke (no GPU stack) pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27595298127/job/81584271785
CHECKS_EXIT=0
```

159. assistant (2026-06-16T05:04:06.789Z)

****All CI checks pass on Linux**** (``CHECKS_EXIT=0``) ? including the full suite replaying my Windows-frozen fixtures on Linux. That empirically confirms ****cross-OS determinism holds**** for the CONTROL path (the big risk, cleared). Merging per "merge as you go":

160. user (2026-06-16T05:04:13.899Z)

failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'

```
=== master now ===
64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization harness (#189)
a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)
```

161. assistant (2026-06-16T05:04:45.131Z)

M1 is ****merged**** ? master is now ``64494ef`` (#189). (The ``gh`` error was only its local-branch cleanup step: it couldn't switch the local checkout to ``master`` because the concurrent session has ``master`` checked out in *its own* worktree ``tier1-windowing`` ? which incidentally confirms we're both safely isolated in worktrees. The merge itself succeeded.) Let me clean up my worktree and confirm the remote branch is gone.

162. user (2026-06-16T05:04:50.604Z)

```
error: failed to delete 'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/golden-fixtures': Permission denied
wt_remove_exit=0
=== remaining worktrees ===
C:/Users/avidu/Projects/badminton-highlight-indexer
95cd154 [docs/golden-regression-fixtures-plan]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-a771ff1d85e18f2ed
7e0881a [perf/wasb-fast-frames]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy
5cb7bbf [bhi-execpolicy]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality
877ab1f [docs/rally-quality-research]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/local-noai
3ea7ad4 [feat/local-no-ai-segmenter]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier0-eval
9e03e5a [feat/tier0-rally-seg-eval]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing
a3b209a [master]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-netcross
effd920 [feat/w2-netcross-gate]
=== remote M1 branch gone? ===
=== local branch cleanup ===
error: cannot delete branch 'docs/golden-regression-fixtures-plan' used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer'
Deleted branch feat/golden-fixtures-m1 (was 84f8428).
```

163. user (2026-06-16T05:05:22.560Z)

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-16 ~11:00 (REAL-FOOTAGE VALIDATION of W1+W2 + ? F: DRIVE
DISCONNECTED). Owner: "push GPU/Gemini"
```

164. assistant (2026-06-16T05:05:21.611Z)

M1 is merged and cleaned up. Notes: the remote branch was deleted; the local worktree dir hit a Windows file-lock on delete but git deregistered it (it's gone from `git worktree list`; the leftover gitignored dir is harmless). The worktree list also shows the concurrent session running several worktrees (`tier1-windowing`, `w2-netcross`, `tier0-eval`) ? confirming the isolation approach was right.

Let me record the checkpoint in memory, then report.

165. user (2026-06-16T05:05:52.877Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

166. assistant (2026-06-16T05:06:13.345Z)

Milestone shipped. Here's the status.

? M1 merged ? [PR #189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) (master `64494ef`)

The video-free regression harness is live:

- `backend/eval/golden\_fixtures.py` ? shared `replay\_fixture()` (CONTROL / pure-stdlib, ****no numpy**** on the replay path) reusing the shipped eval stack.
- 3 ****synthetic**** Tier-0 fixtures (clean rally / dead-time split / occlusion split), tool-generated; manifest + README + `.gitattributes` LF pin.
- `tests/test\_golden\_fixtures.py` ? parse-compare characterization, a positive "no person/audio/pose/gait/face" privacy assertion, perturbation + idempotence tests.

****Verified to the pre-LGTM bar**** (locally ***and*** in CI): 746 passed / 7 skipped, coverage 80.41%
? 70%, ruff + mypy clean.

Your two questions, answered concretely

- ****Concurrency didn't hinder me ? proven, not asserted.**** I worked in an isolated git worktree (the other session is running `tier1-windowing` / `w2-netcross` / `tier0-eval` worktrees ? heavy parallelism). I ****rebased onto master *after* their #187 (W1 bounded windowing) and #188 (W2 net-crossings gate)**** and `--check` still passed 3/3. The pinned preset (`max\_win=0`) + CONTROL-only design make my gate ****robust to their windowing churn****.
- ****Cross-OS determinism confirmed:**** CI (Linux) replayed my Windows-frozen fixtures and went green ? the risk that killed naive snapshotting is cleared for this path.
- ****The template norm is live and enforced:**** #186 merged earlier puts the `CLAUDE.md` convention + `docs/PROJECT\_DOC\_TEMPLATE.md` on master, plus the cross-session memory.

Honest notes

- ****Tracker debt:**** `docs/GOLDEN\_REGRESSION\_FIXTURES.md` §7 still shows M1 as ? ? I should have flipped it ***inside*** #189. I'll flip M1 ? + status ? IN PROGRESS in the M2 PR (and fold each row-flip into its deliverable PR going forward).
- ****P0** (the `eval\_baselines/\*.json` name-scrub) is deliberately held****** ? it would rewrite keys the active windowing/metrics work depends on. It lands once their track settles or in coordination.
- Minor: a Windows file-lock left an orphaned `.claude/worktrees/golden-fixtures` dir (gitignored, harmless).

Next

Per "merge as you go," I'll continue down the §7 tracker: ****M2**** (harden the freeze tool + flip the tracker) ? ****M3/M4**** (the gates) ? ****P1**** (de-attribution tooling) ? each a small, CI-green, self-merged PR in an isolated worktree. ****P2/P3 stay counsel-gated**** (no real-data commit without your sign-off). Say the word if you'd rather I pause, reorder, or coordinate P0 with the windowing session before proceeding.

167. user (2026-06-16T05:43:30.656Z)

any updates?

168. assistant (2026-06-16T05:45:24.092Z)

Let me check the current state before answering ? whether master moved, my open PRs, and anything in flight.

169. user (2026-06-16T05:45:31.708Z)

=== origin/master (last 4) ===

64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization harness (#189)

a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)

1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)

a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0) (#187)

=== my open PRs ===

=== M1 (#189) state ===

MERGED 2026-06-16T05:04:10Z

=== §7 tracker: M1 row status on master ===


```

3:> **Status:** `DRAFT` ? owner-gated . **Owner:** Avi . **Created:** 2026-06-16 . **Last
updated:** 2026-06-16
116:| **TM1 no-key public** | map fixture?source/match/player; decrypt Tier-1; confirm a source
even holding the candidate file | read Tier-0 numbers + run shuttle-only harness (intended;
useless-in-isolation) |
145:| **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper.**
`eval_baselines/fixtures/` + `backend/eval/golden_fixtures.py`
(`load_manifest`/`load_fixture`/`replay_fixture`, schema_version dispatch,
`Path(__file__)`-relative); 3?5 hand-authored synthetic fixtures (clean rally / held-shuttle
Gate-A / occlusion / quiet / multi-window); `manifest.json`; `.gitattributes` LF pin. | P0 | No
(synthetic) | ? | ? |
146:| **M2** | **Freeze/refresh extractor.** `backend/eval/freeze_fixture.py` composing shipped
fns; `--refresh-snapshot[--all]`; `--license/--minors-free` refusal gate; idempotence unit test.
| M1 | No (synthetic) | ? | ? |
149:| **P1** | **De-attribution + minimization pass.** `scripts/deattribute_golden.py`: private
Fork-A `*_golden_features.json` ? de-attributed/minimized/time-origin-reset public fixture;
**random surrogate id** ; person/audio stripped + **positive assertion** no person/audio keys;
quantize fps/frame_width; **person-optional Tier-0 loader** ; provenance hard-fail gate. | M1 |
No (tooling) | ? | ? |

```

170. assistant (2026-06-16T05:47:48.482Z)

Checked the state ? here's the honest update.

Last landing: M1 ([#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189)), merged. **That actually delivers the core ask:** a clean clone now runs the rally-seg regression with **no video / GPU / DB** (synthetic Tier-0, CONTROL-only, cross-OS verified ? CI on Linux replayed my Windows-frozen fixtures green).

Nothing has landed since, and nothing of mine is running in the background. Master is unchanged at `64494ef`, I have no open PRs, and the concurrent windowing session hasn't merged anything new either. I **paused** at the M1 milestone **rather than auto-continue** ? because the headline is met and what's left is either forward-looking or needs your steer (details below). So the true answer to "any updates?" is: M1 is done; I held for your read.

Two honest notes:

- **M1 absorbed more than its row:** the freeze/refresh/`--check` CLI (that was M2's core) and the characterization gate (M3) all shipped inside M1. So \$7's M2/M3 are largely done-in-M1.
- **Tracker debt:** `docs/GOLDEN\_REGRESSION\_FIXTURES.md` \$7 still shows M1 as ?. I'll flip it (and reconcile M2/M3) in whatever lands next.

What remains is genuinely discretionary or coordination-sensitive, so rather than auto-merge plumbing that may not be needed soon, let me confirm direction:

171. user (2026-06-16T05:50:51.217Z)

Your questions have been answered: "M1 delivered the core ask (video-free regression on a clean clone). What should I prioritize next?"="do flip the tracker and prioritize the cross-OS guarantee first and then keep building (assume legal is approved and flip that bit to approved if needed) the privacy plumbing. Im curious what real data you need? We have a few golden annotations, and a lot of videos in the F: drive.". You can now continue with these answers in mind.

172. assistant (2026-06-16T05:52:48.296Z)

Great direction ? let me answer your data question, handle the "legal approved" flip responsibly, then execute in your order (tracker ? cross-OS ? plumbing).

What real data I'd need (and it's modest): for the public tier the fixtures are **shuttle-only**, so per cleared video I need just three numeric things ? **(1)** its detector **trajectory CSV** (`Frame,Visibility,X,Y` from WASB/TrackNet ? these likely already exist in `output/wasb/` or the OneDrive `golden-data/trajectories/` from prior runs), **(2)** its **golden rally-label CSV** (your existing annotations), **(3)** `fps`/`frame\_width`. **No video, frames, person/pose, or audio** ? those never enter the public tier. So what I actually need

from you is the *clearance call*: **which F:/golden videos are owned + minors-free + OK to derive-and-commit**; I (or this machine's detector) produce the trajectory CSVs, and the de-attribution tool strips all identity before anything is committed.

On "assume legal is approved": understood ? I'll record it honestly as **your informed acceptance of the *researched* risk (dated), not as a retained-counsel opinion** (I won't fabricate a lawyer review), and crucially **the conditions stay binding**: owned/Fork-A source, minors excluded, biometric/person streams excluded, de-attributed. That's exactly what the de-attribution tool's refusal gate will enforce in code. So P2 flips to owner-accepted and P3 (real fixtures) unblocks *under those conditions*.

Now executing. Let me set up an isolated worktree and read the CI config + the doc sections I'll touch.

173. user (2026-06-16T05:52:52.587Z)

```
branch 'feat/golden-fixtures-crossos' set up to track 'origin/master'.
HEAD is now at 950106b fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-
empty video (#188 follow-up) (#190)
=== ci.yml ===
name: CI
```

```
on:
  push:
    branches: [master]
  pull_request:
```

```
permissions:
  contents: read
```

```
concurrency:
  group: ci-${{ github.ref }}
  cancel-in-progress: true
```

```
jobs:
  # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle + bugbear.
  # Also the repo-wide syntax guard ? ruff reports invalid-syntax on every
  # tracked .py, including scripts/ and training/, which the backend-only
  # py_compile smoke sweep does not reach.
```

```
  lint:
    name: Ruff lint
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v6
      - uses: actions/setup-python@v6
        with:
          python-version: "3.13"
      - name: Install ruff
        run: python -m pip install ruff
      - name: Ruff check
        run: ruff check . --output-format github
```

```
  # Type gate (mypy.ini at repo root): lenient green baseline with an explicit
  # per-module quarantine list ? ratchet by removing entries as modules clean
  # up. Typed core deps are installed so mypy sees their real signatures
  # (without them ignore_missing_imports would Any-stub half the checks).
```

```
  typecheck:
    name: Mypy (lenient baseline)
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v6
      - uses: actions/setup-python@v6
        with:
          python-version: "3.13"
      - name: Install mypy + typed core deps
```

```

    run: python -m pip install mpy pydantic fastapi uvicorn python-dotenv google-genai
numpy
- name: Mypy
  run: mypy backend training

# Fast guard against the 7fbbeb0 class of breakage: a SyntaxError or
# module-level import error shipping while the suite stays green.
# Deliberately does NOT install cv2/torch/numpy ? the smoke tests must hold
# on machines without the GPU stack (test_import_smoke stubs what's absent).
import-smoke:
  name: Syntax sweep + import smoke (no GPU stack)
  runs-on: ubuntu-latest
  steps:
    - uses: actions/checkout@v6
    - uses: actions/setup-python@v6
      with:
        python-version: "3.13"
    - name: Install minimal deps
      run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-genai
    - name: Run import smoke tests
      run: python -m pytest tests/test_import_smoke.py -v

full-suite:
  name: Full test suite (offline, mocked)
  runs-on: ubuntu-latest
  steps:
    - uses: actions/checkout@v6
    - uses: actions/setup-python@v6
      with:
        python-version: "3.13"
        cache: pip
    - name: System libs for opencv-python
      run: |
        sudo apt-get update
        sudo apt-get install -y libgl1 libgl1.0-0 || sudo apt-get install -y libgl1
libgl1.0-0
    - name: Install CPU-only torch (no CUDA wheels on a CPU runner)

```

174. assistant (2026-06-16T05:53:12.094Z)

Master advanced to `950106b` (#190, another W2 follow-up) ? I branched from it. #190 touches the gate (not CONTROL), so my fixtures should be unaffected; I'll re-verify. Let me see the rest of the full-suite job to model a lightweight cross-OS job's deps.

175. user (2026-06-16T05:53:13.238Z)

```

80     - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
81     # torch/torchvision are NOT in pyproject deps ? their CPU/CUDA wheels
82     # need an out-of-band index that PEP 621 can't express. Keep installing
83     # them separately here, before the editable install pulls the rest.
84     run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
85     - name: Install package (editable) + dev tooling
86     # Editable PEP 621 install: runtime deps + the `dev` group
87     # (pytest, pytest-cov, httpx for the fastapi TestClient, ruff, mypy).
88     run: python -m pip install -e .[dev]
89     # obsreport (private, soft dep) is absent here: reporting tests skip via
90     # pytest.importorskip ? run_all_tests.py prints which. The coverage floor
91     # is therefore calibrated against THIS lane's number (local runs with
92     # obsreport installed measure a little higher).
93     # Floor calibrated 2026-06-11 against this lane's measured 72% (local
94     # with obsreport: 73%). Ratchet it UP as coverage grows; never down
95     # without an owner decision.
96     - name: Run full suite (with coverage floor)
97     run: python run_all_tests.py --cov=backend --cov=training --cov-report=term

```

```
--cov-fail-under=70
```

```
98
```

176. assistant (2026-06-16T05:54:03.394Z)

Let me check whether the golden-fixtures import chain actually needs the heavy GPU deps (torch/cv2) ? that decides whether the cross-OS CI job can be lightweight.

177. user (2026-06-16T05:54:10.853Z)

```
=== top-level imports in the replay chain (cv2/torch/numpy?) ===
--- backend/pipeline/detectors/tracknet_runner.py
--- backend/eval/distill_local.py
--- backend/eval/experiment.py
--- backend/eval/classifier.py
18:import numpy as np
--- backend/pipeline/detectors/rally_gate.py
=== does importing golden_fixtures work with NO cv2/torch? (simulate clean clone) ===
import OK; --check follows
```

178. assistant (2026-06-16T05:54:53.657Z)

Decisive finding: the replay chain imports **no** cv2/torch at module top (only `numpy`, via `classifier`), and `import golden_fixtures` works on a clean clone. So the cross-OS CI job can be **lightweight** (numpy + minimal deps, no GPU stack) ? a *separate fast job*, not a full-suite×3 matrix, so it doesn't bloat everyone's CI. Let me read the doc sections I'll reconcile, then write both changes.

179. user (2026-06-16T05:54:54.701Z)

```
139
```

```
140     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One small PR per row,
branched from origin/master, no stacking. Update the row (status + PR link) as work lands.
```

```
141
```

```
142     | ID | Deliverable | Depends on | Counsel-gated? | Status | PR |
```

```
143     |----|-----|-----|-----|-----|----|
```

```
144     | P0 | Leak remediation + required CI leak-scan. Scrub real names/paths in
eval_baselines/heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json,
backend/eval/eval_partial_labels.py, tests/test_cleanup_caches.py ? opaque surrogate keys +
private map; add CI leak-scan (MD5 / known-name / non-surrogate / plaintext-SOPS) as a
required status check; decide on git history (PR #77, accept vs filter-repo). | ? | No
(owned data hygiene) | ? | ? |
```

```
145     | M1 | Fixture framework + synthetic Tier-0 + shared replay helper.
```

```
eval_baselines/fixtures/ + backend/eval/golden_fixtures.py
(load_manifest/load_fixture/replay_fixture, schema_version dispatch,
Path(__file__)-relative); 3?5 hand-authored synthetic fixtures (clean rally / held-shuttle
Gate-A / occlusion / quiet / multi-window); manifest.json; .gitattributes LF pin. | P0 | No
(synthetic) | ? | ? |
```

```
146     | M2 | Freeze/refresh extractor. backend/eval/freeze_fixture.py composing
shipped fns; --refresh-snapshot[--all]; --license/--minors-free refusal gate; idempotence
unit test. | M1 | No (synthetic) | ? | ? |
```

```
147     | M3 | Characterization gate. tests/test_golden_characterization.py ? parsed-
structure compare, exact-int/approx-float, windowing+label fingerprint guard, CONTROL-only;
schema-drift + row-count-cap tests; multi-OS CI matrix (win/ubuntu/mac). | M1, M2 | No | ? |
? |
```

```
148     | M4 | Quality-floor gate + first regression_baseline.json.
tests/test_golden_quality_floor.py; create eval_baselines/regression_baseline.json with
windowing fingerprint; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for
runtime. | M1, M2 | No | ? | ? |
```

```
149     | P1 | De-attribution + minimization pass. scripts/deattribute_golden.py:
private Fork-A *_golden_features.json ? de-attributed/minimized/time-origin-reset public
fixture; random surrogate id; person/audio stripped + positive assertion no person/audio
keys; quantize fps/frame_width; person-optional Tier-0 loader; provenance hard-fail gate. |
M1 | No (tooling) | ? | ? |
```

```

150 | **P2** | **Counsel sign-off (gating ? not a code PR).** Retained IN + EU/UK counsel
review of the final committed Tier-0 subset + source-provenance log. | P1 | ? **Yes** | ? | ? |
151 | **P3** | **Tier-0 REAL subset.** Commit counsel-cleared, de-attributed,
minors/biometric-excluded real Fork-A subset; unconditional test. | P1, P2 | ? **Yes** | ? | ? |
152 | **O1** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1`
proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
153 | **O2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
154 | **DOC** | **Finalize this doc + cross-links.** Add `eval_baselines/fixtures/` row to
`CODE_MAP $0` + TEST MAP; cross-link `EVALUATION.md` / `EXPERIMENT_HARNESS.md` /
`GOLDEN_DATA_SHARING.md` / `QUALITY_ITERATIONS.md`; update `docs/README.md` index + memory
`current-state`/handoff. | M1..M4 | No | ? | ? |
155
156 **Must-fix-before-any-public-commit (red-team), folded into the rows above:** ? numeric-
tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5); ? scrub names in **all**
files + history decision (P0); ? leak-scan a **required** check, hooks are convenience only
(P0); ? person-optional loader + **positive** no-person assertion, don't trust the silent `0.0`
default (P1); ? random surrogate id not HMAC-over-roster (P1); ? quantize/drop fps/frame_width
(P1); ? document Tier-0 omits the fusion path (DOC); ? counsel + Fork-A + minors/biometric
exclusion as hard preconditions (P2).
157
158 ---
159
160 ## 8. Open owner / counsel questions
161
162 **Counsel (gating ? block P2/P3):**
163 1. Does the EU/UK **sui-generis database right** subsist in our self-derived
shuttle/rally streams on the *Sportradar* "obtained vs created" reasoning, and does
redistributing a de-attributed Tier-0 subset cause *CV-Online* harm?
164 2. Post *Thomson Reuters v. Ross* + USCO-2025, is **US fair use for the extraction
step** defensible if the dataset could compete in a "data/analytics" market ? does publishing
only numbers (never footage) matter?
165 3. India: are anti-scraping ToS enforceable; **IT Act s.43/s.66** exposure for any 3rd-
party ingestion; does pending *ANI v. OpenAI* (~Apr 2026) move it? Confirm **DPDP s.9/Rule 10**
and whether per-player tracking is prohibited s.9(3) monitoring of a minor.
166 4. Sign-off on the **final committed Tier-0 subset + provenance log** ; is non-
attributability-via-private-source a defensible posture to assert given its
conditional/revocable nature?
167
168 **Owner (strategic):**
169 5. Publishing the Fork-A subset **forfeits the trade-secret moat** (Indian breach-of-
confidence framing) ? weighed against community/benchmark/paper aims. Your call.
170 6. **Forward-looking commitment:** do not publish a highlight reel of the *same* source
whose fixtures are committed (without re-clearing), and accept a periodic re-assessment as the
public-video environment grows.
171 7. **Scope of first public commit:** synthetic-only (zero provenance risk, recommended)
vs a counsel-cleared real subset.
172 8. **Tier-1 mechanism:** out-of-repo+token (recommended) vs SOPS+age in-repo. And: is
protobuf (O1) wanted, or is canonical-JSON enough?
173
174 ---
175
176 ## 9. Definition of done
177
178 The project is **DONE** (flip `Status ? DONE`, archive) when:
179 - [ ] **P0, M1?M4, P1, DOC** are ? and merged.
180 - [ ] A clean `git clone` on **win + ubuntu + mac** runs `python run_all_tests.py` and
the **Tier-0 gates pass green with no key / no video / no GPU / no DB**.
181 - [ ] The **CI leak-scan is a required status check** and passes (no MD5 / real-name /
non-surrogate / plaintext-SOPS in the tree).
182 - [ ] Characterization + quality-floor gates are **parse-compared, CONTROL-only,
windowing-fingerprinted**, and demonstrably catch a seeded behaviour change.
183 - [ ] `docs/CODE_MAP.md $0` + TEST MAP updated; this doc cross-linked; `NEXT_STEPS.md` +

```

```
memory `current-state` point here.
184 - [ ] **Either** the public commit is synthetic-only, **or** **P2 counsel sign-off +
P3** are ? for any real-data commit.
185 - [ ] $6 honest-limits caveat rides in the test pass/fail messages, not only the docs.
186
187 *Optional 01/02 may remain ? without blocking DONE (they're earned, not assumed).*
188
189 ---
190
```

180. assistant (2026-06-16T05:56:01.141Z)

I have everything I need. Let me make the changes: the lightweight cross-OS CI job + the doc reconciliation (M1 ?, M2/M3 status, new cross-OS row, the owner-accepted legal flip recorded honestly, status ? IN PROGRESS).

181. user (2026-06-16T05:56:06.873Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\github\workflows\ci.yml has been updated successfully. (file state is current in your context ? no need to Read it back)

182. user (2026-06-16T05:56:13.458Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)

183. user (2026-06-16T05:56:28.019Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)

184. user (2026-06-16T05:56:48.927Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)

185. user (2026-06-16T05:56:58.249Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)

186. user (2026-06-16T05:57:03.921Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-crossos\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)

187. assistant (2026-06-16T05:57:10.439Z)

Now verify locally (confirm #190's gate change didn't disturb my fixtures, and the CI YAML is valid), then commit:

188. user (2026-06-16T05:57:20.075Z)

```
=== --check (post #190) ===
[ok] clean_single_rally: 1 candidates, 1 segments, F1=1.000000
[ok] multi_rally_with_deadtime: 2 candidates, 2 segments, F1=1.000000
[ok] occlusion_break: 2 candidates, 2 segments, F1=1.000000
```

```
[golden-fixtures] 3/3 fixtures match (synthetic-tier = plumbing correctness, not field quality ?
$6)
CHECK=0
=== fixtures test ===
..... [100%]
17 passed in 2.04s
=== ci.yml YAML valid? ===
jobs: ['lint', 'typecheck', 'import-smoke', 'full-suite', 'golden-crossos']
golden-crossos OK; os matrix: ['ubuntu-latest', 'windows-latest', 'macos-latest']
```

189. assistant (2026-06-16T05:57:31.254Z)

Verified: `--check` 3/3 (robust to #190 too), fixtures test 17 passed, `ci.yml` valid with the 3-OS matrix. Committing and pushing:

190. user (2026-06-16T05:57:40.919Z)

```
=== staged ===
.github/workflows/ci.yml
docs/GOLDEN_REGRESSION_FIXTURES.md
=== branch/HEAD ===
feat/golden-fixtures-crossos
=== push ===
branch 'feat/golden-fixtures-crossos' set up to track 'origin/feat/golden-fixtures-crossos'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch]      feat/golden-fixtures-crossos -> feat/golden-fixtures-crossos
```

191. assistant (2026-06-16T05:57:47.648Z)

Pushed. Opening the PR and watching CI ? the new `golden-crossos` jobs on Windows/macOS/Linux are the actual cross-OS validation (the macOS data point I didn't have yet):

192. user (2026-06-16T05:58:01.421Z)

```
https://github.com/avidullu/badminton-highlight-indexer/pull/191
=== watch checks ===
no checks reported on the 'feat/golden-fixtures-crossos' branch
CHECKS_EXIT=0
```

193. assistant (2026-06-16T05:58:11.867Z)

CI hadn't registered the checks at that instant (the watch exited early). Let me re-watch now that the run has been created:

194. user (2026-06-16T06:00:37.187Z)

```
Ruff lint      pass    10s    https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27597416464/job/81590506217
Syntax sweep + import smoke (no GPU stack)      pass    25s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506233
Full test suite (offline, mocked)              pass    2m16s  https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27597416464/job/81590506353
Golden fixtures cross-OS (macos-latest) pass    16s    https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27597416464/job/81590506232
Golden fixtures cross-OS (ubuntu-latest)        pass    20s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506265
Golden fixtures cross-OS (windows-latest)       pass    1m15s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506247
Mypy (lenient baseline) pass    36s    https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27597416464/job/81590506221
Ruff lint      pass    10s    https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27597416464/job/81590506217
Syntax sweep + import smoke (no GPU stack)      pass    25s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506233
```

```
Full test suite (offline, mocked)      pass    2m16s  https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506353
Golden fixtures cross-OS (macos-latest) pass    16s    https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506232
Golden fixtures cross-OS (ubuntu-latest) pass    20s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506265
Golden fixtures cross-OS (windows-latest) pass   1m15s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506247
Mypy (lenient baseline) pass    36s    https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506221
Ruff lint      pass    10s    https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506217
Syntax sweep + import smoke (no GPU stack)      pass    25s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27597416464/job/81590506233
CHECKS_EXIT=0
```

195. assistant (2026-06-16T06:01:20.403Z)

****All checks green ? including all three cross-OS jobs**** (macOS, Ubuntu, Windows all pass). The Windows-frozen fixtures now provably replay identically on macOS and Linux, and that guarantee is ****enforced on every PR going forward****. Merging:

196. user (2026-06-16T06:01:28.153Z)

```
failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'
```

=== master now ===

```
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
```

197. assistant (2026-06-16T06:01:42.642Z)

#191 merged (master `adb88f4`) ? cross-OS guarantee locked. Now continuing to the privacy plumbing (P1 + M2 refusal gate) as you directed. Let me set up a fresh worktree and record the checkpoint.

198. user (2026-06-16T06:01:47.573Z)

```
branch 'feat/golden-fixtures-p1' set up to track 'origin/master'.
HEAD is now at adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile
project tracker (#191)
error: failed to delete 'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-crossos': Permission denied
=== p1 worktree HEAD ===
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
```

199. user (2026-06-16T06:01:47.618Z)

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5        node_type: memory
6        type: project
7        originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-16 (GOLDEN REGRESSION FIXTURES **M1 SHIPPED ? PR #189 MERGED**,
```


master `64494ef`). Executing the [[working-agreement-tracked-project-docs]] project
(`docs/GOLDEN\_REGRESSION\_FIXTURES.md`); #186 (plan + template + CLAUDE.md convention) merged
earlier so the tracked-project-doc NORM is LIVE. M1 = `backend/eval/golden\_fixtures.py` (shared
`replay\_fixture`: parse_trajectory_csv ? distill_local.build_candidates(PINNED preset) ?
VideoCandidates(5 shuttle cols ONLY) ? experiment.predict(**CONTROL**, pure-stdlib, NO numpy) ?
metrics.score) + 3 SYNTHETIC Tier-0 fixtures (clean_single_rally / multi_rally_with_deadtime /
occlusion_break, generated by the freeze tool, NOT hand-authored) + manifest + README +
`.gitattributes` LF pin + `tests/test\_golden\_fixtures.py` (parse-compare characterization,
positive no-person/audio privacy assertion, perturbation + idempotence). Built in an ISOLATED
git worktree (the concurrent quality session runs tier1-windowing/w2-netcross/tier0-eval
worktrees ? heavy parallelism, zero collision). KEY WINS: (1) rebased onto master AFTER #187(W1
bounded windowing)+#188(W2 net-crossings gate) and `--check` STILL 3/3 ? the pinned preset
(max_win=0) + CONTROL-only path make the gate ROBUST to the windowing track's churn; (2) CI
(Linux) replayed my Windows-frozen fixtures green ? **cross-OS determinism confirmed** for the
CONTROL path. Verified locally AND in CI: 746 passed/7 skipped, coverage 80.41%, ruff+mypy
clean. ? TRACKER DEBT: `docs/GOLDEN\_REGRESSION\_FIXTURES.md` \$7 still shows M1 ? ? I forgot to
fold the row-flip into #189; flip to ? + status?IN PROGRESS WITH the M2 PR (lesson: each
deliverable PR updates its own tracker row). ?? NEXT: M2 (freeze-tool hardening + tracker flip)
? M3/M4 gates ? P1 de-attribution tooling; **P0 (eval_baselines name-scrub) HELD** to avoid
colliding with the active windowing/eval work; P2/P3 counsel-gated. Owner authorized "merge as
you go" for this project (2026-06-16). Ties [[gotcha-shared-checkout-branch-collisions]] +
[[decision-rigor-norm]].

200. user (2026-06-16T06:02:54.257Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

201. assistant (2026-06-16T06:02:44.148Z)

Master is `adb88f4`; P1 worktree ready (the gf-crossos dir hit the same harmless Windows lock ? deregistered, gitignored leftover). Let me record the checkpoint, then launch the P1 + M2-refusal-gate build.

202. assistant (2026-06-16T06:02:59.394Z)

Now launching the P1 + M2-refusal-gate build in the isolated worktree (background, so I stay responsive); I'll verify + merge it on completion.

203. user (2026-06-16T06:03:32.021Z)

Async agent launched successfully.
agentId: a04b845757f1b4481 (internal ID - do not mention to user. Use SendMessage with to: 'a04b845757f1b4481' to continue this agent.)
The agent is working in the background. You will be notified automatically when it completes. Do not duplicate this agent's work ? avoid working with the same files or topics it is using. Work on non-overlapping tasks, or briefly tell the user what you launched and end your response.
output_file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a04b845757f1b4481.output
Do NOT Read or tail this file via the shell tool ? it is the full subagent JSONL transcript and reading it will overflow your context. If the user asks for progress, say the agent is still running; you'll get a completion notification.

204. assistant (2026-06-16T06:03:54.379Z)

Done a full slice this turn. Status:

? Cross-OS guarantee locked ? [PR #191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) n

- New lightweight `golden-crossos` CI job (Ubuntu/Windows/macOS) ? **all three green**. Your Windows-frozen fixtures provably replay identically on macOS + Linux, and it's now **enforced on every PR**. It's a *separate fast job* (numpy-only, no GPU stack), so it doesn't slow the rest of the repo's CI.
- **Tracker reconciled** (the debt I owed): \$7 now M1 ?, M2 ?, M3 ?, new MX ?; **Status ? IN

PROGRESS**.

- **Legal flipped to owner-accepted** ? recorded *honestly* as your informed acceptance of the **researched** risk (2026-06-16), **not** a counsel opinion. The §3 conditions stay binding (owned/Fork-A · minors excluded · biometric/person excluded · de-attributed); P2 ? ?, P3 (real fixtures) unblocked **under those conditions**, still hard-gated *in code* by P1's refusal gate.

? Now building (background) ? P1 + M2 refusal gate

Extending `golden\_fixtures.py` with `freeze\_real\_fixture()`: opaque random surrogate slug (never the `video\_id`), metadata strip, a **metric-invariant** time-origin reset, the **refusal gate** (raises unless owned + minors-free + license), and a positive no-person/audio assertion ? plus unit tests on **synthetic input only** (no real data committed). I'll verify (full suite + ruff + mypy + cross-OS) and merge on completion.

Over to you when convenient

The tooling will be ready to mint real fixtures, but it needs **your clearance list**: which F:/golden videos are **owned + minors-free** and OK to derive-and-commit. Per cleared video I only need its **trajectory CSV** (`Frame,Visibility,X,Y`) + **golden label CSV** + `fps`/`frame\_width` ? no video, frames, person, or audio. (P0 ? the existing-name scrub ? stays held until the windowing session's churn settles, to avoid breaking their keys.)

I'll ping you when P1+M2 lands.

205. user (2026-06-16T06:03:54.473Z)

please continue with the diligence and discipline of quantitatively recording progress and merging the PRs with improved coverage and CIs green. so long as the quality iterations and your work can be unhindering to each other please keep making progress

206. assistant (2026-06-16T06:04:29.556Z)

Let me check whether the P1+M2 build has completed (and whether master moved), so I know whether to verify-and-merge or let it finish before starting anything that would touch the same files.

207. user (2026-06-16T06:04:30.503Z)

<retrieval_status>not_ready</retrieval_status>

<task_id>a04b845757f1b4481</task_id>

<task_type>local_agent</task_type>

<status>running</status>

208. user (2026-06-16T06:04:33.681Z)

=== origin/master (last 3) ===

adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker (#191)

950106b fix(fusion): w2 net-crossings gate is do-no-harm ? never empty a non-empty video (#188 follow-up) (#190)

64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization harness (#189)

=== p1 worktree changes so far ===

M backend/eval/golden_fixtures.py

209. assistant (2026-06-16T06:05:46.735Z)

Checked: the P1+M2 build is **still running** (it's mid-edit ? `backend/eval/golden\_fixtures.py` already shows changes), and **master is stable at `adb88f4`** (the windowing session hasn't merged anything since #191, so no rebase churn).

Here's the disciplined read on "keep making progress without hindering each other": right now the in-flight P1 subagent owns `golden\_fixtures.py`, the new test file, and the §7 tracker.

Every remaining golden-fixtures deliverable touches one of those (or the tracker table), and the windowing session owns `windowing.py`/the gate ? so spinning up a *second* concurrent code PR now would collide with my own in-flight work or theirs, which is exactly the hindering you want avoided. So the correct move is to **let P1+M2 finish** (it's close), not force parallel churn.

When it completes (I'll be notified automatically), I'll hold to the bar you set:

- **Verify myself** ? full `run\_all\_tests.py`, `ruff`, `mypy`, the cross-OS `--check`, and confirm **coverage improves or holds** vs master (P1 adds tested code to a module that was at 85%, so it should rise); I'll **record the numbers** (test counts, coverage %, before/after) in the \$7 tracker + memory.
- **Merge only on green CI** (including the new Windows/macOS/Linux cross-OS jobs).
- **Rebase past any windowing merges** and re-run `--check` before merging, so my gate stays robust to their churn.

Then I continue down the tracker: after P1+M2 ? the **DOC** cross-links (CODE_MAP \$0 / EVALUATION / README) and, once you give the **per-video clearance list** (owned + minors-free), the first real Tier-0 fixture via the now-built refusal-gated tool. **P0 stays held** until the windowing churn settles.

No action needed from you ? I'll report each merge with its coverage delta as it lands.

210. user (2026-06-16T06:12:19.708Z)

```
<task-notification>
<task-id>a04b845757f1b4481</task-id>
<tool-use-id>toolu_01WEgFhPtVew647r24SjpJ8S</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a04b845757f1b4481.output</output-
file>
<status>completed</status>
<summary>Agent "Implement P1 de-attribution + M2 refusal gate" completed</summary>
<result>Clean ? only the two intended changes, no real fixture committed, scratch removed. (That
was a read-only `git status` to report changed files, no git mutation.)
```

Report

Implemented P1 (de-attribution + minimization) and M2's `--license`/`--minors-free`/`--owned` refusal gate by extending the M1 module. No real data committed ? tooling + synthetic tests only.

Files changed

- **backend/eval/golden_fixtures.py** (extended, not rewritten ? reuses `replay\_fixture`, `\_write\_json`, `write\_trajectory\_csv`, `write\_labels\_csv`, `FORBIDDEN\_KEYS`, `\_round`, `gt\_hash`):
 - `freeze\_real\_fixture(label, trajectory\_csv\_path, labels\_csv\_path, \*, license, minors\_free, owned, fps, frame\_width, fixtures\_dir=FIXTURES\_DIR, slug=None, time\_origin\_reset=True, source\_note="")` ? the full P1 pipeline: (a) refusal gate, (b) opaque random `fx` & `urandom(8).hex()` slug, (c) reads trajectory via `TrackNetRunner.parse\_trajectory\_csv` + labels via `load\_gt\_intervals(..., strict=True)`, (d) metric-invariant time-origin reset (subtracts `t0\_frame`/`t0\_sec` from both streams), (e) writes `trajectory.csv`/`labels.csv`/`provenance.json` (`kind:"cleared\_real") + `expected.json` via `replay_fixture`, (f) positive privacy assertion re-scanning written text for `FORBIDDEN_KEYS`/MD5/source-path tokens, (g) `_append_manifest_entry` that doesn't clobber synthetic rows.
 - `RefusalError(ValueError)`, `\_new\_surrogate\_slug`, `\_scan\_forbidden`, `\_reset\_trajectory\_rows`, `\_reset\_labels`, `\_manifest\_entry\_for`, `\_append\_manifest\_entry`;
 - `os/re` imports, `TrackNetRunner` import, `MD5\_RE`.
 - CLI `--freeze-real` subcommand with `--trajectory/--labels/--license/--owned/--minors-free/--fps/--frame-width/--slug/--label/--source-note/--no-time-origin-reset`; refusal prints the reason to stderr and exits **2**.
- **tests/test_golden_real_fixtures.py** (new, 12 tests, synthetic input in `tmp\_path`, no network/GPU): refusal-gate parametrization (minors/owned/empty/blank/None license), de-attribution assertions, supplied-slug honored, manifest non-clobber, source-note leak caught, **time-origin-reset metric-invariance**, round-trip `replay\_fixture`+`compare\_expected`==[], and

strict-loader enforcement.

Verification (verbatim)

```
1. **Full suite + coverage:** `759 passed, 7 skipped, 7 warnings in 96.30s` ? `Required test coverage of 70% reached. Total coverage: 80.49%`. `RESULT: PASS - all tests green`.
2. **`--check`:** all 3 synthetic fixtures match ? `[golden-fixtures] 3/3 fixtures match`.
3. **`ruff check`:** `All checks passed!` ? **`mypy backend training`:** `Success: no issues found in 103 source files`.
4. **CLI refusal demo** (no safety flags, exit 2):
...
```

```
[golden-fixtures] REFUSED (--freeze-real provenance gate):
REFUSED to freeze a real Tier-0 fixture ? unsafe provenance (...):
- minors_free must be True (...)
- owned must be True (...)
- license must be a non-blank string (...)
EXIT_CODE=2
...
```

The happy path (with `--owned --minors-free --license owned` into a scratch dir) wrote a clean `fx` fixture with no video_id/path/name in provenance and exit 0. Tmp CSVs were generated under the gitignored `scratch/` and removed afterward.

Spec deviation

One bug I caught and fixed mid-implementation (not a spec deviation, but worth noting): the positive source-path scan initially matched a source file `*stem*` (``traj``) as a bare substring, which false-positived against the unavoidable schema literal `"trajectory.csv"`. Fixed by skipping any source token that is itself a substring of the fixed schema filename literals (``trajectory`/`labels`/`expected`/`provenance` + `.csv`/`.json``), so genuine path/basename leaks are still caught (covered by ``test_positive_privacy_assertion_catches_leak_in_source_note``). ``git status`` confirms only ``backend/eval/golden_fixtures.py`` (M) and ``tests/test_golden_real_fixtures.py`` (??) changed; ``eval_baselines/fixtures/`` still holds only the 3 synthetic M1 fixtures. No git commands run.</result>
<usage><subagent_tokens>100286</subagent_tokens><tool_uses>37</tool_uses><duration_ms>527034</duration_ms></usage>
</task-notification>

211. assistant (2026-06-16T06:12:33.461Z)

P1+M2 reports green with `**improved coverage (80.41%→80.49%)**` and 12 new tests (746→759). Per the pre-LGTM gate I verify myself. Let me review the actual diff and the new tests.

212. user (2026-06-16T06:12:35.700Z)

```
=== changed files (only the 2 intended? no real fixture?) ===
M backend/eval/golden_fixtures.py
?? tests/test_golden_real_fixtures.py
=== golden_fixtures.py diff ===
diff --git a/backend/eval/golden_fixtures.py b/backend/eval/golden_fixtures.py
index ef347e6..6eb6212 100644
--- a/backend/eval/golden_fixtures.py
+++ b/backend/eval/golden_fixtures.py
@@ -39,6 +39,8 @@ from __future__ import annotations
     import argparse
     import json
     import math
+import os
+import re
     import sys
     from dataclasses import dataclass
     from pathlib import Path
@@ -50,6 +52,7 @@ from backend.eval.experiment import CONTROL, VideoCandidates, predict
     from backend.eval.gt_loader import load_gt_intervals
     from backend.eval.metrics import score
     from backend.eval.regression import gt_hash
+from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
```

```

Interval = Tuple[float, float]

@@ -87,6 +90,10 @@ FORBIDDEN_KEYS: Tuple[str, ...] = ("person", "pose", "gait", "face", "audio",
"p
_FLOAT_ROUND_DP = 6 # all written floats rounded to this many decimals (determinism)
_FLOAT_TOL = 1e-6 # abs tolerance when parse-comparing recomputed vs committed floats

+ #: A 32-hex-char MD5 (the ``video_id`` shape, ``compute_video_id``) ? the leak-scan refuses to
+ #: emit a fixture whose written text contains one (it would be an attribution back-channel,
$4c/$5).
+_MD5_RE = re.compile(r"\b[0-9a-fA-F]{32}\b")
+

# =====
# Synthetic trajectory generation (deterministic ? NO random)
@@ -454,6 +461,259 @@ def refresh_snapshot(slug: Optional[str] = None, all_slugs: bool = False,
    return targets

+ #: =====
+ #: P1 ? De-attribution + minimization: OWNED real trajectory+labels ? committable Tier-0
+ #: fixture, reusing the SAME replay_fixture snapshot. (docs/GOLDEN_REGRESSION_FIXTURES.md
+ #: $4c anti-attribution, $5 threat model, $7 row P1, $6 honest limits.)
+ #: =====
+
+
+class RefusalError(ValueError):
+    """The provenance/biometric refusal gate (M2) rejected an unsafe freeze.
+
+    A subclass of ``ValueError`` (so existing ``except ValueError`` paths still catch it) that
+    names why the unsafe public-fixture path was refused. The refusal is the code-enforced
+    provenance gate ($4c "provenance hard-fail gate", $7 ?): the owner-recorded / minors-
+    excluded
+    / license flags are human-asserted, but the gate makes the unsafe path impossible to reach
+    *silently* ? a fixture cannot be written unless all three are affirmatively set.
+    """
+
+
+def _new_surrogate_slug() -> str:
+    """An opaque RANDOM surrogate slug ? ``fx_<16 hex>`` from ``os.urandom`` ($4c).
+
+    Deliberately NOT the MD5 ``video_id`` and NOT ``HMAC(salt, video_id?name)`` ? a random,
+    NON-reversible id with no algebraic link to the source, so it can't be brute-forced over a
+    tiny known roster even if a salt leaked. The surrogate?source map (if kept) lives off-git.
+    """
+    return f"fx_{os.urandom(8).hex()}"
+
+
+ #: The fixed schema filenames the writers ALWAYS emit (as ``paths`` literals). Source-path
+ #: tokens
+ #: that collide with these are NOT a leak ? exclude them so a short source stem like ``traj``
+ #: can't
+ #: false-positive against the literal ``trajectory.csv``/``labels.csv``/``expected.json``.
+_SCHEMA_FILENAME_LITERALS: Tuple[str, ...] = (
+    "trajectory", "trajectory.csv", "labels", "labels.csv", "expected", "expected.json",
+    "provenance", "provenance.json",
+)
+
+def _scan_forbidden(text: str, *, where: str, source_paths: Sequence[str]) -> None:
+    """POSITIVE privacy assertion ($4c, red-team ?): raise if ``text`` leaks anything
+    attributable.
+
+    Scans the written JSON *text* for (a) any ``FORBIDDEN_KEYS`` biometric/identity substring,
+    (b) a 32-hex MD5 (the ``video_id`` shape), or (c) any identifying token of a source path

```

```

+ (full path / basename / stem) ? excluding the fixed schema filename literals the writers
+ always emit. The trajectory is shuttle-only by nature; this guards against a *regression*
+ re-introducing a person/audio/pose stream or an id/path back-channel ? never the silent
default.
+ """
+ low = text.lower()
+ for key in FORBIDDEN_KEYS:
+     if key in low:
+         raise RefusalError(
+             f"{where}: forbidden key substring {key!r} found in written output ? "
+             f"Tier-0 fixtures are shuttle-only; person/pose/gait/face/audio/player MUST NOT
+
+             f"appear (docs/GOLDEN_REGRESSION_FIXTURES.md §4c).")
+ m = _MD5_RE.search(text)
+ if m:
+     raise RefusalError(
+         f"{where}: a 32-hex MD5 ({m.group(0)!r}) found in written output ? that is the "
+         f"video_id shape and an attribution back-channel; never write it (§4c/§5).")
+ for sp in source_paths:
+     if not sp:
+         continue
+     p = Path(sp)
+     for token in (sp, p.name, p.stem):
+         t = (token or "").strip().lower()
+         # Skip a token that is itself a substring of a schema-literal the writers always
emit
+         # (e.g. a source stem ``traj`` ? ``trajectory.csv``) ? that collision is not a
leak.
+         if not t or any(t in lit for lit in _SCHEMA_FILENAME_LITERALS):
+             continue
+         if t in low:
+             raise RefusalError(
+                 f"{where}: source-path token {token!r} found in written output ? the source
+
+                 f"filename/path must never be persisted (§4c metadata strip).")
+
+
+def _reset_trajectory_rows(points: Sequence[Any], t0_frame: int
+
+    ) -> List[Tuple[int, int, float, float]]:
+    """Shift every ``TrajectoryPoint`` frame by ``-t0_frame`` ? ``(Frame,Visibility,X,Y)``
rows.
+
+    Visibility/X/Y are passed through untouched (the shuttle geometry is metric-invariant under
a
+    pure time shift); only the absolute frame index moves toward 0, severing the match
timeline.
+    """
+    rows: List[Tuple[int, int, float, float]] = []
+    for p in points:
+        rows.append((int(p.frame) - t0_frame, 1 if p.visible else 0, float(p.x), float(p.y)))
+    rows.sort(key=lambda r: r[0])
+    return rows
+
+
+def _reset_labels(gts: Sequence[Interval], t0_sec: float) -> List[Interval]:
+    """Shift every label by ``-t0_sec`` (same offset as the trajectory ? metric-invariant)."""
+    return [(s - t0_sec, e - t0_sec) for s, e in gts]
+
+
+def freeze_real_fixture(
+    label: str,
+    trajectory_csv_path: Any,
+    labels_csv_path: Any,
+    *,
+    license: str,

```

```

+ minors_free: bool,
+ owned: bool,
+ fps: float,
+ frame_width: float,
+ fixtures_dir: Path = FIXTURES_DIR,
+ slug: Optional[str] = None,
+ time_origin_reset: bool = True,
+ source_note: str = "",
+) -> Dict[str, Any]:
+ """Turn an OWNED, minors-free real trajectory + golden labels into a committable, de-
attributed
+ Tier-0 fixture ? reusing M1's ``replay_fixture`` for the snapshot. ($4c / $5 / $7 row P1.)
+
+ The refusal gate (M2) makes the unsafe path impossible to reach silently: a fixture is
emitted
+ ONLY when ``minors_free is True`` AND ``owned is True`` AND ``license`` is a non-blank
string.
+ The committed bytes carry NO filename / video_id / match / player / capture-date ? only an
opaque random surrogate slug, the shuttle-only trajectory (time-origin reset by default),
the
+ reset labels, and a ``kind="cleared_real"`` provenance block. A positive privacy assertion
+ re-reads the written ``expected.json`` + ``provenance.json`` and refuses if anything
attributable
+ slipped in.
+
+ NOTE ($6 honest limits): the random surrogate + reset timeline make the fixture non-
attributable
+ only *relative to the unpublished source* ? it is NOT absolutely anonymous (EDPS v. SRB).
De-
+ attribution does not cure provenance/licensing; this sits BEHIND counsel sign-off (P2) and
the
+ owner-asserted Fork-A / minors / biometric-exclusion flags.
+
+ Returns the recomputed ``expected.json`` snapshot dict (identical to ``replay_fixture``).
+ """
+ # --- (a) REFUSAL GATE (M2) ? the code-enforced provenance hard-fail ($4c / $7 ?) -----
+ reasons: List[str] = []
+ if minors_free is not True:
+     reasons.append("minors_free must be True (DPDP child = under 18; behavioural monitoring
"
+
+         "of a minor is prohibited ? $3, guardrail 'exclude minors')")
+ if owned is not True:
+     reasons.append("owned must be True (only owner-recorded Fork-A source is safe to
commit; "
+
+         "broadcast/scraped = Fork B, do NOT commit ? $3 D1)")
+ if not (isinstance(license, str) and license.strip()):
+     reasons.append("license must be a non-blank string (per-record provenance tag ? $4c "
+         "'provenance hard-fail gate')")
+ if reasons:
+     raise RefusalError(
+         "REFUSED to freeze a real Tier-0 fixture ? unsafe provenance (de-attribution does
NOT "
+
+         "cure provenance/licensing; this gate sits behind counsel sign-off, $6/P2):\n - "
+         + "\n - ".join(reasons))
+
+ traj_path = Path(str(trajectory_csv_path))
+ labels_path = Path(str(labels_csv_path))
+
+ # --- (b) opaque RANDOM surrogate slug ? never the video_id MD5, never the source name ---
+ out_slug = slug if slug is not None else _new_surrogate_slug()
+
+ # --- (c) read via the SHIPPED readers; labels via the STRICT loader (never lenient) ----
+ points = TrackNetRunner.parse_trajectory_csv(str(traj_path))
+ if not points:
+     raise ValueError(f"{traj_path}: no parseable trajectory rows (Frame,Visibility,X,Y)")

```

```

+   gts = load_gt_intervals(str(labels_path), strict=True)
+
+   # --- (d) consistent time-origin reset (default ON; MUST be metric-invariant, §4c) -----
+   # Subtract the SAME offset from BOTH the trajectory frames and the label seconds: because
+   # metrics.score is built from pure differences (interval_iou, |start-start|, |end-end|),
the
+   # score is bit-identical with vs without reset ? only the absolute timeline is severed. The
+   # quality-floor / characterization snapshot is therefore unchanged (asserted in the tests).
+   if time_origin_reset:
+       t0_frame = min(int(p.frame) for p in points)
+       t0_sec = t0_frame / float(fps)
+   else:
+       t0_frame, t0_sec = 0, 0.0
+   rows = _reset_trajectory_rows(points, t0_frame)
+   reset_gts = _reset_labels(gts, t0_sec)
+
+   # --- (e) write the de-attributed fixture (REUSE M1 writers + replay_fixture) -----
+   fdir = fixtures_dir / out_slug
+   fdir.mkdir(parents=True, exist_ok=True)
+   write_trajectory_csv(rows, fdir / "trajectory.csv")
+   write_labels_csv(reset_gts, fdir / "labels.csv")
+
+   iou = 0.5
+   prov = {
+       "slug": out_slug,
+       "schema_version": CURRENT_SCHEMA,
+       "kind": "cleared_real",
+       "label": str(label),
+       "license": license,
+       "minors_free": True,
+       "owned": True,
+       "fps": _round(fps),

```

213. assistant (2026-06-16T06:12:43.462Z)

Code quality is strong (clear refusal gate, metric-invariant reset, the false-positive fix for schema-literal collision). Let me see the rest of the diff + the tests, then run my own verification.

214. user (2026-06-16T06:12:45.569Z)

```

+       "fps": _round(fps),
+       "frame_width": _round(frame_width),
+       "iou": _round(iou),
+       "time_origin_reset": bool(time_origin_reset),
+       "windowing": {
+           "vt": PINNED_VT,
+           "mg": PINNED_MERGE_GAP,
+           "mwd": PINNED_MIN_WINDOW,
+           "max_win": PINNED_MAX_WINDOW,
+       },
+       "paths": {
+           "trajectory": "trajectory.csv",
+           "labels": "labels.csv",
+           "expected": "expected.json",
+       },
+       "gt_hash": gt_hash(reset_gts),
+       "source_note": str(source_note),
+       "note": (
+           "De-attributed REAL Fork-A fixture (owner-recorded, owned, minors-excluded,
biometric-"
+           "free, shuttle-only). Opaque RANDOM surrogate slug; absolute timeline reset. NON-"
+           "attributable only relative to the UNPUBLISHED source (EDPS v. SRB) ? never
'anonymous'. "
+           "Tier-0 / CONTROL-only. §6: behaviour-locking, not correctness; sits behind counsel

```



```
(P2)."
```

```
+     ),
```

```
+   }
```

```
+   _write_json(fdir / "provenance.json", prov)
```

```
+ 
```

```
+   expected = replay_fixture(fdir)           # REUSE M1 ? correct-by-construction snapshot
```

```
+   _write_json(fdir / "expected.json", expected)
```

```
+ 
```

```
+   # --- (f) POSITIVE privacy assertion AFTER writing (scan the written text, §4c / ?) -----
```

```
+   source_paths = [str(traj_path), str(labels_path)]
```

```
+   _scan_forbidden((fdir / "expected.json").read_text(encoding="utf-8"),
```

```
+                   where=f"{out_slug}/expected.json", source_paths=source_paths)
```

```
+   _scan_forbidden((fdir / "provenance.json").read_text(encoding="utf-8"),
```

```
+                   where=f"{out_slug}/provenance.json", source_paths=source_paths)
```

```
+ 
```

```
+   # --- (g) append to the manifest with kind="cleared_real" (don't clobber synthetic) -----
```

```
+   _append_manifest_entry(prov, fixtures_dir)
```

```
+   return expected
```

```
+ 
```

```
+ 
```

```
+def _manifest_entry_for(prov: Dict[str, Any]) -> Dict[str, Any]:
```

```
+    """A manifest row for a single fixture, derived from its provenance block."""
```

```
+    slug = prov["slug"]
```

```
+    return {
```

```
+        "slug": slug,
```

```
+        "fps": _round(float(prov["fps"])),
```

```
+        "frame_width": _round(float(prov["frame_width"])),
```

```
+        "windowing": dict(prov.get("windowing", {})),
```

```
+        "kind": prov.get("kind", "synthetic"),
```

```
+        "schema_version": int(prov.get("schema_version", CURRENT_SCHEMA)),
```

```
+        "paths": {
```

```
+            "dir": slug,
```

```
+            "trajectory": f"{slug}/trajectory.csv",
```

```
+            "labels": f"{slug}/labels.csv",
```

```
+            "expected": f"{slug}/expected.json",
```

```
+            "provenance": f"{slug}/provenance.json",
```

```
+        },
```

```
+    }
```

```
+ 
```

```
+ 
```

```
+def _append_manifest_entry(prov: Dict[str, Any], fixtures_dir: Path) -> None:
```

```
+    """Add/replace one fixture's manifest row WITHOUT clobbering the others (e.g.
```

```
+    synthetic)."""
```

```
+    entries = load_manifest(fixtures_dir)
```

```
+    entries = [e for e in entries if e.get("slug") != prov["slug"]]
```

```
+    entries.append(_manifest_entry_for(prov))
```

```
+    entries.sort(key=lambda e: str(e.get("slug")))
```

```
+    _write_json(fixtures_dir / "manifest.json", entries)
```

```
+ 
```

```
+ 
```

```
+ # =====
```

```
+ # Manifest + loaders
```

```
+ # =====
```

```
@@ -653,9 +913,30 @@ def main(argv: Optional[Sequence[str]] = None) -> int:
```

```
    help="re-freeze expected.json from COMMITTED trajectory+labels (no
```

```
regeneration)")
```

```
    g.add_argument("--check", action="store_true",
```

```
                  help="replay all fixtures + parse-compare vs committed expected.json (exit
```

```
0/1)")
```

```
-    ap.add_argument("--slug", help="single slug for --refresh-snapshot")
```

```
+    g.add_argument("--freeze-real", action="store_true",
```

```
                  help="(P1) de-attribute an OWNED real trajectory+labels into a committable
```

```
Tier-0 "
```

```
+ 
```

```
                "fixture; REFUSED without --owned --minors-free and a non-blank
```

```
--license")
```

```

+ ap.add_argument("--slug", help="single slug for --refresh-snapshot, or the surrogate slug
for "
+
+         "--freeze-real (default: a random fx_<hex> id)")
+ ap.add_argument("--all", action="store_true", help="all slugs for --refresh-snapshot")
+ ap.add_argument("--reason", help="recorded reason for --refresh-snapshot (review the
diff!)")
+ # --freeze-real (P1 / M2 refusal gate) options:
+ ap.add_argument("--trajectory", help="--freeze-real) source trajectory CSV
(Frame,Visibility,X,Y)")
+ ap.add_argument("--labels", help="--freeze-real) source golden labels CSV (start,end)")
+ ap.add_argument("--license", dest="license_", default=None,
+         help="--freeze-real) the source license tag (required, non-blank)")
+ ap.add_argument("--owned", action="store_true",
+         help="--freeze-real) assert owner-recorded Fork-A source (required)")
+ ap.add_argument("--minors-free", dest="minors_free", action="store_true",
+         help="--freeze-real) assert the source contains NO minors (required)")
+ ap.add_argument("--fps", type=float, help="--freeze-real) source fps")
+ ap.add_argument("--frame-width", dest="frame_width", type=float, help="--freeze-real)
source frame width (px)")
+ ap.add_argument("--label", default="cleared_real", help="--freeze-real) non-identifying
scenario label")
+ ap.add_argument("--source-note", dest="source_note", default="",
+         help="--freeze-real) non-identifying note (NEVER a path/name/id)")
+ ap.add_argument("--no-time-origin-reset", dest="time_origin_reset", action="store_false",
+         help="--freeze-real) DISABLE the anti-attribution time-origin reset
(default ON)")
+ ap.set_defaults(time_origin_reset=True)
+ ap.add_argument("--fixtures-dir", type=Path, default=FIXTURES_DIR,
+         help="override the fixtures directory (default: repo
eval_baselines/fixtures)")
+ args = ap.parse_args(list(argv) if argv is not None else None)
@@ -679,6 +960,31 @@ def main(argv: Optional[Sequence[str]] = None) -> int:
+     + (f" (reason: {args.reason})" if args.reason else "")
+     return 0

+ if args.freeze_real:
+     missing = [n for n, v in (("--trajectory", args.trajectory),
+                             ("--labels", args.labels),
+                             ("--fps", args.fps),
+                             ("--frame-width", args.frame_width)) if v is None]
+     if missing:
+         ap.error("--freeze-real requires " + ", ".join(missing))
+     try:
+         expected = freeze_real_fixture(
+             args.label, args.trajectory, args.labels,
+             license=args.license_ if args.license_ is not None else "",
+             minors_free=args.minors_free, owned=args.owned,
+             fps=args.fps, frame_width=args.frame_width,
+             fixtures_dir=fixtures_dir, slug=args.slug,
+             time_origin_reset=args.time_origin_reset, source_note=args.source_note)
+     except RefusalError as e:
+         # The code-enforced provenance gate (§4c / §7 ?): refuse loudly, exit non-zero.
+         print(f"[golden-fixtures] REFUSED (--freeze-real provenance gate):\n{e}",
file=sys.stderr)
+         return 2
+         print(f"[golden-fixtures] froze de-attributed REAL fixture '{expected['slug']}' "
+             f"(kind=cleared_real) -> {fixtures_dir / expected['slug']}")
+         print(" S6: behaviour-locking, NOT correctness; non-attributable only relative to the
"
+
+         "UNPUBLISHED source ? never 'anonymous'. Sits behind counsel sign-off (P2).")
+         return 0

+ # --check
+ return _cmd_check(fixtures_dir)

```

215. user (2026-06-16T06:12:45.792Z)

```
1      """Golden Regression Fixtures (P1 + M2) ? de-attribution / minimization + the refusal
gate.
2
3      These tests exercise ``golden_fixtures.freeze_real_fixture`` (the P1 tooling that turns
an OWNED,
4      minors-free real trajectory + golden labels into a committable, de-attributed Tier-0
fixture) and
5      its M2 ``--license`` / ``--minors-free`` / ``--owned`` refusal gate.
6
7      Everything runs on SYNTHETIC input written into ``tmp_path`` ? NO real data, NO network,
NO GPU.
8      We reuse M1's synthetic generators (``builtin_specs`` / ``make_synthetic_trajectory``
and the
9      ``write_trajectory_csv`` / ``write_labels_csv`` writers) to manufacture the "source"
CSVs, then
10     prove:
11         * the refusal gate blocks every unsafe combination (M2);
12         * a successful freeze is de-attributed (random ``fx_`` slug, no id/name/path, no
FORBIDDEN_KEYS);
13         * the time-origin reset is metric-INVARIANT (score identical with/without; only
absolute
14             ``control_segments`` shift toward 0);
15         * the freeze round-trips (``replay_fixture`` + ``compare_expected`` == []).
16
17     $6 honest-limits caveat: a green suite proves only that the deterministic post-
trajectory transforms
18     are unchanged for the frozen cases ? characterization is behaviour-locking, NOT
correctness.
19     """
20     import pytest
21
22     from backend.eval import golden_fixtures as gf
23
24
25     # -----
26     # Helpers ? manufacture a SYNTHETIC "real" source (trajectory + labels CSV) in tmp_path.
27     # -----
28
29
30     def _write_source(tmp_path, *, frame_offset: int = 0):
31         """Write a synthetic source trajectory+labels CSV pair into ``tmp_path``.
32
33         Reuses M1's first built-in scenario (a clean single rally), optionally shifted by
34         ``frame_offset`` frames (and the matching label seconds) to model an arbitrary
absolute
35         capture timeline. Returns ``(traj_path, labels_path, fps, frame_width, gts)``.
36         """
37         spec = gf.builtin_specs()[0] # clean_single_rally
38         fps, fw = spec.fps, spec.frame_width
39         rows = gf.make_synthetic_trajectory(spec)
40         if frame_offset:
41             rows = [(f + frame_offset, v, x, y) for (f, v, x, y) in rows]
42         gts = list(spec.gts)
43         if frame_offset:
44             off_sec = frame_offset / fps
45             gts = [(s + off_sec, e + off_sec) for (s, e) in gts]
46
47         traj = tmp_path / "src_trajectory.csv"
48         labels = tmp_path / "src_labels.csv"
49         gf.write_trajectory_csv(rows, traj)
50         gf.write_labels_csv(gts, labels)
51         return traj, labels, fps, fw, gts
52
```

```

53
54 # -----
55 # M2 ? refusal gate
56 # -----
57
58
59 @pytest.mark.parametrize(
60     "kwargs, why",
61     [
62         ({"minors_free": False, "owned": True, "license": "owned"},
63 "minors_free=False"),
64         ({"minors_free": True, "owned": False, "license": "owned"}, "owned=False"),
65         ({"minors_free": True, "owned": True, "license": ""}, "license=' '"),
66         ({"minors_free": True, "owned": True, "license": " "}, "license=blank"),
67         ({"minors_free": True, "owned": True, "license": None}, "license=None"),
68     ],
69 )
70 def test_refusal_gate_blocks_unsafe(tmp_path, kwargs, why):
71     """freeze_real_fixture refuses (ValueError) unless minors_free AND owned AND a non-
72 blank license.
73
74 The refusal is the code-enforced provenance gate (§4c / §7 ?): the unsafe public-
75 fixture path
76 must be impossible to reach silently.
77 """
78 traj, labels, fps, fw, _ = _write_source(tmp_path)
79 with pytest.raises(ValueError) as ei:
80     gf.freeze_real_fixture(
81         "scenario", traj, labels,
82         fps=fps, frame_width=fw, fixtures_dir=tmp_path / "fixtures", **kwargs)
83 # RefusalError is a ValueError subclass; the message must name a reason.
84 assert isinstance(ei.value, gf.RefusalError), why
85 assert "REFUSED" in str(ei.value), why
86 # Nothing was written for a refused freeze.
87 assert not (tmp_path / "fixtures").exists() or not any((tmp_path /
88 "fixtures").iterdir()), (
89     f"{why}: a refused freeze must not write any fixture")
90
91 # -----
92 # P1 ? de-attribution
93 # -----
94
95 def test_real_fixture_is_deattributed(tmp_path):
96     """A successful safe freeze is de-attributed: kind=cleared_real, fx_-prefixed
97 surrogate slug,
98 no id/name/filename/source-path key, and NO FORBIDDEN_KEYS in expected.json."""
99 fixtures = tmp_path / "fixtures"
100 traj, labels, fps, fw, _ = _write_source(tmp_path)
101 expected = gf.freeze_real_fixture(
102     "clean_rally", traj, labels,
103     license="owned", minors_free=True, owned=True,
104     fps=fps, frame_width=fw, fixtures_dir=fixtures)
105
106 slug = expected["slug"]
107 assert slug.startswith("fx_"), f"surrogate slug must be opaque fx_<hex>, got
108 {slug!r}"
109
110 prov = gf._read_json(fixtures / slug / "provenance.json")
111 assert prov["kind"] == "cleared_real"
112 assert prov["minors_free"] is True and prov["owned"] is True
113 assert prov["license"] == "owned"
114 assert prov["slug"] == slug
115 # No attribution keys / no source filename anywhere in the provenance.

```

```

112         for banned in ("video_id", "name", "filename", "path", "match", "capture_date"):
113             assert banned not in prov, f"provenance must not carry an attribution key
{banned!r}"
114         prov_text = (fixtures / slug /
"provenance.json").read_text(encoding="utf-8").lower()
115         for tok in ("src_trajectory", "src_labels", str(traj).lower(), str(labels).lower()):
116             assert tok not in prov_text, f"source token {tok!r} leaked into provenance"
117
118         # expected.json carries no biometric/identity/non-shuttle stream.
119         exp_text = (fixtures / slug / "expected.json").read_text(encoding="utf-8").lower()
120         for forbidden in gf.FORBIDDEN_KEYS:
121             assert forbidden not in exp_text, f"FORBIDDEN_KEYS leak {forbidden!r} in
expected.json"
122
123         # The fixture is registered in the manifest with kind=cleared_real.
124         manifest = gf.load_manifest(fixtures)
125         rows = [e for e in manifest if e["slug"] == slug]
126         assert len(rows) == 1 and rows[0]["kind"] == "cleared_real"
127
128
129     def test_supplied_slug_is_honored(tmp_path):
130         """An explicitly supplied slug is used verbatim (the surrogate map is the caller's
concern)."""
131         fixtures = tmp_path / "fixtures"
132         traj, labels, fps, fw, _ = _write_source(tmp_path)
133         expected = gf.freeze_real_fixture(
134             "scenario", traj, labels, slug="fx_custom00",
135             license="owned", minors_free=True, owned=True,
136             fps=fps, frame_width=fw, fixtures_dir=fixtures)
137         assert expected["slug"] == "fx_custom00"
138         assert (fixtures / "fx_custom00" / "expected.json").is_file()
139
140
141     def test_freeze_real_does_not_clobber_existing_manifest(tmp_path):
142         """Freezing a second real fixture must APPEND to the manifest, not wipe the
first."""
143         fixtures = tmp_path / "fixtures"
144         traj, labels, fps, fw, _ = _write_source(tmp_path)
145         e1 = gf.freeze_real_fixture("a", traj, labels, slug="fx_aaaa0001",
146                                   license="owned", minors_free=True, owned=True,
147                                   fps=fps, frame_width=fw, fixtures_dir=fixtures)
148         e2 = gf.freeze_real_fixture("b", traj, labels, slug="fx_bbbb0002",
149                                   license="owned", minors_free=True, owned=True,
150                                   fps=fps, frame_width=fw, fixtures_dir=fixtures)
151         slugs = {e["slug"] for e in gf.load_manifest(fixtures)}
152         assert {e1["slug"], e2["slug"]} <= slugs
153
154
155     def test_positive_privacy_assertion_catches_leak_in_source_note(tmp_path):
156         """source_note is non-identifying by contract ? a path/name in it trips the positive
scan."""
157         fixtures = tmp_path / "fixtures"
158         traj, labels, fps, fw, _ = _write_source(tmp_path)
159         with pytest.raises(gf.RefusalError):
160             gf.freeze_real_fixture(
161                 "scenario", traj, labels, slug="fx_leak0001",
162                 license="owned", minors_free=True, owned=True,
163                 fps=fps, frame_width=fw, fixtures_dir=fixtures,
164                 source_note=str(traj)) # leaking the source path ? must be refused
165
166
167     # -----
168     # P1 (d) ? time-origin reset is metric-invariant
169     # -----
170

```

```

171
172     def test_time_origin_reset_is_metric_invariant(tmp_path):
173         """A large-offset source frozen WITH reset has IDENTICAL score to the SAME source
frozen
174         WITHOUT reset ? but its absolute control_segments shift toward 0 (severed
timeline)."""
175         offset_frames = 9000 # +5 minutes at 30fps ? a large absolute capture-timeline
offset
176         traj, labels, fps, fw, _ = _write_source(tmp_path, frame_offset=offset_frames)
177
178         with_reset = gf.freeze_real_fixture(
179             "scenario", traj, labels, slug="fx_reset_on",
180             license="owned", minors_free=True, owned=True,
181             fps=fps, frame_width=fw, fixtures_dir=tmp_path / "with",
182             time_origin_reset=True)
183         without_reset = gf.freeze_real_fixture(
184             "scenario", traj, labels, slug="fx_reset_off",
185             license="owned", minors_free=True, owned=True,
186             fps=fps, frame_width=fw, fixtures_dir=tmp_path / "without",
187             time_origin_reset=False)
188
189         # (1) Score is bit-identical ? metrics.score is built from pure differences.
190         assert with_reset["score"] == without_reset["score"], (
191             "time-origin reset must be metric-invariant (score built from pure
differences)")
192
193         # (2) Absolute timeline differs ? the reset pulls control_segments toward 0.
194         seg_on = with_reset["control_segments"]
195         seg_off = without_reset["control_segments"]
196         assert len(seg_on) == len(seg_off) and len(seg_on) >= 1
197         assert seg_on[0][0] < seg_off[0][0], "reset must shift the first segment start
toward 0"
198
199         # The shift equals the offset in seconds (within rounding).
200         shift = seg_off[0][0] - seg_on[0][0]
201         assert abs(shift - offset_frames / fps) < 1e-3, (
202             f"segment shift {shift} should equal the frame offset in seconds {offset_frames
/ fps}")
203
204         # Reset trajectory starts at frame 0.
205         first_frame = int((tmp_path / "with" / "fx_reset_on" / "trajectory.csv")
206             .read_text(encoding="utf-8").splitlines()[1].split(",")[0])
207         assert first_frame == 0, "with reset, the trajectory must start at Frame 0"
208
209         # -----
210         # P1 (e/f) ? round-trip / correct-by-construction
211         # -----
212
213     def test_freeze_real_round_trips_check(tmp_path):
214         """After freeze_real_fixture, replay_fixture + compare_expected vs the written
expected.json
215         returns [] ? idempotent / correct-by-construction (the same path M1's --check
uses)."""
216         fixtures = tmp_path / "fixtures"
217         traj, labels, fps, fw, _ = _write_source(tmp_path)
218         expected = gf.freeze_real_fixture(
219             "scenario", traj, labels, slug="fx_rt000001",
220             license="owned", minors_free=True, owned=True,
221             fps=fps, frame_width=fw, fixtures_dir=fixtures)
222
223         fdir = fixtures / expected["slug"]
224         committed = gf._read_json(fdir / "expected.json")
225         recomputed = gf.replay_fixture(fdir)
226         assert gf.compare_expected(recomputed, committed) == [], (
227             "a just-frozen real fixture must replay-compare clean (correct-by-

```

```

construction)")
228
229
230     def test_freeze_real_uses_strict_label_loader(tmp_path):
231         """The labels are read through the STRICT gt_loader ? a malformed label row fails
loudly,
232         never silently shrinks the GT set."""
233         fixtures = tmp_path / "fixtures"
234         traj, labels, fps, fw, _ = _write_source(tmp_path)
235         # Corrupt one label row (end <= start) ? strict loader must raise.
236         labels.write_text("start,end\n5.0,1.0\n", encoding="utf-8", newline="\n")
237         with pytest.raises(ValueError):
238             gf.freeze_real_fixture(
239                 "scenario", traj, labels, slug="fx_strict01",
240                 license="owned", minors_free=True, owned=True,
241                 fps=fps, frame_width=fw, fixtures_dir=fixtures)
242

```

216. assistant (2026-06-16T06:13:13.343Z)

Code and tests are high quality ? refusal gate, opaque random slug, strict loader, metric-invariant reset (shifts both streams), positive privacy scan, manifest non-clobber; 12 well-targeted synthetic tests. Now my own verification gate:

217. user (2026-06-16T06:15:06.153Z)

```

=== origin/master ===
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
=== no real fixture committed? (only 3 synthetic dirs) ===
README.md
clean_single_rally
manifest.json
multi_rally_with_deadtime
occlusion_break
=== --check ===
[ok]   occlusion_break: 2 candidates, 2 segments, F1=1.000000
[golden-fixtures] 3/3 fixtures match (synthetic-tier = plumbing correctness, not field quality ?
$6)
=== full suite + coverage ===
-----
TOTAL                                     7674   1497   80%
Required test coverage of 70% reached. Total coverage: 80.49%
759 passed, 7 skipped, 7 warnings in 91.30s (0:01:31)
PYTEST=0
=== golden_fixtures.py module coverage ===
29 passed in 3.69s
=== ruff + mypy ===
All checks passed!
Success: no issues found in 103 source files

```

218. assistant (2026-06-16T06:15:27.698Z)

Verified green myself: ****759 passed / 7 skipped****, coverage ****80.49%**** (? from 80.41% ? improved), `--check` 3/3`, no real fixture committed, ruff + mypy clean, golden_fixtures module tests 29 passed. Now I'll flip the tracker rows (M2 ?, P1 ?) and add a quantitative progress log, in this same PR. Let me read the current tracker rows to edit precisely.

219. user (2026-06-16T06:15:28.939Z)

```

142     | ID | Deliverable | Depends on | Counsel-gated? | Status | PR |
143     |----|-----|-----|-----|-----|----|
144     | **P0** | **Leak remediation + required CI leak-scan.** Scrub real names/paths in
`eval_baselines/heuristic_lovo_n6.json`, `gemini_wrapper_2026-06-10.json`,

```

```

`backend/eval/eval_partial_labels.py`, `tests/test_cleanup_caches.py` ? opaque surrogate keys +
private map; add CI leak-scan (MD5 / known-name / non-surrogate / plaintext-SOPS) as a
**required status check** ; decide on git history (PR #77, accept vs `filter-repo`). | ? | No
(owned data hygiene) | ? | ? |
145 | **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper** (also
delivered M2-core + M3 ? see below). `backend/eval/golden_fixtures.py` (`replay_fixture`, schema
dispatch, `Path(__file__)`-relative) + 3 **tool-generated** synthetic fixtures + manifest +
README + `.gitattributes` LF pin + `tests/test_golden_fixtures.py`. CONTROL-only / pure-stdlib;
cross-OS verified. | ? | No (synthetic) | ? | [#189](https://github.com/avidullu/badminton-
highlight-indexer/pull/189) |
146 | **M2** | **Freeze/refresh extractor.** Core (`freeze_synthetic` / `refresh_snapshot` /
`--check` CLI + idempotence test) **shipped in M1 (#189)**. Remaining: the
`--license` / `--minors-free` **refusal gate** for real fixtures ? folded into **P1**. | M1 | No |
? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |
147 | **M3** | **Characterization gate** ? `tests/test_golden_fixtures.py`: parse-compared
(exact-int / approx-float 1e-6), **CONTROL-only**, positive no-person/audio assertion,
perturbation + idempotence. **Shipped in M1 (#189)**. (Windowing/label fingerprint guard + row-
count cap ? fold into M4.) | M1 | No | ? | [#189](https://github.com/avidullu/badminton-
highlight-indexer/pull/189) |
148 | **MX** | **Cross-OS determinism guard (CI).** Lightweight `golden-crossos` CI job
(ubuntu/windows/macOS) running `--check` + the fixtures test ? locks the cross-OS guarantee
(CONTROL replay is numpy-only, no GPU stack; a *separate fast job*, not full-suite×3). | M1 | No
| ? | _this PR_ |
149 | **M4** | **Quality-floor gate + first `regression_baseline.json`.**
`tests/test_golden_quality_floor.py`; create `eval_baselines/regression_baseline.json` with
**windowing fingerprint** ; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for
runtime. | M1, M2 | No | ? | ? |
150 | **P1** | **De-attribution + minimization pass.** `scripts/deattribute_golden.py`:
private Fork-A `_golden_features.json` ? de-attributed/minimized/time-origin-reset public
fixture; **random surrogate id** ; person/audio stripped + **positive assertion** no person/audio
keys; quantize fps/frame_width; **person-optional Tier-0 loader** ; provenance hard-fail gate. |
M1 | No (tooling) | ? | ? |
151 | **P2** | **Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the
researched IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source .
minors excluded . biometric/person streams excluded . de-attributed). Owner's informed risk
acceptance ? **NOT a retained-counsel opinion** ; optional future counsel confirmation per §8. |
P1 | Owner-accepted | ? | ? |
152 | **P3** | **Tier-0 REAL subset.** Commit de-attributed, minors/biometric-excluded,
owned Fork-A real fixtures (unblocked by P2). Hard-gated **in code** by P1's refusal gate (owned
+ minors-free + license required) ? needs the owner's per-video clearance list. | P1, P2 |
conditions §3 | ? | ? |
153 | **O1** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1`
proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
154 | **O2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()` ; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
155 | **DOC** | **Finalize this doc + cross-links.** Add `eval_baselines/fixtures/` row to
`CODE_MAP §0` + TEST MAP; cross-link `EVALUATION.md` / `EXPERIMENT_HARNESS.md` /
`GOLDEN_DATA_SHARING.md` / `QUALITY_ITERATIONS.md` ; update `docs/README.md` index + memory
`current-state`/handoff. | M1..M4 | No | ? | ? |
156
157 | **Must-fix-before-any-public-commit (red-team), folded into the rows above:** ? numeric-
tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5); ? scrub names in **all**
files + history decision (P0); ? leak-scan a **required** check, hooks are convenience only
(P0); ? person-optional loader + **positive** no-person assertion, don't trust the silent `0.0`
default (P1); ? random surrogate id not HMAC-over-roster (P1); ? quantize/drop fps/frame_width
(P1); ? document Tier-0 omits the fusion path (DOC); ? counsel + Fork-A + minors/biometric
exclusion as hard preconditions (P2).
158

```

220. assistant (2026-06-16T06:16:07.514Z)

All green. Now flipping the tracker (M2 ?, P1 ?, MX?#191) and adding a quantitative progress log:

221. user (2026-06-16T06:16:11.142Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-p1\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)

222. user (2026-06-16T06:16:19.973Z)

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223. user (2026-06-16T06:16:27.233Z)

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224. user (2026-06-16T06:16:41.980Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-p1\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)

225. assistant (2026-06-16T06:16:53.281Z)

Tracker updated with the quantitative log. Committing (explicit paths), then push:

226. user (2026-06-16T06:17:06.318Z)

```
warning: in the working copy of 'tests/test_golden_real_fixtures.py', LF will be replaced by
CRLF the next time Git touches it
=== staged ===
backend/eval/golden_fixtures.py
docs/GOLDEN_REGRESSION_FIXTURES.md
tests/test_golden_real_fixtures.py
=== base check ===
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
=== only my commit? ===
955e902 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
=== push ===
branch 'feat/golden-fixtures-p1' set up to track 'origin/feat/golden-fixtures-p1'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch]      feat/golden-fixtures-p1 -> feat/golden-fixtures-p1
```

227. assistant (2026-06-16T06:17:10.066Z)

Committed (only my commit atop `adb88f4`) and pushed. Opening the PR and watching CI:

228. user (2026-06-16T06:17:26.057Z)

```
https://github.com/avidullu/badminton-highlight-indexer/pull/192
=== watch checks ===
no checks reported on the 'feat/golden-fixtures-p1' branch
CHECKS_EXIT=0
```

229. assistant (2026-06-16T06:17:30.373Z)

PR #192 opened; checks hadn't registered yet. Re-watching:

230. user (2026-06-16T06:19:44.453Z)

```

Mypy (lenient baseline) pass 31s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27598234111/job/81593017818
Ruff lint pass 9s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27598234111/job/81593017774
Syntax sweep + import smoke (no GPU stack) pass 17s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598234111/job/81593017676
Full test suite (offline, mocked) pass 2m6s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27598234111/job/81593017757
Golden fixtures cross-OS (macos-latest) pass 16s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27598234111/job/81593017844
Golden fixtures cross-OS (ubuntu-latest) pass 23s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598234111/job/81593017816
Golden fixtures cross-OS (windows-latest) pass 53s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598234111/job/81593017725
Mypy (lenient baseline) pass 31s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27598234111/job/81593017818
Ruff lint pass 9s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27598234111/job/81593017774
Syntax sweep + import smoke (no GPU stack) pass 17s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598234111/job/81593017676
CHECKS_EXIT=0

```

231. assistant (2026-06-16T06:20:57.296Z)

****All checks pass ? including the 3 cross-OS jobs.**** Merging #192:

232. user (2026-06-16T06:21:04.153Z)

failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'

=== master ===

8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
(#192)
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
950106b fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-empty video (#188
follow-up) (#190)

233. assistant (2026-06-16T06:21:30.583Z)

#192 merged ? master `8d3e185`. Let me record the checkpoint, then report.

234. user (2026-06-16T06:21:31.627Z)

```

10      # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12      **Checkpoint:** 2026-06-16 ~11:30 (RALLY-QUALITY: ? W2 SAFEGUARD **#190 MERGED** + FULL
6-VIDEO GPU VALIDATION RUNNING; F: back, GPU free). Owner: "GPU free, steamroll; the SIBLING
session is hardening eval+privacy (sealed **structured-video fixtures** ? the [[golden-set-
vondoring]] / #186+#189+#191 GOLDEN_REGRESSION_FIXTURES track: repo testable WITHOUT the videos
+ hide video privacy); **~10 MORE golden videos within days** ? till then n=6 (+ these unlabeled
real clips vs Gemini) is a directional **'Hail Mary'**, not conclusive. Defaults stay OFF ? flip
all together as a milestone." (My #190 is in master's history; sibling advanced master to
`adb88f4` via #191 + is building P1 in worktree `gf-p1` ? INDEPENDENT of my work.)

```

235. user (2026-06-16T06:22:00.809Z)

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md has been updated successfully. (file state is current in your
context ? no need to Read it back)

236. assistant (2026-06-16T06:22:19.225Z)

****P1 + M2 merged ? [PR #192](https://github.com/avidullu/badminton-highlight-indexer/pull/192)****
(master `8d3e185`), verified to the bar you set.

Quantitative progress (this session, golden-fixtures project)

	PR	Deliverable	Suite	Coverage	CI	
		----		-----		-----
#186		Plan + template + `CLAUDE.md` convention		? ?		green
#189		M1 framework + synthetic Tier-0 + characterization		746?/7s		80.41% green
#191		MX cross-OS CI guard + tracker/legal reconcile		full?	80.41%	green (3-OS)
#192		P1 de-attribution tool + M2 refusal gate		759?/7s		**80.49%** ? green (3-OS)

Tracker now: ****M1 ? · M2 ? · M3 ? · MX ? · P1 ? · P2 ? (owner-accepted)****. Coverage trending ****up****, every merge green including the Windows/macOS/Linux cross-OS job, and each verified by me (full suite + ruff + mypy + `--check`) before merge. All built in isolated worktrees ? and the sibling quality session's own checkpoint confirms our two tracks are ****independent**** (no collision).

Where I'm headed (and the one thing I need from you)

- ****Next safe step (no input needed):**** ****DOC**** ? make the framework discoverable (README index + `CODE\_MAP \$0` fixtures row + cross-link `GOLDEN\_DATA\_SHARING`). I'll do this next.
- ****The high-value step is P3 (real Tier-0 fixtures), and it's blocked on you:**** the refusal-gated tool is built and tested, so minting a real fixture just needs your ****per-video clearance list**** ? which F:/golden videos are ****owned + minors-free**** and OK to derive-and-commit. Per cleared video I need only its ****trajectory CSV**** + ****golden label CSV**** + `fps`/`frame\_width` (no video, frames, person, or audio). I saw the sibling note ****~10 more golden videos landing within days**** ? that's a great input pipeline for this.
- ****Deferred with reasons:**** M4 quality-floor (trivial on synthetic ? meaningful only once real fixtures land), ****P0**** name-scrub (still held ? it rewrites `eval\_baselines` keys the active windowing session depends on), O1/O2 (optional).

I'll proceed with ****DOC**** next. Whenever you drop the clearance list (or it's easier to just point me at the golden trajectory/label CSVs), I'll mint the first real Tier-0 fixtures through the refusal-gated tool ? green and de-attributed.

237. user (2026-06-16T06:22:56.861Z)

please do go ahead and merge any PRs which pass the QA and due diligence of this repo

238. assistant (2026-06-16T06:24:32.028Z)

On it. The DOC cross-links are trivial mechanical edits (so I'll do those solo, per the ultracode "trivial edits" exception), but ****"due diligence"** on already-merged code is exactly where a workflow earns its keep

? so I'll run an adversarial audit of the shipped golden-fixtures feature in parallel, then fix anything real it surfaces as follow-up PRs. Let me set up the worktree first.

239. user (2026-06-16T06:24:37.948Z)

branch 'feat/golden-fixtures-doc' set up to track 'origin/master'.
HEAD is now at 8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate (#192)
=== master ===
8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate (#192)
=== CODE_MAP \$0 hotness (recent commits touching it) ===
4942d51 feat(tools): batch highlight RUNNER ? clean, non-destructive, Drive-safe reel generation (#182)
e69e23d docs: fix doc drift across 16 docs (#173)
c143f40 feat(packaging): PEP 621 pyproject + editable install + entry point (DEFERRED #7)
=== CODE_MAP \$0 file-placement table head ===
1:# CODE_MAP.md
7:? ****2026-06-10 audit-hardening sweep changed several behaviors documented below ? trust git history over stale lines until the next full regen:**** `add\_video` is now an ****UPSERT**** (no FK-

```

cascade wipe on re-ingest; segmentation is destroy-AFTER-confirm via
`segment_watermarks`/`prune_segments`); detector artifacts are keyed by video_id, not
basename; indexing.wasb.* is now a real typed config (was silently dropped); timeout_sec
defaults 1800s (was 0/hang-forever); SQLite opens with WAL + busy_timeout;
match_intervals requires real overlap (iou>0); run_regression baseline lives in tracked
eval_baselines/ and exits 3 on a missing baseline; the API validates
video_path/output_filename. New tests: test_api_endpoints, test_compiler,
test_wasb_runner, test_reingest_safety, test_detector_keying, test_config_wasb,
test_path_safety, test_reliability_hardening (+ collector test_licensing_gate,
test_import_guard, test_cvat_robustness).
20:## 0. REPO LAYOUT ? where new files go (READ BEFORE committing a new file)
32:| A test | tests/test_*.py | pytest auto-discovers; the full-suite lane has a 70%
coverage floor. Pure/offline + mocked (no GPU/WSL/network). |
36:| Eval baseline / leaderboard JSON | eval_baselines/ | Tracked (survives a clean checkout);
the no-regression gate reads it. |
42:(grandfathered ? imported by tests/test_eval_harness.py). The old diagnostic setup.py
moved to
48:## 1. PIPELINE
50:### 1A. Runtime: video file -> highlight reel
78:### 1B. WASB shuttle substrate (P2 of wasb_hybrid)
90:### 1C. Calibration / eval flow (NO pipeline run unless told)
108:## 2. SUBSYSTEMS
110:### 2.0 Orchestration & config
136:### 2.1 Segmenters (under pipeline/segmenters) ?
141:- @backend/pipeline/segmenters/trajectory_hybrid.py::TrajectoryHybridSegmenter ? SHARED
ENGINE for the trajectory hybrids (2026-06-11 collapse of the copy-paste twins; the "fix BOTH"
footgun is retired): healthcheck gate -> P2 trajectory->windows (+persist) -> P3 provider
handoff -> P4 DB/link; single ::shot_count_from (old wasb/tracknet shot_count drift gone).
NOT registered itself. ? Subclass contract: family (config section
indexing.<family>.velocity_threshold(0.02), output/<family> trajectory dir,
<family>-hybrid-<model> metadata tag), display_label (log lines), name/description
properties, _build_runner(idx_cfg) -> (DetectorRunner, detection_params extras). Two fusion
seams (IDENTITY by default ? twins byte-identical, pinned by the equivalence test):
::enrich_candidate_stats(cw,points,fps,frame_width,video_path,idx_cfg) (base = the 4
motion-key stats dict; computed once into window_stats, used for persist + the handoff gate)
and ::select_for_handoff(windows>window_stats,idx_cfg) (base = all indices). ? shared
knobs still YOLO-named (log_yolo_candidates/store_yolo_candidates); whole-video single pass;
needs {PROVIDER}_API_KEY (missing -> []). Equivalence pinned by
tests/test_trajectory_hybrid_equivalence.py (detector identity, shot_count, offsets, + the
base-hook-identity tests ? parametrized over both twins).
142:- @backend/pipeline/segmenters/fusion_hybrid.py::FusionSegmenter (name 'fusion_hybrid',
default-OFF multi-signal fusion P2) ? subclasses the engine; family='fusion',
display_label='Fusion'. ::build_runner selects the substrate via
indexing.fusion.substrate ('wasb'|'tracknet') reusing the existing runner+config.
::enrich_candidate_stats overrides to ALWAYS add net_crossings (Gate A signal, GPU-free via
rally_gate.gate_windows(..., estimate=self._axis_estimate(points)) ? axis/net cached per
trajectory) and, only when fusion.cue_flags.person, the ~5 person features
(::person_features ? lazy PersonDetector.detections_per_frame over the window's ORIGINAL-
video frame range ? fusion_features.compute_person_features). ::select_for_handoff
overrides to apply served Gate A (fusion.min_crossings) ? Gate B (fusion.min_players,
person-only) ? default 0/0 ? all windows kept. ? persisted candidates are ALWAYS the full
superset (offline substrate intact); only the handoff/play_segments are gated. ? to A/B vs a
tuned wasb run, set fusion.velocity_threshold = wasb.velocity_threshold (the engine reads
idx_cfg['fusion'].velocity_threshold). ? person features eager even for gated windows (P3
cleanup). Tests: tests/test_fusion_hybrid.py.
149:### 2.2 Detectors & windowing (WSL-quarantined; ABC abstraction complete)
158:- @backend/pipeline/detectors/fusion_features.py ? pure per-window person-cue feature
math (no I/O/torch/cv2; mirrors rally_gate discipline). Inputs: per-frame
{frame_idx:[(tid,bbox)]} + the in-window shuttle points + optional normalized court_polygon
+ net=(axis,pos). ::point_in_polygon (ray-cast), ::person_in_court (foot-point; None
polygon=count-all; bad dims=False), ::in_court_counts, ::players_in_court (median),
::in_court_ratio, ::mean_player_motion (per-track centre displacement; x by width, y by
height), ::two_sided_motion (both net-halves occupied), ::shuttle_player_colocation (shuttle
inside a person box  $\pm$  pad_frac halo over aligned visible frames ? the held-shuttle
discriminator), ::compute_person_features (aggregator ? the 5 PERSON_FEATURE_NAMES). All

```

```

scale/translation-invariant, graceful to None/empty. Tests: `tests/test_fusion_features.py`.
161:### 2.3 Eval & calibration (all pure core; I/O in CLI drivers)
178:- `@backend/eval/regression.py` ? CI gate. `::DEFAULT_TOLERANCE=0.05`,
`::DEFAULT_BASELINE_PATH=eval_baselines/regression_baseline.json` (tracked, NOT under gitignored
output/), `::regress`, `::regress_with_provider` (GT via ?`annotation_registry.get(tier)`;
"file"->FileProvider/"vendor"->HumanVendor/"auto"->oracle),
`::save_baseline`/`::load_baselines`, `::gt_hash` (label-set fingerprint, reused by
experiment.py), `::RegressionResult`. ? imports `annotation_registry` from
`*`backend.annotations`*` NOT `backend.eval.annotations`; gate fires on precision OR recall drop
(F1 reported, not gated); no baseline -> `regressed=False` (silent pass). CLI
`run_regression.py` exit 0 ok / 1 regress / 2 no-DB / 3 no-baseline = the promotion gate.
179:- `@backend/eval/experiment.py` ? A/B + shadow + **SxS** variant layer (the thin
"experiment" glue per `docs/EXPERIMENT_HARNESS.md`; composes
metrics/calibration/classifier/training/regression ? NO new scoring math).
`::Variant`(segmenter,config_overrides,cue_flags,`min_crossings`[Gate A],`min_players`[Gate
B],classifier_model,feature_names,threshold,merge_gap;
`to_dict/from_dict/spec_hash/is_learned`), `::CONTROL` (pinned zero-gate baseline).
`::VideoCandidates`(name,intervals,features{col-major},gts), `::predict` (gate?learned-
filter?`gemini_refine.merge_overlaps`), `::evaluate_variant` (static / pinned-model / **honest-
LOVO** train-on-others ? mirrors distill_local), `::compare` (labeled **B?A** deltas +
`label_set_hash`), `::compare_unlabeled` (**NEW-footage SxS**: agreed/a_only/b_only via
`match_intervals`, no GT ? the owner's "SxS on new videos"), `::record_experiment` (?
`eval_baselines/experiment_leaderboard.json`), `::load_video_candidates` (DB bridge via
`query_candidate_segments`). ? enabling a gate needs its feature persisted (absent?0.0?fails
gate, warns); learned variant on an unlabeled video needs a pinned `classifier_model`. **No-reg
invariant (tested): all cues OFF ? byte-identical to CONTROL.**
181:### 2.3a Audio (E1 probe; NOT adopted for fusion ? ADR-007)
185:### 2.4 Stitcher / tools / utils
193:### 2.5 Validators ?
203:### 2.6 Sports / Providers / Annotations ?
211:### 2.7 Infrastructure (DB) ?
214:### 2.8 Reporting / observability ? (per-video, soft-dep `obsreport`)
216:- `@backend/reporting/harvest.py` ? records raw FACTS into `obsreport.RunRecorder` (the
footgun RULES live in spec.yaml, not here). `::record_run(...)` ? top-level assembler (does NOT
write); builds RunRecorder, reads `config_default = config["indexing"]["default_segmenter"]`,
runs 4 stage builders + `_highlights`, `rec.finalize()`.
`::AI_SEGMENTERS={yolo_hybrid,tracknet_hybrid,wasb_hybrid,gemini}` ? (add new AI segmenters
here). `::video_meta_from_context` (derives VideoMeta from `val_context['ffprobe_meta']`; now
also `audio_present` for Input Quality reports extension; $ `fps_source` ?
{unknown,probed,fallback}), `::_validation_stage`, `::_segmentation_stage` ($ `details.{segment
er_requested,segmenter_actual,config_default,matches_config_default,segmenter_explicitly_request
ed,was_reingest,detector,metadata_source}`; ? marks `motion` output "synthetic/heuristic"),
`::_ai_stage` ($ `details.{ai_based,provider,model,api_key_present,oversize_clips_skipped}`;
metric `confirmed_segments`), `::highlights` (coverage_pct). `record_run(run_signals=)` carries
`run_signals.pop(video_id)` counters into `_ai_stage`. ? `record_run` fps fallback:
`fps_source=='unknown'` + ran -> promotes to "fallback", forces fps=30.0 (mirrors runners).
`::SPEC_PATH`/`::load_spec` resolve `spec.yaml` next to module. (Soft import of obsreport so
pure helpers like video_meta are importable for tests/coverage even without the dep.)
222:## 3. DATA CONTRACTS (canonical shapes ? one place)
260:## 4. WASB WSL CONTRACT (external dep `~/models/WASB-SBDT`, MIT; NOT in this repo ? claims
below are unverified against external source, only the repo-side caller wasb_infer.py was
checked)
270:## 5. CROSS-REPO (sibling private repo `sports-data-collector` ? external, unverified here)

```

240. assistant (2026-06-16T06:25:26.250Z)

CODE_MAP is **cold** (last touched by #182, not by the windowing session) ? safe to edit. Let me read the \$0 table to place the fixtures row, then launch the audit and do the DOC PR.

241. user (2026-06-16T06:25:26.478Z)

```

20      ## 0. REPO LAYOUT ? where new files go (READ BEFORE committing a new file)
21
22      Top-level tree and the **placement rule** for each kind of file. When unsure, prefer an
existing

```

```

23      **registry / extension point** (`?`) over a new top-level file.
24
25      | You're adding? | Put it in? | Notes |
26      |---|---|---|
27      | Backend / runtime code | `backend/<subsystem>/` |
`pipeline/{segmenters,detectors,validators}`, `providers/`, `sports/`, `annotations/`,
`infrastructure/`, `reporting/`, `stitcher/`, `utils/`, `config/`, `features/`, `api/`. Most
additions are **register a class in the relevant `?` registry**, not a new module tree. |
28      | **Eval LIBRARY** (importable, pure ? metrics, calibration, a scorer, a feature) |
`backend/eval/` | Pure core, no side-effects at import; CLI/I-O lives in a separate driver.
These are **imported widely** (e.g. `calibrate_wasb` by 8 sites) ? they are library, not
scripts; do NOT move them out. |
29      | **Standalone run-driver / one-off entry CLI** (has `__main__`, NOT imported by app
code) | **`scripts/`** | e.g. `scripts/run_phase3_eval.py`, `scripts/run_process_and_stitch.py`.
Run as `python scripts/<name>.py`. If it needs `load_dotenv()/sys.path` before importing
backend, add `scripts/<name>.py" = ["E402"]` to `ruff.toml`. |
30      | **Ops / maintenance tool** (not part of the served app ? cache GC, batch admin) |
`tools/` | Run as `python -m tools.<name>` (it's an importable package). e.g.
`tools/cleanup_caches.py`. Keep the destructive decision path pure + unit-tested. |
31      | Training code (model fitting) | `training/` | Behind the CI-enforced import wall
(ADR-008) ? `backend.main` must not import it. Own `requirements-train.txt`. |
32      | A test | `tests/test_*.py` | pytest auto-discovers; the full-suite lane has a **70%
coverage floor**. Pure/offline + mocked (no GPU/WSL/network). |
33      | A doc (design, plan, living log) | `docs/` | Index it in `docs/README.md`. Only
**terminal-state** (DONE/REJECTED) work moves to `docs/archives/`. |
34      | A throwaway repro / debug script | `scratch/` | **Gitignored** + excluded from
ruff/pytest. Never imported by app code. |
35      | A model artifact (weights + manifest) | `artifacts/<name>/<ver>/` | ADR-008:
**manifest committed** (provenance), **weights gitignored**. |
36      | Eval baseline / leaderboard JSON | `eval_baselines/` | Tracked (survives a clean
checkout); the no-regression gate reads it. |
37      | Pipeline run outputs (reels, trajectories, DBs, frames) | `output/` | **Gitignored.**
Per-video isolation is planned in `PER_VIDEO_OUTPUT_LAYOUT.md`. |
38
39      **Canonical files that stay at the repo ROOT** (referenced by CI / packaging / docs ? do
not move):
40      `pyproject.toml` (PEP 621 packaging, hatchling; `indexer` entry point) .
`run_all_tests.py` (CI: `ci.yml`) .
41      `run_regression.py` (the no-regression gate CLI, cited in `EXPERIMENT_HARNESS.md`) .
`run_eval.py`
42      (grandfathered ? imported by `tests/test_eval_harness.py`). The old diagnostic
`setup.py` moved to
43      `scripts/doctor.py` (it shadowed the build system); invoke via `python
scripts/doctor.py` or `indexer --setup`.
44      New standalone run-drivers go in **`scripts/`**, not the root.
45
46      ---

```

242. assistant (2026-06-16T06:26:18.668Z)

CODE_MAP \$0 table is clear ? I'll add a fixtures row after the `eval\_baselines/` row. First, launching the adversarial due-diligence audit of the shipped feature (ultracode ? this is the substantial piece), then the DOC PR runs in parallel.

243. user (2026-06-16T06:26:59.382Z)

Workflow launched in background. Task ID: w8qso4xrr

Summary: Adversarial due-diligence audit of the SHIPPED golden-regression-fixtures feature (4 merged PRs)

Transcript dir: C:\Users\avidu\claudu\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_ef85c152-821

Script file: C:\Users\avidu\claudu\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-fixtures-shipped-audit-wf_ef85c152-821.js

(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:

"C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-fixtures-shipped-audit-wf_ef85c152-821.js"} to iterate without resending the script.)
Run ID: wf_ef85c152-821
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\workflows\scripts\golden-fixtures-shipped-audit-wf_ef85c152-821.js", resumeFromRunId: "wf_ef85c152-821"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

244. assistant (2026-06-16T06:27:25.748Z)

Audit is running in the background. Now the DOC PR. Let me grab the exact anchors for the CODE_MAP eval-section entry and the GOLDEN_DATA_SHARING cross-link.

245. user (2026-06-16T06:27:27.187Z)

```
179 - `@backend/eval/experiment.py` ? A/B + shadow + **SxS** variant layer (the thin
"experiment" glue per `docs/EXPERIMENT_HARNESS.md`; composes
metrics/calibration/classifier/training/regression ? NO new scoring math).
`::Variant`(segmenter,config_overrides,cue_flags,`min_crossings`[Gate A],`min_players`[Gate
B],classifier_model,feature_names,threshold,merge_gap;
`to_dict/from_dict/spec_hash/is_learned`), `::CONTROL` (pinned zero-gate baseline).
`::VideoCandidates`(name,intervals,features{col-major},gts), `::predict` (gate?learned-
filter?`gemini_refine.merge_overlaps`), `::evaluate_variant` (static / pinned-model / **honest-
LOVO** train-on-others ? mirrors distill_local), `::compare` (labeled **B?A** deltas +
`label_set_hash`), `::compare_unlabeled` (**NEW-footage SxS**: agreed/a_only/b_only via
`match_intervals`, no GT ? the owner's "SxS on new videos"), `::record_experiment` (?
`eval_baselines/experiment_leaderboard.json`), `::load_video_candidates` (DB bridge via
`query_candidate_segments`). ? enabling a gate needs its feature persisted (absent?0.0?fails
gate, warns); learned variant on an unlabeled video needs a pinned `classifier_model`. **No-reg
invariant (tested): all cues OFF ? byte-identical to CONTROL.**
180
181 ### 2.3a Audio (E1 probe; NOT adopted for fusion ? ADR-007)
182 - `@backend/pipeline/audio/hit_detector.py` ? classical-onset shuttle-hit detector. Pure
numpy + stdlib `wave` (NO scipy); ffmpeg only for extraction. `::HitEvent`(t,strength),
`::extract_audio` (ffmpeg -> mono 16k PCM WAV), `::load_wav`, `::onset_envelope` (pure-numpy
STFT spectral flux; `band=(lo,hi)` band-limit), `::detect_hits` (adaptive `mean+k.std`;
`k`=sensitivity knob), `::detect_hits_in_video`. ? separate signal ? WASB never modified by
this; per audio-e1-findings recall 100% but discrimination ~2.2x, band-pass null -> NOT adopted
(PR #35).
```

246. user (2026-06-16T06:27:28.027Z)

```
48 ## Guardrails
49 - The shared folder lives **outside the repo** ? never committed. Keep `output/`
gitignored and do **not** symlink the shared folder into a tracked path.
50 - The features derive from the **Proprietary** corpus ? keep the OneDrive folder
**private** (no public share links).
51
52 See also: `docs/NEXT_STEPS.md` (the "Next-session pickup" anchor),
`docs/MULTI_SIGNAL_FUSION_PLAN.md` (P3), `docs/TRACKNET_WSL_SETUP.md` (the GPU/WSL detector
setup).
```

247. assistant (2026-06-16T06:28:05.646Z)

Audit running. Now the DOC edits in the worktree:

248. user (2026-06-16T06:28:09.105Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\claude\worktrees\gf-
doc\docs\CODE_MAP.md has been updated successfully. (file state is current in your context ? no

need to Read it back)

249. user (2026-06-16T06:28:19.739Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-doc\docs\CODE_MAP.md has been updated successfully. (file state is current in your context ? no need to Read it back)

250. user (2026-06-16T06:28:27.837Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-doc\docs\GOLDEN_DATA_SHARING.md has been updated successfully. (file state is current in your context ? no need to Read it back)

251. user (2026-06-16T06:28:35.536Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

252. assistant (2026-06-16T06:28:46.632Z)

Right ? that file wasn't read in *this* worktree path. Let me read the DOC row there (master includes #192's tracker edits, so line numbers shifted), then edit.

253. user (2026-06-16T06:28:48.369Z)

```
155:| **DOC** | **Finalize this doc + cross-links.** Add `eval_baselines/fixtures/` row to
`CODE_MAP $0` + TEST MAP; cross-link `EVALUATION.md` / `EXPERIMENT_HARNESS.md` /
`GOLDEN_DATA_SHARING.md` / `QUALITY_ITERATIONS.md`; update `docs/README.md` index + memory
`current-state`/handoff. | M1..M4 | No | ? | ? |
```

254. user (2026-06-16T06:28:53.499Z)

```
153      | **01** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1`
proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
154      | **02** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
155      | **DOC** | **Finalize this doc + cross-links.** Add `eval_baselines/fixtures/` row to
`CODE_MAP $0` + TEST MAP; cross-link `EVALUATION.md` / `EXPERIMENT_HARNESS.md` /
`GOLDEN_DATA_SHARING.md` / `QUALITY_ITERATIONS.md`; update `docs/README.md` index + memory
`current-state`/handoff. | M1..M4 | No | ? | ? |
156
```

255. user (2026-06-16T06:29:01.109Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-doc\docs\GOLDEN_REGRESSION_FIXTURES.md has been updated successfully. (file state is current in your context ? no need to Read it back)

256. assistant (2026-06-16T06:29:13.486Z)

All 4 DOC edits in. Docs-only change. Committing and pushing:

257. user (2026-06-16T06:29:24.159Z)

```
=== origin/master ===
8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
(#192)
=== staged (docs only) ===
docs/CODE_MAP.md
docs/GOLDEN_DATA_SHARING.md
docs/GOLDEN_REGRESSION_FIXTURES.md
```



```
=== only my commit? ===
854db56 docs(eval): golden-fixtures discoverability ? CODE_MAP $0/$2.3 + GOLDEN_DATA_SHARING
cross-link (DOC)
=== push ===
branch 'feat/golden-fixtures-doc' set up to track 'origin/feat/golden-fixtures-doc'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      feat/golden-fixtures-doc -> feat/golden-fixtures-doc
```

258. assistant (2026-06-16T06:29:29.808Z)

Pushed (only my commit; master unchanged). Opening the DOC PR and watching CI:

259. user (2026-06-16T06:31:54.887Z)

```
https://github.com/avidullu/badminton-highlight-indexer/pull/193
=== watch ===
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615692
Full test suite (offline, mocked)               pass    1m58s  https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615786
Golden fixtures cross-OS (macos-latest)         pass    16s   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615626
Golden fixtures cross-OS (ubuntu-latest)        pass    23s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615638
Golden fixtures cross-OS (windows-latest)       pass    51s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615667
Mypy (lenient baseline) pass    30s   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615610
Ruff lint      pass    10s   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615627
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27598751082/job/81594615692
CHECKS_EXIT=0
```

260. assistant (2026-06-16T06:32:00.442Z)

DOC checks all green. Merging #193:

261. user (2026-06-16T06:32:07.857Z)

```
failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'

---master---
085c07b docs(eval): golden-fixtures discoverability ? CODE_MAP $0/$2.3 + GOLDEN_DATA_SHARING
cross-link (DOC) (#193)
```

262. user (2026-06-16T06:32:07.866Z)

```
<retrieval_status>not_ready</retrieval_status>

<task_id>w8qso4xrr</task_id>

<task_type>local_workflow</task_type>

<status>running</status>
```

263. assistant (2026-06-16T06:32:27.581Z)

****DOC merged ? #193**** (master `085c07b`). The audit is still running. Let me record the checkpoint and report.

264. user (2026-06-16T06:32:28.681Z)

```

10      # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12      **Checkpoint:** 2026-06-16 (GOLDEN FIXTURES **P1 + M2 SHIPPED ? PR #192 MERGED**, master
`8d3e185`). Project [[working-agreement-tracked-project-docs]]
(`docs/GOLDEN_REGRESSION_FIXTURES.md`) ? **4 PRs merged this session**: #186
(plan+template+CLAUDE.md convention), #189 (M1 framework + synthetic Tier-0 + characterization),
#191 (MX cross-OS CI guard ? all 3 OSes green ? + tracker/legal reconcile), #192 (P1 de-
attribution tool + M2 refusal gate). Tracker: M1? M2? M3? MX? P1? P2?(owner-accepted).
**Coverage 80.41%?80.49%; all CI green incl. the 3-OS `golden-crossos` job.** P1 =
`golden_fixtures.freeze_real_fixture()`: refusal gate (RefusalError unless
owned+minors_free+license), opaque random `fx<hex>` slug, metric-invariant time-origin reset,
strict GT loader, positive no-person/audio/MD5/source-path scan, manifest non-clobber,
`--freeze-real` CLI (exit 2 on refuse); 12 synthetic tests, **NO real data committed**. Built
via background subagents in ISOLATED worktrees; verified each MYSELF (full
suite+cov+ruff+mypy+--check) before merge per the pre-LGTM gate. ROBUST to the concurrent
quality track (#187/#188/#190 windowing/gate) ? pinned preset + CONTROL-only; the sibling's own
~11:30 checkpoint confirms the two tracks are INDEPENDENT. ?? REMAINING: **P3 (real Tier-0
fixtures) = next high-value step, BLOCKED on the owner's per-video CLEARANCE LIST** (which
F:/golden videos are owned + minors-free); per cleared video need only the trajectory CSV
(Frame,Visibility,X,Y) + golden label CSV + fps/fw ? no video/frames/person/audio. (Sibling
notes **~10 more golden videos within days** ? good P3 input.) **DOC** (discoverability cross-
links: README + CODE_MAP $0 + GOLDEN_DATA_SHARING) = next safe non-blocked step. **M4** quality-
floor = synthetic-trivial ? defer to real fixtures. **P0** name-scrub still HELD (collides with
active eval_baselines/windowing churn). 01/02 optional. Owner directives: "keep making progress,
merge as you go, so long as unhindering". Ties [[golden-set-vendoring]] + [[gotcha-shared-
checkout-branch-collisions]].

```

265. user (2026-06-16T06:32:48.692Z)

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

266. assistant (2026-06-16T06:33:07.636Z)

DOC merged ? [#193](https://github.com/avidullu/badminton-highlight-indexer/pull/193)
(master `085c07b`). That's **5 PRs merged this session**, all green on the bar you set:

PR	Deliverable	Coverage	CI
#186	Plan + template + convention	?	green
#189	M1 framework + synthetic Tier-0	80.41%	green
#191	Cross-OS CI guard + reconcile	80.41%	green (3-OS)
#192	P1 de-attribution + M2 refusal gate	**80.49%**	green (3-OS)
#193	DOC discoverability cross-links	80.49%	green (3-OS)

Tracker: M1 ? . M2 ? . M3 ? . MX ? . P1 ? . P2 ? . DOC ?. The golden-fixtures **core is complete** ? a clean clone runs the rally-seg regression with no video/GPU/DB, cross-OS enforced on every PR, the de-attribution tool + refusal gate are in, coverage trended up, every merge verified by me then green in CI.

Final due-diligence in flight: an adversarial audit of the *shipped* code is running (5 lenses ? determinism / privacy / correctness+coverage / robustness / honesty ? adversarial verify ? triage). This catches what per-PR checks might miss (e.g. the real-fixture tests not being in the cross-OS job, privacy-scan completeness). **I'll fix any confirmed must-fix as follow-up green PRs**, or record a clean audit. I'll report its verdict when it lands.

Then I'm at the honest edge of what I can merge without you:

- **P3 (real Tier-0 fixtures)** ? the tooling is built and refusal-gated; it just needs your **per-video clearance list** (which F:/golden videos are owned + minors-free). That's the next real value, and it's the one thing blocked on you. The sibling's "~10 more golden videos within days" is a great input.
- **M4** quality-floor ? meaningful only once real fixtures land (trivial on synthetic).
- **P0** name-scrub ? still held (rewrites `eval\_baselines` keys the active windowing session depends on).

So: I'll keep merging anything that passes ? right now that's whatever the audit surfaces. Drop me the clearance list whenever and I'll mint the first real fixtures through the gate.

267. user (2026-06-16T06:37:48.951Z)

```
<task-notification>
<task-id>w8qso4xrr</task-id>
<tool-use-id>toolu_01S1Dv5bFSJ76KTSKH0DaMwn</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\w8qso4xrr.output</output-file>
<status>completed</status>
<summary>Dynamic workflow "Adversarial due-diligence audit of the SHIPPED golden-regression-
fixtures feature (4 merged PRs)" completed</summary>
<result>{"audits":[{"lens":"DETERMINISM / cross-OS (Golden Regression Fixtures, PRs
#186/#189/#191/#192, worktree gf-doc @ 8d3e185)","summary":"The COMMITTED synthetic CONTROL path
is genuinely cross-OS deterministic and the per-PR verification of it holds: (frame/fps)*fps
round-trips exactly for these fixtures so net_crossings is a stable integer count; JSON writers
use sort_keys=True; compare_expected is dict/set-order-independent and covers every written
field; CRLF is fully handled by csv.reader(newline='')+json.load+the eol=lf .gitattributes pin.
The lens's specific net_crossings/frame*fps and iteration-order suspicions are clean ? I say so
honestly. BUT the lens's central suspicion is correct and uncovered a real defect cluster on the
OTHER path. The de-attribution path (freeze_real_fixture / time-origin reset / M2) is exercised
ONLY by tests/test_golden_real_fixtures.py, which the golden-crossos CI matrix does NOT run (it
runs --check + test_golden_fixtures.py only); that test file runs solely on the ubuntu full-
suite lane, so the reset path has ZERO cross-OS coverage. And inside that under-covered path the
documented 'provably metric-invariant / bit-identical scores' guarantee for the time-origin
reset is FALSE for any non-whole-second t0: I empirically broke it on 26 of 40 sub-second frame
offsets, with a worst score delta of 1.0000000000001e-06 ? landing exactly past the 1e-6
compare_expected tolerance cliff. The committed invariance test only passes because it hard-
codes a whole-second (9000-frame/300.0s) offset that dodges the rounding cliff. Separately, a
real fixture whose GT label starts before its first trajectory frame produces a negative reset
label-start that the STRICT re-parse rejects, crashing freeze_real_fixture AFTER it has already
written trajectory/labels/provenance to disk (orphaned partial fixture). Finally, the 'pure-
stdlib / no numpy on the CONTROL path' framing over-claims for IMPORT: importing golden_fixtures
fails on a clean clone without numpy (experiment-&gt;classifier imports numpy at module top) ?
only numpy EXECUTION is avoided."},"findings":[{"title":"Time-origin reset is NOT metric-
invariant for non-whole-second t0 ? contradicts the 'provably metric-invariant / bit-identical
scores' claim, and the worst delta lands past the 1e-6
cliff","severity":"high","evidence":["docs/GOLDEN_REGRESSION_FIXTURES.md:94 ('Time-origin reset
... Provably metric-invariant ... bit-identical scores')";
backend/eval/golden_fixtures.py:619-623 + :583-586 assert the same. But the reset subtracts
t0_sec in float and writes labels/trajectory at 6dp (_fmt, golden_fixtures.py:195-207,274-279),
then the STRICT gt_loader re-parses the rounded text ? so when t0_sec is non-terminating, the
reset vs no-reset scores diverge. I swept frame offsets 0..39 (fps=30) through
freeze_real_fixture(time_origin_reset=True vs False): score equality BROKE on 26/40 offsets;
e.g. offset=7 gives mean_start_error 0.033333 (reset) vs 0.033334 (no reset). Worst measured
|delta|=1.0000000000001e-06, i.e. STRICTLY greater than _FLOAT_TOL=1e-6 (golden_fixtures.py:91)
used by compare_expected.approx (golden_fixtures.py:786-789) ? so this can flip a parse-compare
digit/result. The committed test only passes because it hard-codes offset_frames=9000 ?
9000/30=300.0s exactly (tests/test_golden_real_fixtures.py:175), a whole-second offset that
never hits the cliff."},"recommendation":"Either (a) weaken the doc/code claim to 'metric-
invariant up to the 6dp serialization rounding' and prove the bound is < _FLOAT_TOL, or (b)
make it actually invariant: compute t0 so the shift is an exact 6dp / exact-frame quantity (e.g.
reset by an integer number of frames only and derive t0_sec consistently, or round t0_sec to 6dp
before subtracting from both streams). Add a parametrized test over fractional-second offsets
(e.g. 7,11,13 frames) asserting score equality AND |delta| <= _FLOAT_TOL,
replacing/augmenting the single 9000-frame case."},"title":"De-attribution / time-origin-reset
path has ZERO cross-OS coverage ? golden-crossos CI omits tests/test_golden_real_fixtures.py
(the lens's KEY suspicion,
confirmed)","severity":"medium","evidence":[".github/workflows/ci.yml:119-122 ? the golden-
crossos matrix (ubuntu/windows/macOS) runs only `python -m backend.eval.golden_fixtures --check`
and `pytest tests/test_golden_fixtures.py`. tests/test_golden_real_fixtures.py
(freeze_real_fixture, M2 refusal gate, time-origin-reset invariance, strict-loader) is run ONLY
by the ubuntu-only full-suite lane via run_all_tests.py pytest discovery (ci.yml:67-97). So
every de-attribution/reset assertion ? including the metric-invariance claim above ? is verified
```

on ubuntu alone. The doc markets this exact path as the safe-to-publish real-fixture pipeline (P1/P3) and §9 DoD requires 'win+ubuntu+mac' green, but the float-sensitive reset arithmetic that path performs is never run on win/mac.", "recommendation": "Add tests/test_golden_real_fixtures.py to the golden-crossos matrix step (it is fully synthetic/tmp_path, no GPU/network ? see its module docstring), so the reset round-trip is exercised on all three OSes. This is also what surfaces finding #1 cross-OS before a real P3 fixture is committed."}, {"title": "freeze_real_fixture crashes after partial on-disk writes when a GT label precedes the first trajectory frame (negative reset start rejected by STRICT re-parse)", "severity": "low", "evidence": "backend/eval/golden_fixtures.py:624-630 sets t0_frame=min(frame) and t0_sec=t0_frame/fps, then _reset_labels subtracts t0_sec from every label (golden_fixtures.py:552-554). If a label starts before the first trajectory frame, the reset start goes negative; the writes happen at golden_fixtures.py:635-636 (trajectory.csv, labels.csv) and :671 (provenance.json) BEFORE replay_fixture(fdir) at :673, whose strict load_gt_intervals raises ValueError('negative start') (gt_loader.py:78-79). I reproduced it: a synthetic source with trajectory shifted to frame 100 + a label at t=0 fails with 'labels.csv:2: negative start (-3.333333)', leaving trajectory.csv/labels.csv/provenance.json orphaned in the output dir. No cleanup-on-failure and no pre-write validation that all reset labels stay >= 0.", "recommendation": "Validate before writing: after computing reset_gts, raise RefusalError/ValueError if any reset start < 0 (label timeline predates the trajectory ? a labeling/time-base mismatch), and/or write into a temp dir that is only promoted on success so a failed freeze leaves no partial fixture directory."}, {"title": "Pure-stdlib / no numpy on the CONTROL path' over-claims for IMPORT ? golden_fixtures cannot be imported on a clean clone without numpy", "severity": "low", "evidence": "golden_fixtures.py:34-35 docstring ('pulls in no numpy/cv2 on the CONTROL path') and eval_baselines/fixtures/README.md ('CONTROL / pure-stdlib only ? no numpy'). But golden_fixtures.py:51 does `from backend.eval.experiment import CONTROL, VideoCandidates, predict`, experiment.py:49 does `from backend.eval.classifier import LogisticRegression`, and classifier.py:18 does `import numpy as np` at module top. I simulated a clean clone by blocking numpy import: `from backend.eval import golden\_fixtures` raises ModuleNotFoundError: No module named 'numpy'. CI stays green only because golden-crossos explicitly `pip install ... numpy` (ci.yml:118) and even comments 'CONTROL replay is numpy-only'. So the accurate statement is the already-verified one (numpy is not EXECUTED), not 'no numpy' / 'pure-stdlib'.", "recommendation": "Correct the docstring/README to 'avoids numpy EXECUTION on the replay path (numpy is still imported transitively via experiment->classifier and must be installed)', matching ci.yml's own numpy install, OR break the import chain (import LogisticRegression lazily inside experiment.predict's learned branch) if a truly numpy-free import is desired. Low risk today since CI installs numpy, but the doc/README claim is false as written."}], "coverageGa ... (truncated 71744 chars, full result in C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\w8qso4xrr.output)</result>
<usage><agent_count>9</agent_count><subagent_tokens>980200</subagent_tokens><tool_uses>214</tool_uses><duration_ms>649481</duration_ms></usage>
</task-notification>

268. assistant (2026-06-16T06:38:05.653Z)

The audit earned its keep ? it confirmed real defects, including a genuine **metric-invariance bug** in the time-origin reset (off by exactly past the 1e-6 cliff for sub-second offsets) that the committed test masks by using a whole-second offset. Let me read the full triage (synthesis + the other lenses' confirmed findings) before fixing.

269. user (2026-06-16T06:38:08.014Z)

```
10:      "lens": "DETERMINISM / cross-OS (Golden Regression Fixtures, PRs #186/#189/#191/#192,
worktree gf-doc @ 8d3e185)",
47:      "lens": "PRIVACY / de-attribution ? completeness of `_scan_forbidden`, surrogate non-
reversibility, refusal-gate bypass, raw-trajectory fingerprinting, and honesty of the doc's
bounded claims, for the shipped `freeze_real_fixture` de-attribution path.",
90:      "lens": "CORRECTNESS + COVERAGE",
135:     "lens": "ROBUSTNESS to the concurrent quality track + future windowing/gate change
(audit of shipped Golden Regression Fixtures, PRs #186/#189/#191/#192, worktree gf-doc @
8d3e185)",
178:     "lens": "HONESTY / over-claim",
249:[Omitted long matching line]
250:     "mustFix": [
```

```
253:     "niceToHave": [  
268:     "nonIssues": [  

```

270. user (2026-06-16T06:38:12.938Z)

```
248     "synth": {  
249         "overallVerdict": "SHIPPED feature PASSES due-diligence for what is actually  
committed today (the 3 synthetic Tier-0 fixtures) but is NOT clean for the de-attribution /  
real-fixture (P3) path it was built to enable. Nothing currently relied on is broken: the  
synthetic CONTROL replay is genuinely cross-OS deterministic (integer-exact net_crossings,  
sort_keys, parse-compare, CRLF/.gitattributes handled), the refusal gate (M2) and os.urandom  
surrogate are sound, the time-origin reset is exact for the committed whole-second case, the  
suite is green (759 pass, 80.49%, ruff+mypy clean), and `--check` is self-consistent. However,  
the verifications and audits CONFIRM one real privacy hole that bites the moment the freeze-real  
tooling is used on real data, plus a cluster of confirmed-but-deferrable  
correctness/coverage/doc gaps and one forward-trap test. The single hard blocker before ANY P3  
real (`cleared_real`) fixture or counsel sign-off is the unscanned free-text  
`source_note`/`label` leak. The remaining items (raw per-frame CSV vs doc, reset non-invariance,  
compare_expected blind spot, scan substring over-skip, W1/max_win pinned-but-ignored, doc over-  
claims, test_synthetic_provenance_tags forward-trap, missing negative tests) are real but either  
do not touch committed data or are guardrails for an as-yet-unbuilt path; they must be resolved  
before the real-data path is depended on, but do not block the synthetic ship. Recommend gating  
P3/counsel sign-off on the mustFix item plus reconciling the doc's de-attribution claims (raw-  
CSV exclusion, metric-invariance, minimization-to-5-scalars) with what the code actually does,  
since the owner's 2026-06-16 risk acceptance was made against a doc that overstates the de-  
attribution implemented.",  
250         "mustFix": [  
251             "backend/eval/golden_fixtures.py `scan_forbidden` (lines 501-535) does NOT  
catch free-text identifiers in the operator-supplied `source_note` (line 663) and `label` (line  
643) fields ? it only matches FORBIDDEN_KEYS biometric substrings, a 32-hex MD5, and source-PATH  
tokens. Both fields are written verbatim into the committed provenance.json and then scanned,  
but a player/match/event name or date passes silently (empirically: label='mens singles final' +  
source_note='Lin Dan vs Lee, India Open 2024, Court 1' froze with NO refusal). The only test  
(tests/test_golden_real_fixtures.py:155) masks this by passing a PATH token. Fix before any P3  
real fixture: restrict `--source_note`/`--label` to an allow-list charset/controlled vocabulary  
OR drop them from committed provenance (move human notes to the off-git surrogate map), and add  
a test that a free-text identifying note is rejected or never persisted. This is the one hole  
the refusal gate exists to prevent and is the hard gate on the counsel-signoff/P3 path."  
252         ],  
253         "niceToHave": [  
254             "Reconcile docs/GOLDEN_REGRESSION_FIXTURES.md with the shipped code on the raw  
trajectory: $4c:96 ('never commit the per-frame (Frame,Visibility,X,Y) CSV'), $2 D4:45 and  
$4b/TM5 ('only ~5 aggregate scalars/window') are contradicted by freeze_real_fixture  
(golden_fixtures.py:635 write_trajectory_csv) and the synthetic fixtures, which DO commit the  
full per-frame trajectory.csv (the de Montjoye fingerprint) next to expected.json. No real  
fixture is committed yet so this is not a live leak, but the owner's 2026-06-16 risk acceptance  
was made against a doc that overstates minimization. Either update $4c/$4b/$3/$5 to state the  
per-frame CSV IS committed (de-attributed via time-origin reset + random slug + metadata strip)  
and re-do the non-attributability analysis on that basis with owner re-confirmation, or change  
the freeze to commit only the aggregate snapshot. Block P3 on whichever is chosen.",  
255             "Make the time-origin reset actually metric-invariant for non-whole-second t0 in  
freeze_real_fixture (golden_fixtures.py:619-630): the reset subtracts full-precision  
t0_sec=t0_frame/fps while labels/trajectory serialize at 6dp, so reset-vs-no-reset scores  
diverge on 26/40 sub-second frame offsets with a worst |delta|=1.000000000001e-06 ? strictly  
past _FLOAT_TOL=1e-6 (line 91). The 'provably metric-invariant / bit-identical' claim (docs $94,  
golden_fixtures.py:583-586,619-623) is therefore false, and the only invariance test  
(test_golden_real_fixtures.py:175) hard-codes a whole-second 9000-frame offset that dodges the  
cliff. Round t0_sec to 6dp before subtracting from both streams (or reset by integer frames and  
derive t0_sec consistently), and replace the single 9000-frame test with a parametrized  
fractional-offset test asserting score equality AND |delta|<=_FLOAT_TOL. Fix before the metric-  
invariant de-attribution guarantee is depended on (P3); --check itself is self-consistent  
today.",  
256             "Add tests/test_golden_real_fixtures.py to the golden-crossos CI matrix step  
(.github/workflows/ci.yml:119-122) ? it currently runs only `--check` + test_golden_fixtures.py,  
so the freeze_real/M2-gate/time-origin-reset/strict-loader paths have ZERO cross-OS coverage
```

(run only on the ubuntu full-suite lane). The file is fully synthetic/tmp_path (no GPU/network). This also surfaces the reset 1e-6 cliff cross-OS before a real P3 fixture is frozen.",

257 "Fix tests/test_golden_fixtures.py:70-77 test_synthetic_provenance_tags before P3: it parametrizes over ALL manifest slugs and hard-asserts kind=='synthetic'/license=='synthetic', but freeze_real_fixture writes kind='cleared_real' into the same manifest ? committing the first real fixture (P1's intended output) will red the cross-OS suite. Split the assertion by kind (synthetic-tag check for kind=='synthetic'; a parallel cleared_real check for owned/minors_free/non-blank-license) and rename to reflect it gates all Tier-0 fixtures.",

258 "Tighten compare_expected (golden_fixtures.py:823-830): the `score` block and top-level keys (793-796) are compared against FIXED field lists, so an EXTRA key present on only one side is silently ignored (verified: injecting score['bogus']=99 returns []), weakening the 'catches ANY behaviour change' claim (docs §4d). Mirror the shuttle block's `set(rsc)!=set(csc)` key-set equality check on the score dict and top-level keys before the per-field loops, and add a negative test.",

259 "Change the source-path skip in _scan_forbidden (golden_fixtures.py:530) from substring to exact match: `any(t in lit for lit in \_SCHEMA\_FILENAME\_LITERALS)` over-skips tokens like 'label', 'traj', 'expected', 'prov' that are substrings of schema literals, so a real source named label.csv/trajectory.csv leaking its stem via --source-note is MISSED (verified: /secret/label.csv + 'label arena' text raised no RefusalError). Use `t in \_SCHEMA\_FILENAME\_LITERALS` (exact equality) instead.",

260 "Thread max_window_duration (W1 bounded windowing, PR #187) through the replay or stop pinning it: golden_fixtures.py:75 pins PINNED_MAX_WINDOW=0.0 into provenance.windowing.max_win and the manifest, but _pinned_proposal (287-293) returns only (vt,mg,mwd) and distill_local.build_candidates takes no max_window param, so a future SERVED W1 flip or _split_overlong_groups change passes every fixture SILENTLY. Either extend _pinned_proposal + build_candidates to consume max_win (with a regression test that flipping it changes segments), or drop max_win from the pinned schema and document that W1 is deliberately uncharacterized.",

261 "Add the missing FAILURE-semantics negative tests: no committed test asserts compare_expected returns a NON-EMPTY list for a regression (changed float >1e-6 / changed int / candidate reorder / extra-or-missing key), and _scan_forbidden's FORBIDDEN_KEYS branch (513) and MD5 branch (519) and refresh_snapshot (452-461, incl. --all and the no-target ValueError) have no negative test. The gate's raison d'etre ? that it FIRES on a regression/leak ? is currently unexercised.",

262 "Correct the 'pure-stdlib / no numpy on the CONTROL path' over-claim in golden_fixtures.py docstring (lines 8-9, 34-35), eval_baselines/fixtures/README.md, and the self-contradictory ci.yml:117 comment: importing golden_fixtures requires numpy (golden_fixtures.py:51 -> experiment.py:49 -> classifier.py:18 `import numpy`). The accurate statement is that numpy EXECUTION (BLAS-nondeterministic LogisticRegression) is avoided on the replay path, not numpy import ? which is why golden-crossos pip-installs numpy.",

263 "Wrap the freeze_real_fixture write+scan in try/finally (golden_fixtures.py:632-684) to remove the partially-written fixture dir on a RefusalError: the four artifacts are written BEFORE _scan_forbidden runs, so a refused freeze leaves trajectory.csv/labels.csv/expected.json/provenance.json (potentially leak-bearing) orphaned on disk where a later `git add` could sweep them in. Also add a pre-write validation that all reset label starts are >=0 (a GT label preceding the first trajectory frame yields a negative reset start that the strict re-parse rejects AFTER the partial writes ? gt_loader.py:78-79). Add a regression test asserting the fixture dir does not exist after a scan/validation refusal.",

264 "Fix doc §4a (docs/GOLDEN_REGRESSION_FIXTURES.md:80-81) which over-claims the replay reuses windowing.build_windows / WindowingPreset / 'default SERVED' ? the shipped replay imports none of windowing.py and calls distill_local.build_candidates with a raw 3-tuple. The module docstring (golden_fixtures.py:11-16) has the correct map; the doc is the stale cold-start map for the next contributor.",

265 "Reconcile the two refresh_snapshot reason claims: golden_fixtures.py:451 docstring and docs §6:134 say the CLI 'requires' a recorded reason, but main() (golden_fixtures.py:955-957) only WARNS and exits 0. Either make --reason mandatory (ap.error if absent, like the --freeze-real required-args check) or correct the docstring/doc to 'WARNS if missing'.",

266 "Harden load_fixture schema-version guard (golden_fixtures.py:761-764): it raises only on max(expected, prov-fallback) > CURRENT_SCHEMA, so a forward-incompatible provenance.json (sv=99) when expected.json carries the current version is read silently. Guard both files independently (max of the two) and add a negative test; also plan a lower-bound/migration message for when CURRENT_SCHEMA is first bumped so a stale committed fixture fails with 'needs re-freeze' rather than a confusing schema_version exact-compares mismatch."

```

267         ],
268         "nonIssues": [
269             "'Raw per-frame trajectory is committed verbatim for REAL fixtures, fully
retaining the de Montjoye fingerprint' as a LIVE leak ? REFUTED by verification (verdict:
partial). No cleared_real fixture exists in the tree; all three committed fixtures are
kind='synthetic'/license='synthetic' (manifest.json, provenance.json), and the cited
trajectory.csv coords are machine-generated synthetic parabolas (clean_single_rally starts at
the exact frame-center 640,360), not real-person footage ? so the re-ID argument does not apply
to anything committed. freeze_real_fixture is tooling-only (P1, 'no real data committed'); P3 is
Todo. The REAL, kept part of this finding is the doc-vs-code raw-CSV contradiction, captured in
niceToHave.",
270             "Time-origin reset non-invariance flips/flakes the green suite or the cross-OS
--check gate ? REFUTED. The shipped --check recomputes from the SAME already-reset committed
bytes via replay_fixture (which never re-applies the reset), so it is self-consistent; verified
--check returns [] even for a fixture frozen at the worst-case offset. There are no committed
cleared_real fixtures exercising the reset, and freeze_real is owner-gated and not run in CI.
The defect is a false documented invariant + a masking test (in niceToHave), not an active
breakage.",
271             "Candidate reordering / gt_hash / shuttle-key / candidate-count / segment-count
/ net_crossings drift goes uncaught ? REFUTED. compare_expected uses positional candidate
compare, set-equality on shuttle keys, count checks on candidates/control_segments, and EXACT-
int compare on net_crossings (golden_fixtures.py:798-821); the per-PR verification of these
holds. (The blind spot is narrowly the EXTRA score/top-level key ? in niceToHave.)",
272             "Signature/algorithm changes to the reused
distill_local/trajectory_to_action_windows/rally_gate stack pass silently ? REFUTED (positive
finding): FEATURE_NAMES drift raises at import (assert, golden_fixtures.py:82-84), a
build_candidates proposal-shape change breaks the explicit call with TypeError, and math changes
diff loudly via parse-compare. The loud-fail guarantee for the (vt,mg,mwd) knobs and CONTROL
decision path is real. (Only the W1/max_win axis is the exception ? in niceToHave.)",
273             "Refusal gate (M2) bypass via truthy-but-not-True, surrogate-slug reversibility,
or an on-git surrogate->source map ? REFUTED across audits: the `is not True` checks resist
truthy bypass and fire BEFORE any write, the fx_<16hex> slug is os.urandom-derived with no
algebraic link to the source, and no surrogate->source map is written on-git. These are
genuinely sound.",
274             "net_crossings gate (#188/#190) regressing the CONTROL path ? REFUTED: CONTROL
gates at min_crossings=0, so net_crossings is stored only as a feature, not a segment filter,
and is exact-int compared ? correctly characterized; the control decision path is unaffected.",
275             "fps/frame_width committed un-quantized as a camera/venue quasi-identifier ? BY
DESIGN / disclosed: doc §7 ? honestly admits quantization is a deferred P3 option ('fps/fw
retained as scale constants; quantize is a P3 option'). The only fix needed is reconciling §4c
wording with §7 ? so a reader does not think quantization is active (a doc-consistency nit,
folded into the broader doc reconciliation), and gating P3 on quantization if real fixtures land
first.",
276             "'cross-OS enforced' wording and the '[ok] ... F1=1.000000' per-line lacking an
inline caveat ? minor wording over-reaches. 'enforced' depends on server-side branch-protection
(unverifiable from repo) vs 'runs on every PR'; the F1=1.0 caveat IS carried on the trailing
summary line and in surrounding docs/test messages. Low impact; soften wording
opportunistically, not blocking.",
277             "Schema-literal exclusion causing the 'traj' ? 'trajectory.csv' FALSE-POSITIVE ?
correctly avoided by design (golden_fixtures.py:495-535); the audit's own coverage note confirms
this works as intended. (The opposite direction ? over-skipping real leaks ? is the real bug, in
niceToHave.)"
278         ]
279     }
280 }
281 }

```

271. assistant (2026-06-16T06:40:16.116Z)

The audit delivered real value. ****Verdict: the *committed* synthetic feature passes due-diligence (nothing relied on today is broken), but the de-attribution/real-fixture path has confirmed gaps that must be fixed before P3**** ? and one is owner-facing. Triage:

- ? ****Must-fix (blocks P3):**** `\_scan\_forbidden` doesn't catch identifying free-text in the operator's `--source-note`/`--label` (only biometric/MD5/path tokens) ? a player/match name

passes silently. The one hole the gate exists to prevent.

- ? ****Owner-facing honesty issue:**** the doc claims "never commit the per-frame CSV / only ~5 scalars," but the code ****does**** commit the full per-frame `trajectory.csv`. No live leak (only synthetic committed), ****but your 2026-06-16 risk acceptance was made against a doc that overstates the minimization**** ? you need to re-confirm on the corrected basis. I'll fix the doc and flag it to you.

- ? ****Confirmed hardening:**** reset metric-invariance bug (sub-second `t0` lands past 1e-6), add the real-fixture tests to the cross-OS matrix, `compare\_expected` extra-key blind spot, scan substring?exact, schema-guard both files, partial-write cleanup, the `test\_synthetic\_provenance\_tags` forward-trap, missing failure-semantics negative tests, and the "no-numpy"/\$4a doc over-claims.

I'll fix all of these in one audit-hardening PR. Let me set up the worktree and delegate the implementation (then verify + merge myself).

272. user (2026-06-16T06:40:20.650Z)

branch 'feat/golden-fixtures-audit-hardening' set up to track 'origin/master'.
HEAD is now at 085c07b docs(eval): golden-fixtures discoverability ? CODE_MAP \$0/\$2.3 + GOLDEN_DATA_SHARING cross-link (DOC) (#193)
=== audit worktree HEAD ===
085c07b docs(eval): golden-fixtures discoverability ? CODE_MAP \$0/\$2.3 + GOLDEN_DATA_SHARING cross-link (DOC) (#193)

273. assistant (2026-06-16T06:40:27.826Z)

Worktree at `085c07b`. Delegating the audit-hardening implementation with a precise, itemized spec (then I verify + merge):

274. user (2026-06-16T06:41:12.795Z)

Async agent launched successfully.
agentId: a175e8d3638e28d46 (internal ID - do not mention to user. Use SendMessage with to: 'a175e8d3638e28d46' to continue this agent.)
The agent is working in the background. You will be notified automatically when it completes.
Do not duplicate this agent's work ? avoid working with the same files or topics it is using.
Work on non-overlapping tasks, or briefly tell the user what you launched and end your response.
output_file: C:\Users\avidu\AppData\Local\Temp\claudu\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a175e8d3638e28d46.output
Do NOT Read or tail this file via the shell tool ? it is the full subagent JSONL transcript and reading it will overflow your context. If the user asks for progress, say the agent is still running; you'll get a completion notification.

275. assistant (2026-06-16T06:41:37.046Z)

The audit paid off ? it surfaced a real bug and an honesty gap that per-PR verification missed. Here's the verdict and what I'm doing.

Audit verdict

****The committed synthetic feature passes due-diligence ? nothing relied on today is broken**** (the synthetic CONTROL replay is genuinely cross-OS deterministic, the refusal gate and random surrogate are sound, suite green, `--check` self-consistent). The gaps are all on the ****de-attribution / real-fixture (P3) path**** that isn't exercised by committed data yet.

Fixing now (one audit-hardening PR, building in an isolated worktree)

- ? ****The must-fix:**** `\_scan\_forbidden` lets an identifying `--source-note`/`--label` (e.g. a player/match name) through ? I'm restricting `label` to a strict slug charset and dropping operator free-text `source\_note` from committed provenance.

- ? ****Confirmed hardening:**** make the time-origin reset ***actually*** metric-invariant for sub-second offsets (it was off by exactly past 1e-6); add the real-fixture tests to the cross-OS matrix; close the `compare\_expected` extra-key blind spot; scan substring?exact; guard both schema files; temp-dir-then-promote so a refused freeze leaves no partial files; fix the `test\_synthetic\_provenance\_tags` forward-trap (it'd red the suite when the first real fixture

lands); add the missing failure-semantics negative tests; correct the "no-numpy"/\$4a doc over-claims.

I'll verify it myself (synthetic `--check` unchanged + full suite + coverage ? 80.4% + ruff + mypy + cross-OS) and merge on green.

? One thing that needs ***you*** (honesty issue affecting your risk acceptance)

The doc said **"never commit the per-frame CSV ? only ~5 aggregate scalars."** That's ****false****: the code commits the ****full per-frame `trajectory.csv`**** (de-attributed via reset + random slug + metadata strip). There's ****no live leak**** (only synthetic fixtures are committed), ****but your 2026-06-16 risk acceptance was made against a doc that overstated the minimization**** ? the per-frame stream is the de Montjoye fingerprint, not 5 scalars.

I'm correcting the doc to the truth, but ****this is your call before any real (P3) fixture****:
- ****a)**** Accept committing the de-attributed ****per-frame**** trajectory (re-confirm on the corrected basis ? non-attributability still rests on the source staying private), ****or****
- ****b)**** Have me change the freeze to commit ****only the ~5 aggregate scalars**** per window (true minimization ? a bigger change, removes the raw fingerprint entirely).

My recommendation: ****b)**** if these will ever go public, since it removes the fingerprint outright; ****a)**** is fine if the repo stays private and the source stays unpublished. Either way I won't mint a real fixture until you pick.

I'll report the hardening-PR merge when it lands, and hold P3 on (1) your clearance list and (2) this (a)/(b) decision.

276. user (2026-06-16T07:07:03.151Z)

```
<task-notification>
<task-id>a175e8d3638e28d46</task-id>
<tool-use-id>toolu_01UNNbSUHRsX4cawKiWTVCTS</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\tasks\a175e8d3638e28d46.output</output-
file>
<status>completed</status>
<summary>Agent "Implement golden-fixtures audit hardening" completed</summary>
<result>All work complete and verified. Here is the report.
```

Summary

Hardened the Golden Regression Fixtures feature per the adversarial audit. All 17 confirmed fixes implemented. The 3 committed synthetic fixtures' `expected.json` (and CSVs/provenance) remain ****byte-identical**** ? I hardened the freeze-real / edge paths + docs only.

Files changed

- `backend/eval/golden\_fixtures.py` ? code fixes #1?#9.
- `tests/test\_golden\_fixtures.py` ? fix #11 (by-kind provenance tags).
- `tests/test\_golden\_real\_fixtures.py` ? fix #2 (parametrized invariance) + fixes #1/#3/#4/#5/#6/#7/#10 negative tests.
- `.github/workflows/ci.yml` ? fix #12 (added `test\_golden\_real\_fixtures.py` to `golden-crossos`).
- `docs/GOLDEN\_REGRESSION\_FIXTURES.md` ? fixes #13?#17.
- `eval\_baselines/fixtures/README.md` ? fix #9 (numpy wording).

Fix notes worth flagging

- ****#1**** label now validated against `^[a-z0-9][a-z0-9_\-]*$`` (free-text like "mens singles final" ? `RefusalError`); `source\_note` is accepted-but-NOT-persisted (dropped from committed provenance, still scanned defensively).
- ****#3**** real-fixture freeze now writes to a temp dir and `os.replace`-promotes only after the privacy scan passes; negative reset labels are refused up-front ? a refused/failed freeze leaves ****no**** partial dir.
- ****#2** (important caveat): ****Bit-identical reset-vs-no-reset scores are not achievable for arbitrary sub-second offsets****, and I confirmed this is mathematically irreducible: the

trajectory shifts by integer frames while labels live on the 6dp CSV grid, and 6dp-rounding is non-additive (`round6(8/30) ? round6(7/30) ? round6(1/30)`). I implemented the prompt's prescribed ``t0_sec`+label` 6dp-rounding (eliminates double-rounding), which keeps the `**genuine/raw**` metric delta at the float-error floor ( $\sim 3.3e-7$ , well within ``_FLOAT_TOL``); it is exactly bit-identical for whole-second offsets. The only residual is a one-6dp-ULP ( $1e-6$ ) difference in the `*serialized*` boundary error on sub-second offsets ? the serialization grid, not a metric divergence. The test asserts raw-score invariance within ``_FLOAT_TOL`` (and exact snapshot equality for whole-second offsets), and I corrected the doc/docstring claims from "bit-identical" to "metric-invariant within ``_FLOAT_TOL``; bit-identical for whole-second offsets" (honesty norm).

Verification (verbatim)

1. ``--check``: ``3/3 fixtures match`` ? synthetic replay unchanged. Re-freeze confirms all synthetic ``expected.json`/`provenance.json`/CSV` bytes reproduce exactly.
2. Full suite: ``784 passed, 7 skipped`` ? total coverage `**80.71%**` (? 80.4% floor; +0.22% vs prior 80.49%). ``golden_fixtures.py`` at 89%. The 7 warnings are pre-existing ``ResourceWarning: unclosed database`` from unrelated sqlite tests.
3. ``ruff check .`` ? ``All checks passed!``; ``mypy backend training`` ? ``Success: no issues found in 103 source files``.
4. Fix #1 demonstrated into a tmp dir (not ``eval_baselines/fixtures/``): ``label="mens singles final"`` rejected with no partial dir; a safe ``source_note`` left no ``source_note`` key in committed provenance (boilerplate ``note`` kept, ``label`` committed verbatim); a path-bearing ``source_note`` tripped the defensive scan.

No real fixtures were committed; ``eval_baselines/fixtures/`` contains only the 3 synthetic fixtures + manifest + README. No git commands were run.

Two items I want to be explicit about: the `**#2 "bit-identical" claim is not literally achievable**` for sub-second offsets (I implemented the rounding and documented the honest within-tolerance guarantee instead); and the doc's coverage row now reads 80.71% (the prior ``_this PR_ 80.49%`` row was left intact above the new audit-hardening row).</result>
<usage><subagent_tokens>202425</subagent_tokens><tool_uses>103</tool_uses><duration_ms>1550355</duration_ms></usage>
</task-notification>

277. assistant (2026-06-16T07:07:25.114Z)

The hardening subagent reports all 17 fixes done, synthetic fixtures byte-identical, `**coverage 80.71% (?)**`, and ? notably ? it was `**honest about fix #2**`: bit-identical reset for arbitrary sub-second offsets is mathematically irreducible (integer-frame shift vs the 6dp label grid), so it implemented the rounding (raw delta at the float-error floor $\sim 3.3e-7$, well within tolerance) and corrected the doc claim to "metric-invariant within ``_FLOAT_TOL``" rather than over-claiming. That's exactly the integrity I want. Now I verify myself. Let me review the changed-files stat + the owner-facing doc change first.

278. user (2026-06-16T07:07:28.855Z)

```
=== status (no real fixture?) ===
M .github/workflows/ci.yml
M backend/eval/golden_fixtures.py
M docs/GOLDEN_REGRESSION_FIXTURES.md
M eval_baselines/fixtures/README.md
M tests/test_golden_fixtures.py
M tests/test_golden_real_fixtures.py
=== diffstat ===
.github/workflows/ci.yml          | 5 +-
backend/eval/golden_fixtures.py   | 242 ++++++-----
docs/GOLDEN_REGRESSION_FIXTURES.md | 40 +---
eval_baselines/fixtures/README.md | 8 +-
tests/test_golden_fixtures.py     | 17 +-
tests/test_golden_real_fixtures.py | 311 ++++++-----
6 files changed, 514 insertions(+), 109 deletions(-)
=== eval_baselines/fixtures still only synthetic? ===
README.md
```

```

clean_single_rally
manifest.json
multi_rally_with_deadtime
occlusion_break
=== DOC: the owner-facing minimization correction (§4c/§3/§5 + ? re-confirm) ===
-2. **Privacy/legal filter (SEALED-FIXTURES)** ? only **de-attributed, minimized, owner-source, biometric-free** numbers are ever committed; richer/sensitive streams stay key-gated or out-of-repo. Non-attributability comes from **deletion-at-source**, not encryption.
+2. **Privacy/legal filter (SEALED-FIXTURES)** ? only **de-attributed, owner-source, biometric-free** numbers are ever committed; richer/sensitive streams stay key-gated or out-of-repo. Non-attributability comes from **deletion-at-source**, not encryption. ? "Minimized" means **stream minimization** (shuttle-only, no person/audio/pose) ? **NOT** reduction of the shuttle stream to ~5 scalars: the de-attributed per-frame `trajectory.csv` IS committed (see §4c + its risk-acceptance re-confirmation note).
-| D4 | **Freeze seam = post-detector trajectory** *(design)* | workflow-1 synthesis | Widest deterministic surface; reuse the shipped eval stack verbatim; the raw per-frame CSV is **never committed** (only ~5 aggregate, fps/res-invariant shuttle scalars/window). |
+| D4 | **Freeze seam = post-detector trajectory** *(design; SHIPPED differs ? see ?)* | workflow-1 synthesis | Widest deterministic surface; reuse the shipped eval stack. ?
**Correction (audit 2026-06-16):** the shipped freeze **commits the de-attributed per-frame `(Frame,Visibility,X,Y)` `trajectory.csv`** (shuttle-only, time-origin reset) ? it is **NOT** minimized to ~5 aggregate scalars. See §4c (and its risk-acceptance re-confirmation note). |
-Freeze at the **trajectory CSV ? pure-Python replay**. One shared helper
`backend/eval/golden_fixtures.py::replay_fixture()` is the *single* code path used by both the freeze tool and both gates:
+Freeze at the **trajectory CSV ? CONTROL-path replay**. One shared helper
`backend/eval/golden_fixtures.py::replay_fixture()` is the *single* code path used by both the freeze tool and both gates. As **shipped**, that path is (per the module docstring):
+> ? **Correction (audit 2026-06-16):** the replay does **NOT** import `windowing.py` ? there is no `windowing.build_windows(preset)` / `WindowingPreset` / "default SERVED" call. The windowing params travel as a **raw 3-tuple `(vt, mg, mwd)** pinned in each fixture's `provenance.json` and passed straight to `build_candidates`; the harness deliberately avoids the named `windowing.SERVED` constant so a future SERVED retune can't silently churn fixtures.
(`segmentation_metrics`, `experiment.compare/Variant.spec_hash`, `regression.save_baseline`, `eval_stats` are **not** on the M1 replay path either ? the quality-floor gate M4 will add the baseline/stats pieces.)
-| **Tier-0** | `eval_baselines/fixtures/` (tracked) | none | opaque `fixture_id`, fps/frame_width (quantized), relative `start`/`end`, **5 shuttle scalars**, label, relative `gts`. **No person/audio/pose.** | **Unconditional** ? runs on every PR/fork; turns today's skipped litmus into a real public gate |
+| **Tier-0** | `eval_baselines/fixtures/` (tracked) | none | opaque surrogate slug, fps/frame_width (retained, NOT quantized), the **de-attributed per-frame `trajectory.csv`** `(Frame,Visibility,X,Y)` (shuttle-only, time-origin reset), the 5 shuttle feature columns inside `expected.json`, strict-slug `label`, relative `gts`. **No person/audio/pose.** ? NOT minimized to 5 scalars ? see §4c. | **Unconditional** ? runs on every PR/fork; turns today's skipped litmus into a real public gate |
+> ? **RISK-ACCEPTANCE RE-CONFIRMATION REQUIRED (audit 2026-06-16).** An earlier draft of this doc claimed Tier-0 commits "only ~5 aggregate shuttle scalars" and "**never**" the per-frame CSV". **That is not what ships.** The shipped freeze (`freeze_real_fixture`) **commits the FULL per-frame `(Frame,Visibility,X,Y)` `trajectory.csv`** ? de-attributed (time-origin reset + opaque random slug + metadata strip), but **not minimized to 5 scalars**. The per-frame stream is the de Montjoye fingerprint (4 points ? 95% re-ID against a published reference). The owner's 2026-06-16 risk acceptance (§7 P2) was recorded against the inaccurate "5 scalars" basis and **MUST be RE-CONFIRMED on this corrected basis** before any **real** Tier-0 fixture is committed (P3). Synthetic fixtures (current state) carry zero provenance risk regardless.
-- **Stream minimization** ? Tier-0 keeps shuttle block only; person/audio excluded; pose/gait/face have **no field in the schema by design**.
-- **Raw-trajectory exclusion** ? never commit the per-frame `(Frame,Visibility,X,Y)` CSV.
+- **Per-frame trajectory IS committed (de-attributed), NOT minimized to scalars** ? Tier-0 commits the de-attributed per-frame `trajectory.csv` (shuttle stream only). It is **not** reduced to ~5 aggregate scalars; the 5 shuttle *feature* columns appear (computed) inside `expected.json`, but the raw per-frame `(Frame,Visibility,X,Y)` stream is part of the committed fixture. (This is the corrected basis the ? note above gates on.)
+- **Stream minimization (streams, not resolution)** ? Tier-0 keeps the shuttle block only; person/audio excluded; pose/gait/face have **no field in the schema by design** (positive

```

```
`_scan_forbidden` assertion on the written bytes).
-| **TM5 re-ID adversary** | source unpublished + only ~5 invariant scalars/window + reset
timeline ? no back-match | becomes live if a Fork-A source is ever published |
+| **TM5 re-ID adversary** | source unpublished + reset timeline ? no back-match against a
public reference | ? becomes live if a Fork-A source is ever published ? **and the committed
per-frame trajectory IS a fingerprint** (de Montjoye 4-point re-ID), NOT just ~5 scalars
(corrected §4c). The protection is the unpublished source, not minimization |
+- The committed Tier-0 stream is the **de-attributed per-frame `trajectory.csv`**, NOT ~5
scalars ? a per-frame trajectory is a de Montjoye fingerprint; the only protection is the
**unpublished source** (§4c ?, §5 TM5).
+| _this PR (audit-hardening)_ | adversarial-audit fixes #1?#17 | 784 pass / 7 skip | **80.71%**
(+0.22) | strict-slug `label` + dropped `source_note` (#1); 6dp-rounded reset within
`_FLOAT_TOL` (#2); temp-dir-then-promote + negative-label refusal (#3); exact-membership source-
path scan (#4); key-set-equality in `compare_expected` (#5); `--reason` mandatory (#6);
independent schema guard (#7); `max_win` recorded-not-applied note (#8); numpy-execution wording
(#9); +failure-semantics & by-kind tests (#10/#11); `test_golden_real_fixtures.py` added to
`golden-crossos` CI (#12); doc reconciled to shipped reality incl. per-frame-CSV honesty + risk
re-confirm (#13?#17). Synthetic `--check` byte-identical (unchanged). |
```

279. assistant (2026-06-16T07:07:48.800Z)

The doc reconciliation is exactly right ? a prominent ? "RISK-ACCEPTANCE RE-CONFIRMATION REQUIRED" note, the TM5/TL;DR corrections, all honest. No real fixtures committed. Now my verification gate: review the must-fix code + run the full suite myself.

280. user (2026-06-16T07:07:52.791Z)

```
=== #1 (label validation + source_note drop) & #5 (compare_expected key-set) & #2 (reset
rounding) code ===
+import tempfile
+#: Strict slug charset for a real-fixture ``--label`` ? lowercase alnum, ``_``/``-`` (no
spaces,
+#: no punctuation, no uppercase). A free-text label ("Lin Dan vs Lee, India Open 2024") is an
+_LABEL_SLUG_RE = re.compile(r"^[a-z0-9][a-z0-9_\-]*$")
+#: a real source named ``label.csv`` (stem ``label`` ? ``labels.csv``) leak unnoticed (red-team
#4).
+
+    # skip on substring containment: a token like ``label`` or ``traj`` is a SUBSTRING
of
+
+    # ``labels.csv``/``trajectory.csv``, so substring-skip would silently miss a real
source
+
+    # named ``label.csv`` leaking its stem (red-team #4). Exact membership only.
+    """Shift every label by ``t0_sec`` (same offset as the trajectory ? metric-invariant).
+    ``t0_sec`` MUST already be quantized to ``_FLOAT_ROUND_DP`` decimals by the caller. Each
label
+
+    return [(_round(s) - t0_sec, _round(e) - t0_sec) for s, e in gts]
+    AND ``label`` is a strict slug (lowercase alnum + ``_``/``-`` ? a free-text identifier like
a
+
+    shuttle-only trajectory (time-origin reset by default), the reset labels, and a
+
+    ``source_note`` is **accepted but NOT persisted** ? operator free-text is an attribution
+
+    The fixture is written into a temp dir and promoted (`os.replace`) to the final slug dir
ONLY
+
+    # label is committed verbatim into provenance ? it MUST be a strict slug, never operator
free
+
+    if not (isinstance(label, str) and _LABEL_SLUG_RE.match(label)):
+
+        f"label must match the strict slug charset {_LABEL_SLUG_RE.pattern!r} (lowercase "
+
+        f"name) is an attribution leak; got {label!r} (§4c metadata strip, red-team #1)"
+
+    # Subtract the SAME offset from BOTH the trajectory frames and the label seconds:
metrics.score
+
+    # t0_sec is ROUNDED to 6dp BEFORE subtracting (and _reset_labels rounds each endpoint
first), so
+
+    # HONEST LIMIT (not a bug): the trajectory shifts by an INTEGER frame count while labels
shift
+
+        t0_sec = _round(t0_frame / float(fps))
+
+    # --- (d2) negative-label guard BEFORE any write (red-team #3a) -----
+
+    # A reset label that starts < 0 means the label timeline predates the earliest trajectory
```

```
+ # frame ? an inconsistent input that would crash the strict gt_loader on replay (negative
+ raise RefusalError(
+ f"REFUSED: label timeline predates the trajectory ? {len(neg)} reset label(s) have
a "
+ f"negative start after the t0={t0_sec:.6f}s time-origin reset (e.g. {neg[0]!r}).
The "
+ f"earliest label must not precede the earliest trajectory frame ($4c time-origin
reset).")
```

281. user (2026-06-16T07:09:28.611Z)

```
=== --check (synthetic byte-identical?) ===
[ok] occlusion_break: 2 candidates, 2 segments, F1=1.000000
[golden-fixtures] 3/3 fixtures match (synthetic-tier = plumbing correctness, not field quality ?
$6)
=== full suite + coverage ===
-----
TOTAL                                7697   1485   81%
Required test coverage of 70% reached. Total coverage: 80.71%
784 passed, 7 skipped, 7 warnings in 84.57s (0:01:24)
PYTEST=0
=== ruff + mypy ===
All checks passed!
Success: no issues found in 103 source files
```

282. assistant (2026-06-16T07:09:46.973Z)

Verified myself ? all green, **coverage 80.71% (? from 80.49%)**, 784 passed, `--check`
 synthetic byte-identical, ruff + mypy clean. The must-fix is correctly implemented (strict slug
 `label`, `source\_note` not persisted, exact-membership scan, negative-label guard + temp-dir-
 promote, 6dp-rounded reset). Committing and pushing:

283. user (2026-06-16T07:10:05.254Z)

```
=== origin/master ===
085c07b docs(eval): golden-fixtures discoverability ? CODE_MAP $0/$2.3 + GOLDEN_DATA_SHARING
cross-link (DOC) (#193)
=== staged ===
.github/workflows/ci.yml
backend/eval/golden_fixtures.py
docs/GOLDEN_REGRESSION_FIXTURES.md
eval_baselines/fixtures/README.md
tests/test_golden_fixtures.py
tests/test_golden_real_fixtures.py
=== only my commit? ===
7414446 fix(eval): golden-fixtures audit hardening ? close source_note/label leak + 16 confirmed
audit fixes
=== push ===
branch 'feat/golden-fixtures-audit-hardening' set up to track 'origin/feat/golden-fixtures-
audit-hardening'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      feat/golden-fixtures-audit-hardening -> feat/golden-fixtures-audit-
hardening
```

284. assistant (2026-06-16T07:10:11.575Z)

Pushed (only my commit; master unchanged). Opening the PR and watching CI ? now the cross-OS
 jobs also run the real-fixture tests (validating the reset fix on all 3 OSes):

285. user (2026-06-16T07:13:01.923Z)

```
https://github.com/avidullu/badminton-highlight-indexer/pull/194
=== watch ===
Mypy (lenient baseline) pass    30s    https://github.com/avidullu/badminton-highlight-
```

```

indexer/actions/runs/27600597530/job/81600374069
Ruff lint      pass    9s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27600597530/job/81600374003
Syntax sweep + import smoke (no GPU stack)      pass    19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27600597530/job/81600374036
Full test suite (offline, mocked)              pass    2m8s      https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27600597530/job/81600374162
Golden fixtures cross-OS (macos-latest) pass    14s      https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27600597530/job/81600374104
Golden fixtures cross-OS (ubuntu-latest)      pass    24s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27600597530/job/81600374107
Golden fixtures cross-OS (windows-latest)      pass    1m15s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27600597530/job/81600374031
Mypy (lenient baseline) pass    30s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27600597530/job/81600374069
Ruff lint      pass    9s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27600597530/job/81600374003
Syntax sweep + import smoke (no GPU stack)      pass    19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27600597530/job/81600374036
CHECKS_EXIT=0

```

286. assistant (2026-06-16T07:13:11.920Z)

****All green, including the cross-OS jobs now running the real-fixture tests**** (the reset fix validated on Windows/macOS/Linux). Merging #194:

287. user (2026-06-16T07:13:18.668Z)

failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'

```

---master---
7d84319 fix(eval): golden-fixtures audit hardening ? close source_note/label leak + 16 confirmed
audit fixes (#194)

```

288. assistant (2026-06-16T07:13:31.644Z)

#194 merged ? master `7d84319`. That's the audit-hardening landed. Let me record the checkpoint and give the wrap-up.

289. user (2026-06-16T07:13:32.802Z)

```

10      # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12      **Checkpoint:** 2026-06-16 (GOLDEN FIXTURES: **DOC merged ? PR #193**, master `085c07b`;
final due-diligence AUDIT running). **5 PRs merged this session** (#186/#189/#191/#192/#193);
tracker now M1? M2? M3? MX? P1? P2? DOC? ? coverage 80.49%, all CI green incl. the 3-OS `golden-
crossos` job. DOC = CODE_MAP $0 fixtures row + $2.3 `golden_fixtures` code-nav entry +
GOLDEN_DATA_SHARING cross-link (EVALUATION/QUALITY_ITERATIONS cross-links DEFERRED ? hot in the
quality track). ?? NOW: an adversarial DUE-DILIGENCE audit workflow (`wf_ef85c152-821`; 5 lenses
determinism/privacy/correctness+coverage/robustness/honesty ? adversarial verify ? triage) is
auditing the SHIPPED feature for gaps the per-PR verification missed; will fix any must-fix as
follow-up green PRs and record a clean audit otherwise. REMAINING after audit: **P3 real
fixtures = BLOCKED on the owner's per-video clearance list** (owned + minors-free ? just the
trajectory CSV + golden label CSV + fps/fw); **M4** quality-floor (synthetic-trivial ?
meaningful only with real fixtures); **P0** name-scrub (HELD ? windowing churn); 01/02 optional.
The golden-fixtures project CORE is COMPLETE; the rest is owner-blocked or held. Isolated
worktrees throughout ? unhindering w/ the concurrent quality track. Owner: "merge any PRs that
pass QA + due diligence".

```

290. user (2026-06-16T07:13:57.698Z)

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your

context ? no need to Read it back)

291. assistant (2026-06-16T07:14:19.974Z)

? Golden-fixtures project: non-owner-blocked work complete + audited

****#194** (audit hardening) merged **? master `7d84319`**. That's ****6 PRs** this session ******, every one verified by me then green in CI:

PR	What	Coverage
#186	Plan + reusable template + `CLAUDE.md` convention	?
#189	M1: framework + synthetic Tier-0 + characterization	80.41%
#191	Cross-OS CI guard (3-OS) + tracker/legal reconcile	80.41%
#192	P1 de-attribution tool + M2 refusal gate	80.49%
#193	DOC discoverability (CODE_MAP, GOLDEN_DATA_SHARING)	80.49%
#194	**Audit hardening** ? 17 confirmed fixes	**80.71%**

The ultracode due-diligence audit earned its keep: it caught a ****real reset bug**** (off by exactly past 1e-6), a ****privacy hole**** (free-text leak via `--source-note`/`--label`), and a ****doc honesty gap**** ? all now fixed and green, coverage ****up****. The hardening subagent was commendably honest (it flagged that bit-identical reset for sub-second offsets is mathematically irreducible rather than fake it).

****I've reached the honest edge of "merge anything that passes" ? there are no more *non-blocked* PRs to produce.**** Everything left genuinely needs you or the other session:

? P3 (real fixtures) ? needs ****two decisions from you****

1. ****Clearance list:**** which F:/golden videos are ****owned + minors-free**** and OK to derive-and-commit. (Per video I need only its trajectory CSV + golden label CSV + fps/fw.)
2. ****Risk-acceptance re-confirm:**** the audit found the doc had overstated the minimization ? the committed fixture includes the de-attributed ****per-frame**** `trajectory.csv` (a fingerprint), not just 5 scalars. Pick ****a)**** accept that (re-confirm; fine if the repo/source stay private) or ****b)**** I switch the freeze to commit ****only the 5 aggregate scalars**** (removes the fingerprint ? better if these ever go public).

Deferred (with reasons)

- ****M4**** quality-floor ? meaningful only once real fixtures land (trivial on synthetic).
- ****P0**** name-scrub ? still held; it rewrites `eval\_baselines` keys the active windowing session depends on.
- ****O1/O2**** (protobuf, Tier-1 key-gated) ? optional.

The refusal-gated tool is built, audited, and cross-OS-tested ? the moment you drop the clearance list + the (a)/(b) call, I'll mint the first real Tier-0 fixtures, green and de-attributed. I'll also pick P0 back up as soon as the windowing track settles.

292. user (2026-06-17T14:12:10.707Z)

hello. Can you sync to remote head, ensure you are clean and give me a tldr of next steps you recommend or gated on me?

293. assistant (2026-06-17T14:13:39.762Z)

Hello! Let me sync, check cleanliness, and clean up my session's worktrees ? then give you the TL;DR. First, gathering state (read-only):

294. user (2026-06-17T14:13:45.799Z)

```
=== origin/master HEAD ===
f5f5e52 feat(eval): R5 ? blob-fallback splits the continuously-active blob (default-OFF) (#203)
=== main checkout: branch + status ===
docs/golden-regression-fixtures-plan
?? artifacts/
```

```

?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
=== any UNMERGED commits on this branch vs origin/master? (empty = fully merged) ===
95cd154 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template
=== worktrees ===
C:/Users/avidu/Projects/badminton-highlight-indexer
95cd154 [docs/golden-regression-fixtures-plan]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-a771ffd85e18f2ed
7e0881a [perf/wasb-fast-frames]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy
5cb7bbf [bhi-execpolicy]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/boxhill17
34b9caa [run/boxhill17-highlights]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-golden
c33cf1d [docs/golden-videos-tracker]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality
877ab1f [docs/rally-quality-research]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-today
ce7eb47 [docs/quality-iterations-realfootage]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit
7414446 [feat/golden-fixtures-audit-hardening]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1
955e902 [feat/golden-fixtures-p1]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hardchop
614325d [feat/w1-hard-cap-fallback]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/local-noai
3ea7ad4 [feat/local-no-ai-segmenter]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/r2-impl
a450c53 [feat/r2-tolerance-metric]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/r4-endrule
e248d2d [feat/r4-inactivity-end]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier0-eval
9e03e5a [feat/tier0-rally-seg-eval]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing
a3b209a [master]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-netcross
effd920 [feat/w2-netcross-gate]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-safeguard
5ba6955 [feat/w2-never-empty-safeguard]
=== my feat/golden-fixtures-* branches still present? ===
* docs/golden-regression-fixtures-plan
+ feat/golden-fixtures-audit-hardening
  feat/golden-fixtures-crossos
  feat/golden-fixtures-doc
+ feat/golden-fixtures-p1

```

295. assistant (2026-06-17T14:14:42.185Z)

Got the picture. **origin/master is now `f5f5e52` (#203)** ? the concurrent quality session has merged a lot since my last work (#195?#203: R2/R4/R5 + W1/W2 iterations). All **my** golden-fixtures PRs (#186/#189/#191/#192/#193/#194) are in that history. The main checkout is on a stale, fully-merged leftover branch, and I have two leftover worktrees (gf-p1, gf-audit) + four merged branches to clean up. Syncing + cleaning:

296. user (2026-06-17T14:14:49.531Z)

```

=== sync main checkout to origin/master (current branch is fully-merged #186 leftover) ===
HEAD is now at f5f5e52 feat(eval): R5 ? blob-fallback splits the continuously-active blob
(default-OFF) (#203)
=== remove MY merged worktrees (leave the concurrent session's untouched) ===
=== delete MY merged branches ===
Deleted branch feat/golden-fixtures-p1 (was 955e902).
Deleted branch feat/golden-fixtures-audit-hardening (was 7414446).
Deleted branch feat/golden-fixtures-crossos (was 75403a5).
Deleted branch feat/golden-fixtures-doc (was 854db56).
=== VERIFY: clean + synced ===

```



```
f5f5e52 feat(eval): R5 ? blob-fallback splits the continuously-active blob (default-OFF) (#203)
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
--- my golden-fixtures PRs all merged? (gh) ---
197 docs(eval): golden-labeled video registry (the growing golden-finalized set)
docs/golden-videos-tracker MERGED 2026-06-17T08:29:10Z
193 docs(eval): golden-fixtures discoverability (CODE_MAP + GOLDEN_DATA_SHARING) ? DOC
feat/golden-fixtures-doc MERGED 2026-06-16T06:29:38Z
191 ci(eval): cross-OS determinism guard for golden fixtures + reconcile tracker
feat/golden-fixtures-crossos MERGED 2026-06-16T05:57:59Z
140 feat(eval): wire run-telemetry black box into fusion_golden extractor feat/telemetry-
wire-fusion-golden MERGED 2026-06-14T08:26:44Z
176 docs: next-session handoff + shared golden-data plan (#175) docs/next-session-
handoff MERGED 2026-06-15T04:29:44Z
121 feat(eval): fusion P3 measurement ? golden person-feature extractor + LOVO compare
feat/fusion-p3 MERGED 2026-06-13T20:21:54Z
```

297. assistant (2026-06-17T14:15:25.824Z)

Synced and clean. ****Main checkout is now at `origin/master` `f5f5e52` (#203)****; my two leftover worktrees and four merged branches are removed; the only untracked items are `artifacts/` and `docs/KHELSUTRA\_PRODUCT\_OVERVIEW.md` (not mine ? left alone). All six of my golden-fixtures PRs are confirmed merged.

Let me do one quick check to make my recommendation accurate ? whether the `eval\_baselines` files (P0's target) are still being churned by the quality session, and whether the new golden-video registry (#197) feeds P3:

298. user (2026-06-17T14:15:31.471Z)

```
=== recent commits touching P0's eval_baselines name files (still hot?) ===
595a5a1 ci: add ruff lint lane + fix all violations; remove never-compiled flyer stub (#92)
d19ef36 eval: record the Gemini-wrapper baselines (the commoditization/moat reference) (#84)
d51efe3 feat(training): Gen-0 TAL harness ? one-command train -> honest LOVO -> ship-gate ->
artifact (M0.4 scaffold) (#77)
90fb8b3 Human-GT partial-label eval harness + tuning + iteration doc (#60)
=== real player/venue names STILL committed? (P0 still relevant?) ===
origin/master:backend/eval/eval_partial_labels.py
origin/master:eval_baselines/gemini_wrapper_2026-06-10.json
origin/master:eval_baselines/heuristic_lovo_n6.json
origin/master:eval_baselines/nightly_baseline.json
origin/master:tests/test_cleanup_caches.py
=== golden-video registry (#197) ? feeds P3 clearance? ===
docs/GOLDEN_SET_PHASE1_VIDEOS.md
docs/GOLDEN_VIDEOS.md
```

299. assistant (2026-06-17T14:16:18.211Z)

Done ? ****synced to `f5f5e52` (#203) and clean****; my worktrees/branches cleaned up. Here's the TL;DR.

Where the Golden Regression Fixtures project stands

****Core complete + audited**** ? 6 PRs merged (#186/#189/#191/#192/#193/#194). A clean clone runs the rally-seg regression with no video/GPU/DB, cross-OS enforced on every PR (Win/macOS/Linux), de-attribution tool + refusal gate shipped, all hardened by an adversarial audit, coverage ~80.7%. Nothing of mine is in flight; the parallel quality track (R2?R5, W1/W2) is the **other** session and isn't mine to touch.

Next steps

****? Recommended, and now unblocked (no owner input needed ? I can start on your go):****
- ****P0 ? scrub the committed real names + add a leak-scan CI gate.**** I'd held this because it edits `eval\_baselines`, but those files are ****no longer hot**** (the quality session's recent

churn is all in windowing/gate code, not these). And it's still real:
`adarsh/kushagra/mahadevpura/gbaaddy` are committed today in `eval\_partial\_labels.py`,
`gemini\_wrapper\_2026-06-10.json`, `heuristic\_lovo\_n6.json`, **`nightly\_baseline.json` (new)**,
and `test\_cleanup\_caches.py` (+ a personal `C:/Users/avidu/...` path). ? opaque surrogate keys +
a CI guard that fails on names/MD5.

****? Gated on you (the high-value step ? P3 real fixtures):****

1. ****Clearance list**** ? which golden videos are ****owned + minors-free**** (the new
`docs/GOLDEN\_VIDEOS.md` registry may already list candidates ? point me at it and I'll cross-
check).
2. ****Minimization re-confirm**** (the audit's honesty flag): commit the de-attributed ****per-
frame**** trajectory CSV ? ****a)**** accept it (fine if repo/source stay private), or ****b)**** I
switch the freeze to commit ****only the ~5 aggregate scalars**** (removes the fingerprint; better
if ever public).

****? Deferred (with reason):**** M4 quality-floor (meaningful only once real fixtures land); 01
protobuf / 02 Tier-1 key-gated (optional).

****One sub-decision inside P0 is yours:**** whether to rewrite git history (`filter-repo`) to purge
the names from *past* commits, or just scrub going forward + guard. I'd default to ****scrub-
forward + leak-scan now****, and leave the history rewrite as a separate owner-approved step (it's
destructive and rewrites shared history).

Want me to ****start P0**** now, or would you rather hand me the ****P3 clearance list + a)/(b)**** so
I go straight for the real fixtures?